Algorithme OA*(Liste,Sol)

Input : le graphe
Output : **Si** $f(u_0) = \infty$ pas solution **sinon** solution ds $G(u_0)$ & coût $f(u_0)$.

```
P ← Premier(Liste), Q ← ∅, f(u0) ← h(u0), u ← u0;
while (u ≠ null) et (f(u0) ≠ ∞) do
    P ← P \ {u}, Q ← {u};
    for i ∈ I(u) do
        v ← Si(u) ;
        while v ≠ ∅ do
            if non(v ∈ P ∪ Q) then P ← P ∪ {v};
            f(v) ← h(v) ;
            v ← Si(u);
        fi(u) ← k(u,i) + Σ_{v∈Si(u)} f(v);
    if I(u) = ∅ then f(u) ← ∞;
    else f(u) ← min{fi(u) | i ∈ I(u)};
    Imin(u) ← indice_du_minimum ;
    A ← S^{-1}(u);
    for w ∈ S^{-1}(u) do
        I0(w) ← { i ∈ I(w) | u ∈ Si(w)};
    while A ≠ ∅ do
        Choisir v ∈ A tel que A ∩ S(v) = ∅;
        A ← A \ {v};
        for i ∈ I0(v) do
            fi(v) ← k(v,i) + Σ_{w∈Si(u)} f(w);
        if f_{Imin}(v) ≠ min{fi(v) | i ∈ I(v)} then
            Imin(v) ← indice_du_minimum ;
        if f(v) ≠ f_{Imin}(v) then
            f(v) ← f_{Imin}(v), A ← A ∪ S^{-1}(v) ;
            for w ∈ S^{-1}(v) do
                I0(w) ← I0(w) ∪ {i ∈ I(w) | v ∈ Si(w)}
        I0(v) ← ∅ ;
    G(u0) ← Définir_Solution(u0),
    u ← Choisir_Etat(G(u0));
```

Algorithme OA*(Liste,Sol)

Input : le graphe
Output : **Si** $f(u_0) = \infty$ pas solution **sinon** solution ds $G(u_0)$ & coût $f(u_0)$.

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            f(v) ← h(v) ;
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    while A ≠ ∅ do
        Choisir v ∈ A tel que A ∩ S(v) = ∅;
        A ← A \ {v};
        for i ∈ I0(v) do
            fi(v) ← k(v,i) + Σ_{w∈Si(u)} f(w);
        if f_{Imin}(v) ≠ min{fi(v) | i ∈ I(v)} then
            Imin(v) ← indice_du_minimum ;
        if f(v) ≠ f_{Imin}(v) then
            f(v) ← f_{Imin}(v), A ← A ∪ S^{-1}(v) ;
            for w ∈ S^{-1}(v) do
                I0(w) ← I0(w) ∪ {i ∈ I(w) | v ∈ Si(w)}
        I0(v) ← ∅ ;
    G(u0) ← Définir_Solution(u0),
    u ← Choisir_Etat(G(u0));
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Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;

while $u \neq \perp$ ***et non*** *fini* **do**

Algorithme OA*(Liste,Sol)

Input : le graphe
Output : **Si** $f(u_0) = \infty$ pas solution **sinon** solution ds $G(u_0)$ & coût $f(u_0)$.

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P ← Premier(Liste), Q ← ∅, f(u0) ← h(u0), u ← u0;
while (u ≠ null) et (f(u0) ≠ ∞) do
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            f(v) ← h(v) ;
            v ← Si(u);
        fi(u) ← k(u,i) + Σ_{v∈Si(u)} f(v);
    if I(u) = ∅ then f(u) ← ∞;
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    A ← S^{-1}(u);
    for w ∈ S^{-1}(u) do
        I0(w) ← { i ∈ I(w) | u ∈ Si(w)};
    while A ≠ ∅ do
        Choisir v ∈ A tel que A ∩ S(v) = ∅;
        A ← A \ {v};
        for i ∈ I0(v) do
            fi(v) ← k(v,i) + Σ_{w∈Si(u)} f(w);
        if f_{Imin}(v) ≠ min{fi(v) | i ∈ I(v)} then
            Imin(v) ← indice_du_minimum ;
        if f(v) ≠ f_{Imin}(v) then
            f(v) ← f_{Imin}(v), A ← A ∪ S^{-1}(v) ;
            for w ∈ S^{-1}(v) do
                I0(w) ← I0(w) ∪ {i ∈ I(w) | v ∈ Si(w)}
        I0(v) ← ∅ ;
    G(u0) ← Définir_Solution(u0),
    u ← Choisir_Etat(G(u0));
```

Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;

while $u \neq \perp$ ***et non*** *fini* **do**

Algorithme OA*(Liste,Sol)

Input : le graphe
Output : **Si** $f(u_0) = \infty$ pas solution **sinon** solution ds $G(u_0)$ & coût $f(u_0)$.

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P ← Premier(Liste), Q ← ∅, f(u0) ← h(u0), u ← u0;
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        while v ≠ ∅ do
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            v ← Si(u);
        fi(u) ← k(u,i) + Σ_{v∈Si(u)} f(v);
    if I(u) = ∅ then f(u) ← ∞;
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    Imin(u) ← indice_du_minimum ;
    A ← S^{-1}(u);
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    while A ≠ ∅ do
        Choisir v ∈ A tel que A ∩ S(v) = ∅;
        A ← A \ {v};
        for i ∈ I0(v) do
            fi(v) ← k(v,i) + Σ_{w∈Si(u)} f(w);
        if f_{Imin}(v) ≠ min{fi(v) | i ∈ I(v)} then
            Imin(v) ← indice_du_minimum ;
        if f(v) ≠ f_{Imin}(v) then
            f(v) ← f_{Imin}(v), A ← A ∪ S^{-1}(v) ;
            for w ∈ S^{-1}(v) do
                I0(w) ← I0(w) ∪ {i ∈ I(w) | v ∈ Si(w)}
        I0(v) ← ∅ ;
    G(u0) ← Définir_Solution(u0),
    u ← Choisir_Etat(G(u0));
```

Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u);
        garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
```

Algorithme OA*(Liste,Sol)

Input : le graphe
Output : **Si** $f(u_0) = \infty$ pas solution **sinon** solution ds $G(u_0)$ & coût $f(u_0)$.

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        I0(v) ← ∅ ;
    G(u0) ← Définir_Solution(u0),
    u ← Choisir_Etat(G(u0));
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Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u);
        garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
```

Algorithme OA*(Liste,Sol)

Input : le graphe
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Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u);
        garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A);
        A ← A \ {u};
        miseAJour(u);
```

Algorithme OA*(Liste,Sol)

Input : le graphe
Output : **Si** $f(u_0) = \infty$ pas solution **sinon** solution ds $G(u_0)$ & coût $f(u_0)$.

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            for w ∈ S^{-1}(v) do
                I0(w) ← I0(w) ∪ {i ∈ I(w) | v ∈ Si(w)}
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    G(u0) ← Définir_Solution(u0),
    u ← Choisir_Etat(G(u0));
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Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
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        garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A);
        A ← A \ {u};
        miseAJour(u);
```

Algorithme OA*(Liste,Sol)

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        v ← Si(u) ;
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            f(v) ← f_{Imin}(v), A ← A ∪ S^{-1}(v) ;
            for w ∈ S^{-1}(v) do
                I0(w) ← I0(w) ∪ {i ∈ I(w) | v ∈ Si(w)}
        I0(v) ← ∅ ;
    G(u0) ← Définir_Solution(u0),
    u ← Choisir_Etat(G(u0));
```

Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
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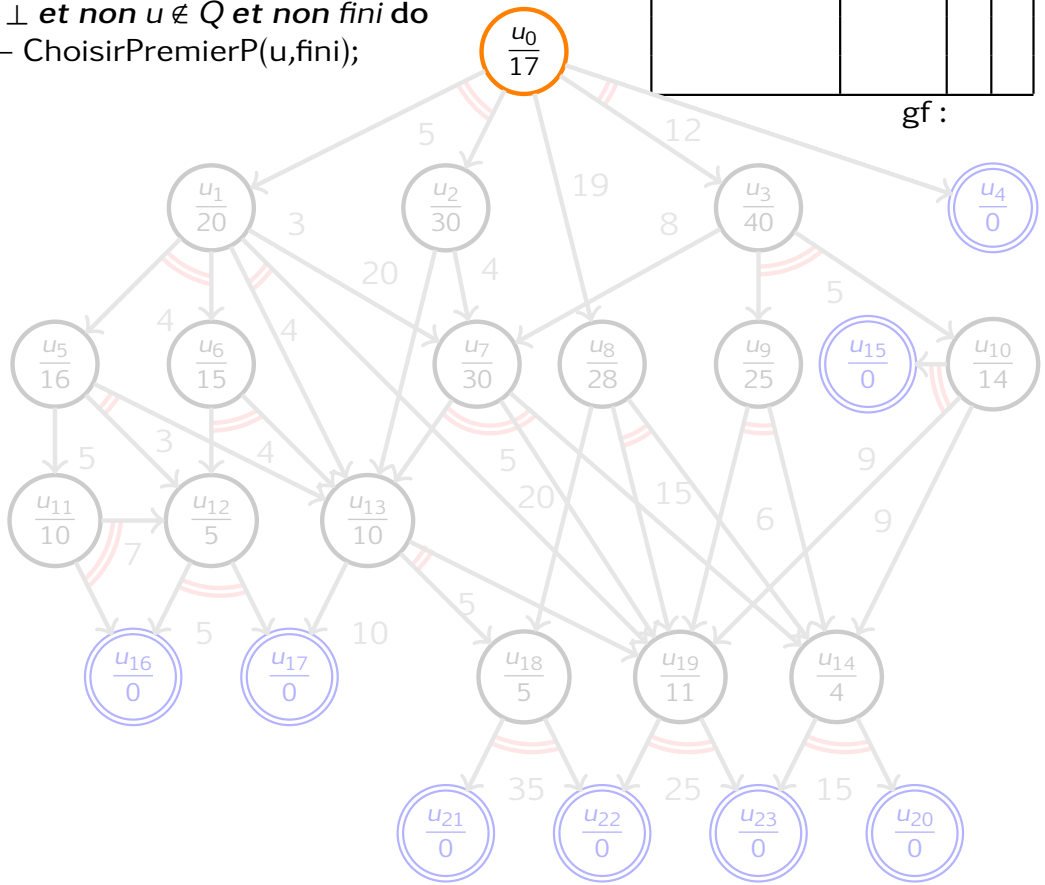
    while gf ≠ ∅ do
        calculerCout(gf,u);
        garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A);
        A ← A \ {u};
        miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ D et non fini do
        u ← ChoisirPremierP(u,fini);
```

Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P

gf :



Algorithme OA*(Liste,Sol)

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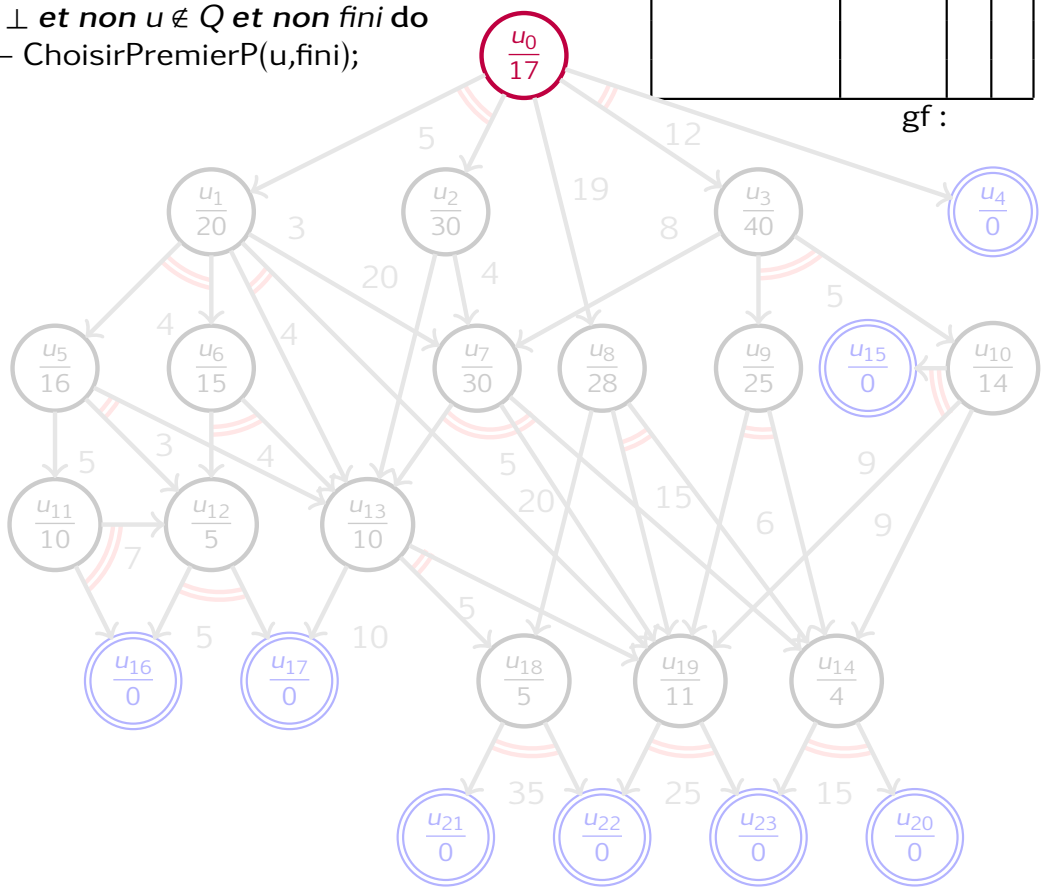
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,17)	(-,-,-)	{}	

gf :



Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
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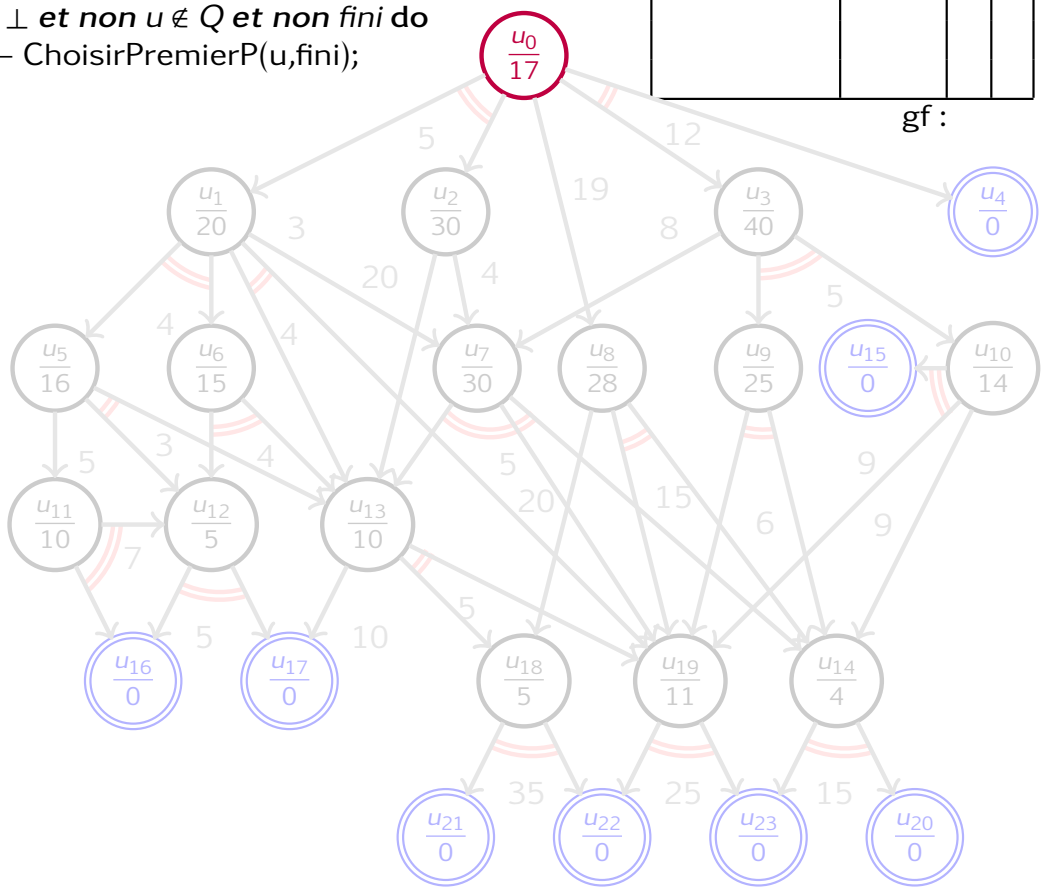
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,17)	(-,-,-)	{}	

gf :



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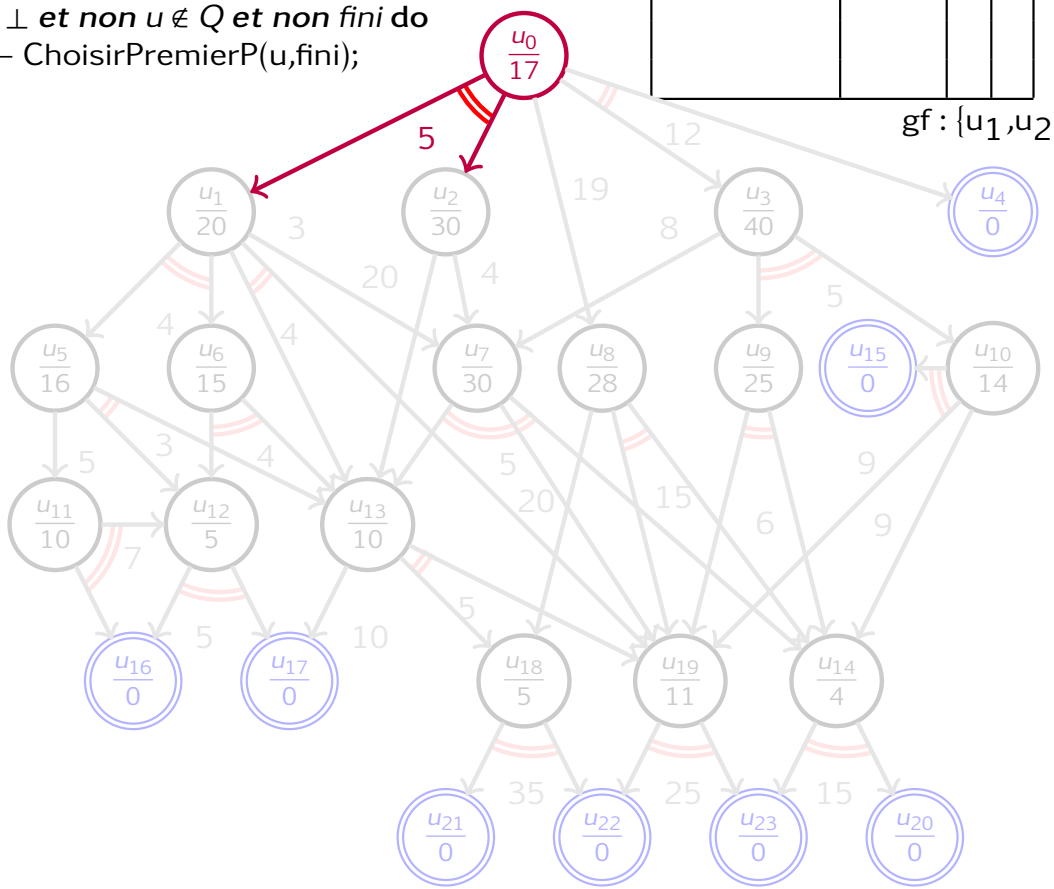
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,17)	(-,-,-)	{}	

gf : $\{u_1, u_2\}$

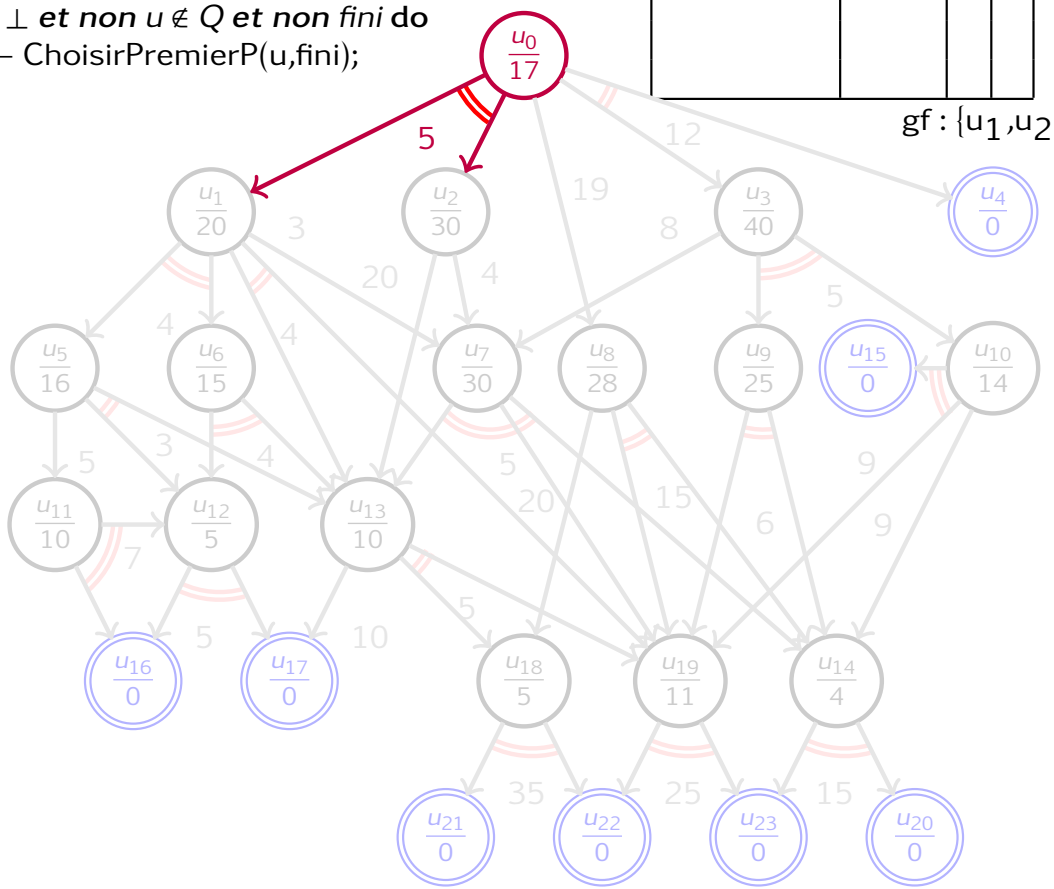


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while $u \neq \perp$ ***et non*** *fini* **do**
  gf ← groupeFils(u) ;
  while $gf \neq \emptyset$ **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while $A \neq \emptyset$ **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while $u \neq \perp$ ***et non*** $u \notin Q$ ***et non*** *fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,17)	(-,-,-)	{}	

gf : $\{u_1, u_2\}$



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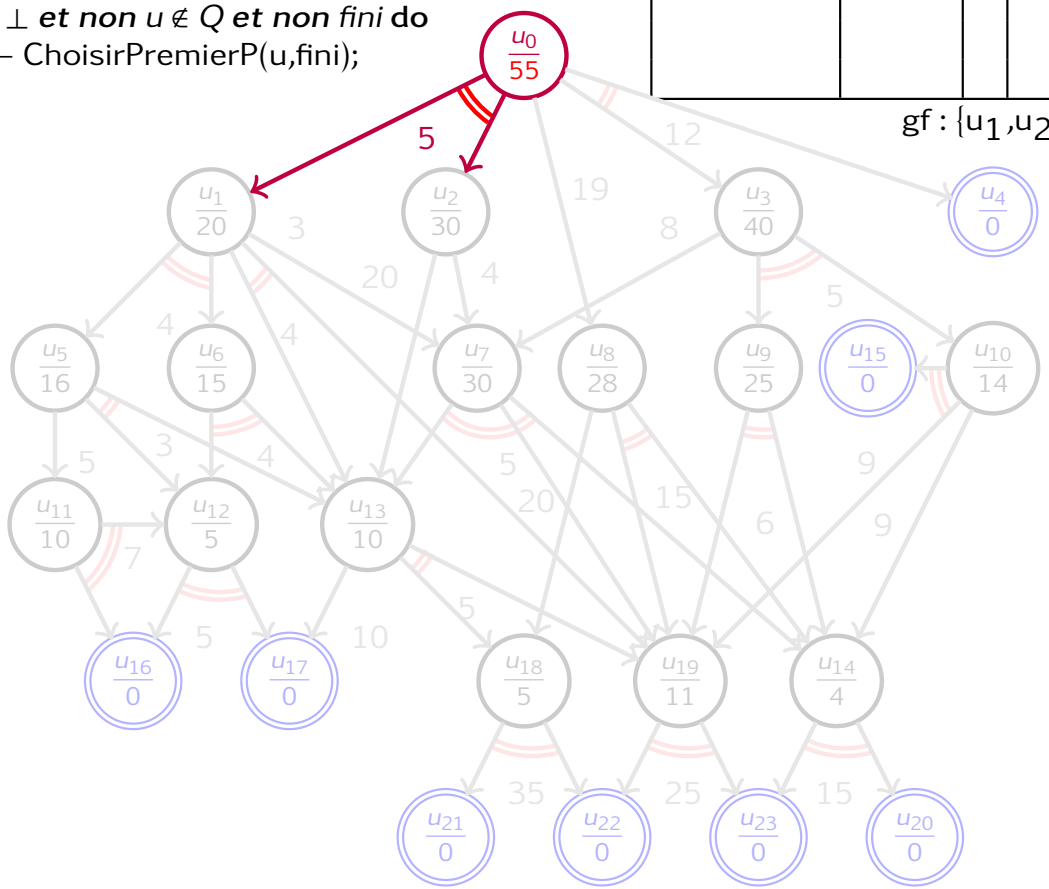
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,55)	(55,-,-)	{}	u_1(5,20), u_2(5,30)

gf : $\{u_1,u_2\}$ = 5+20+30 = 55

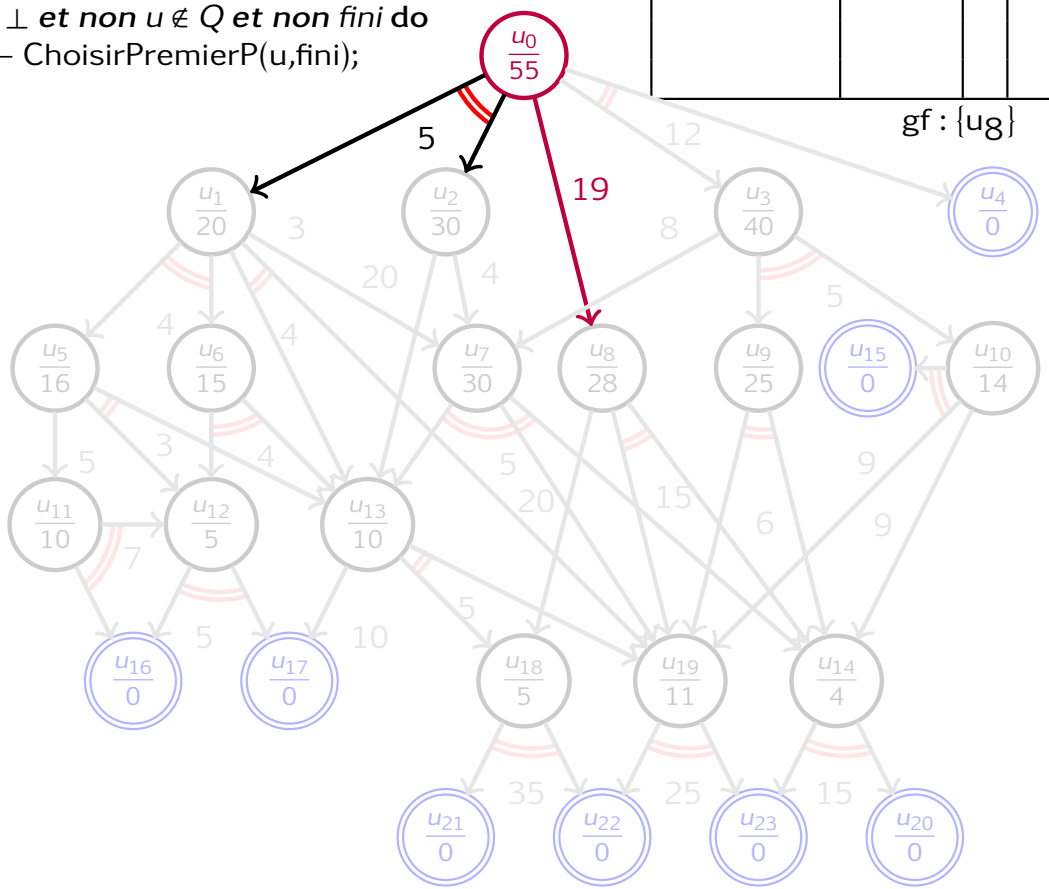


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,55)	(55,-,-)	{}	u_1(5,20), u_2(5,30)

gf : $\{u_8\}$

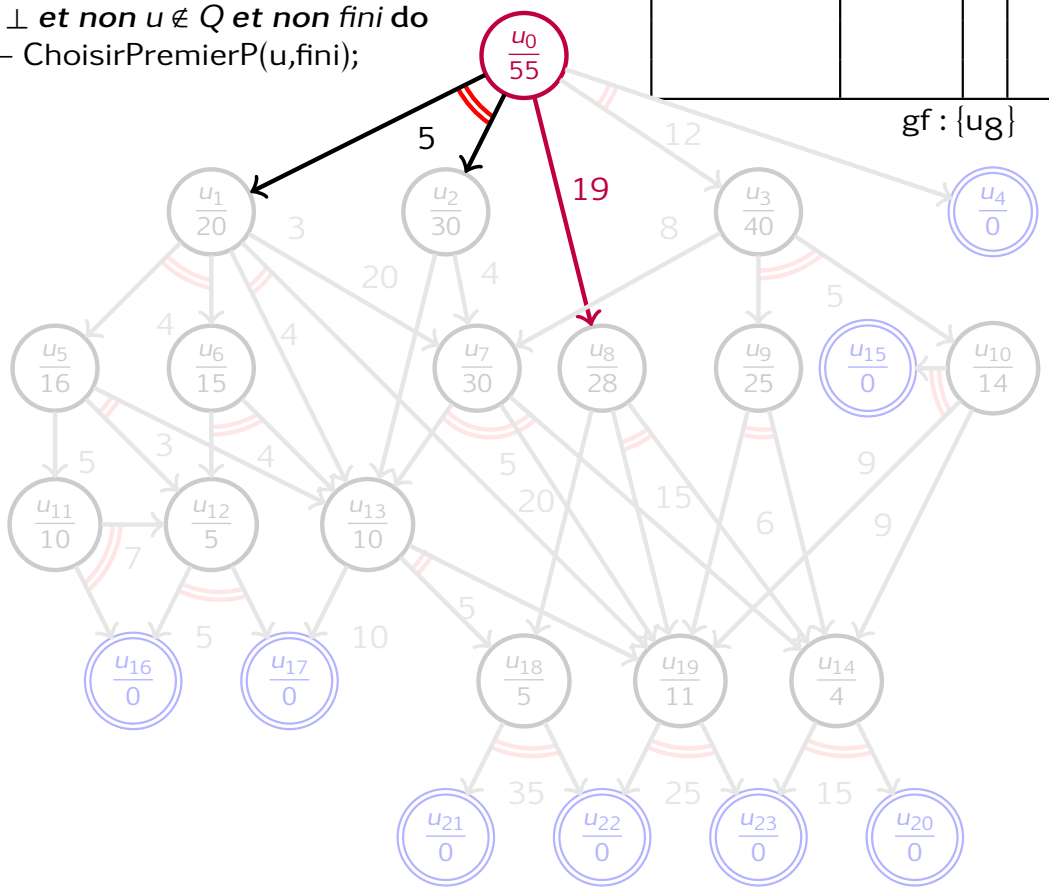


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,55)	(55,-,-)	{}	u_1(5,20), u_2(5,30)

gf : $\{u_8\}$

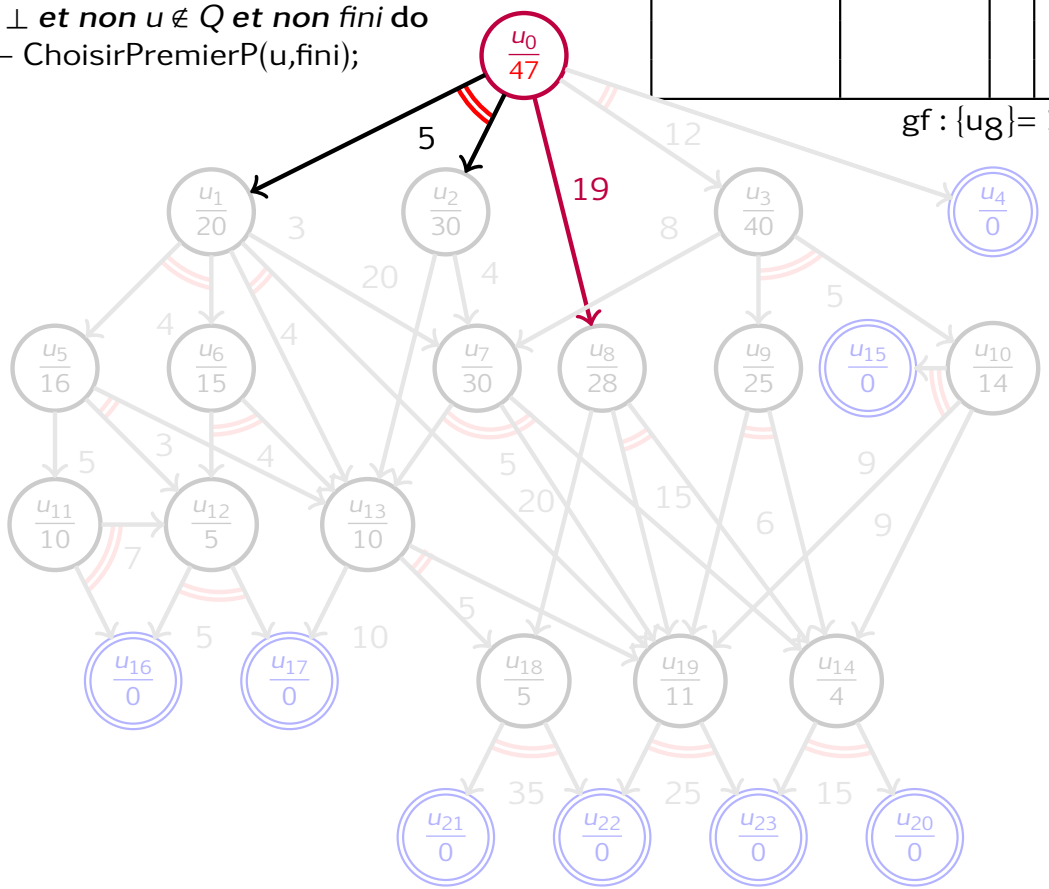


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while $u \neq \perp$ ***et non*** *fini* **do**
  gf ← groupeFils(u) ;
  while $gf \neq \emptyset$ **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while $A \neq \emptyset$ **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while $u \neq \perp$ ***et non*** $u \notin Q$ ***et non*** *fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,47)	(55,47,-)	{}	u_8(19,28)

gf : $\{u_8\}$= 19+28 = 47

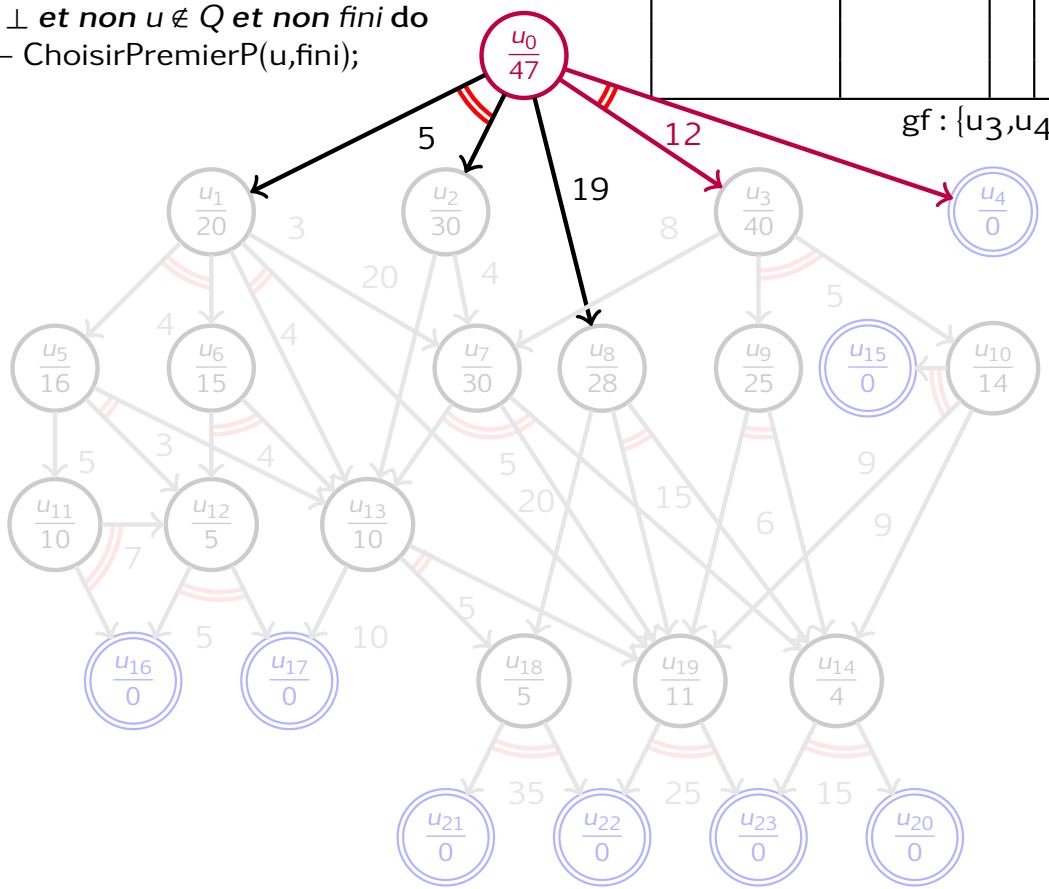


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,47)	(55,47,-)	{}	u_8(19,28)

gf : $\{u_3,u_4\}$

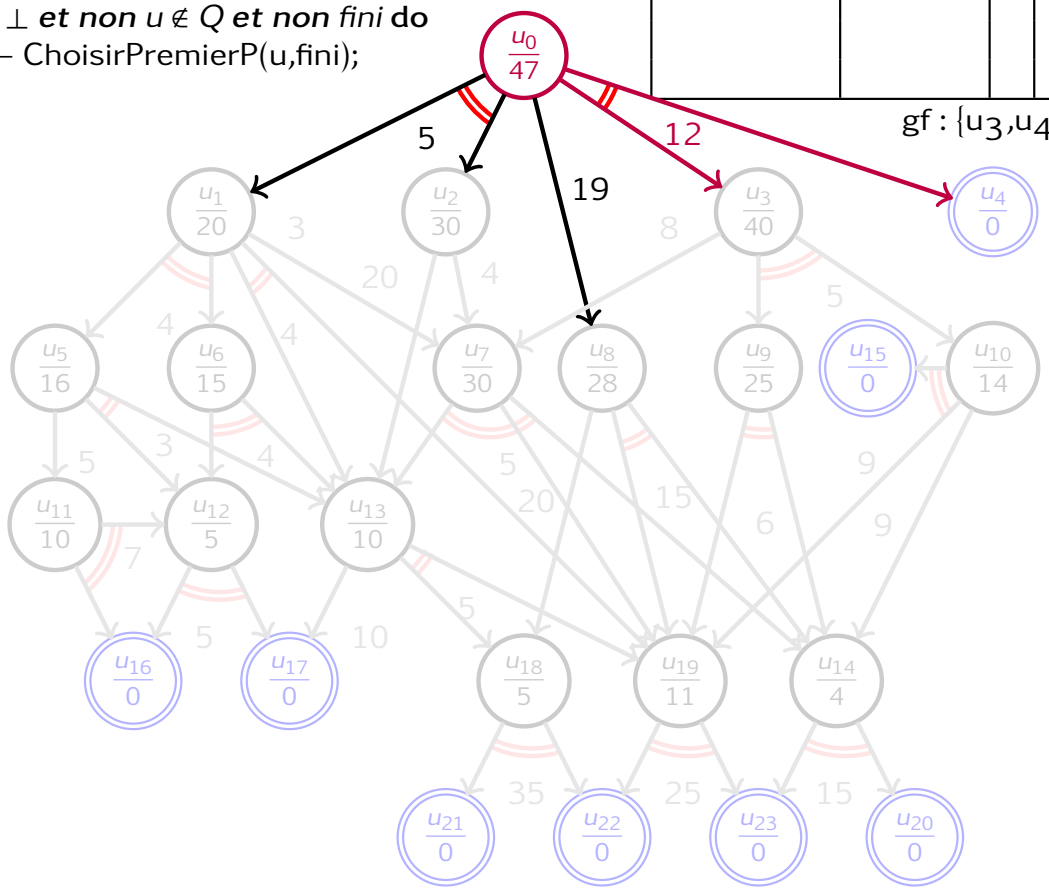


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,47)	(55,47,-)	{}	u_8(19,28)

gf : $\{u_3,u_4\}$

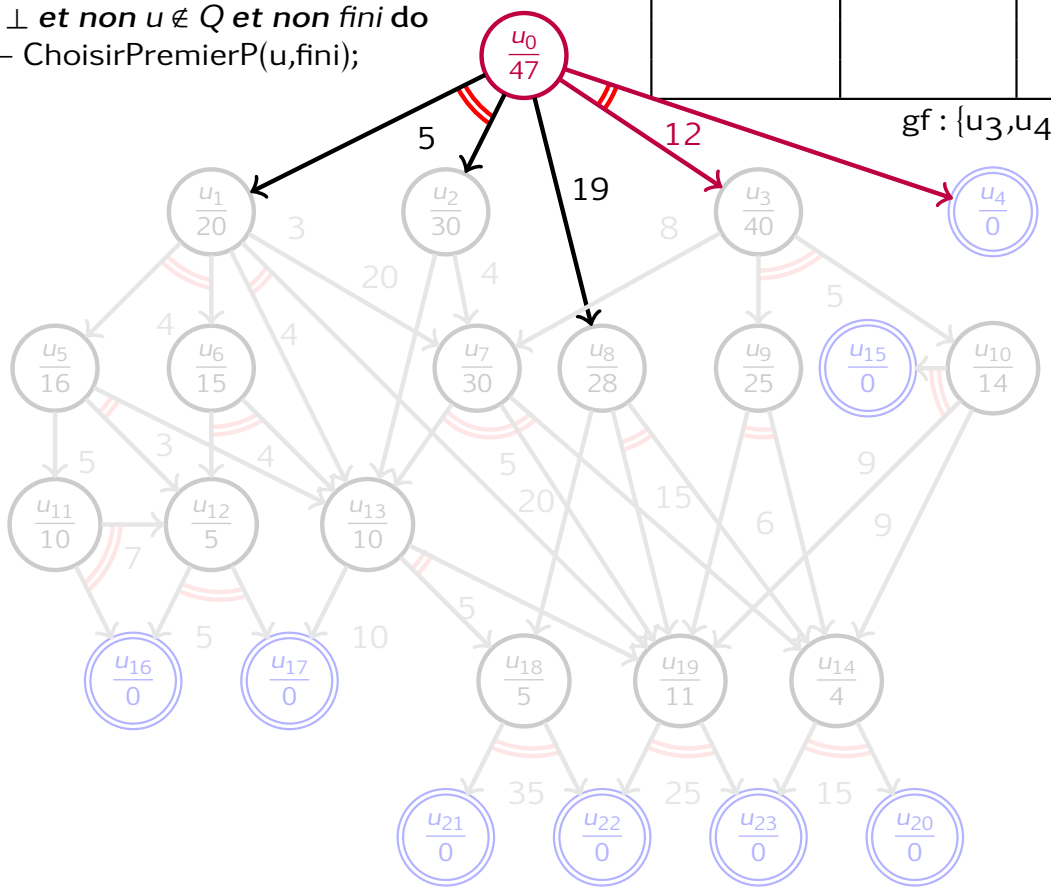


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,47)	(55,47,52)	{}	u_8(19,28)

gf : $\{u_3,u_4\}$= 12+40+0 = 52

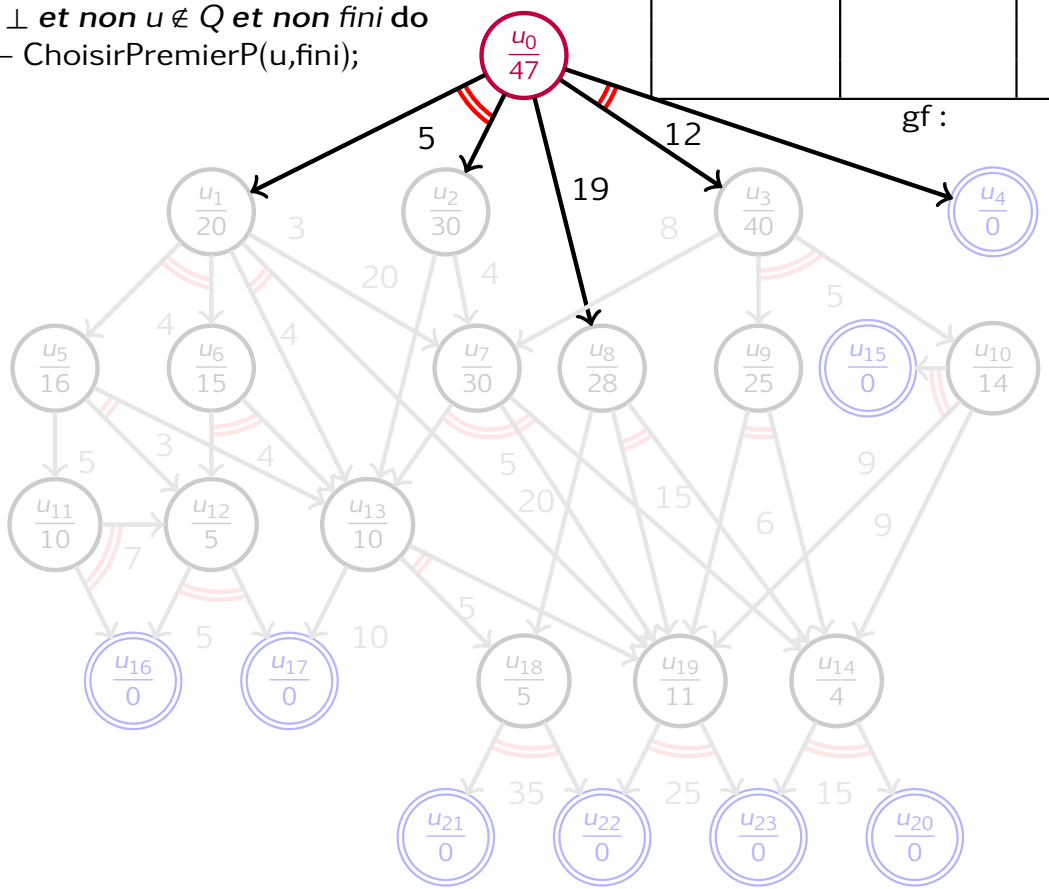


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,47)	(55,47,52)	{}	u_8(19,28)

gf :

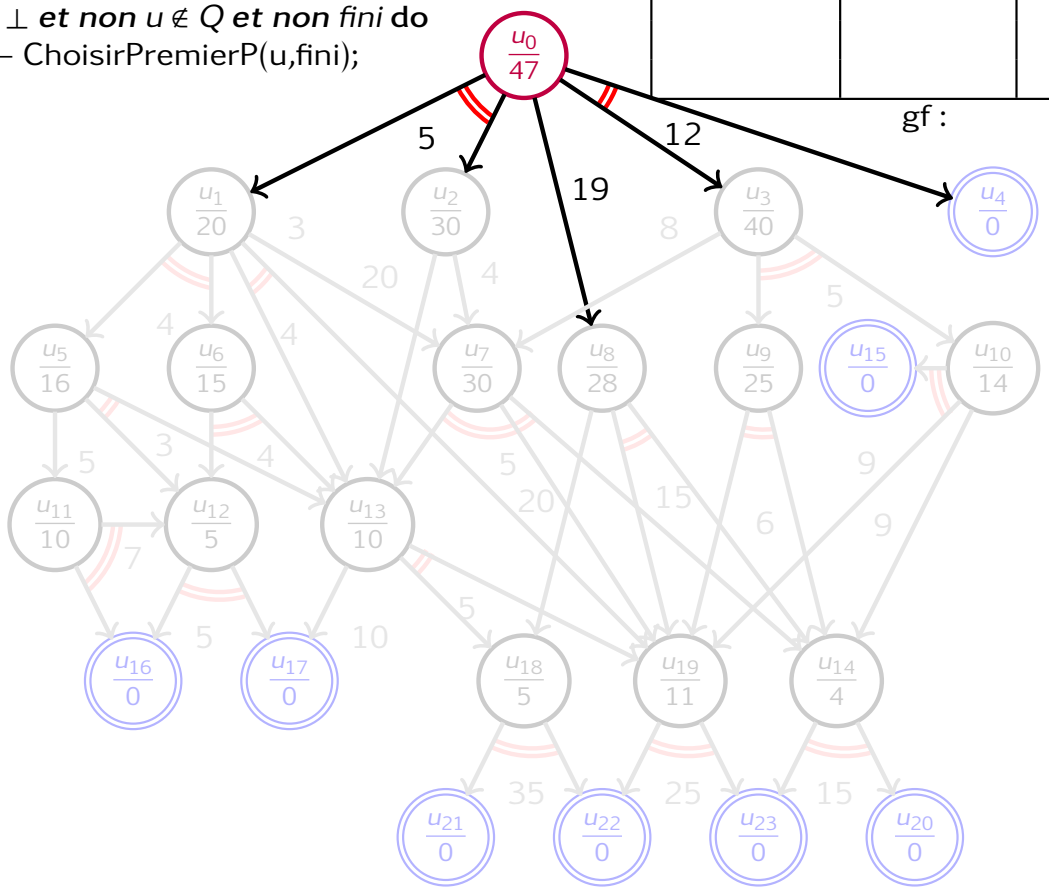


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,47)	(55,47,52)	{}	u_8(19,28)

gf :

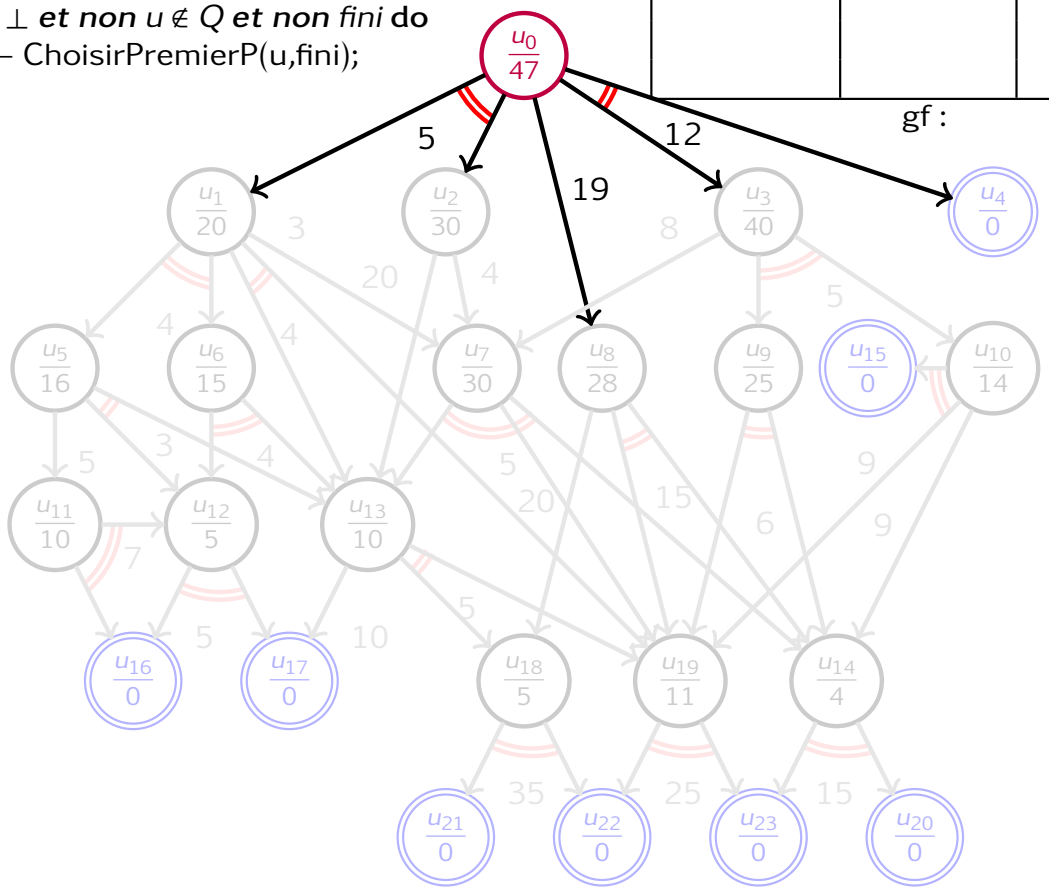


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,47)	(55,47,52)	{}	u_8(19,28)

gf :

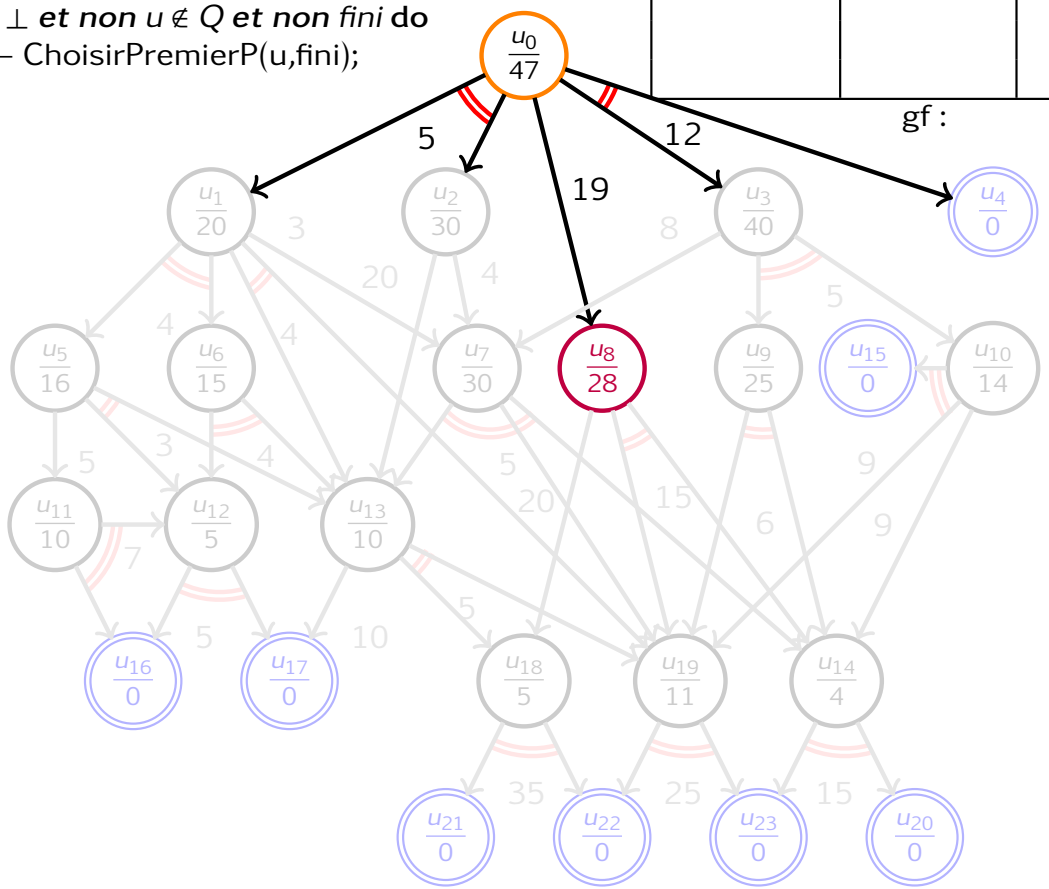


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,47)	(55,47,52)	{}	u_8(19,28)
u_8(19,28)	(-,-)	{}	

gf :



Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
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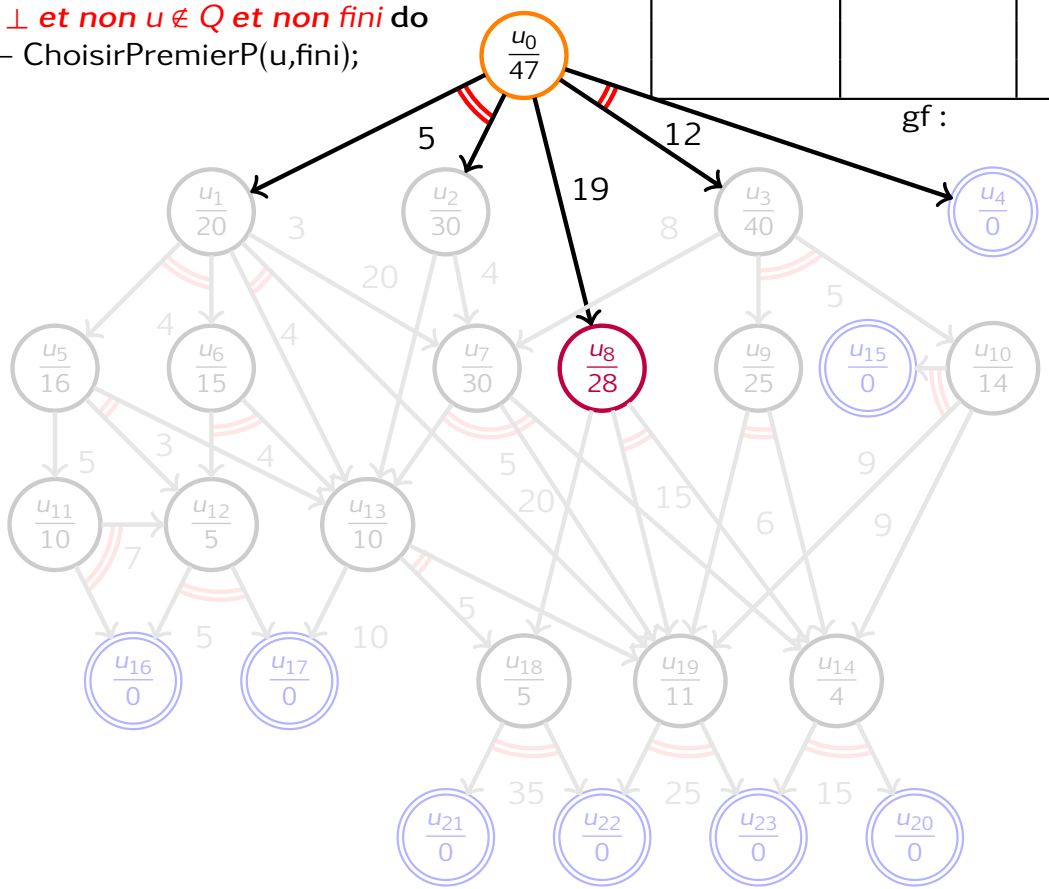
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,47)	(55,47,52)	{}	u_8(19,28)
u_8(19,28)	(-,-)	{}	

gf :

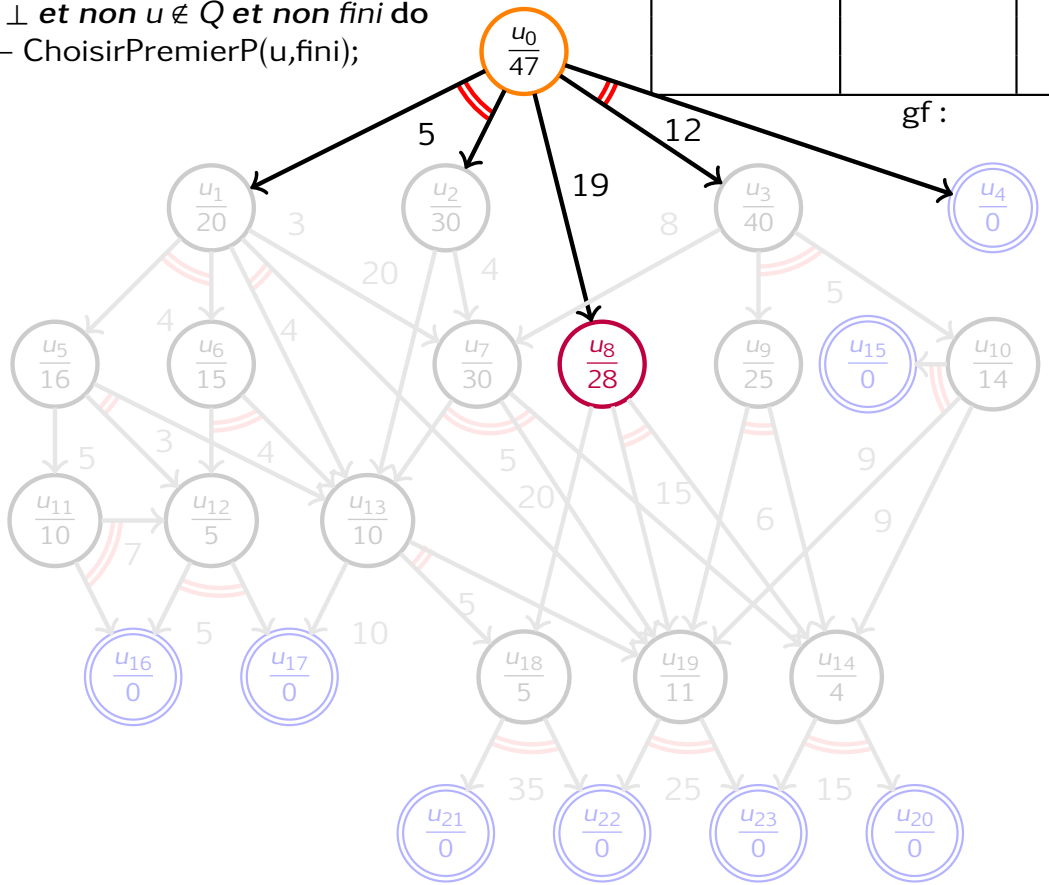


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,47)	(55,47,52)	{}	u_8(19,28)
u_8(19,28)	(-,-)	{}	

gf :

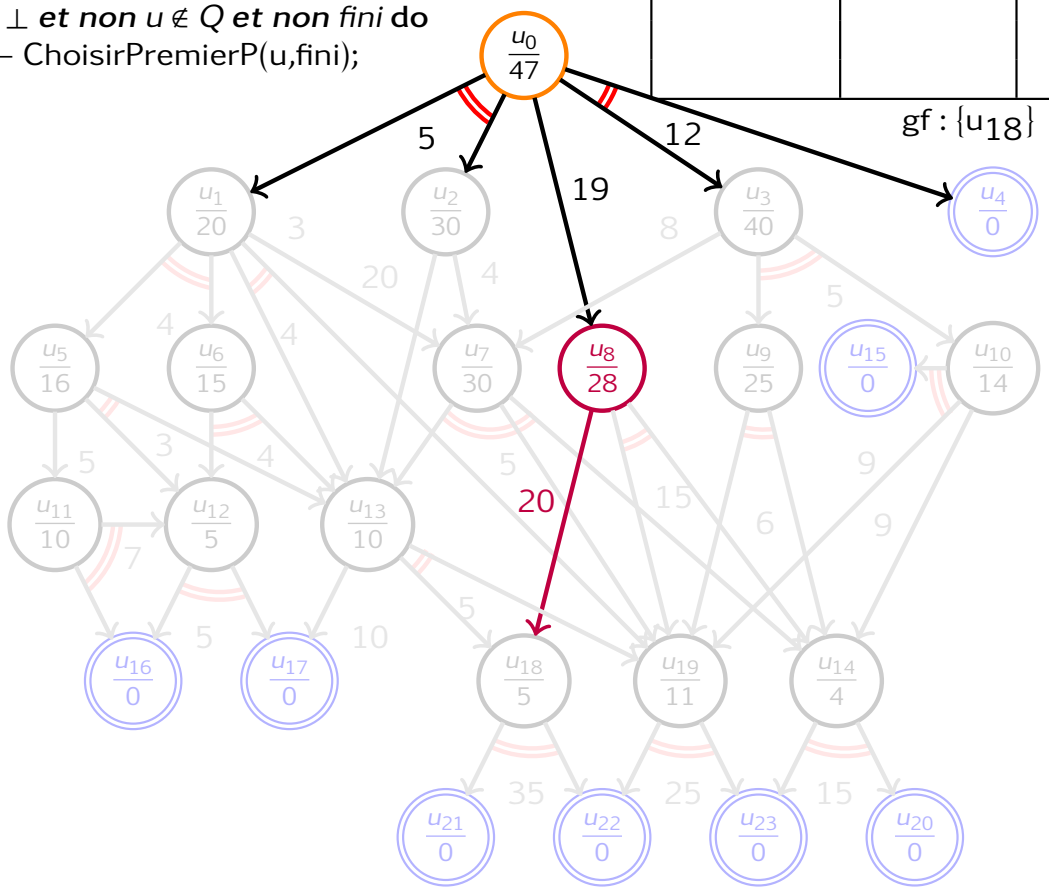


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,47)	(55,47,52)	{}	u_8(19,28)
u_8(19,28)	(-,-)	{}	

gf : {u_{18}}

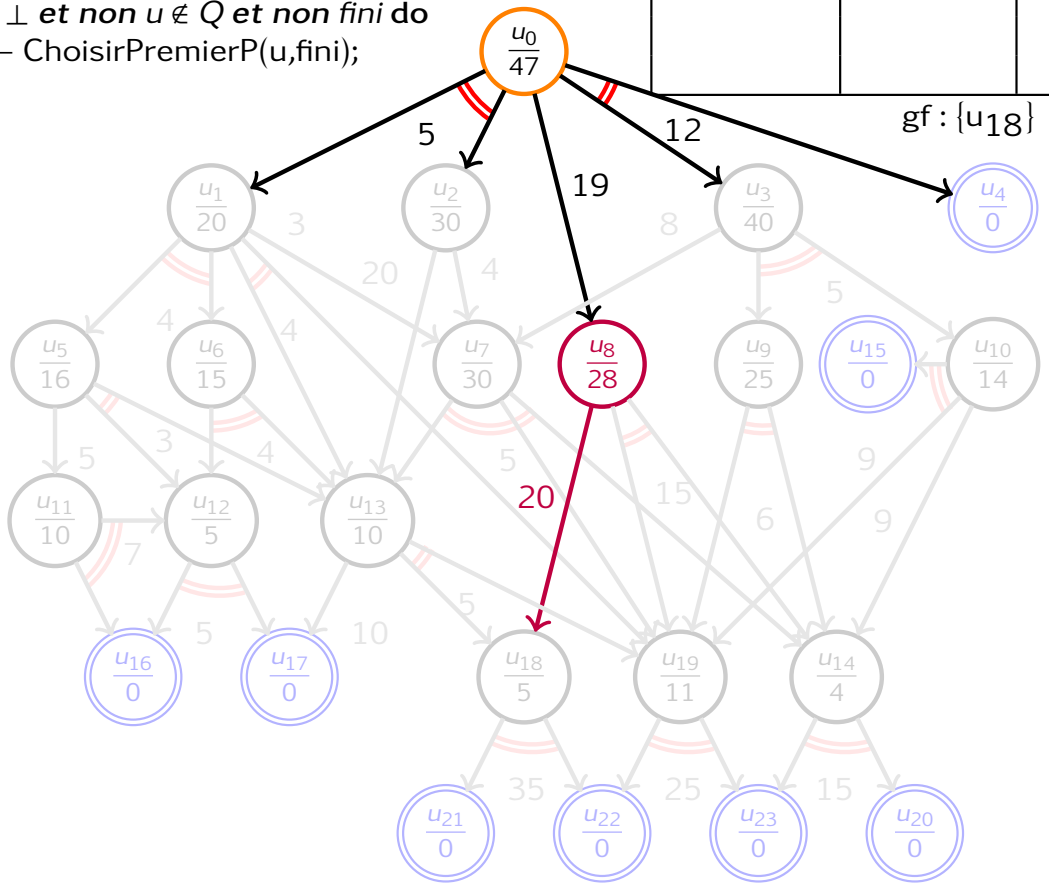


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,47)	(55,47,52)	{}	u_8(19,28)
u_8(19,28)	(-,-)	{}	

gf : $\{u_{18}\}$

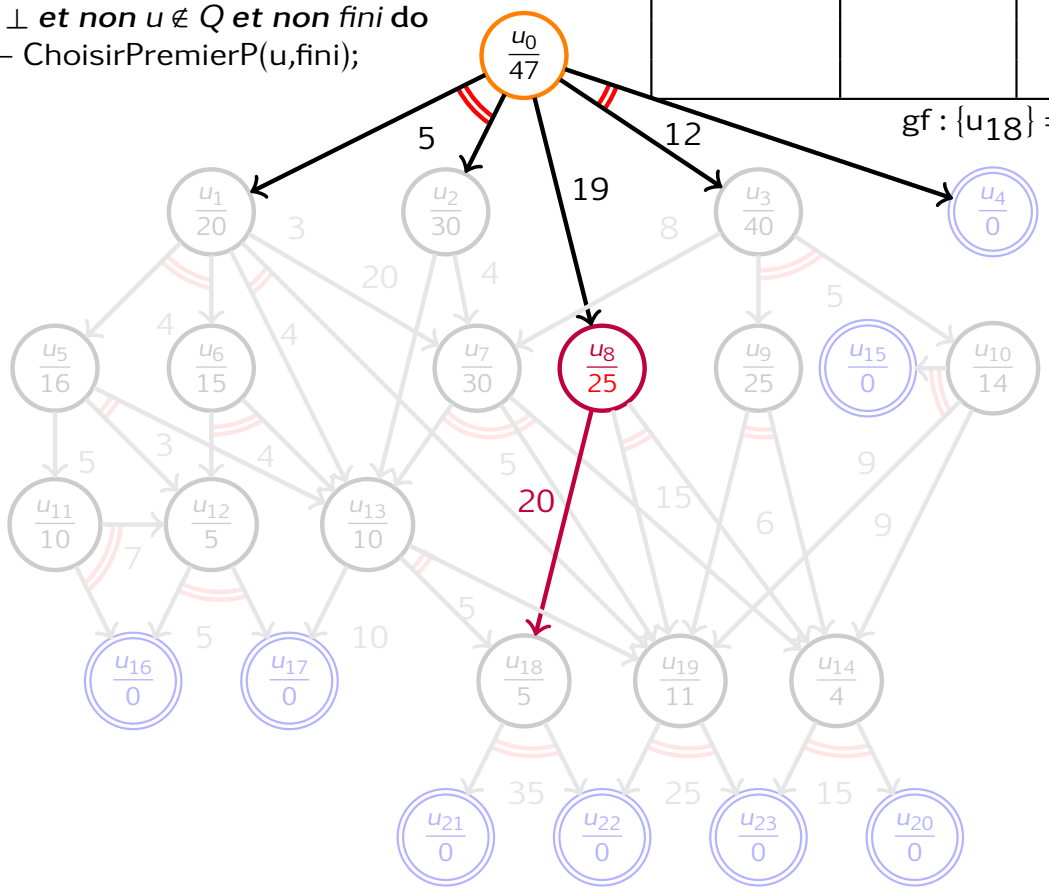


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,47)	(55,47,52)	{}	u_8(19,25)
u_8(19,25)	(25,-)	{u_0}	u_{18}(39,5)

gf : {u_{18}} = 20+5 = 25



Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
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        gf ← gfSuivant(u);
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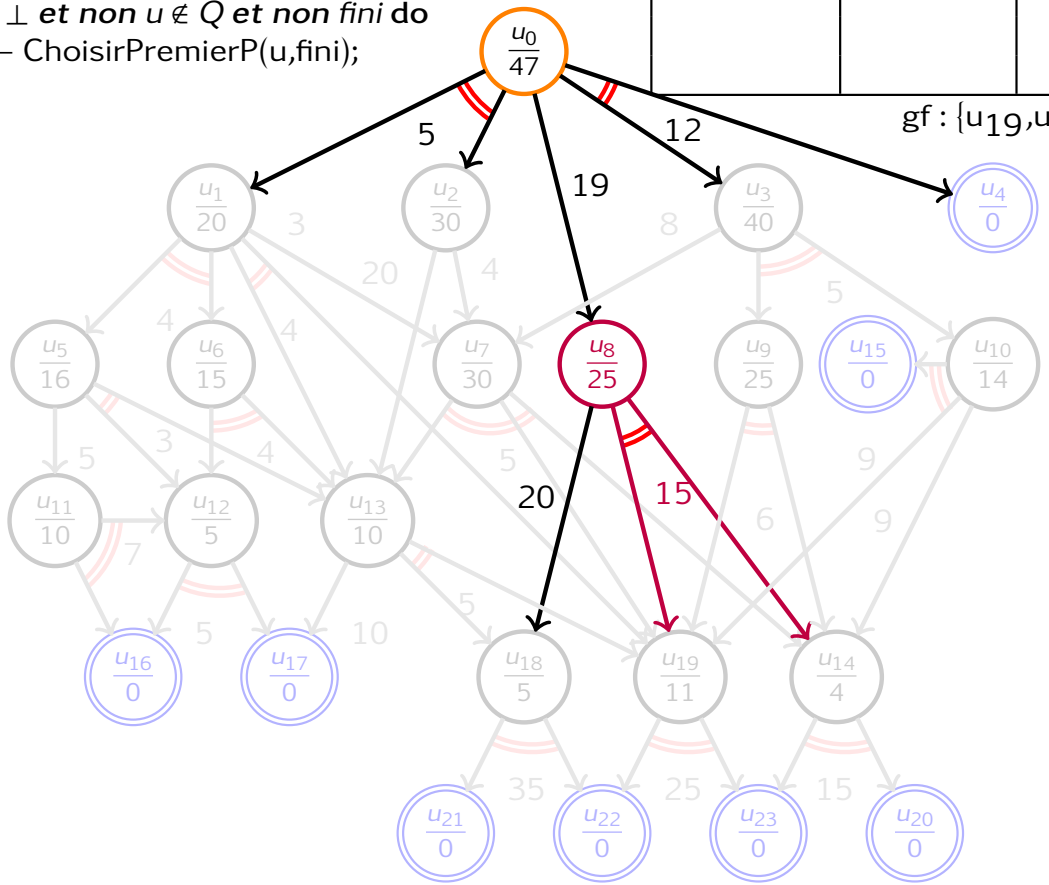
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,47)	(55,47,52)	{}	u_8(19,25)
u_8(19,25)	(25,-)	$\{u_0\}$	u_{18}(39,5)

gf : $\{u_{19}, u_{14}\}$

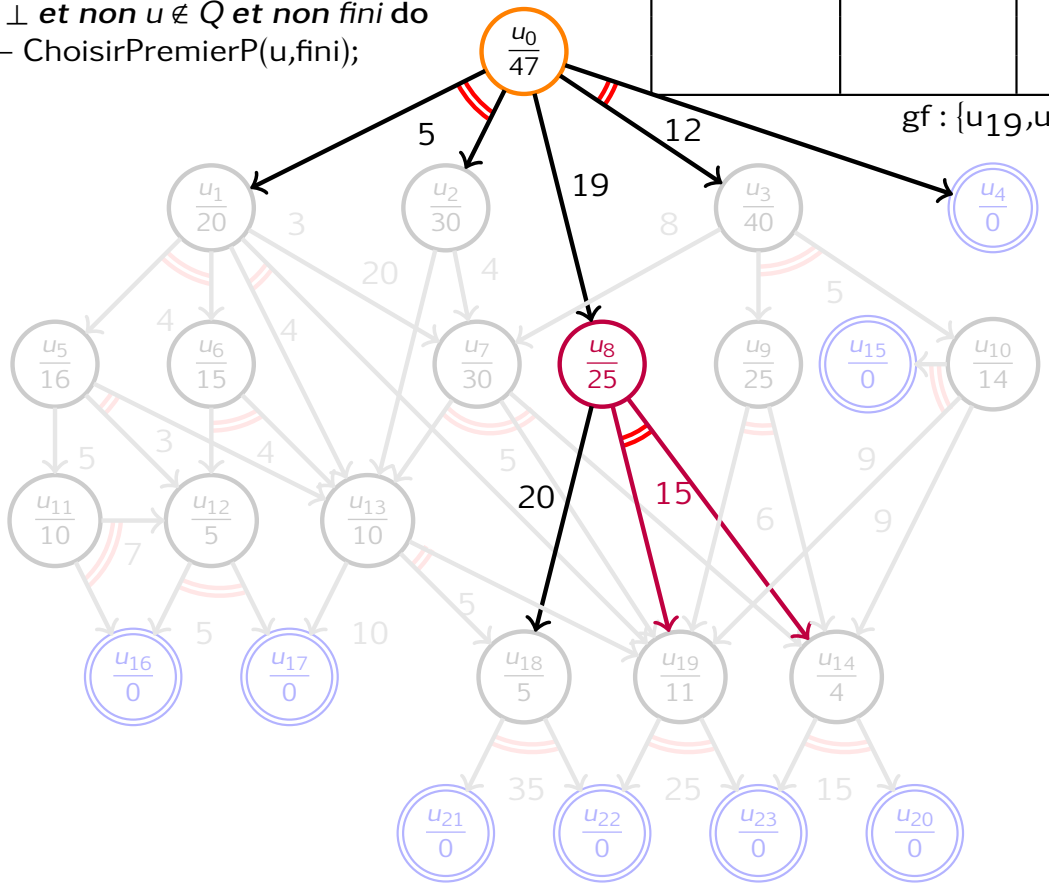


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,47)	(55,47,52)	{}	u_8(19,25)
u_8(19,25)	(25,-)	$\{u_0\}$	u_{18}(39,5)

gf : $\{u_{19}, u_{14}\}$



Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
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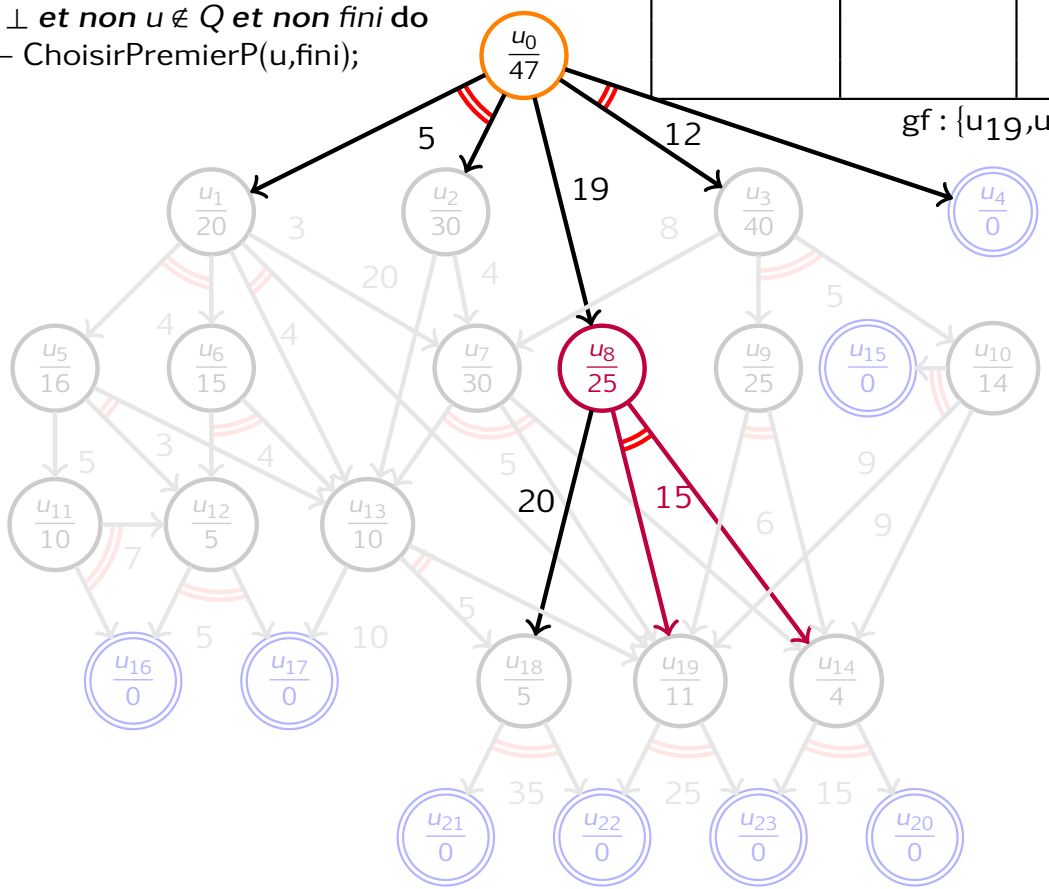
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,47)	(55,47,52)	{}	u_8(19,25)
u_8(19,25)	(25,30)	$\{u_0\}$	u_{18}(39,5)

gf : $\{u_{19}, u_{14}\}$ = 15+11+4 = 30



Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
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        gf ← gfSuivant(u);
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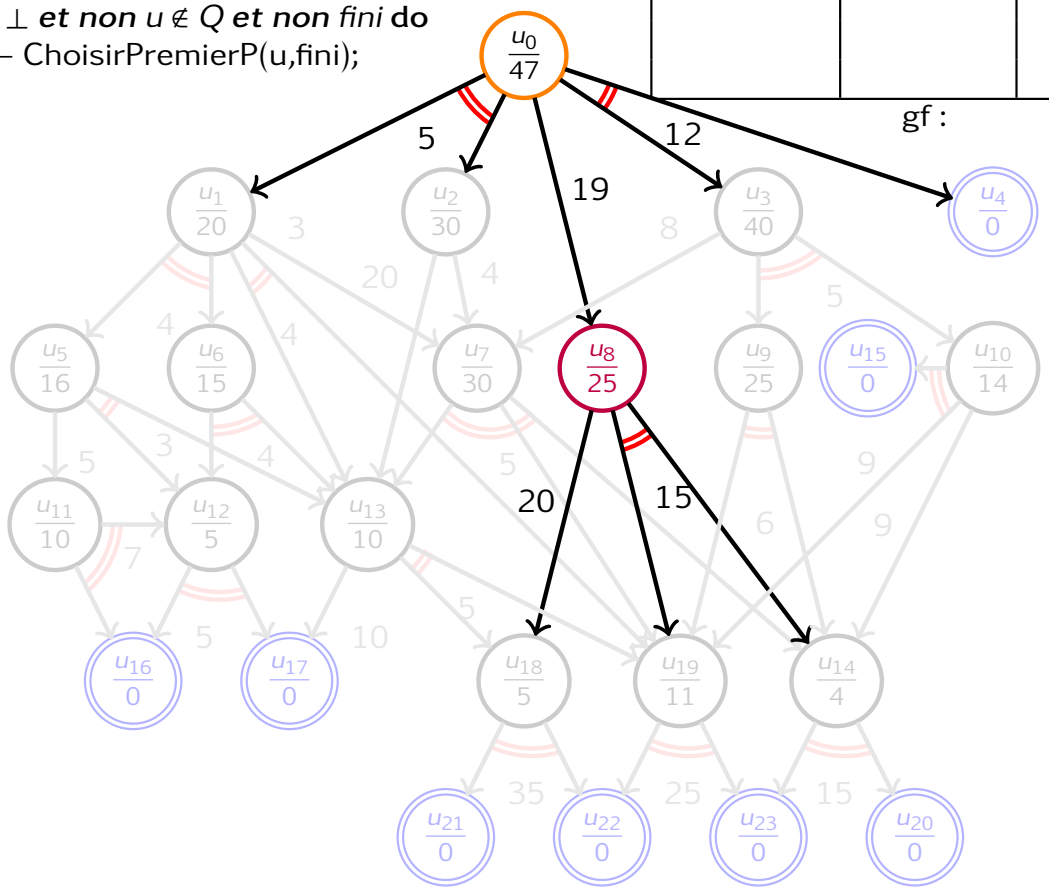
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,47) u_8(19,25)	(55,47,52) (25,30)	{} {u_0}	u_8(19,25) u_{18}(39,5)

gf :



Algorithme OA*(Liste,Sol)

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while u ≠ ⊥ et non fini do
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    while gf ≠ ∅ do
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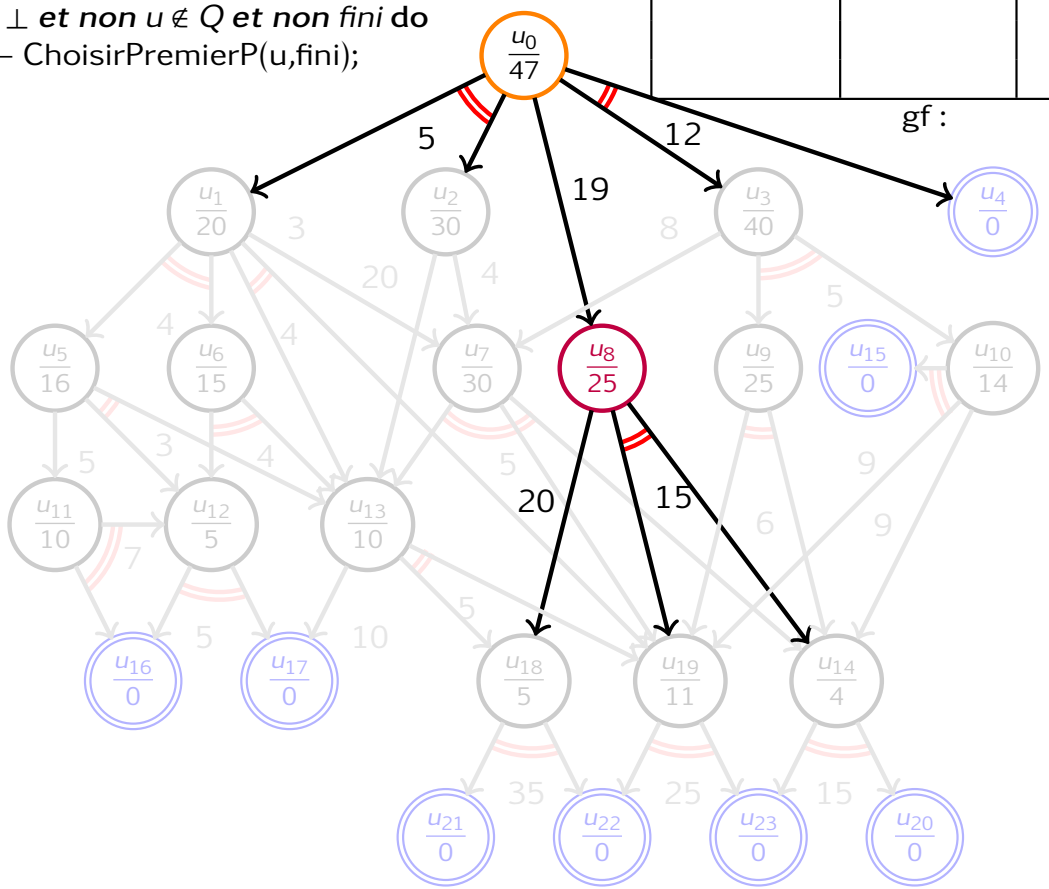
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,47)	(55,47,52)	{}	u_8(19,25)
u_8(19,25)	(25,30)	{u_0}	u_{18}(39,5)

gf :



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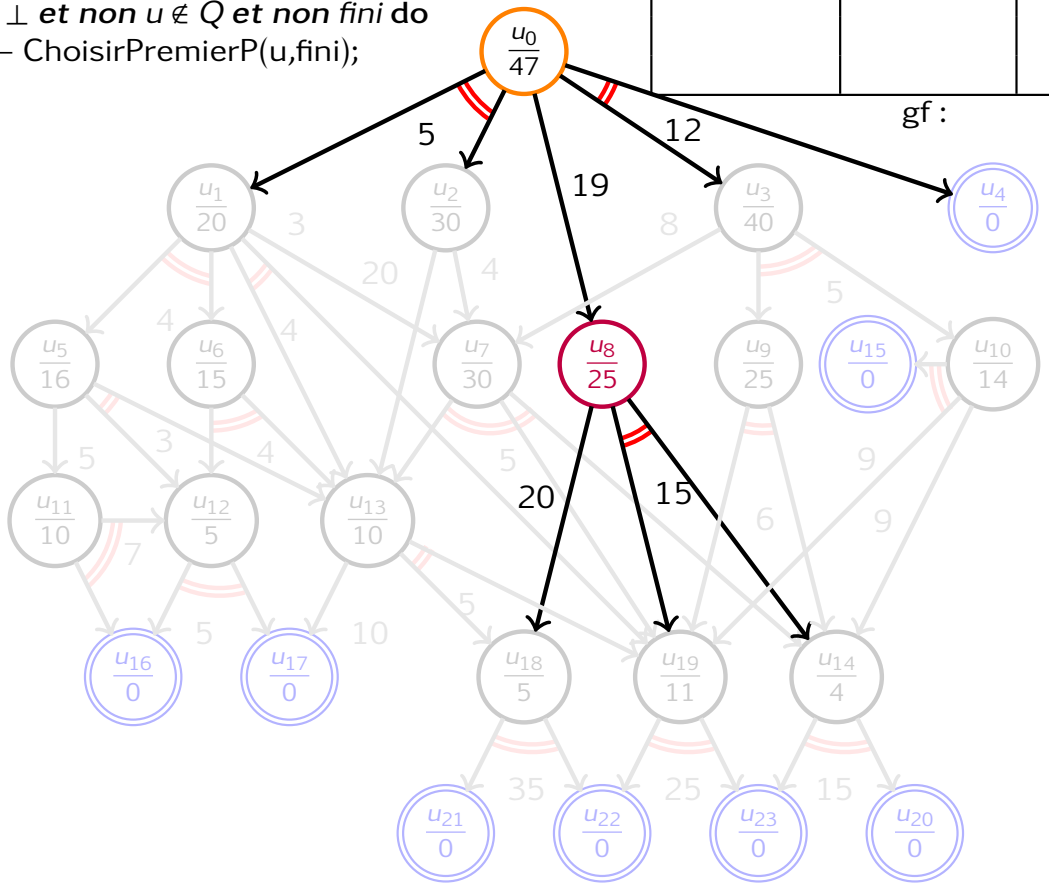
u(g(u),h(u))	fils(u)	A	P
u_0(0,47)	(55,47,52)	{}	u_8(19,25)
u_8(19,25)	(25,30)	$\{u_0\}$	u_{18}(39,5)

gf :



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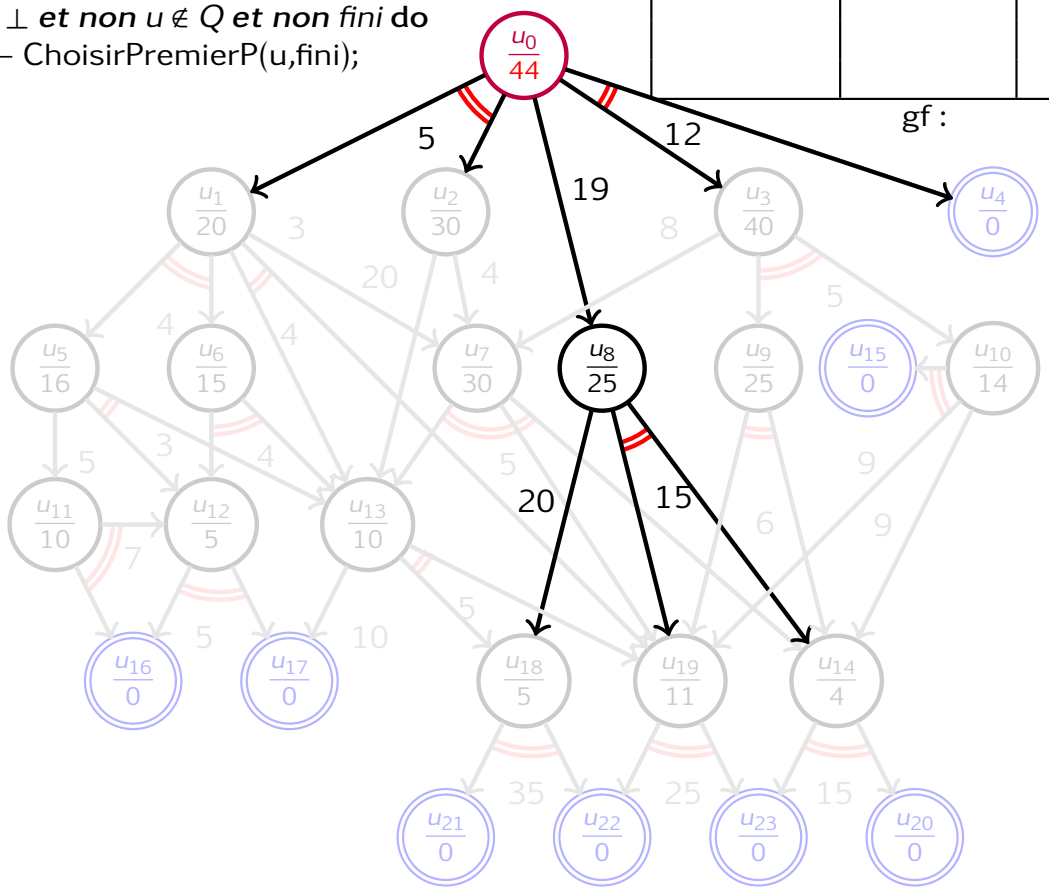
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,44)	(55,44,52)	{}	u_8(19,25)
u_8(19,25)	(25,30)	{u_0}	u_{18}(39,5)

gf :



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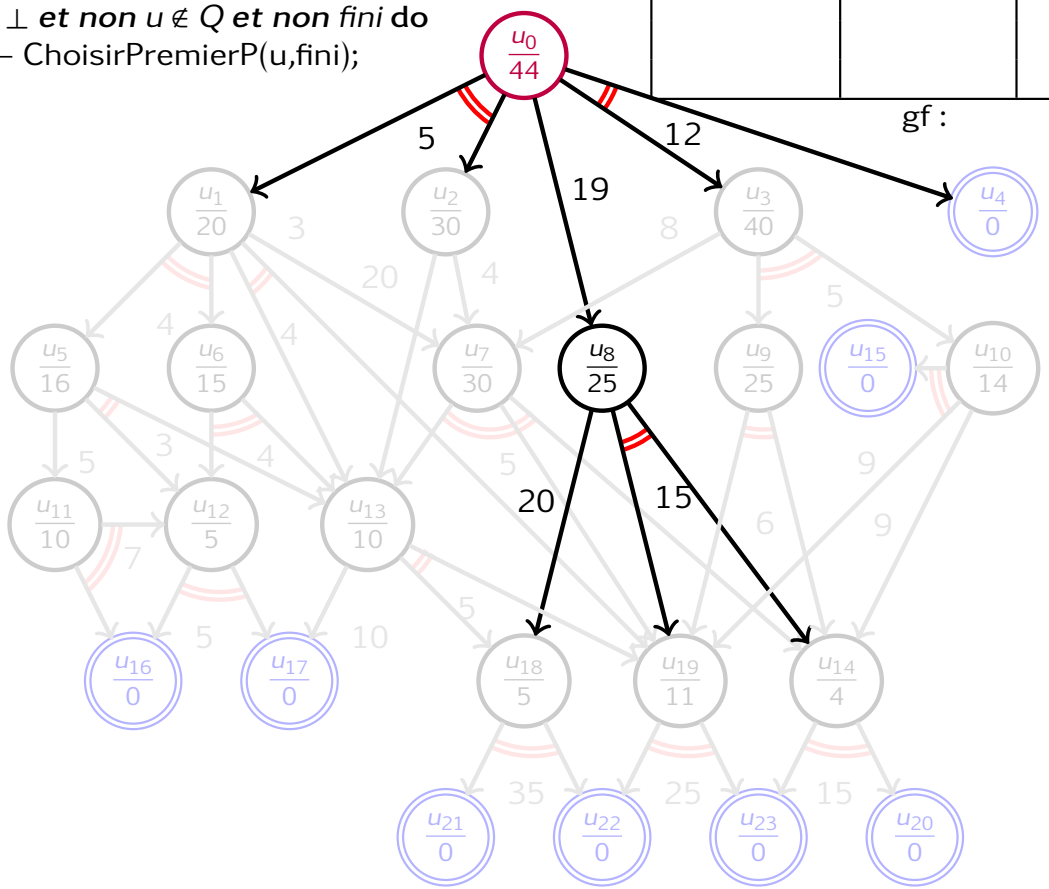
u(g(u),h(u))	fils(u)	A	P
u_0(0,44)	(55,44,52)	{}	u_8(19,25)
u_8(19,25)	(25,30)	{}	u_{18}(39,5)

gf :



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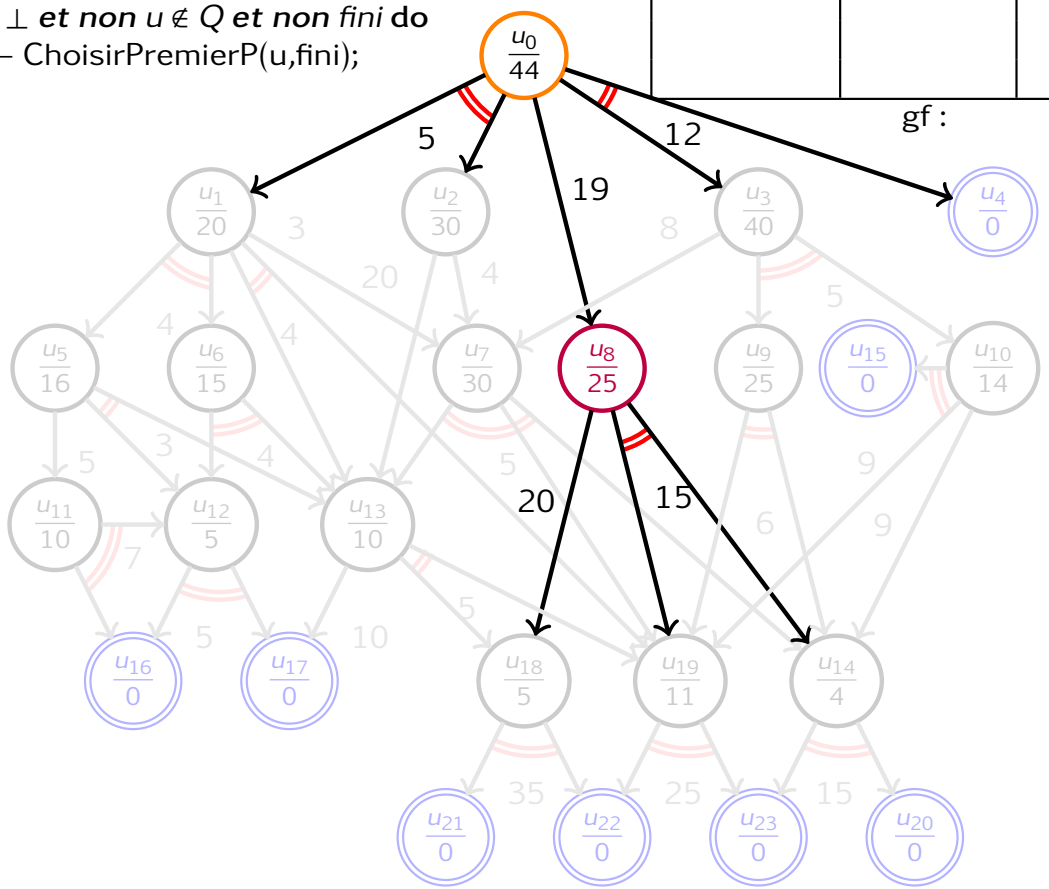
u(g(u),h(u))	fils(u)	A	P
u_0(0,44)	(55,44,52)	{}	u_8(19,25)
u_8(19,25)	(25,30)	{}	u_{18}(39,5)

gf :



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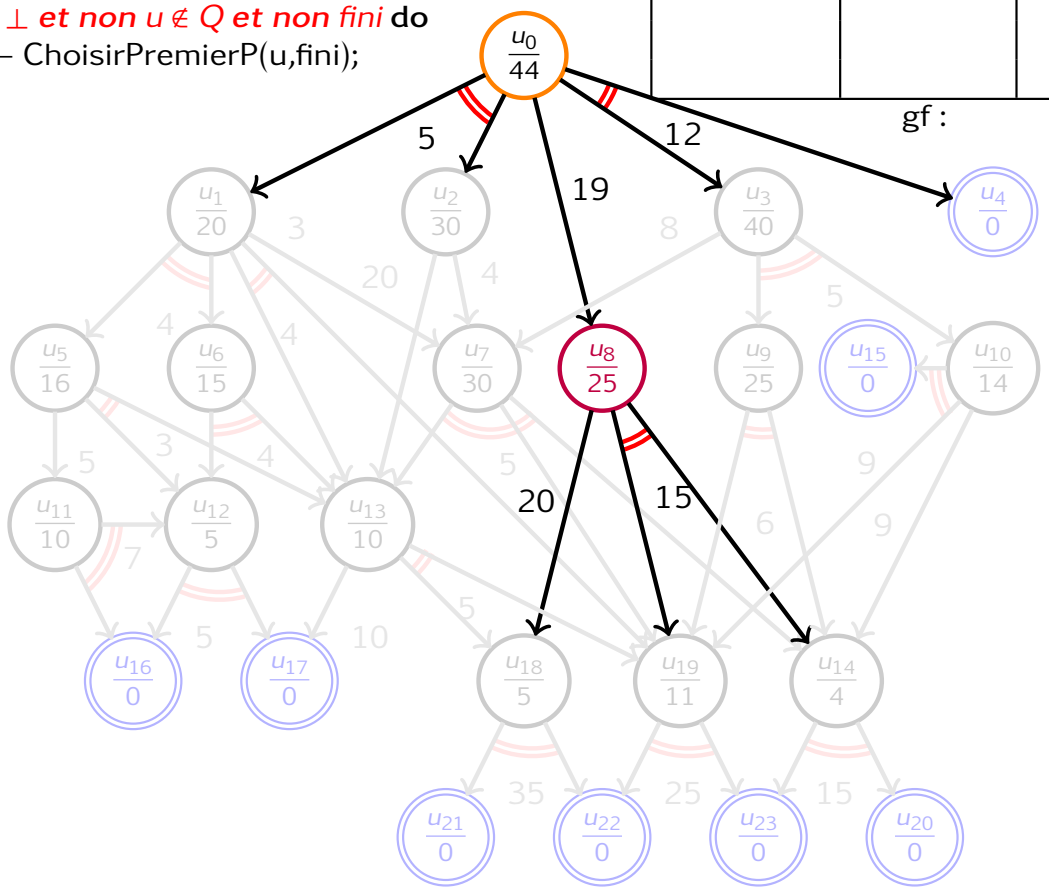
u_0(0,44)	(55,44,52)	{}	u_8(19,25)
u_8(19,25)	(25,30)	{}	u_{18}(39,5)

gf :



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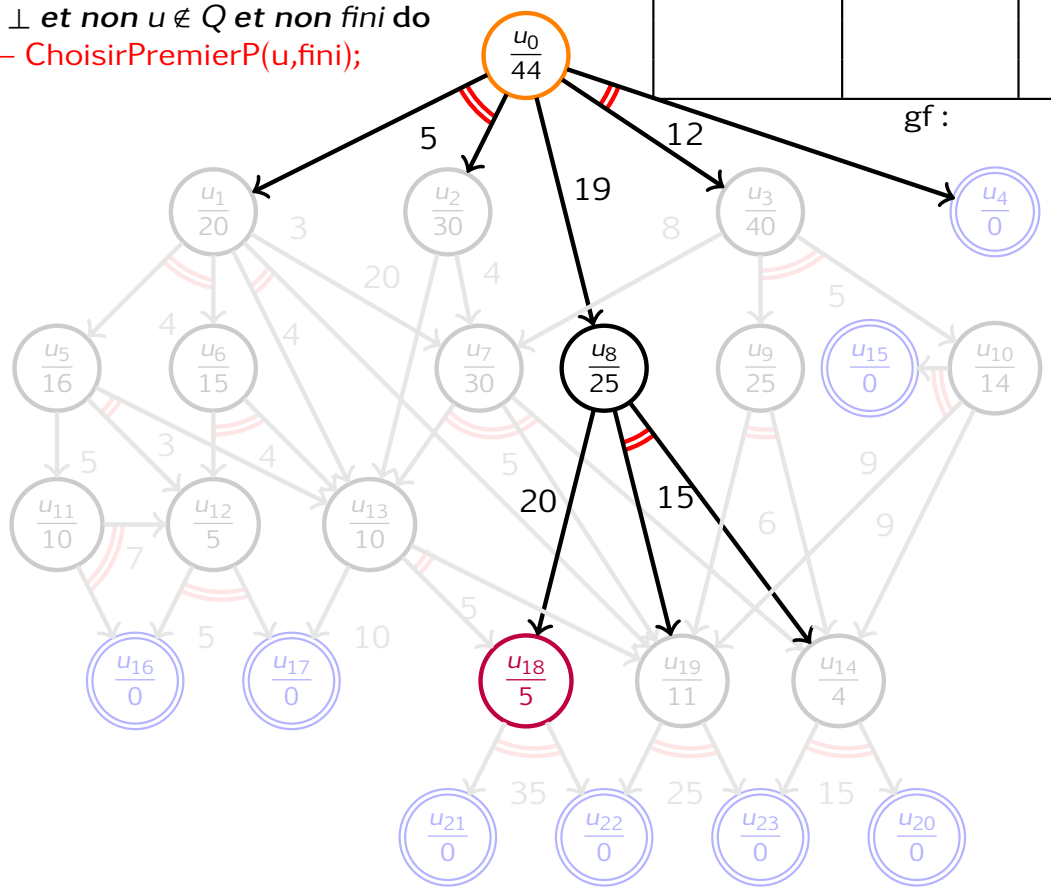
u_0(0,44)	(55,44,52)	{}	u_8(19,25)
u_8(19,25)	(25,30)	{}	u_{18}(39,5)
u_{18}(39,5)	(-)	{}	

gf :



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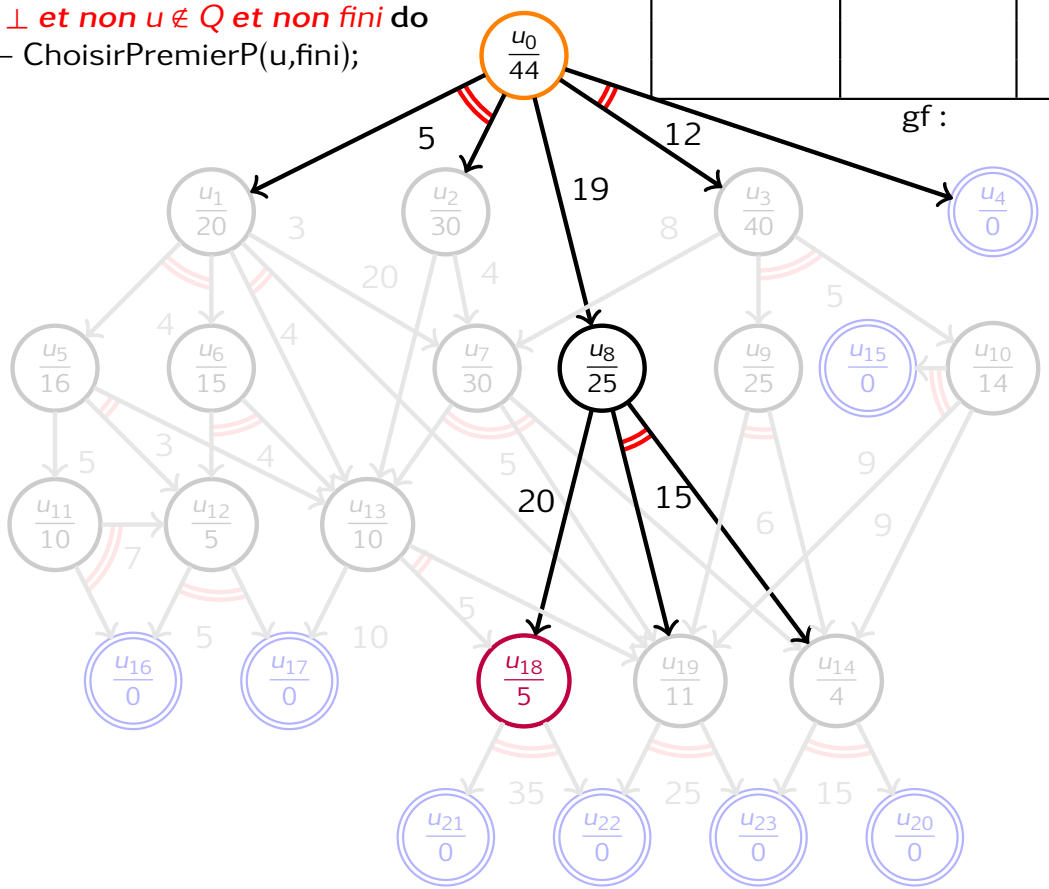

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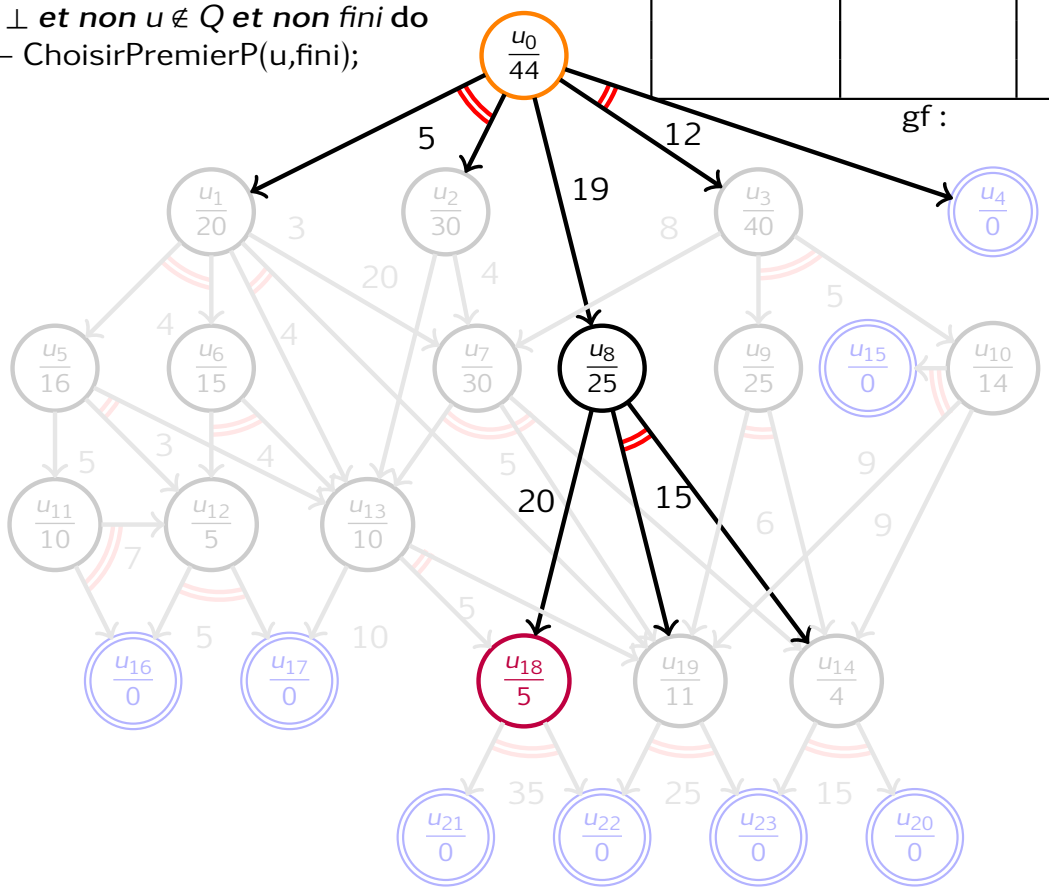
u_8(19,25)	(25,30)	{}	u_{18}(39,5)
u_{18}(39,5)	(-)	{}	

gf :



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u(g(u),h(u))	fils(u)	A	P
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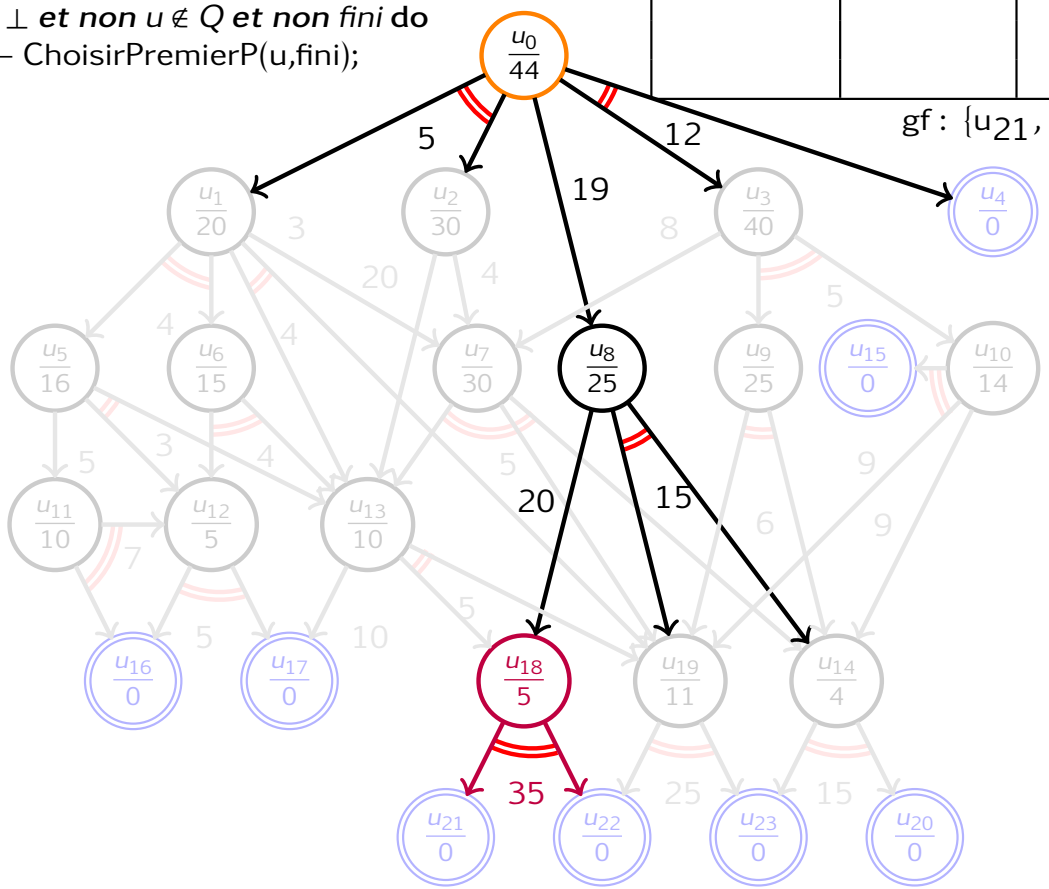
u_8(19,25)	(25,30)	{}	u_{18}(39,5)
u_{18}(39,5)	(-)	{}	

gf : $\{u_{21}, u_{22}\}$



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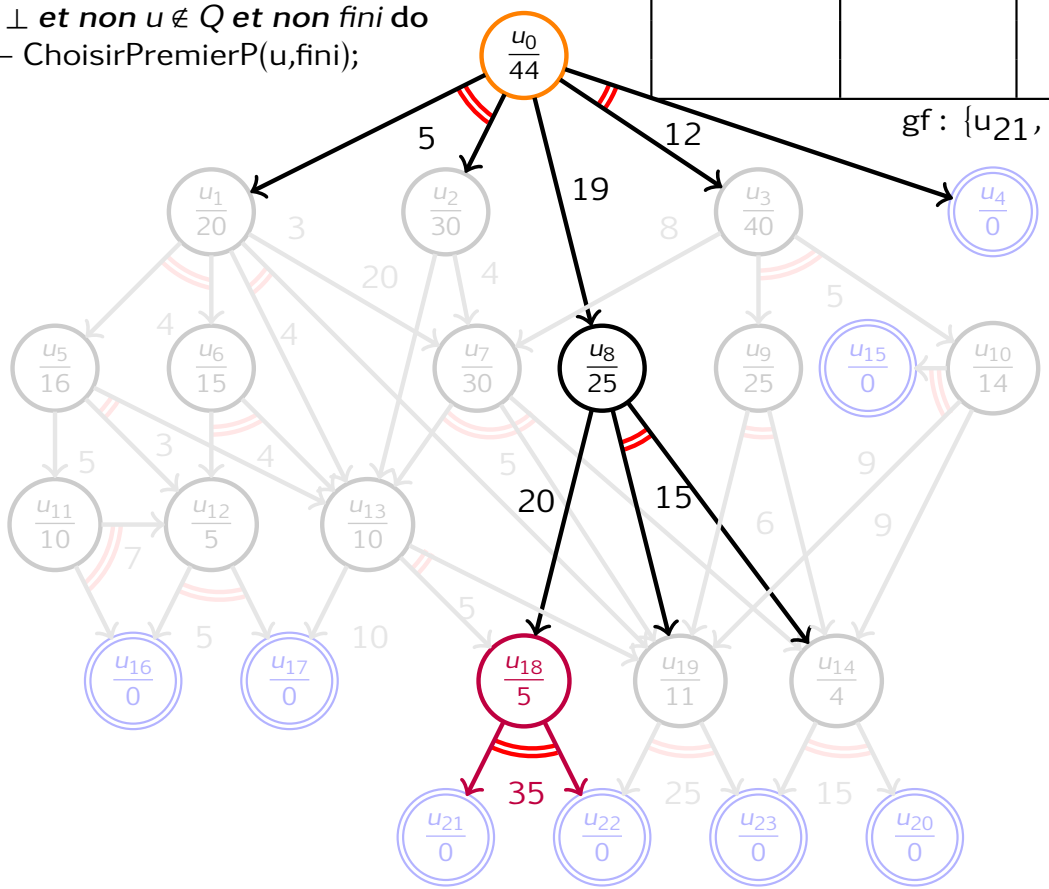
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,44)	(55,44,52)	{}	u_8(19,25)
u_8(19,25)	(25,30)	{}	u_{18}(39,5)
u_{18}(39,5)	(-)	{}	

gf : $\{u_{21}, u_{22}\}$



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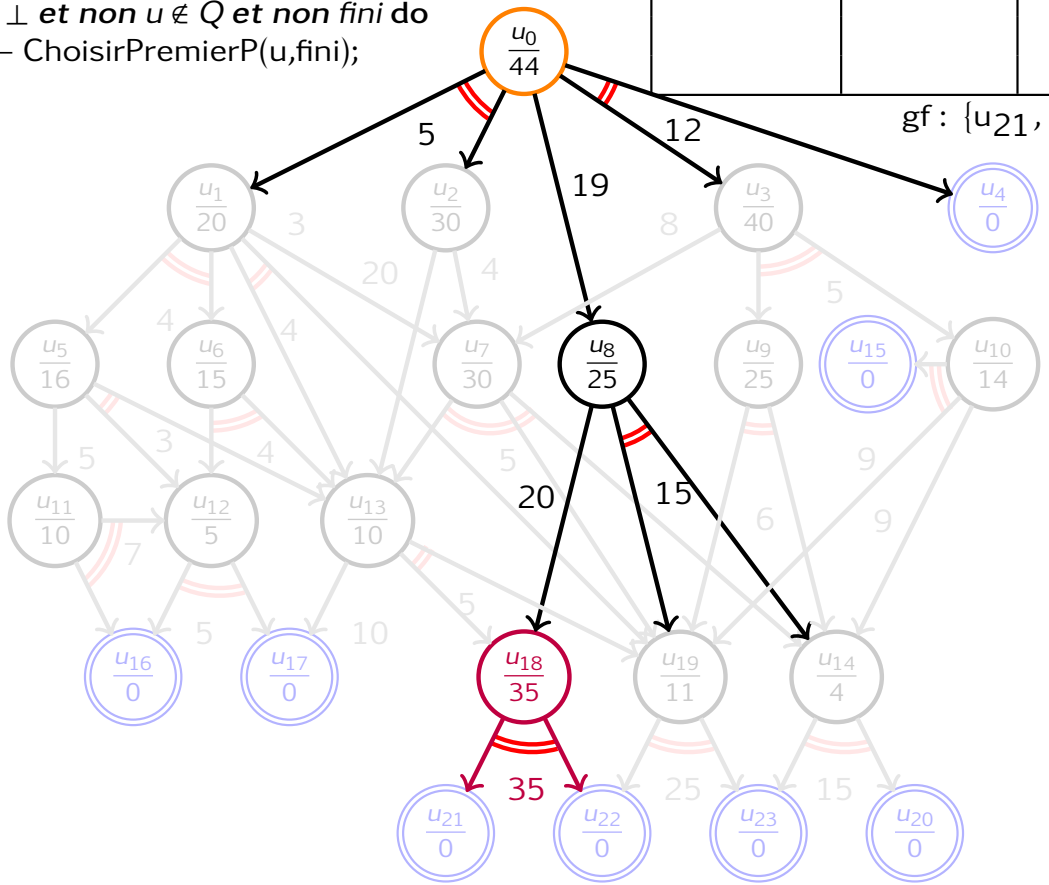
u(g(u),h(u))	fils(u)	A	P
u_0(0,44)	(55,44,52)	{}	u_8(19,25)
u_8(19,25)	(25,30)	{}	u_{18}(39,35)
u_{18}(39,35)	(35)	{u_8}	u_{21}(74,0) u_{22}(74,0)

gf : $\{u_{21}, u_{22}\}$ = 35+0+0 = 35



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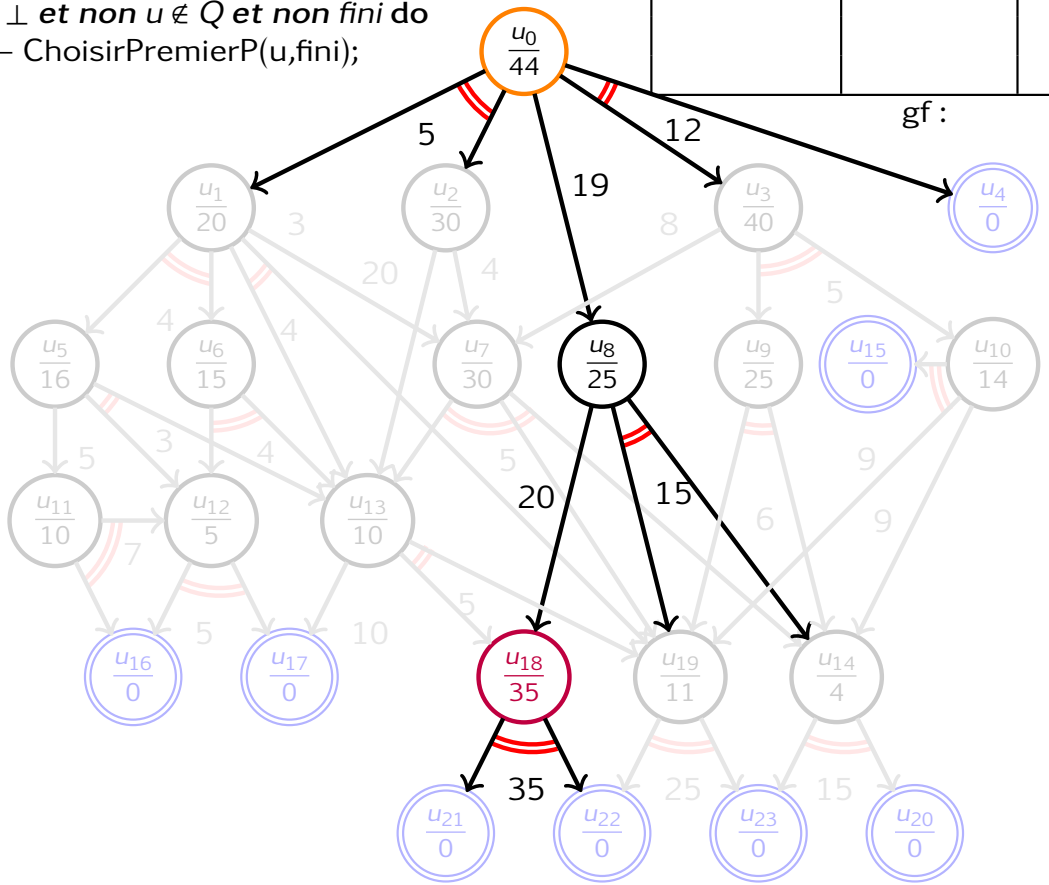
u(g(u),h(u))	fils(u)	A	P
$u_0(0,44)$	(55,44,52)	{}	$u_8(19,25)$
$u_8(19,25)$	(25,30)	{}	$u_{18}(39,35)$
$u_{18}(39,35)$	(35)	$\{u_8\}$	$u_{21}(74,0)$ $u_{22}(74,0)$

gf :



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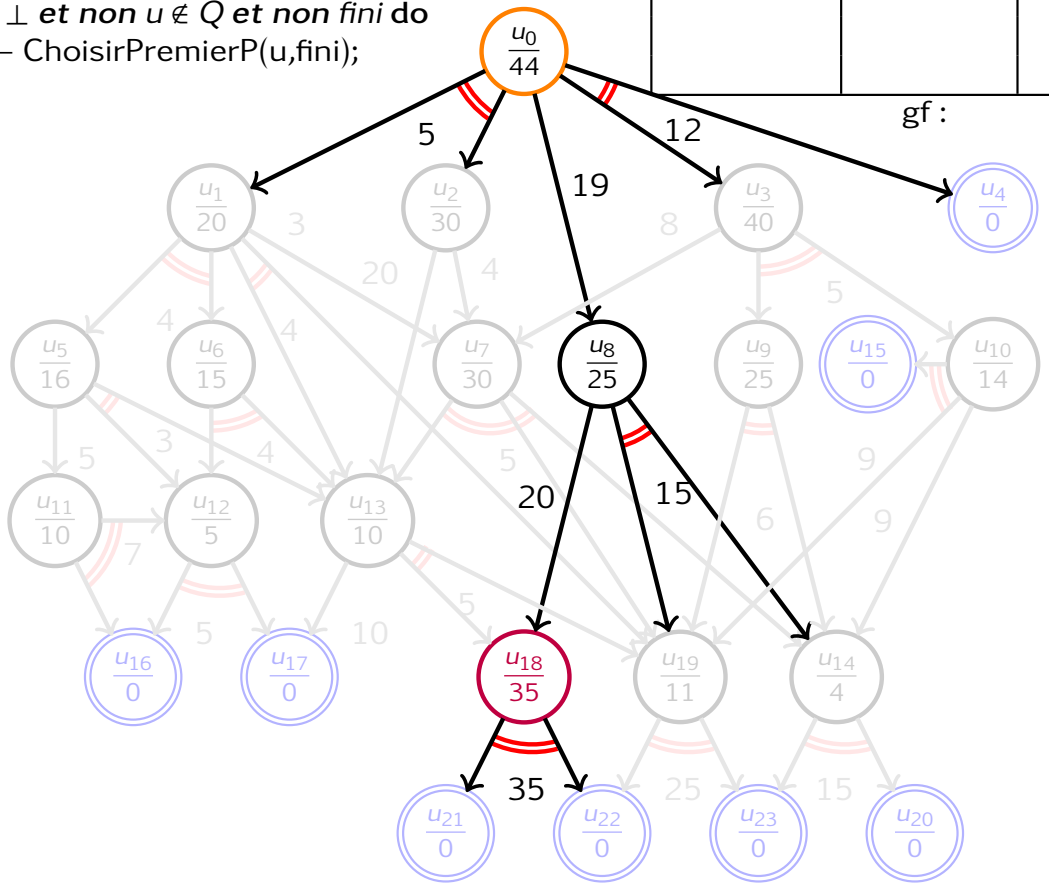
u(g(u),h(u))	fils(u)	A	P
$u_0(0,44)$	(55,44,52)	{}	$u_8(19,25)$
$u_8(19,25)$	(25,30)	{}	$u_{18}(39,35)$
$u_{18}(39,35)$	(35)	$\{u_8\}$	$u_{21}(74,0)$ $u_{22}(74,0)$

gf :



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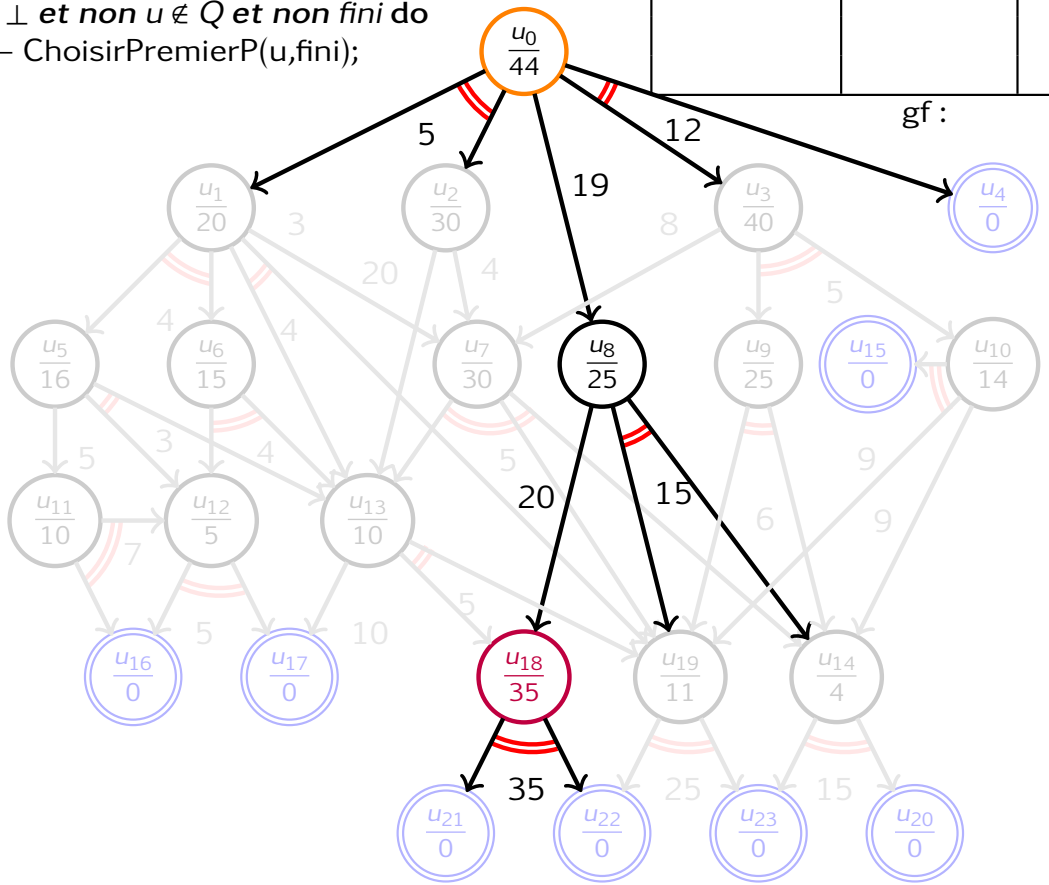
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,44)$	(55,44,52)	{}	$u_8(19,25)$
$u_8(19,25)$	(25,30)	{}	$u_{18}(39,35)$
$u_{18}(39,35)$	(35)	$\{u_8\}$	$u_{21}(74,0)$ $u_{22}(74,0)$

gf :



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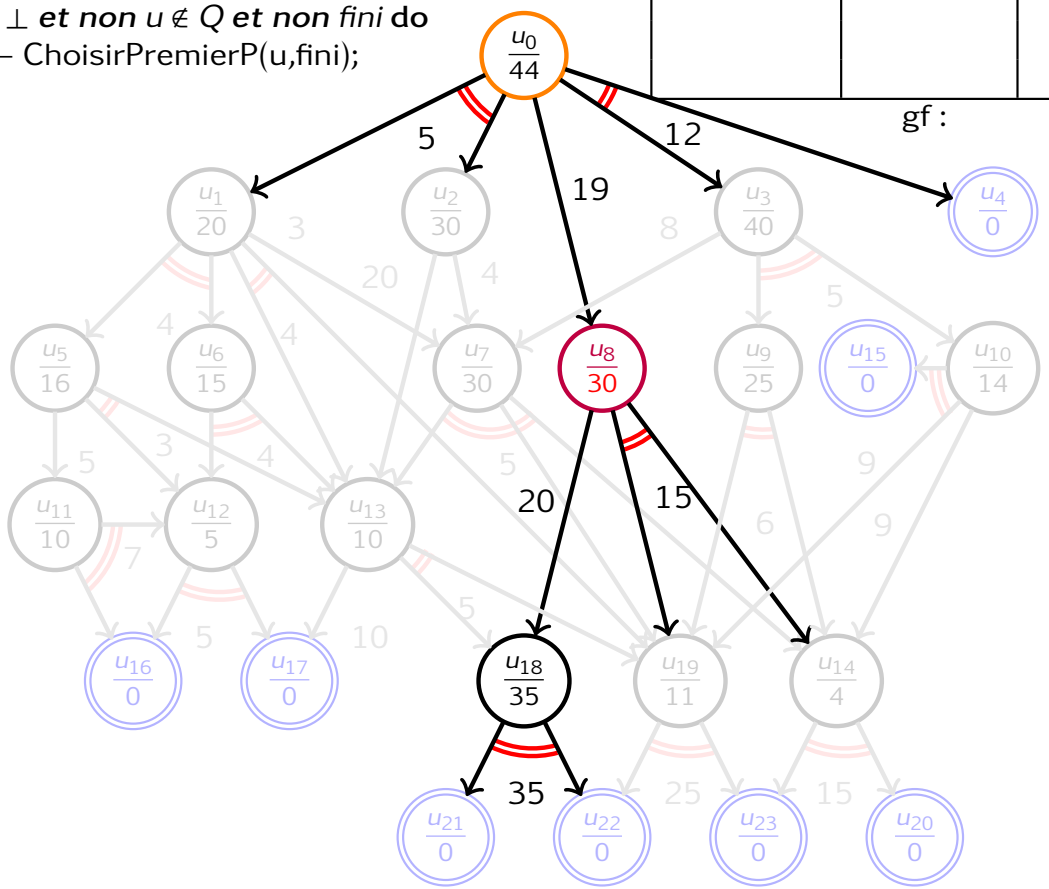
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,44)	(55,44,52)	{}	u_8(19,30)
u_8(19,30)	(55,30)	{u_0}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)

gf :



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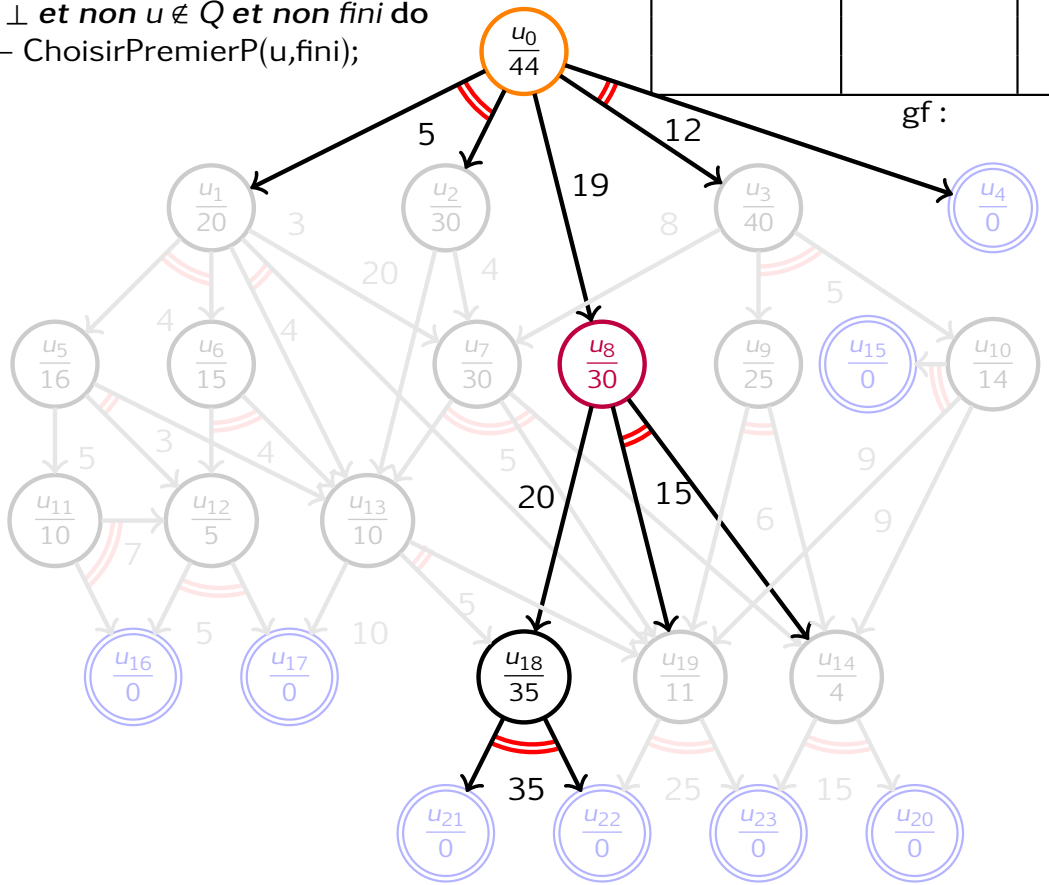
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,44)$	(55,44,52)	{}	$u_8(19,30)$
$u_8(19,30)$	(55,30)	$\{u_0\}$	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$

gf :



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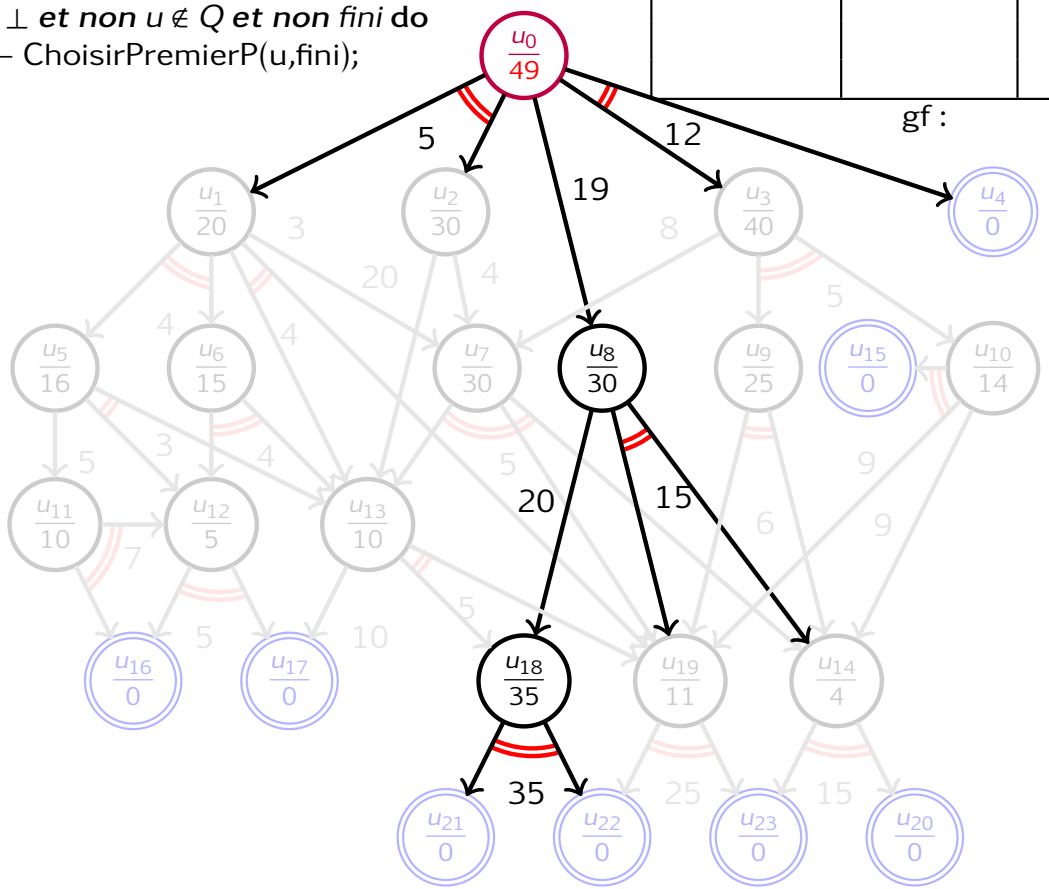
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,49)	(55,49,52)	{}	u_8(19,30)
u_8(19,30)	(55,30)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)

gf :



Algorithme OA*(Liste,Sol)

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u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
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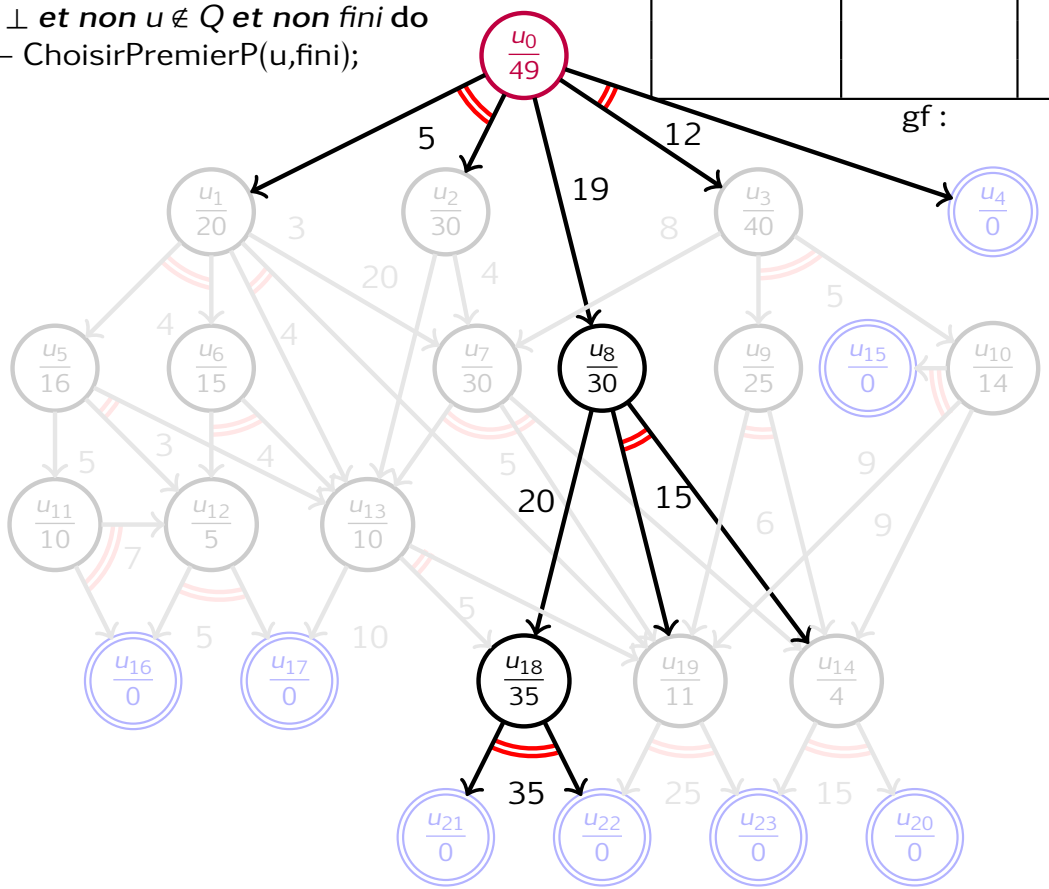
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,49)$	(55,49,52)	{}	$u_8(19,30)$
$u_8(19,30)$	(55,30)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$

gf :



Algorithme OA*(Liste,Sol)

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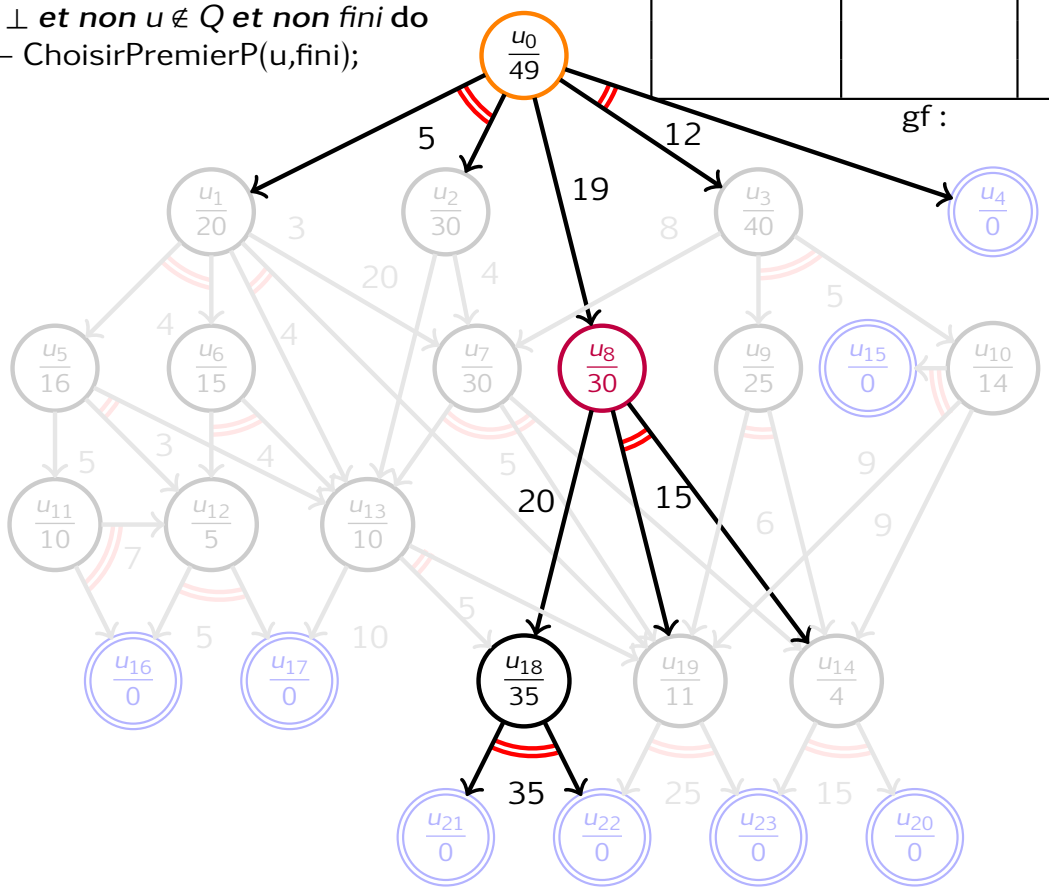
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,49)$	(55,49,52)	{}	$u_8(19,30)$
$u_8(19,30)$	(55,30)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$

gf :



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```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,49)$	(55,49,52)	{}	$u_8(19,30)$
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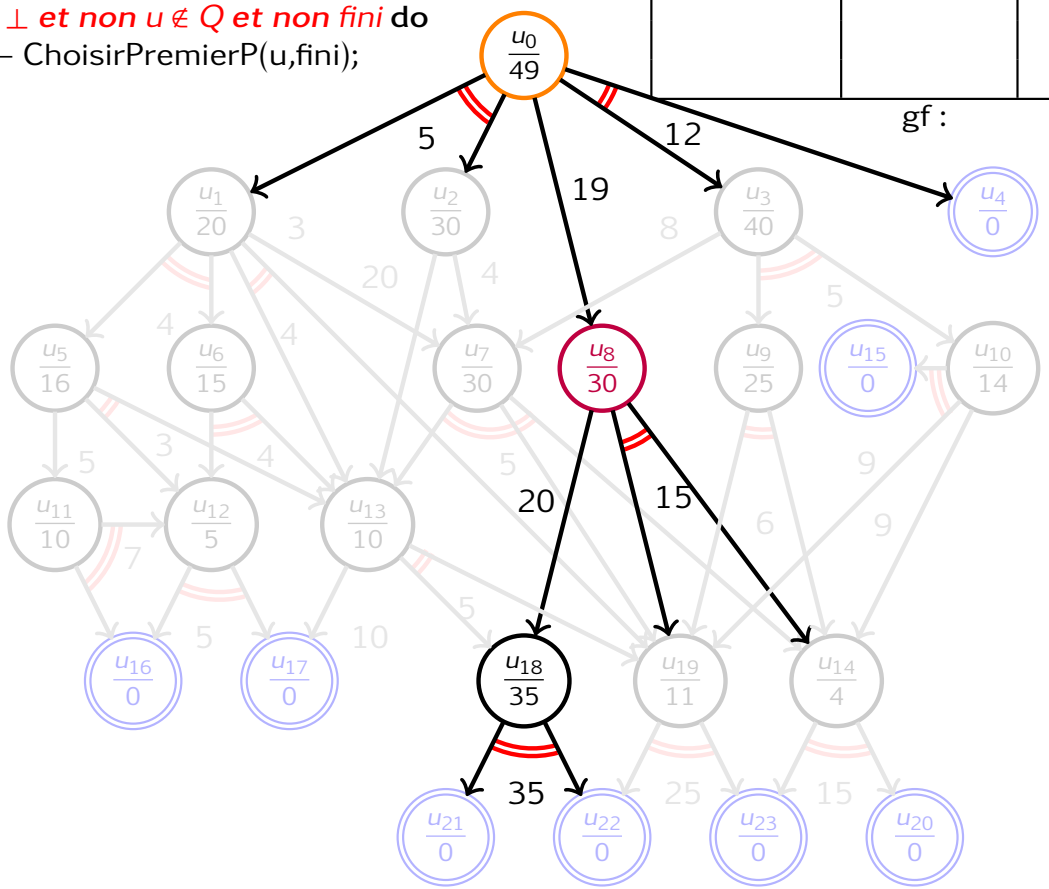
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$

gf :



Algorithme OA*(Liste,Sol)

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```

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$u_0(0,49)$	(55,49,52)	{}	$u_8(19,30)$
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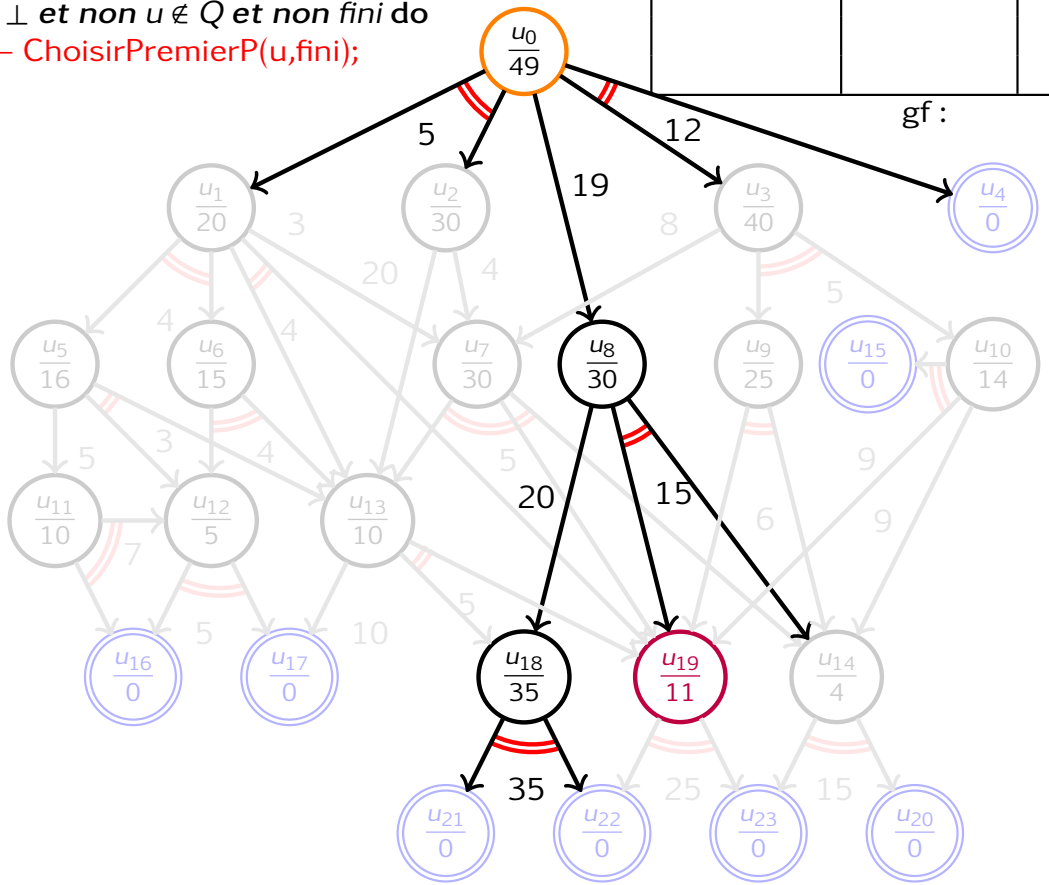
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,11)$	(-)	{}	$u_{14}(34,38)$

gf :



Algorithme OA*(Liste,Sol)

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```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,49)$	(55,49,52)	{}	$u_8(19,30)$
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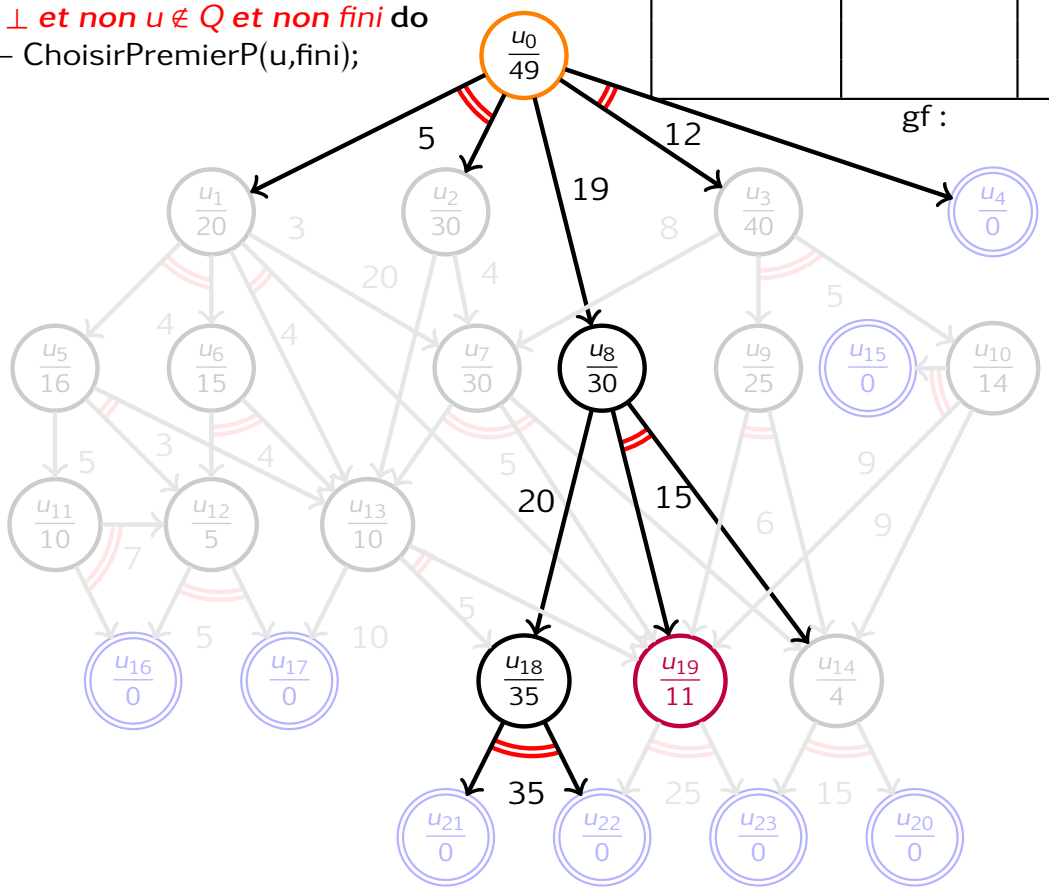
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,11)$	(-)	{}	$u_{14}(34,38)$

gf :



Algorithme OA*(Liste,Sol)

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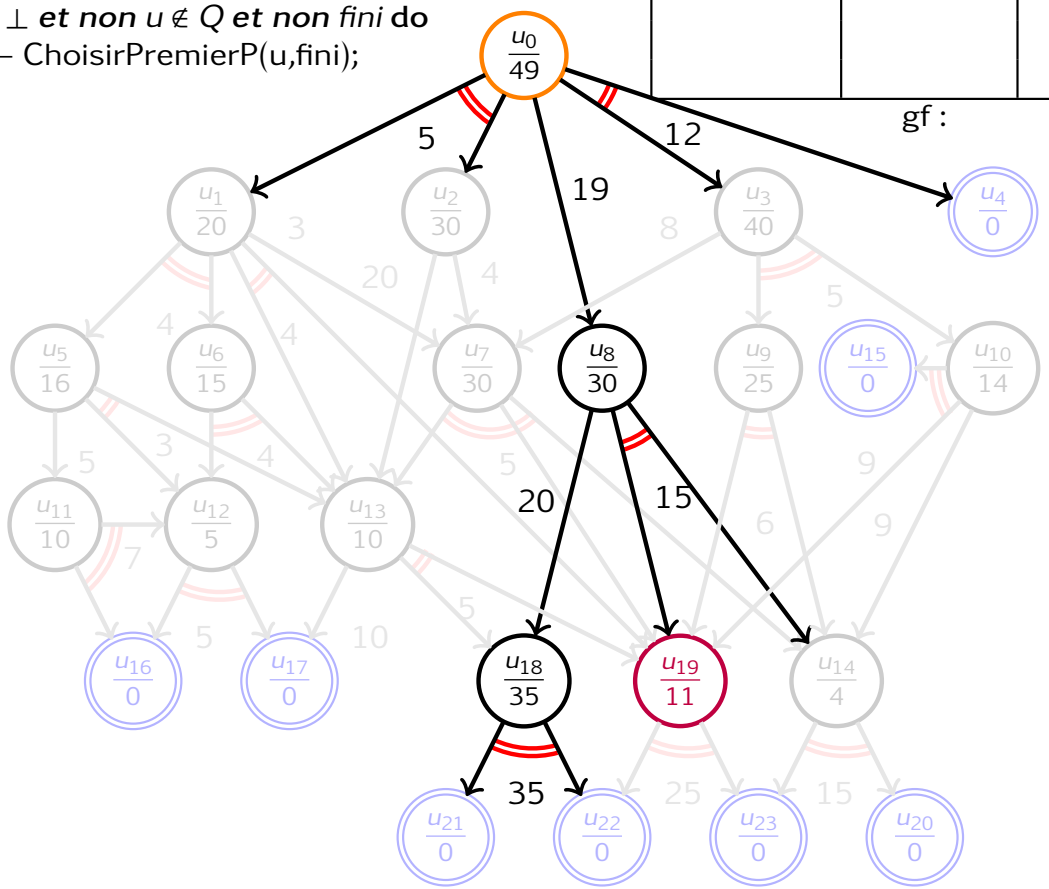
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,49)$	(55,49,52)	{}	$u_8(19,30)$
$u_8(19,30)$	(55,30)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,11)$	(-)	{}	$u_{14}(34,38)$

gf :



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u(g(u),h(u))	fils(u)	A	P
$u_0(0,49)$	(55,49,52)	{}	$u_8(19,30)$
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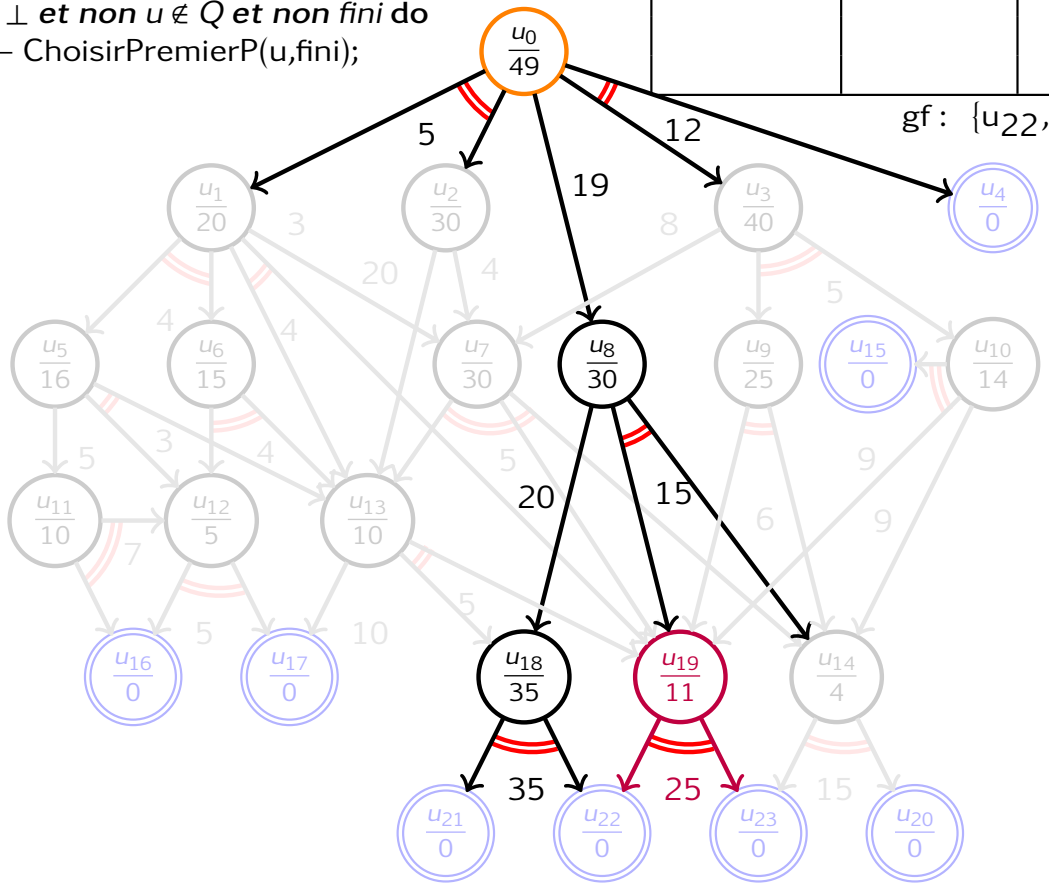
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,11)$	(-)	{}	$u_{14}(34,38)$

gf : $\{u_{22}, u_{23}\}$



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```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,49)$	(55,49,52)	{}	$u_8(19,30)$
$u_8(19,30)$	(55,30)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
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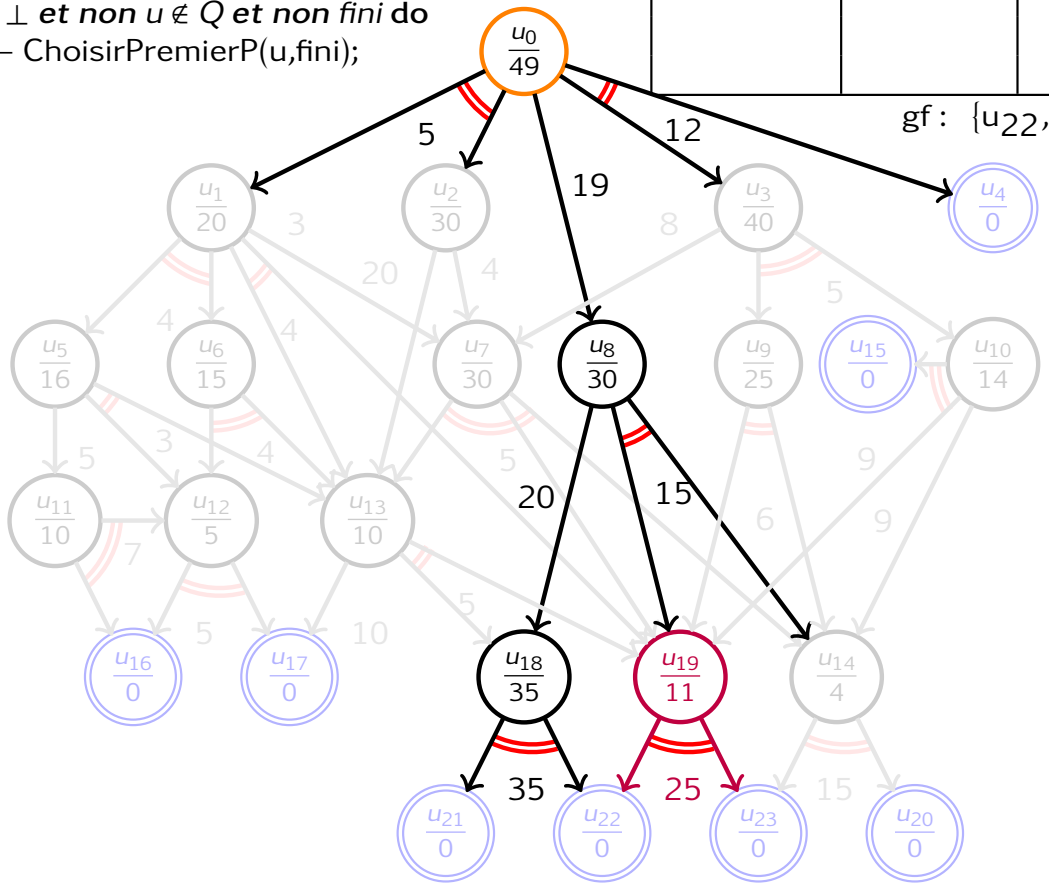
$u_{19}(34,11)$	(-)	{}	$u_{14}(34,38)$

gf : $\{u_{22}, u_{23}\}$



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$u_0(0,49)$	(55,49,52)	{}	$u_8(19,30)$
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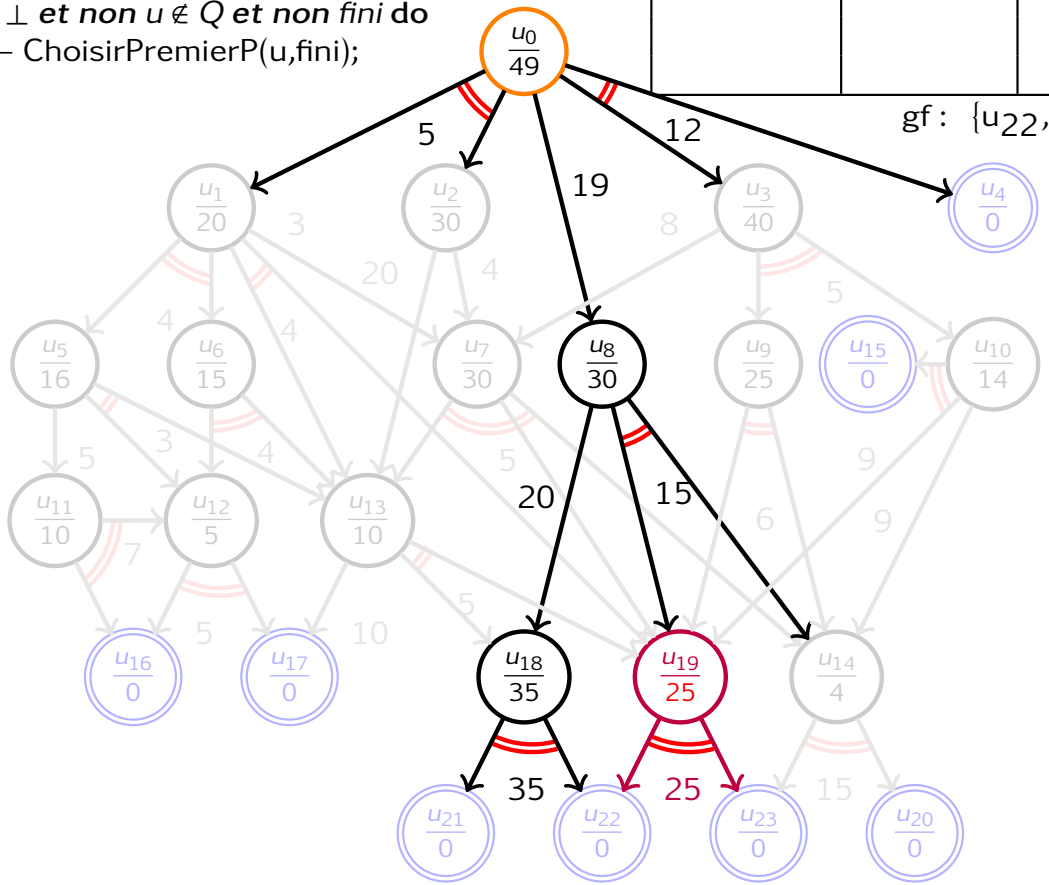
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	$\{u_8\}$	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$

gf : $\{u_{22}, u_{23}\}$ = 25+0+0 = 25



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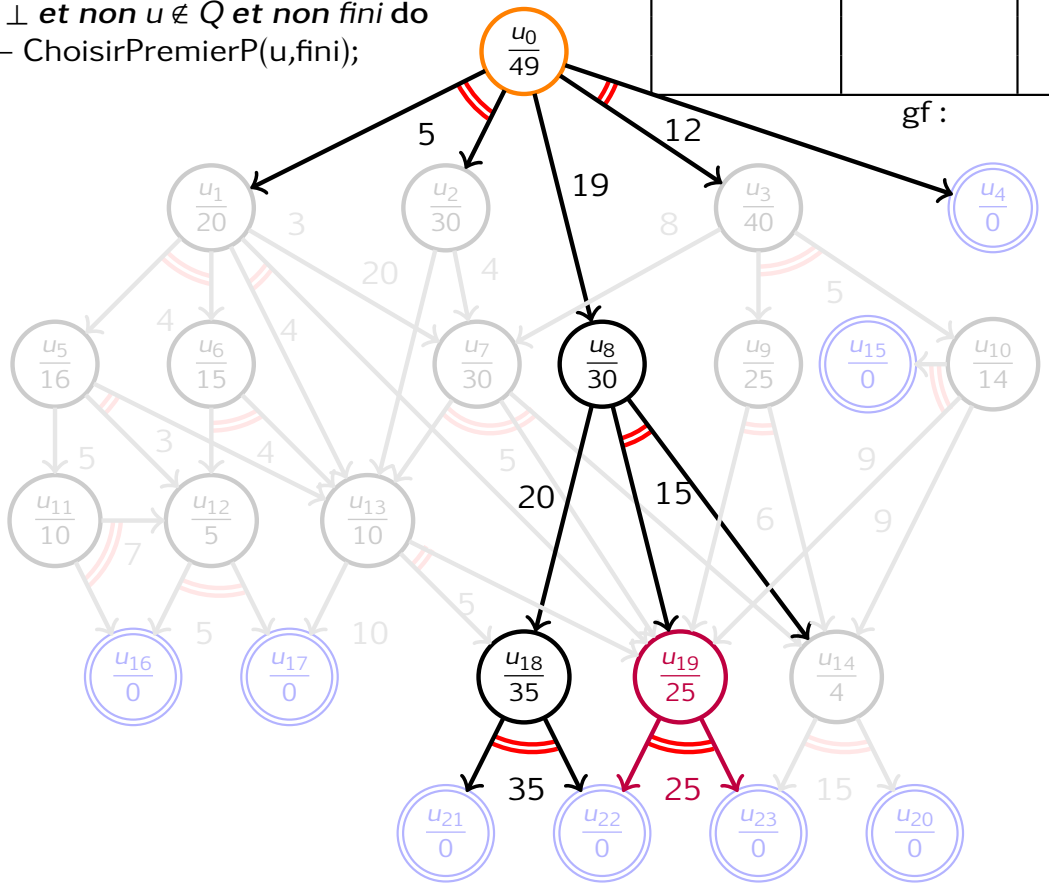
gf :



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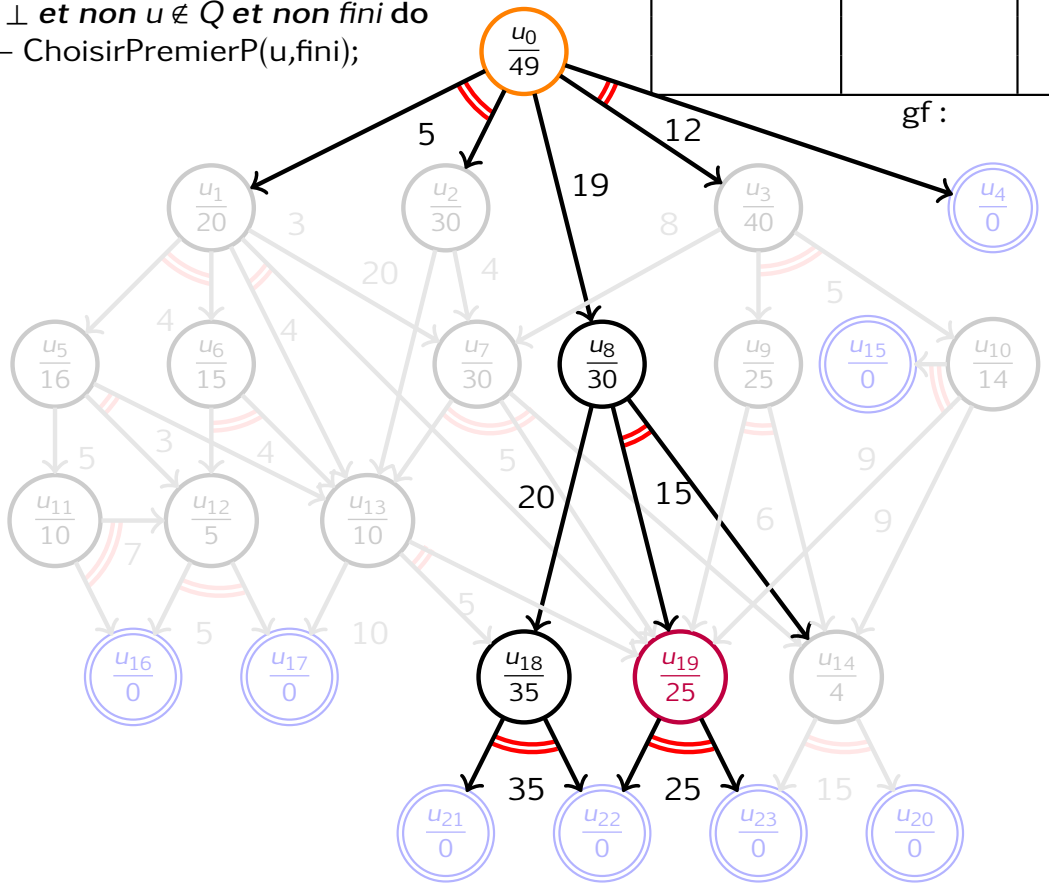

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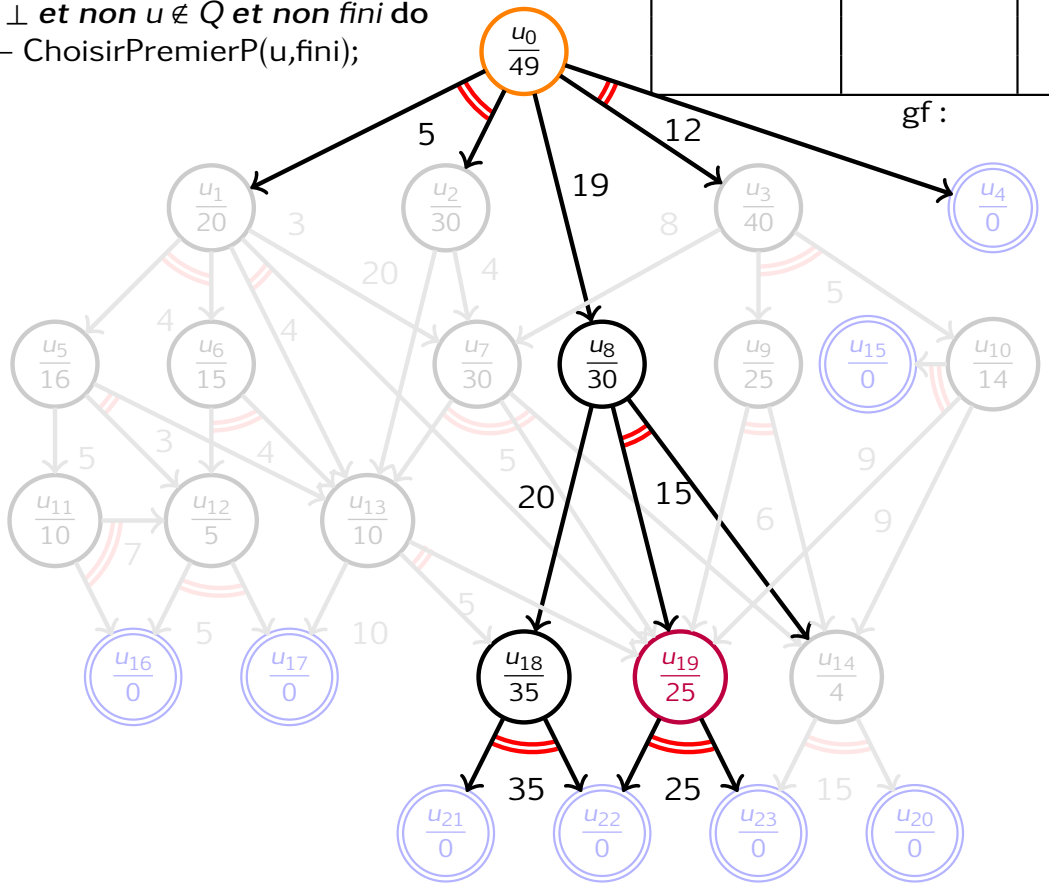

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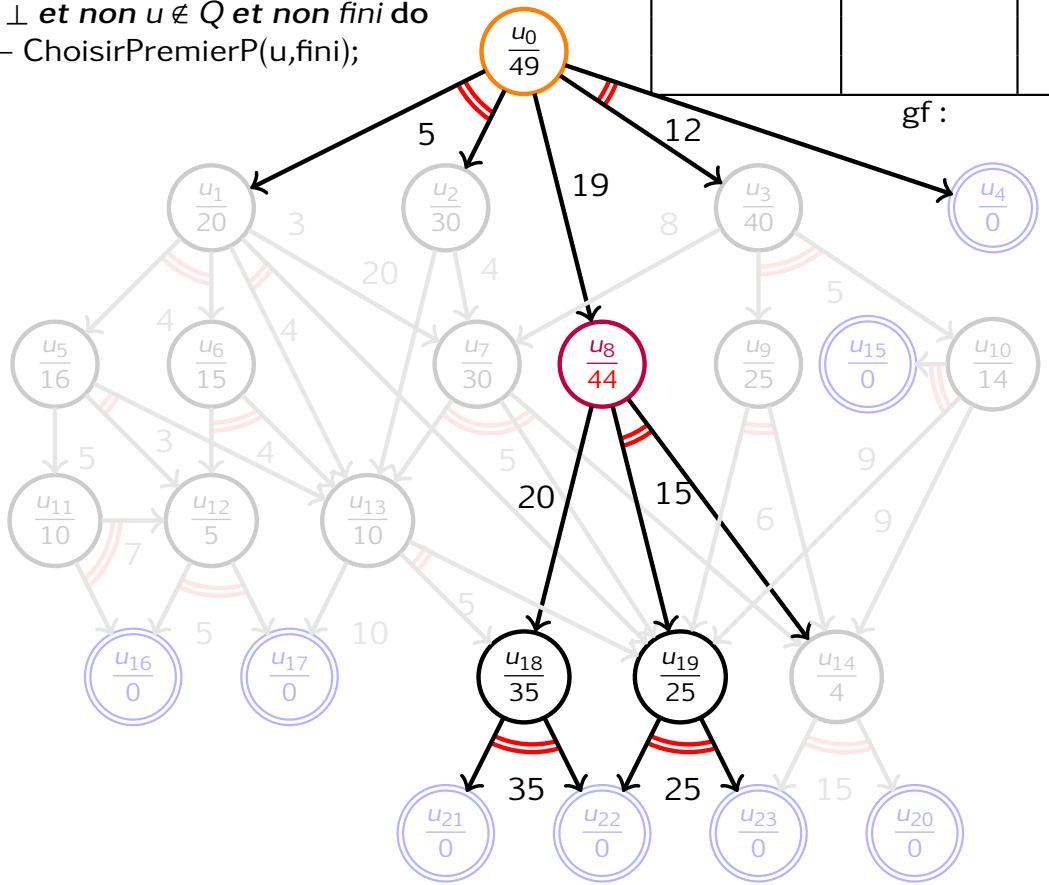
gf :



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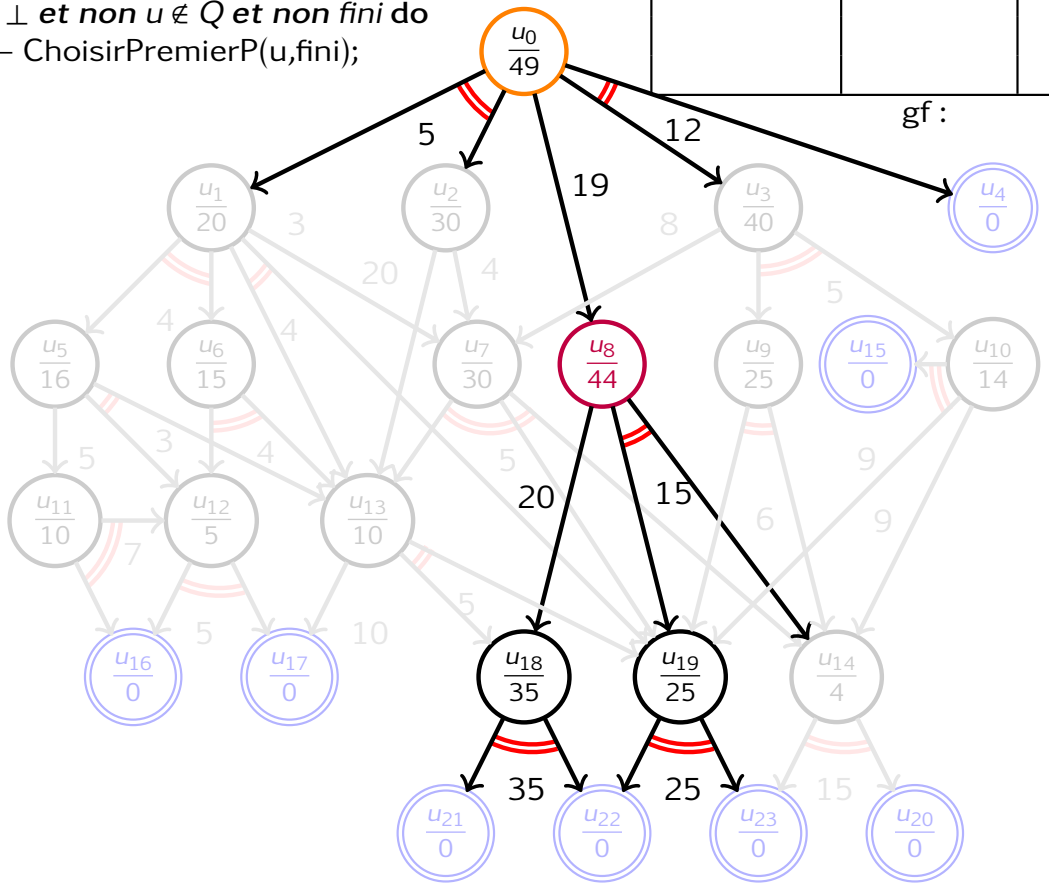
gf :



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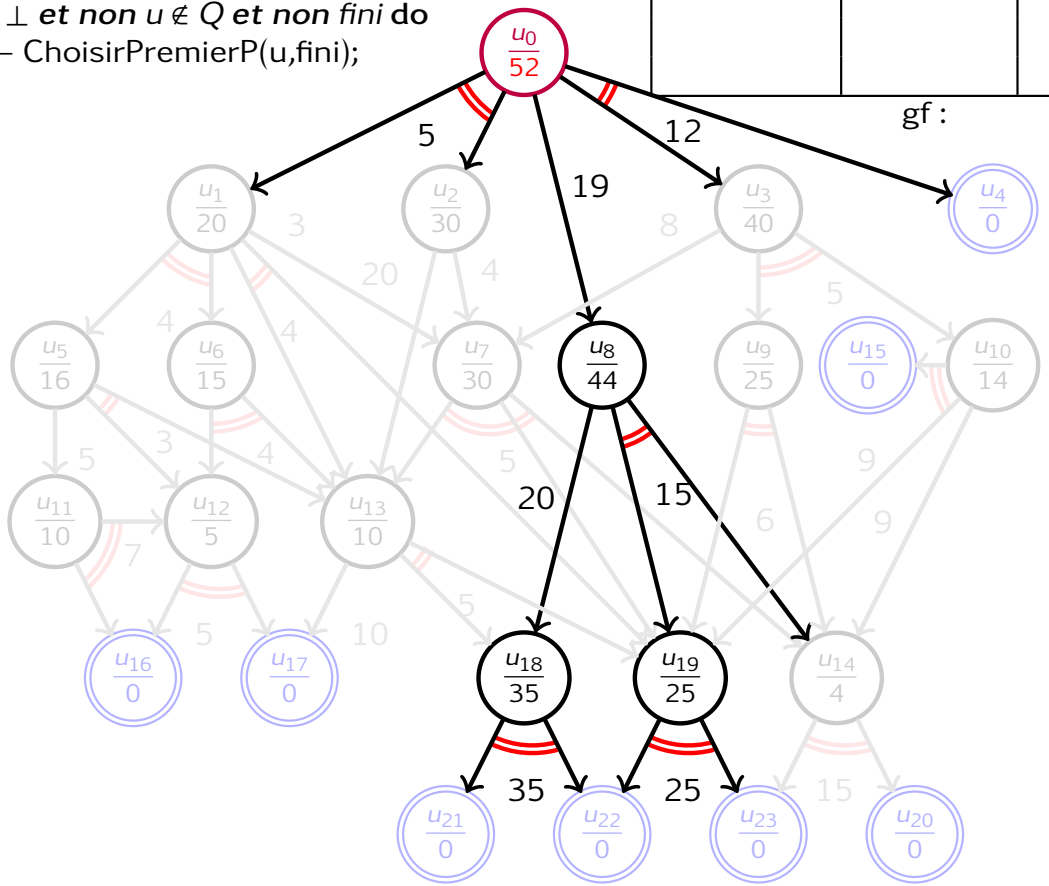
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,52)	(55,63,52)	{}	u_3(12,40) u_4(12,0)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)

gf :



Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
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  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
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  u ← ChoisirPremierP(u,fini);
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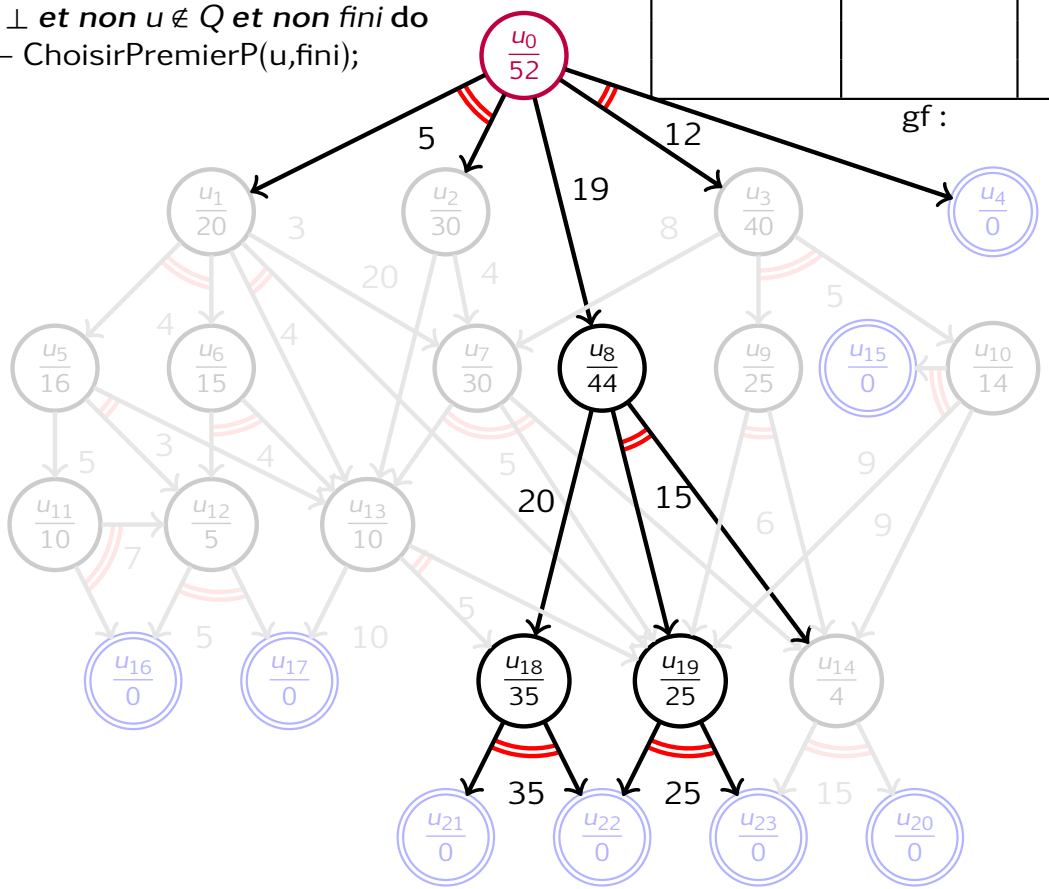
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
$u_0(0,52)$	(55,63,52)	{}	$u_3(12,40)$ $u_4(12,0)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$

gf :

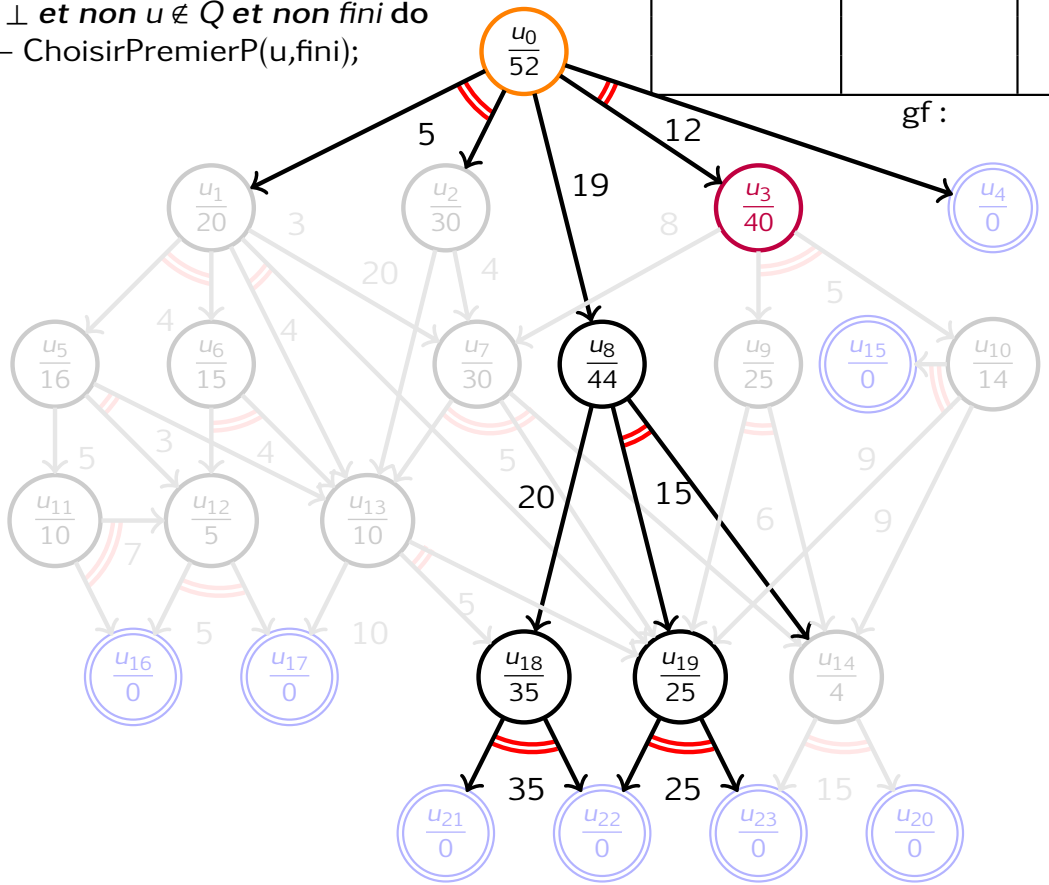


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
$u_0(0,52)$	(55,63,52)	{}	$u_3(12,40)$ $u_4(12,0)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,40)$	(-,-)	{}	$u_4(12,0)$

gf :



Algorithme OA*(Liste,Sol)

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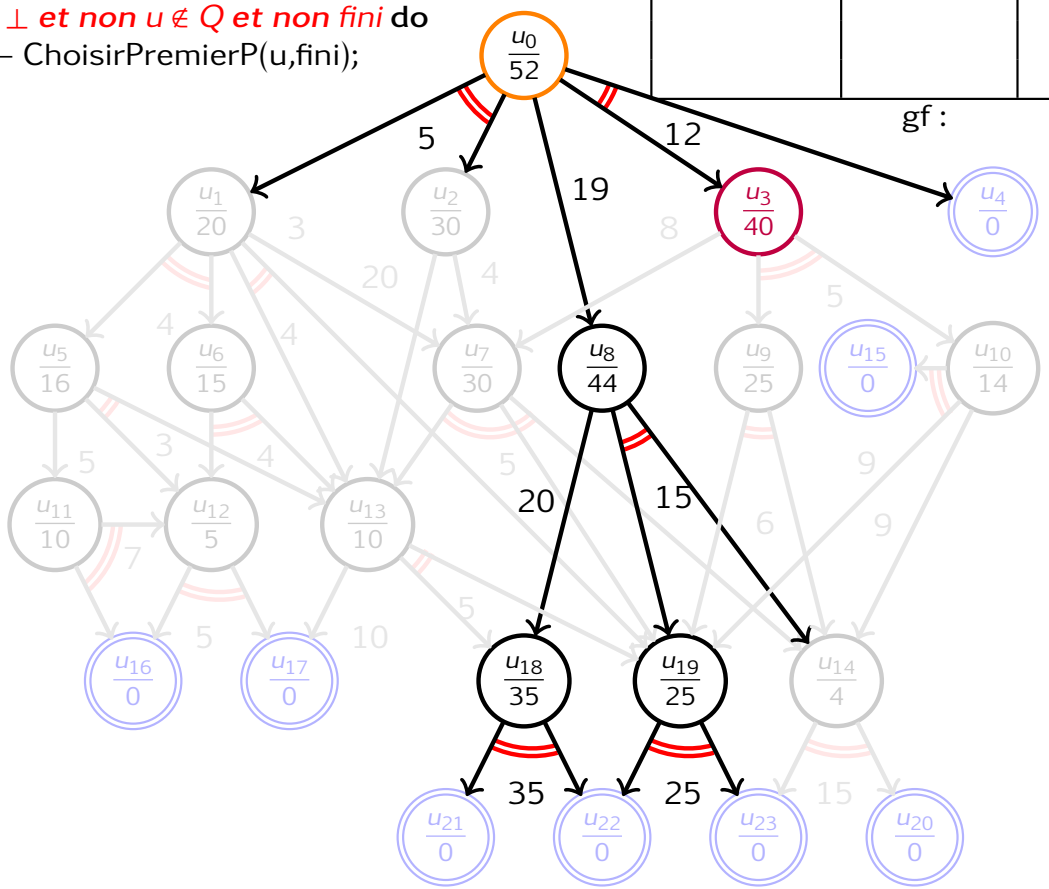
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,52)$	(55,63,52)	{}	$u_3(12,40)$ $u_4(12,0)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,40)$	(-,-)	{}	$u_4(12,0)$

gf :



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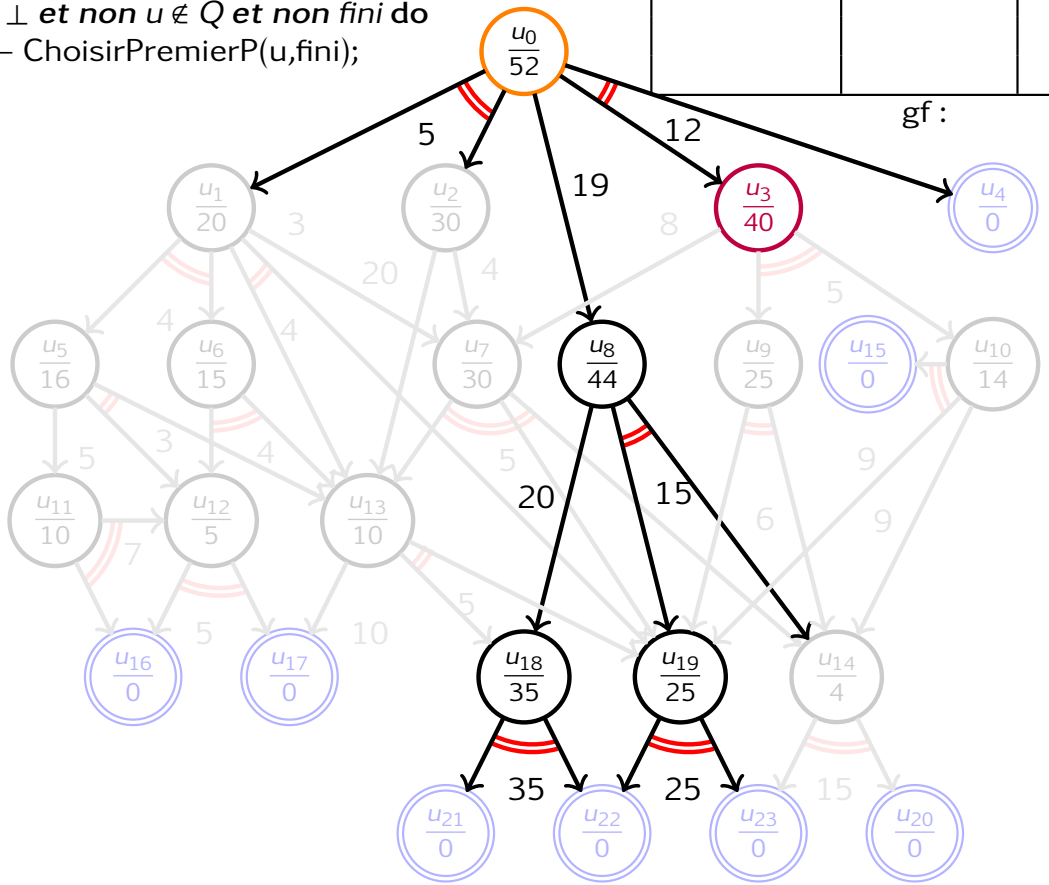
u(g(u),h(u))	fils(u)	A	P
$u_0(0,52)$	(55,63,52)	{}	$u_3(12,40)$ $u_4(12,0)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,40)$	(-,-)	{}	$u_4(12,0)$

gf :



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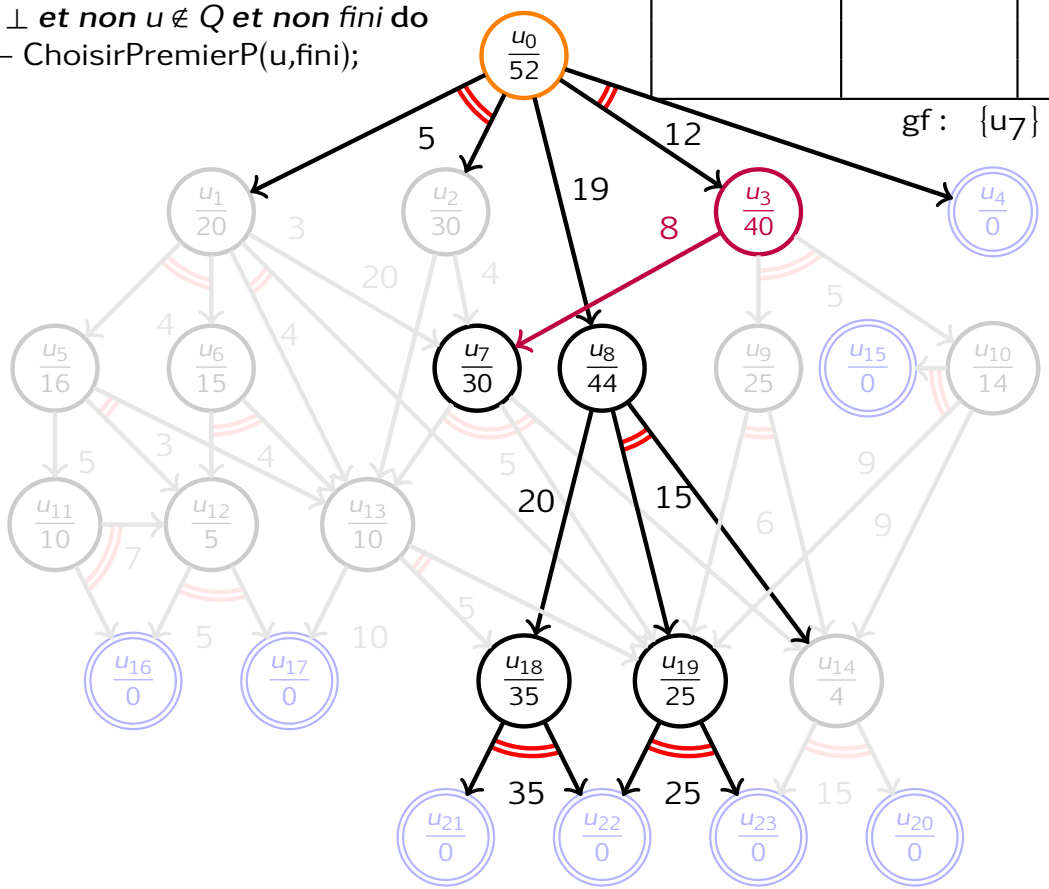
u(g(u),h(u))	fils(u)	A	P
$u_0(0,52)$	(55,63,52)	{}	$u_3(12,40)$ $u_4(12,0)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,40)$	(-,-)	{}	$u_4(12,0)$

gf : $\{u_7\}$



Algorithme OA*(Liste,Sol)

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u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
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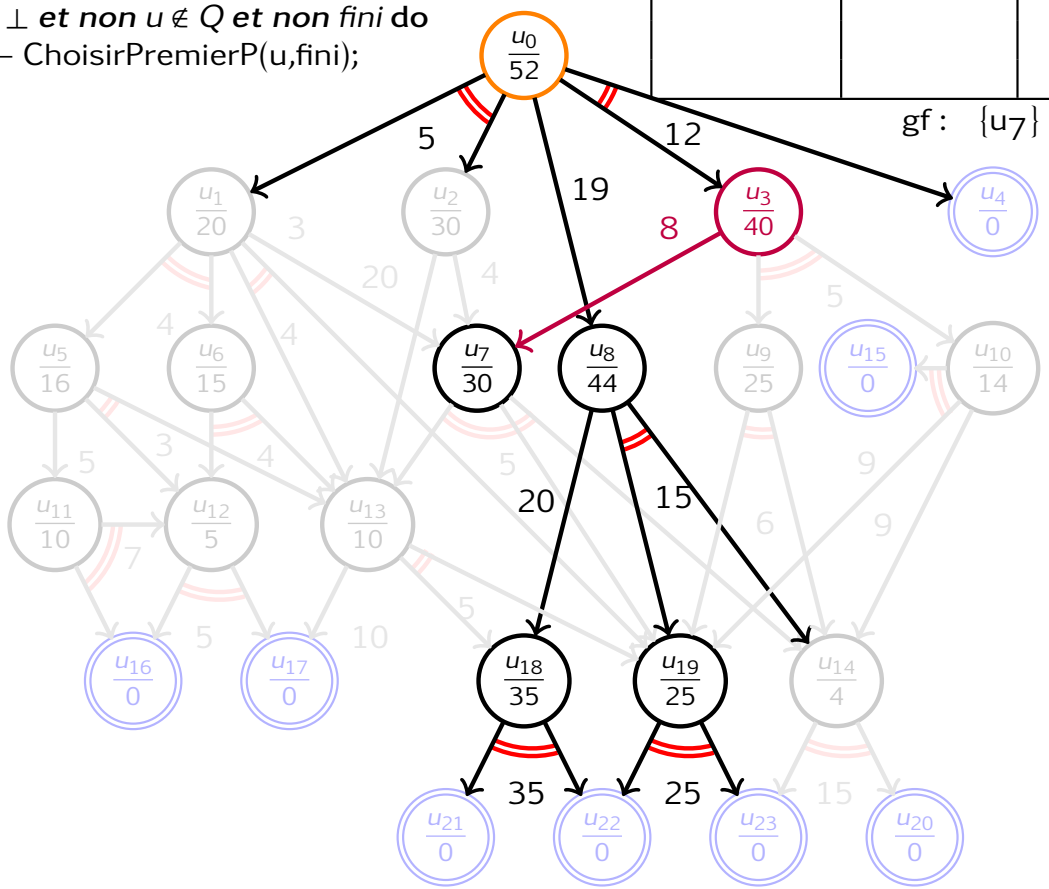
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,52)$	(55,63,52)	{}	$u_3(12,40)$ $u_4(12,0)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,40)$	(-,-)	{}	$u_4(12,0)$

gf : $\{u_7\}$



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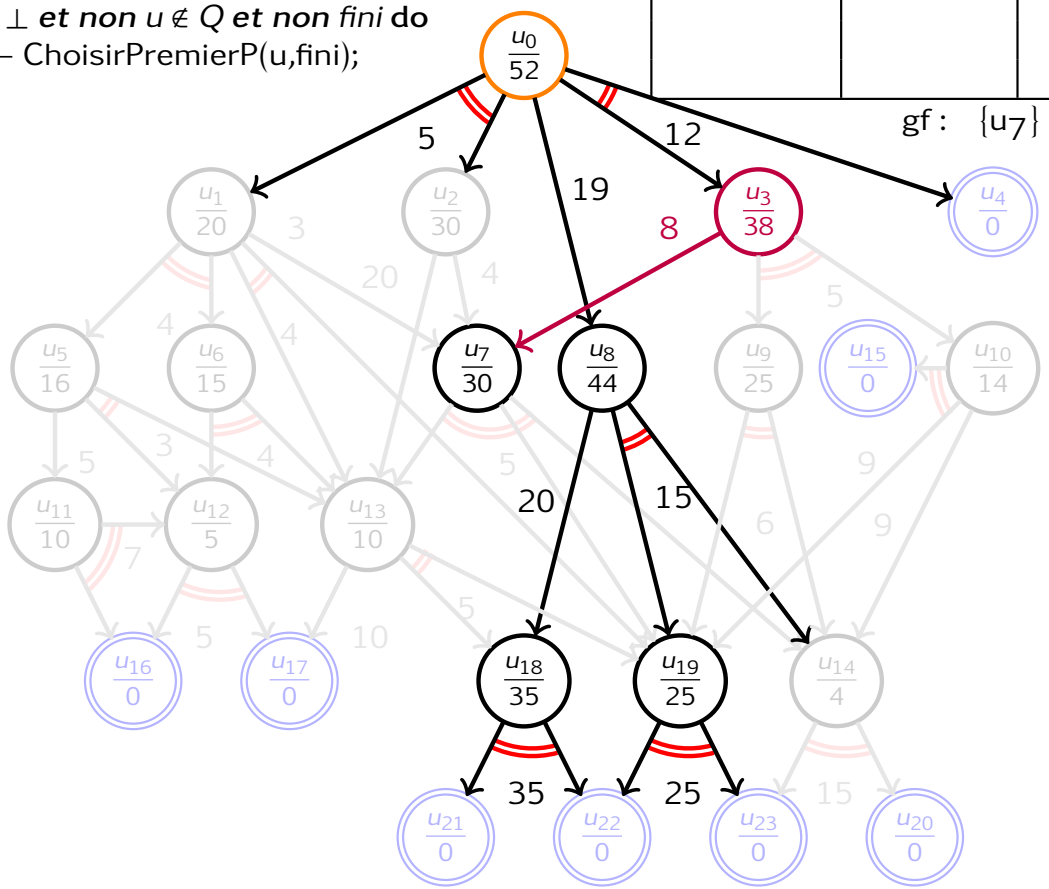
u(g(u),h(u))	fils(u)	A	P
u_0(0,52)	(55,63,52)	{}	u_3(12,38) u_4(12,0)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,38)	(38,-)	{u_0}	u_7(20,30) u_4(12,0)

gf : {u_7} = 8+30 = 38



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u(g(u),h(u))	fils(u)	A	P
$u_0(0,52)$	(55,63,52)	{}	$u_3(12,38)$ $u_4(12,0)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
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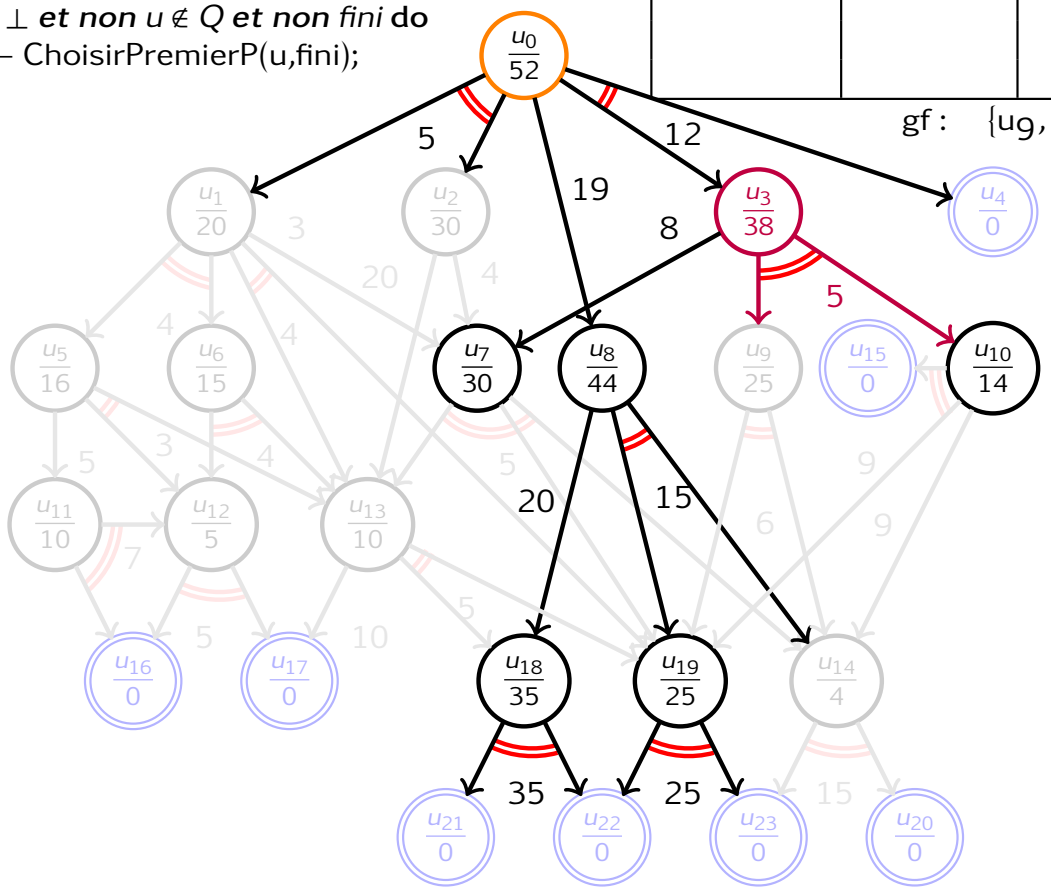
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,38)$	(38,-)	$\{u_0\}$	$u_7(20,30)$ $u_4(12,0)$

gf : $\{u_9, u_{10}\}$



Algorithme OA*(Liste,Sol)

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        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,52)$	(55,63,52)	{}	$u_3(12,38)$ $u_4(12,0)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
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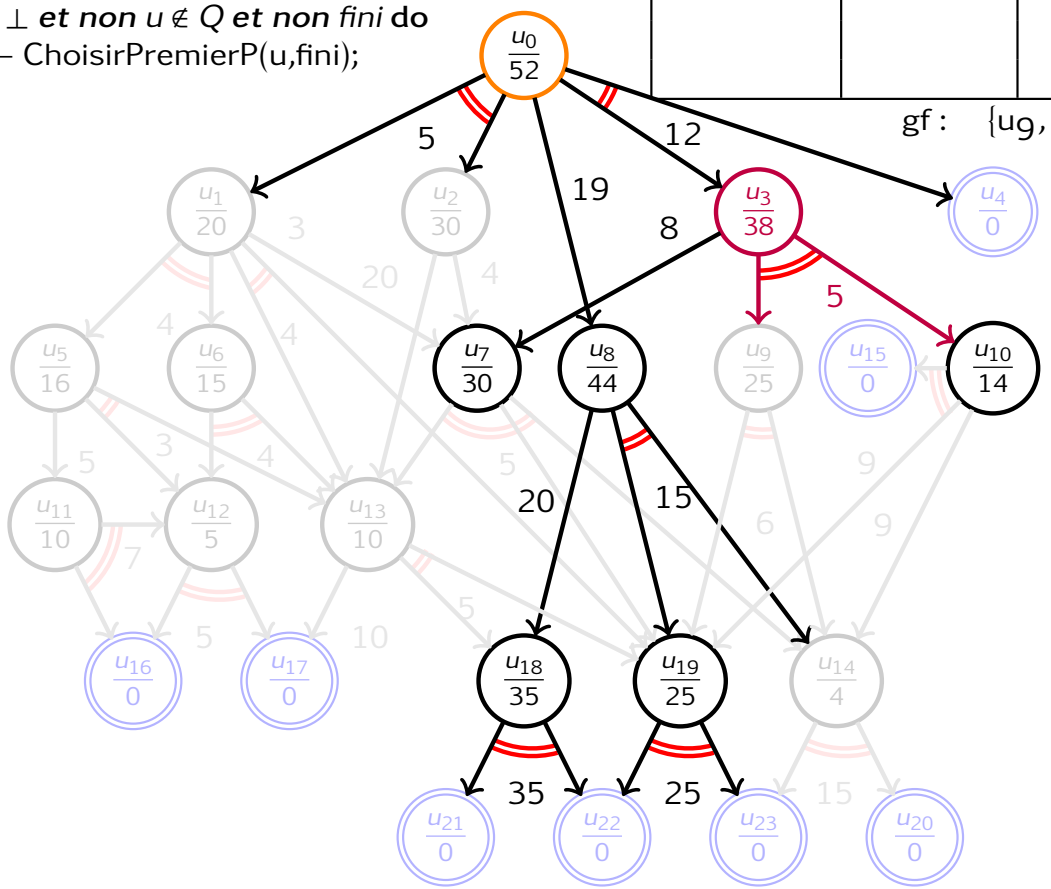
$u_3(12,38)$	(38,-)	$\{u_0\}$	$u_7(20,30)$ $u_4(12,0)$

gf : $\{u_9, u_{10}\}$



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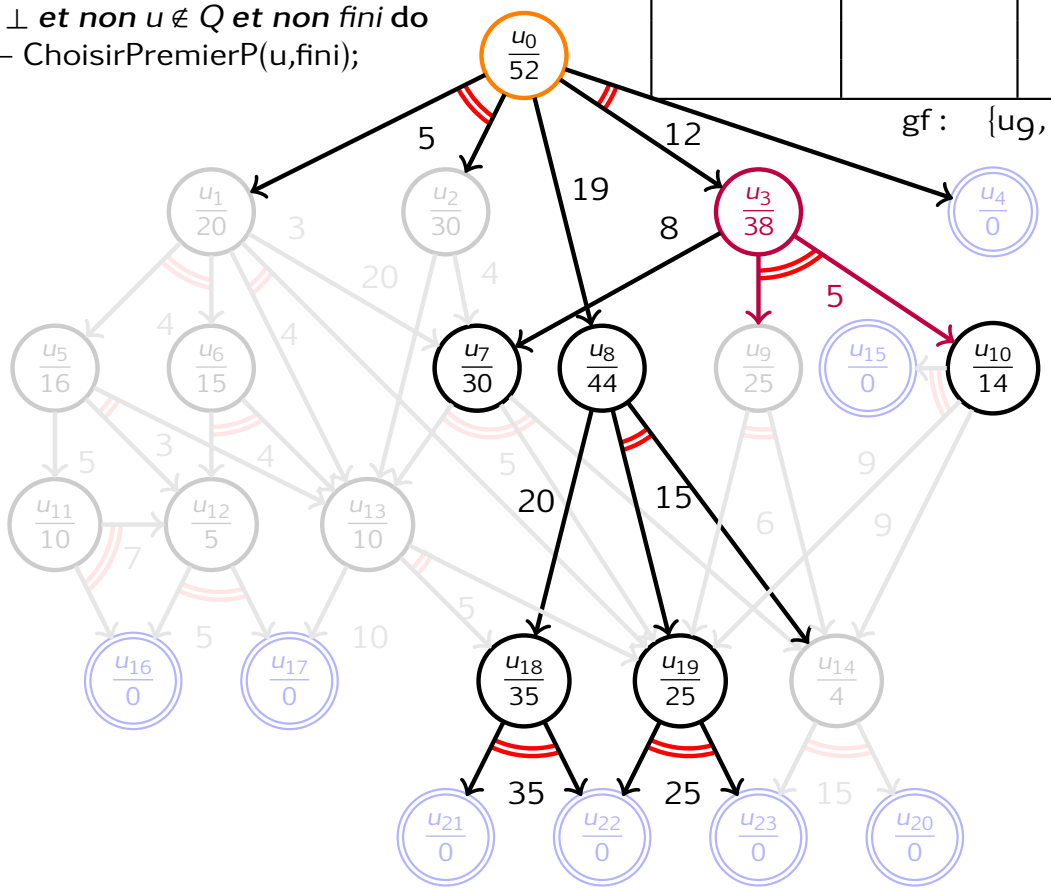
u(g(u),h(u))	fils(u)	A	P
$u_0(0,52)$	(55,63,52)	{}	$u_3(12,38)$ $u_4(12,0)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,38)$	(38,44)	$\{u_0\}$	$u_7(20,30)$ $u_4(12,0)$

gf : $\{u_9, u_{10}\}$ = 5+25+14 = 44



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```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,52)$	(55,63,52)	{}	$u_3(12,38)$ $u_4(12,0)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
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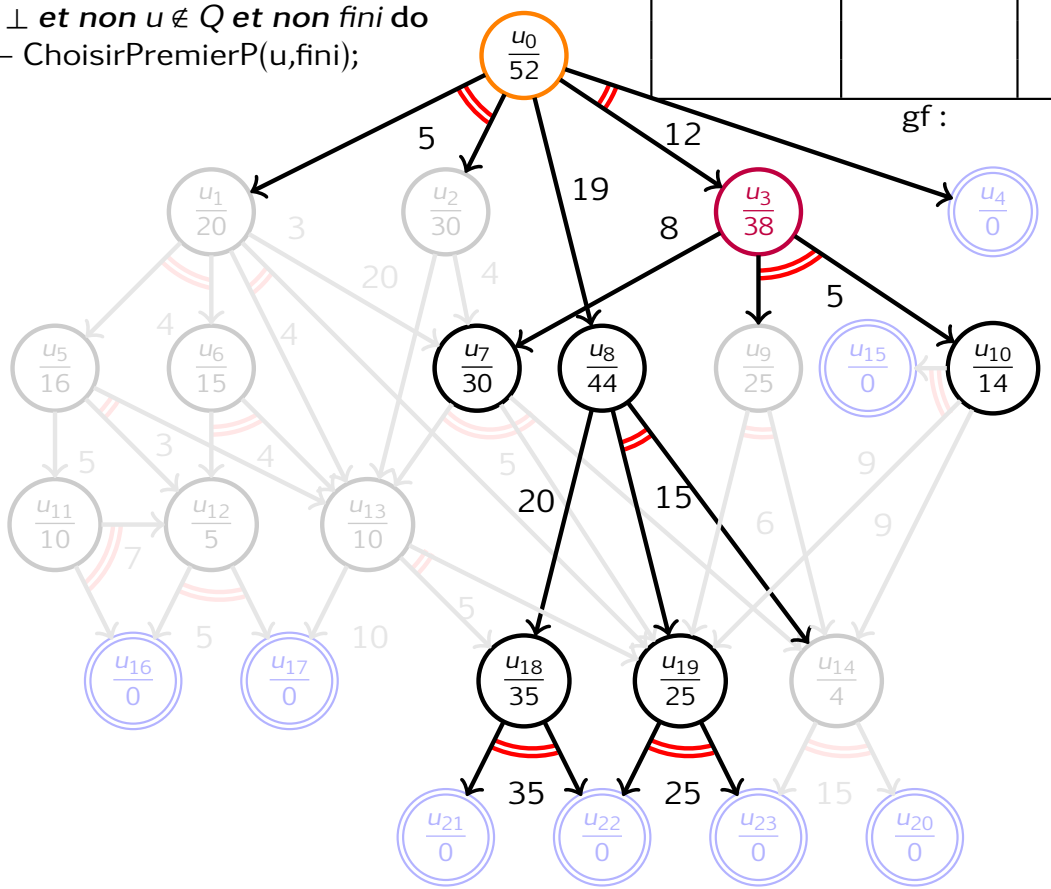
$u_3(12,38)$	(38,44)	$\{u_0\}$	$u_7(20,30)$ $u_4(12,0)$

gf :



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```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,52)$	(55,63,52)	{}	$u_3(12,38)$ $u_4(12,0)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
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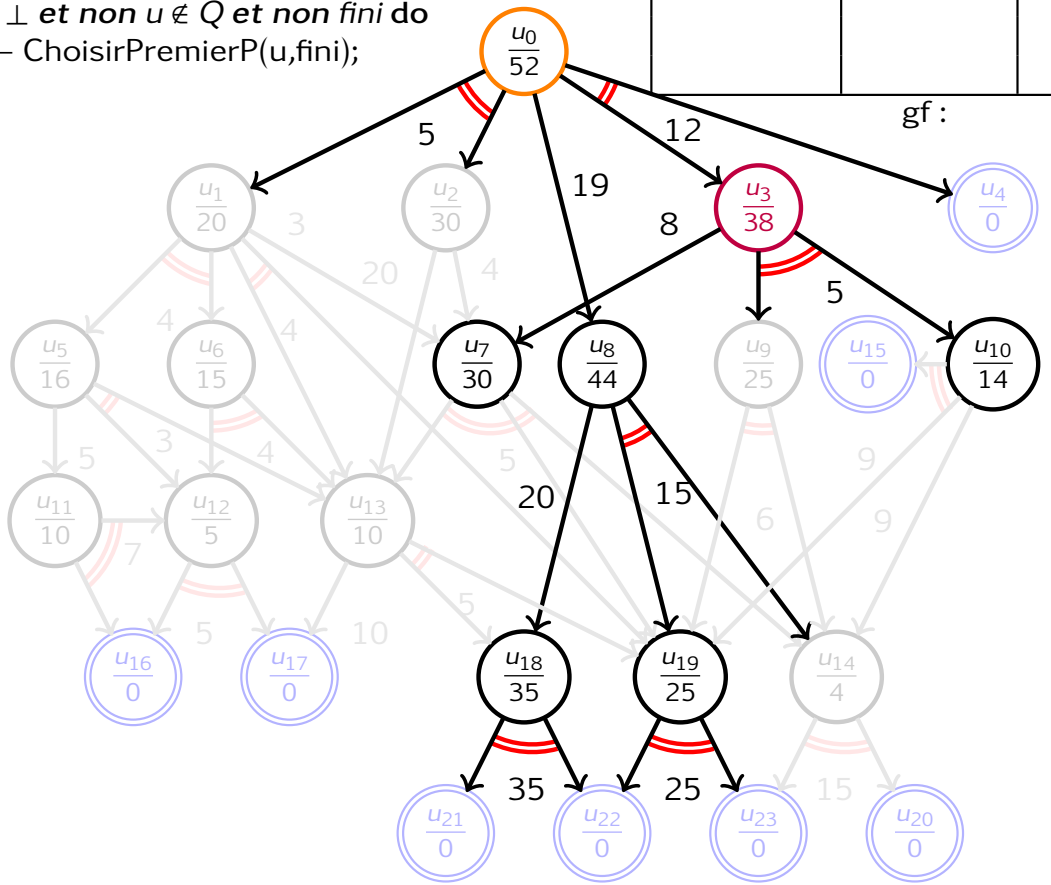
$u_3(12,38)$	(38,44)	$\{u_0\}$	$u_7(20,30)$ $u_4(12,0)$

gf :



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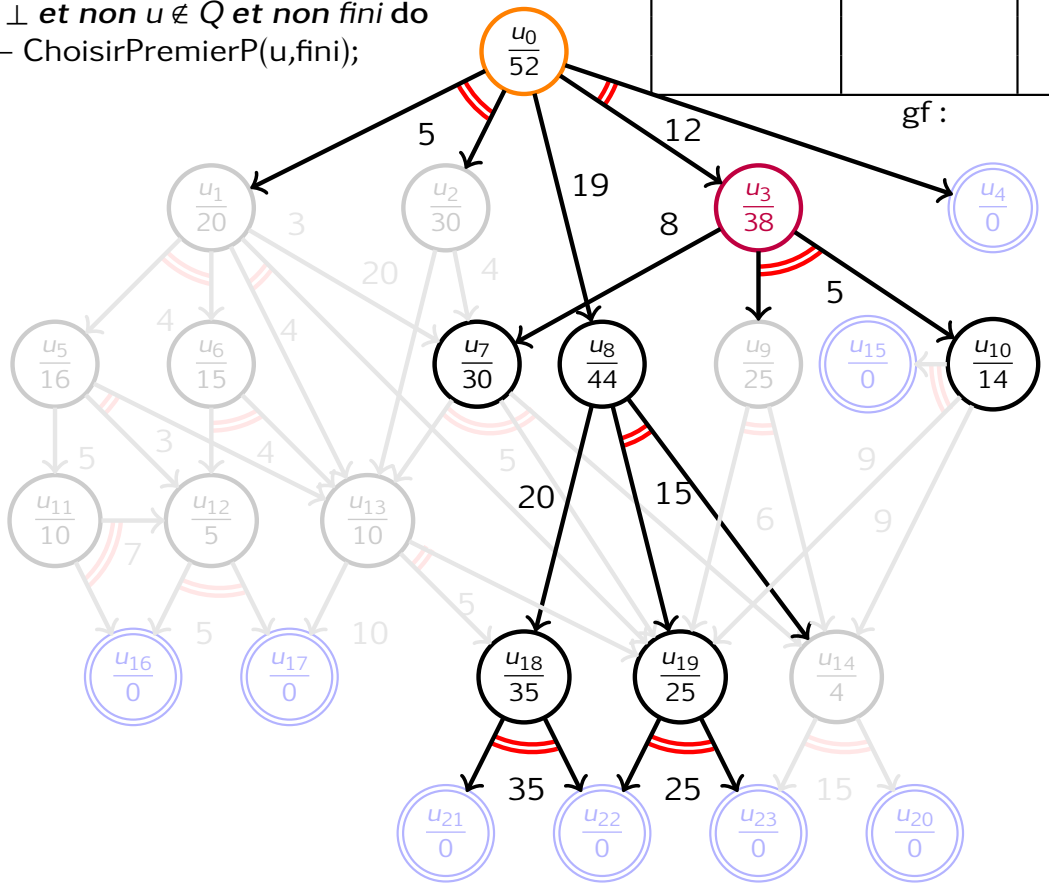
$u_3(12,38)$	(38,44)	$\{u_0\}$	$u_7(20,30)$ $u_4(12,0)$

gf :



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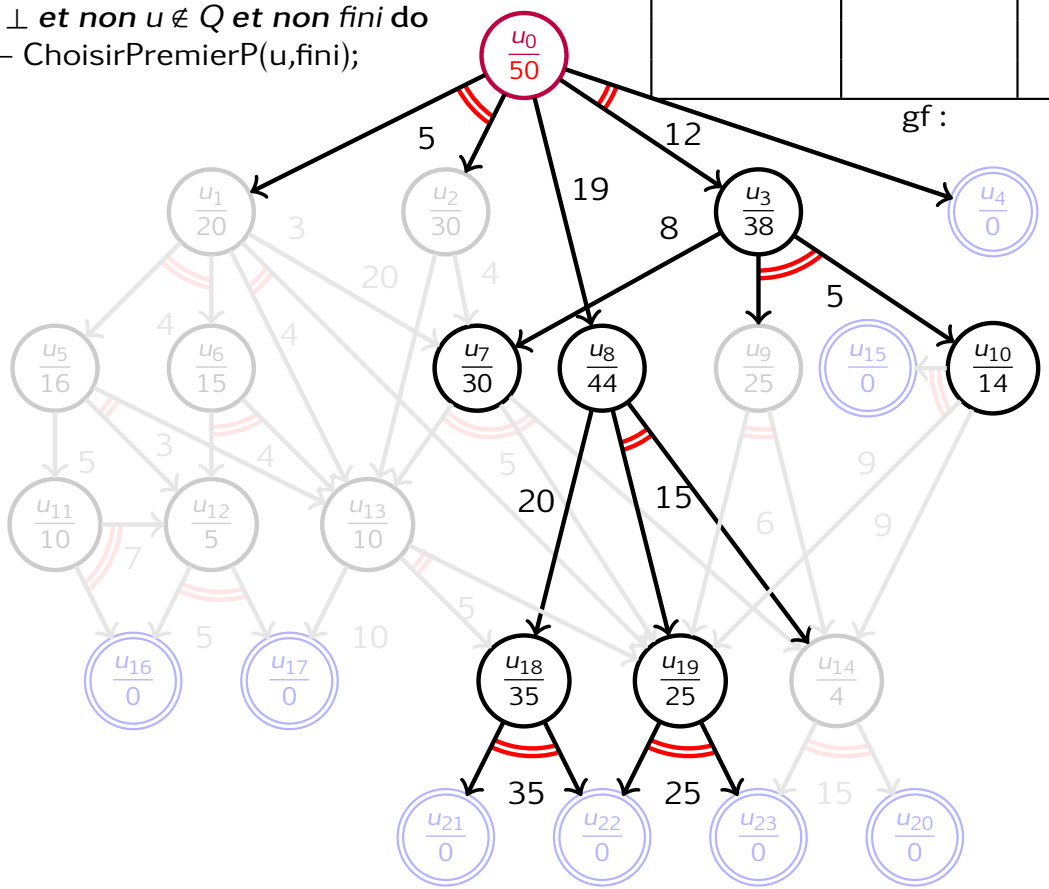
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
$u_0(0,50)$	(55,63,50)	{}	$u_3(12,38)$ $u_4(12,0)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,38)$	(38,44)	$\{u_0\}$	$u_7(20,30)$ $u_4(12,0)$

gf :



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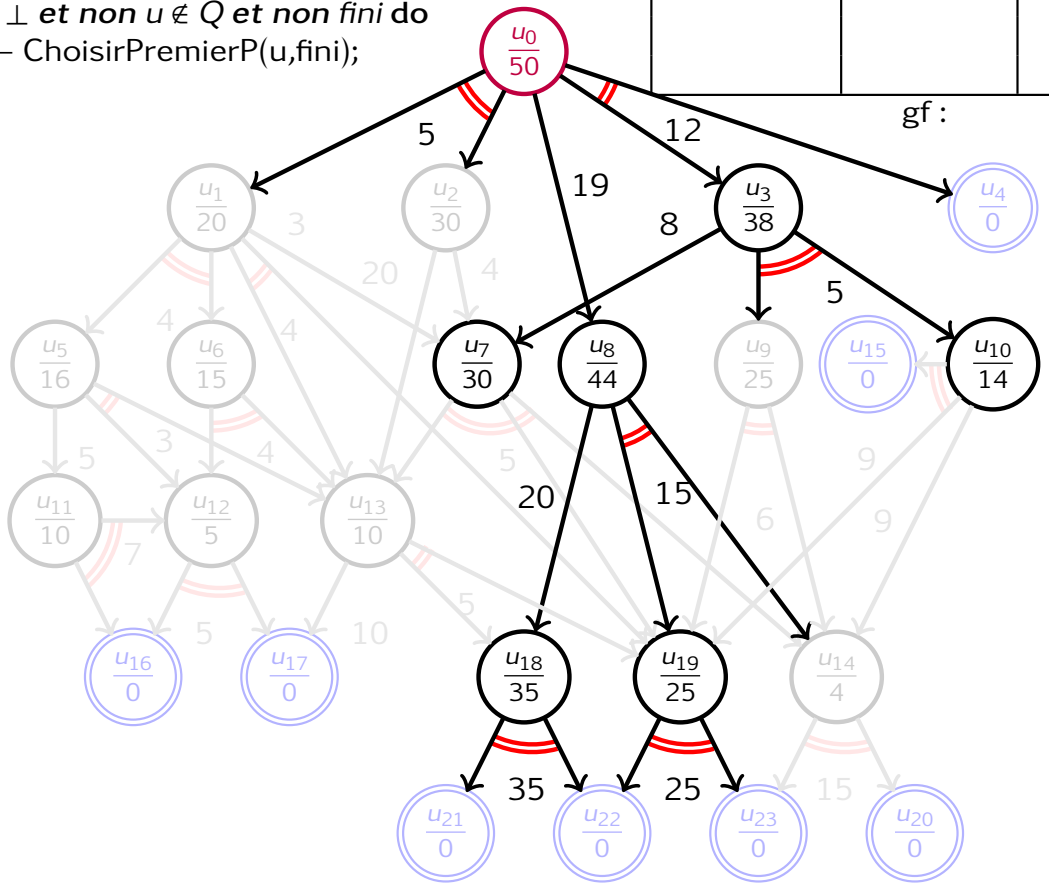
$u_3(12,38)$	(38,44)	{}	$u_7(20,30)$ $u_4(12,0)$

gf :



Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
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  while *gf ≠ ∅* **do**
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    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
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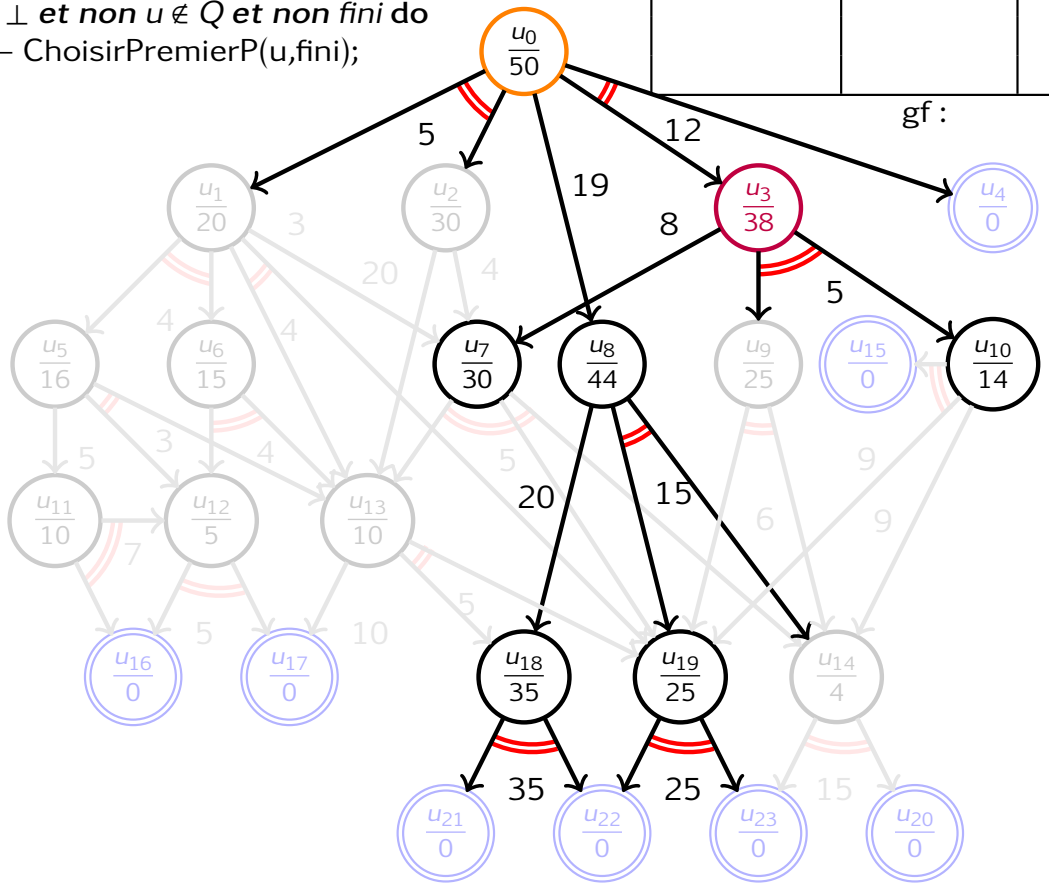
u(g(u),h(u))	fils(u)	A	P
$u_0(0,50)$	(55,63,50)	{}	$u_3(12,38)$ $u_4(12,0)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,38)$	(38,44)	{}	$u_7(20,30)$ $u_4(12,0)$

gf :



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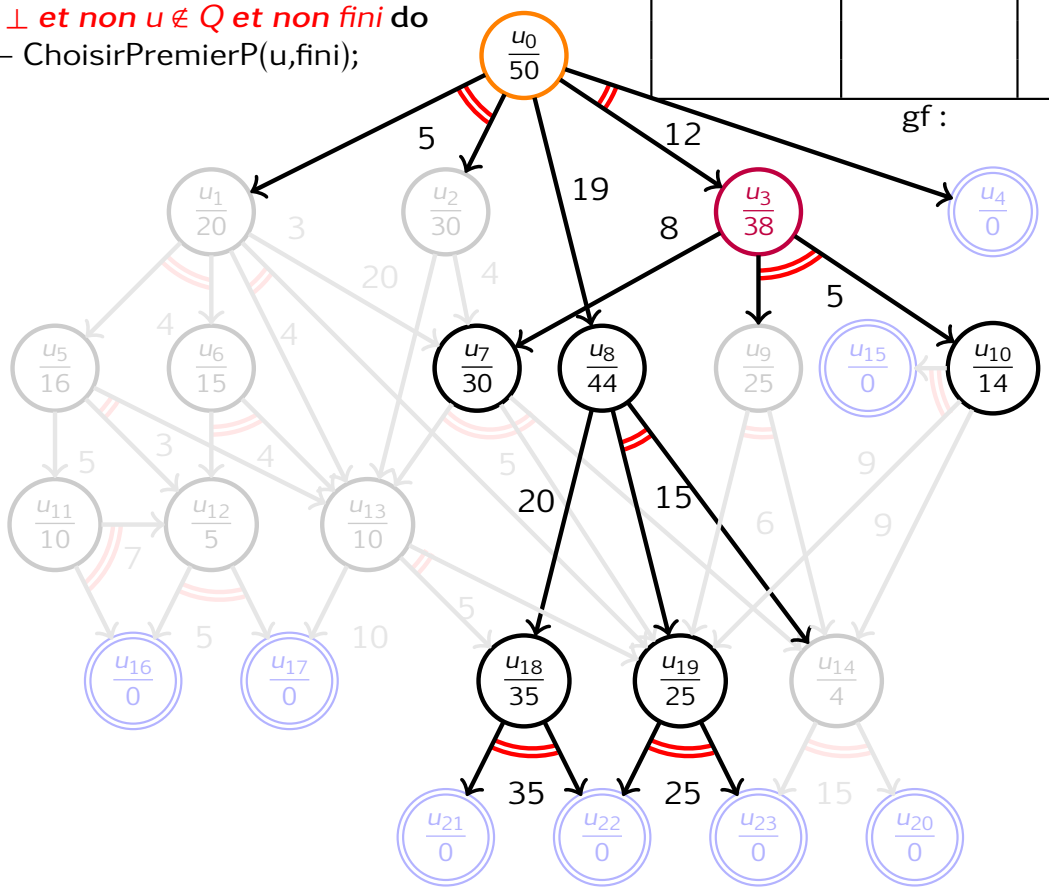

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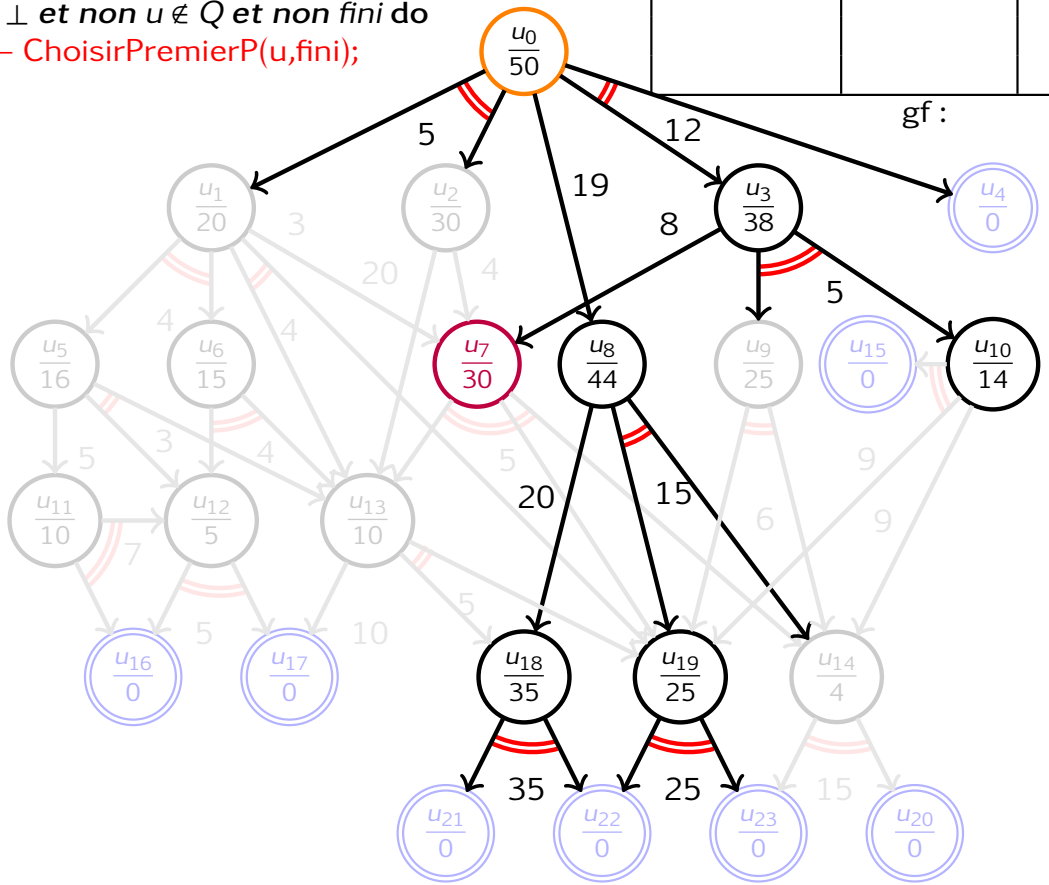
$u_3(12,38)$	(38,44)	{}	$u_7(20,30)$ $u_4(12,0)$
$u_7(20,30)$	(-)	{}	

gf :



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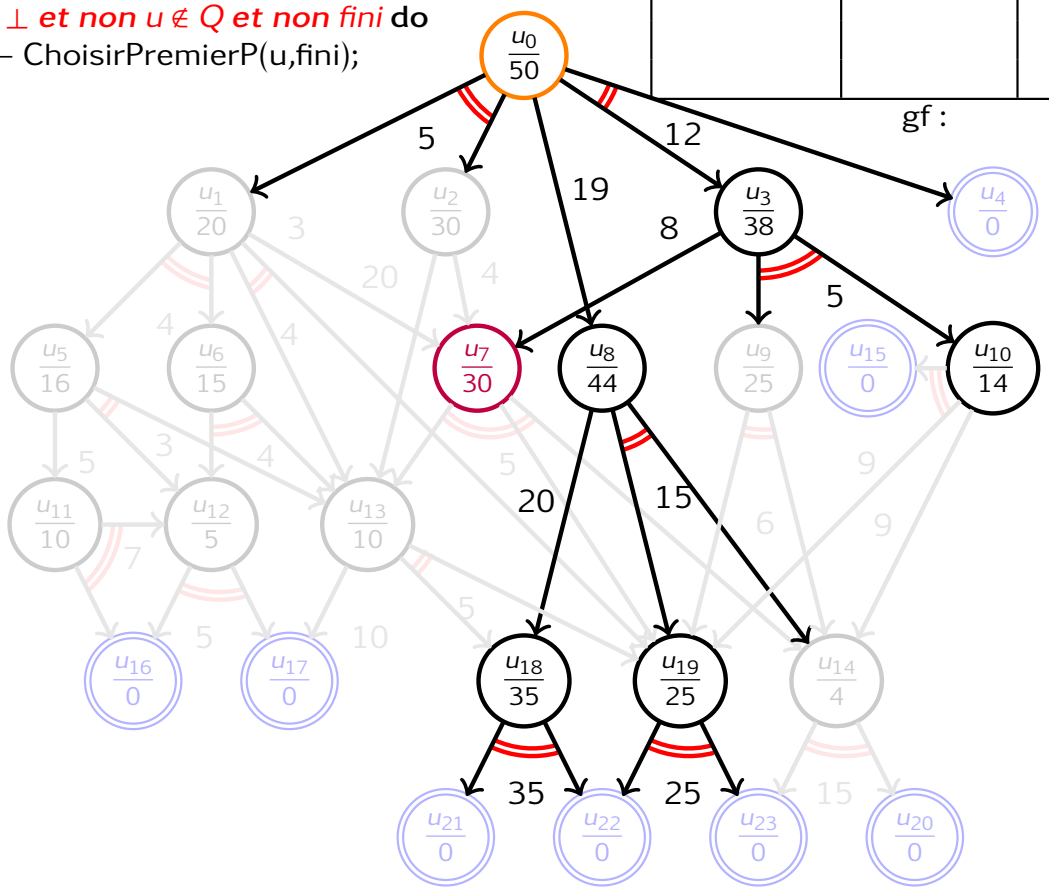

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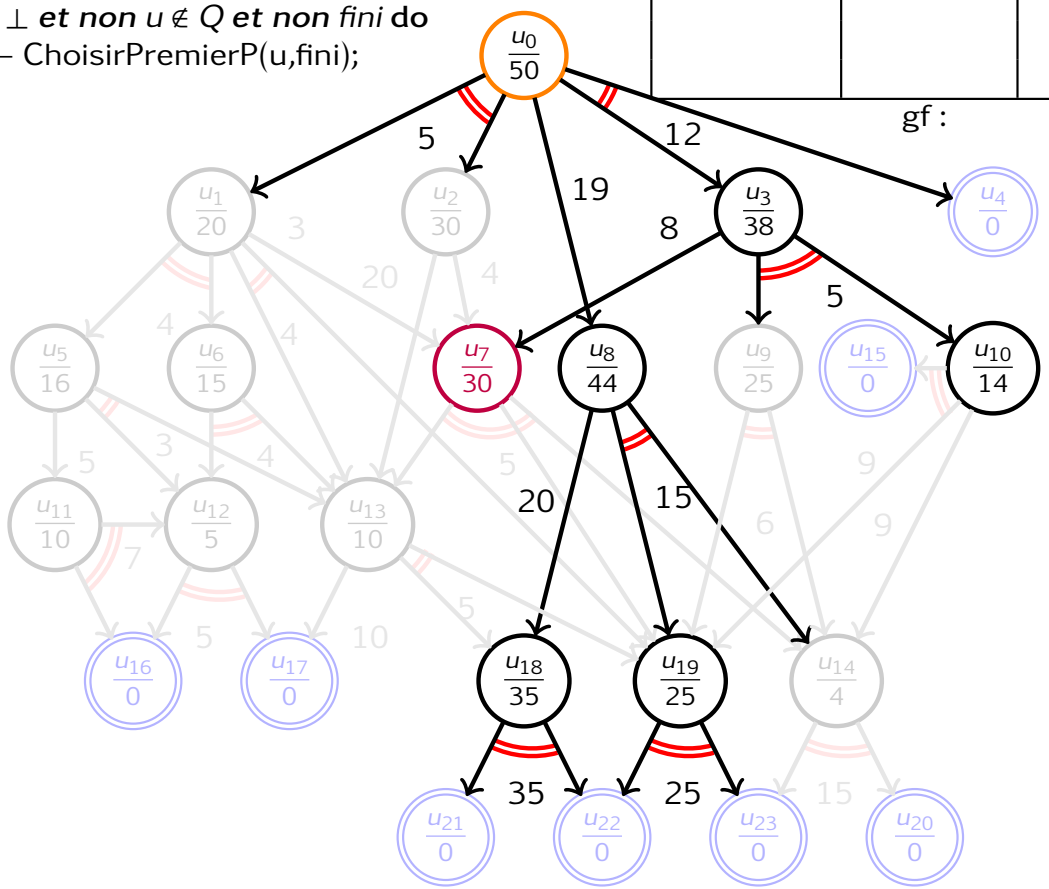

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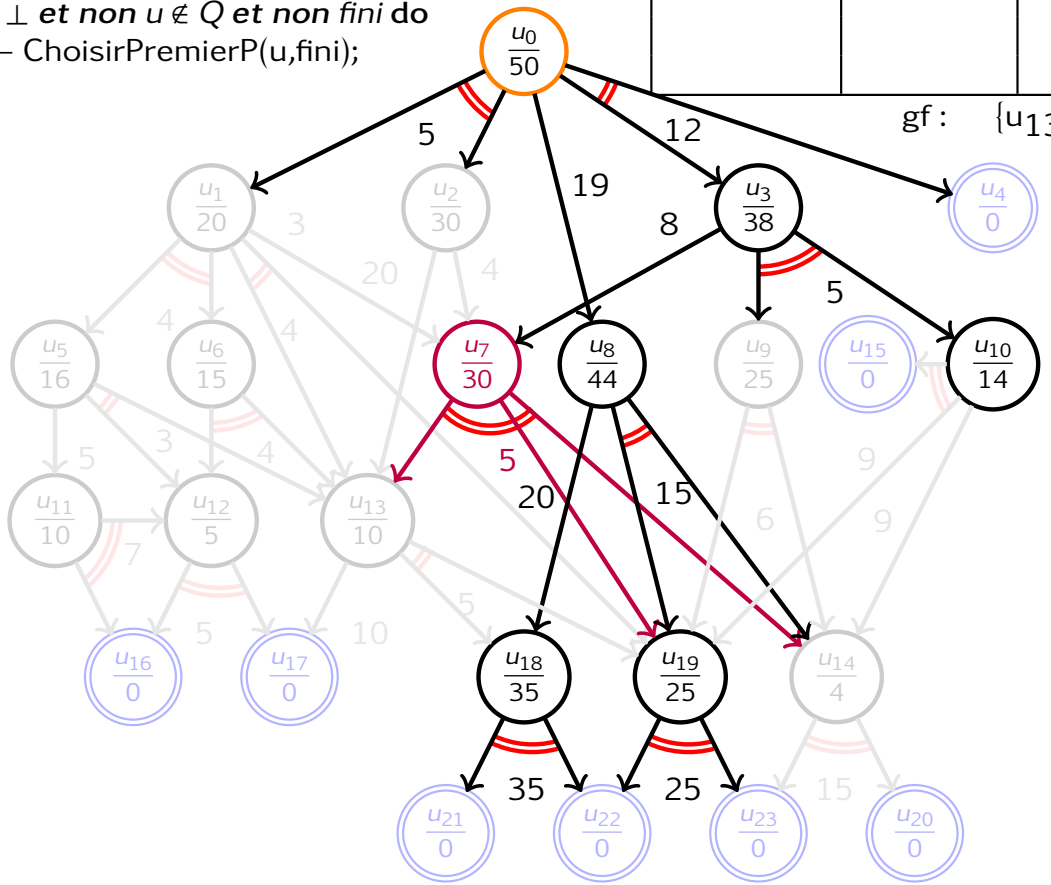
gf : $\{u_{13}, u_{19}\ u_{14}\}$



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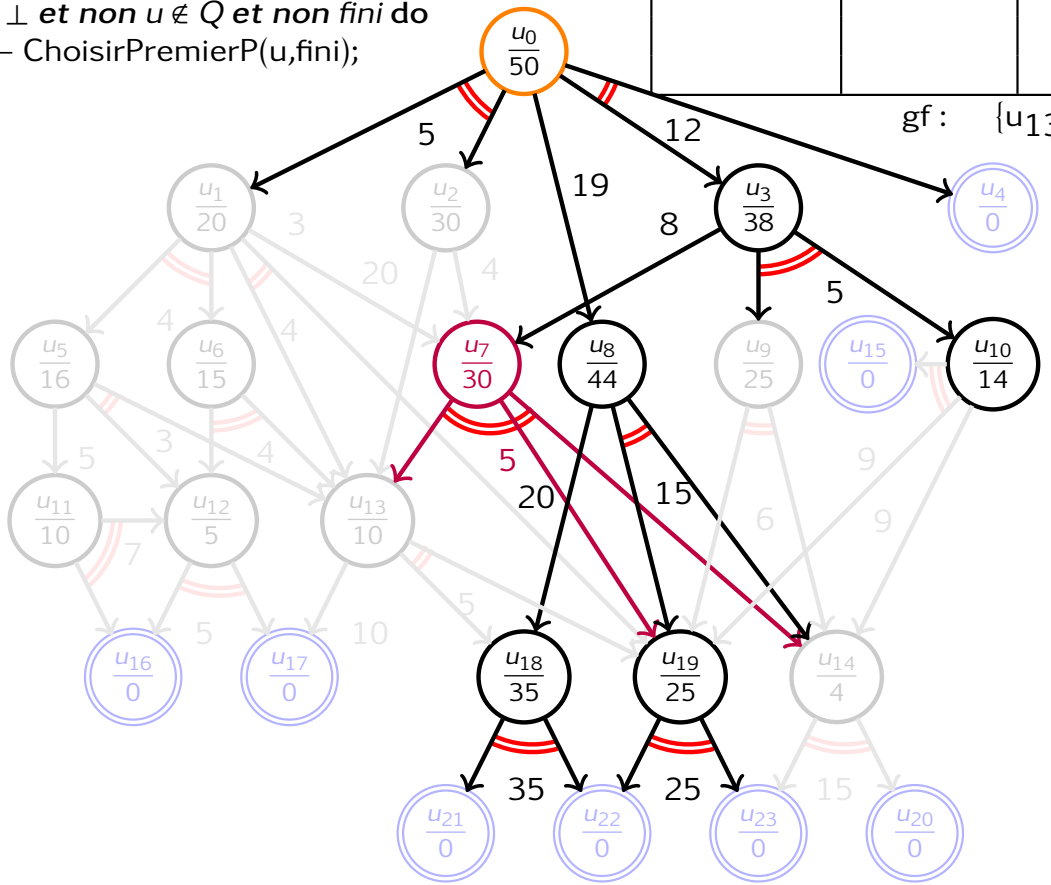

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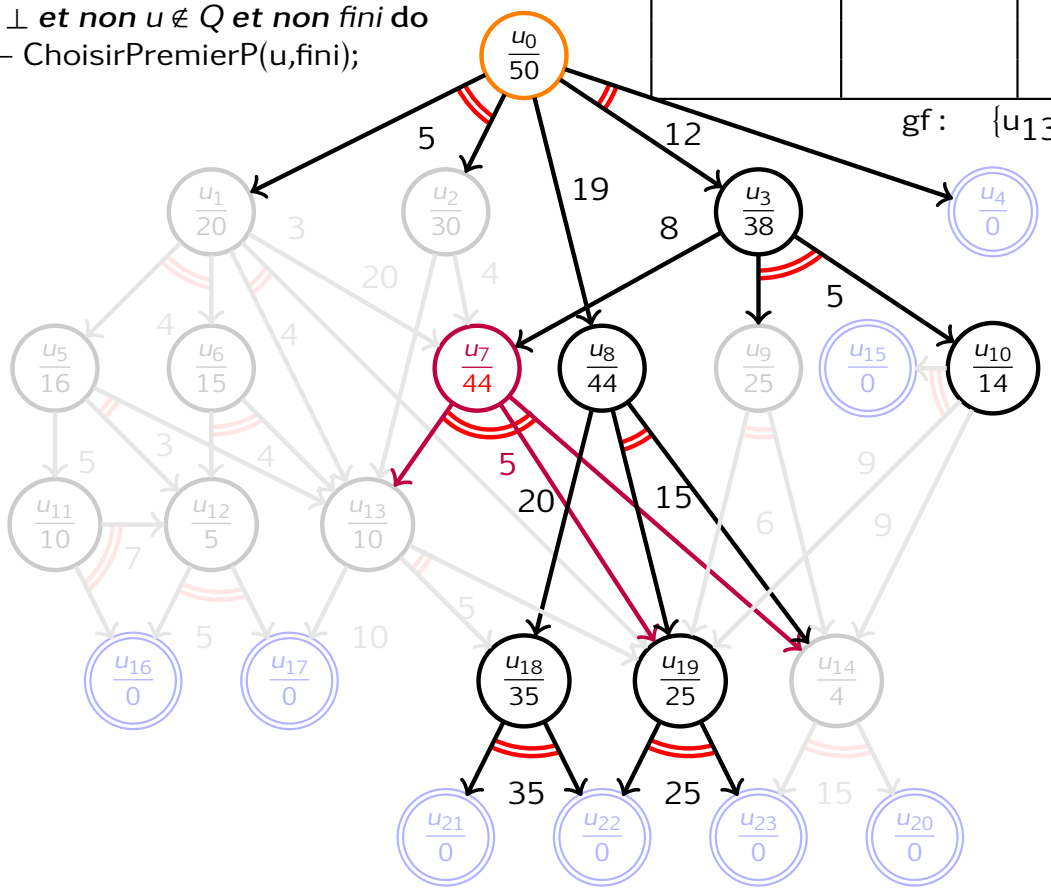
gf : $\{u_{13}, u_{19}\ u_{14}\}$ = 5+10+25+4 = 44



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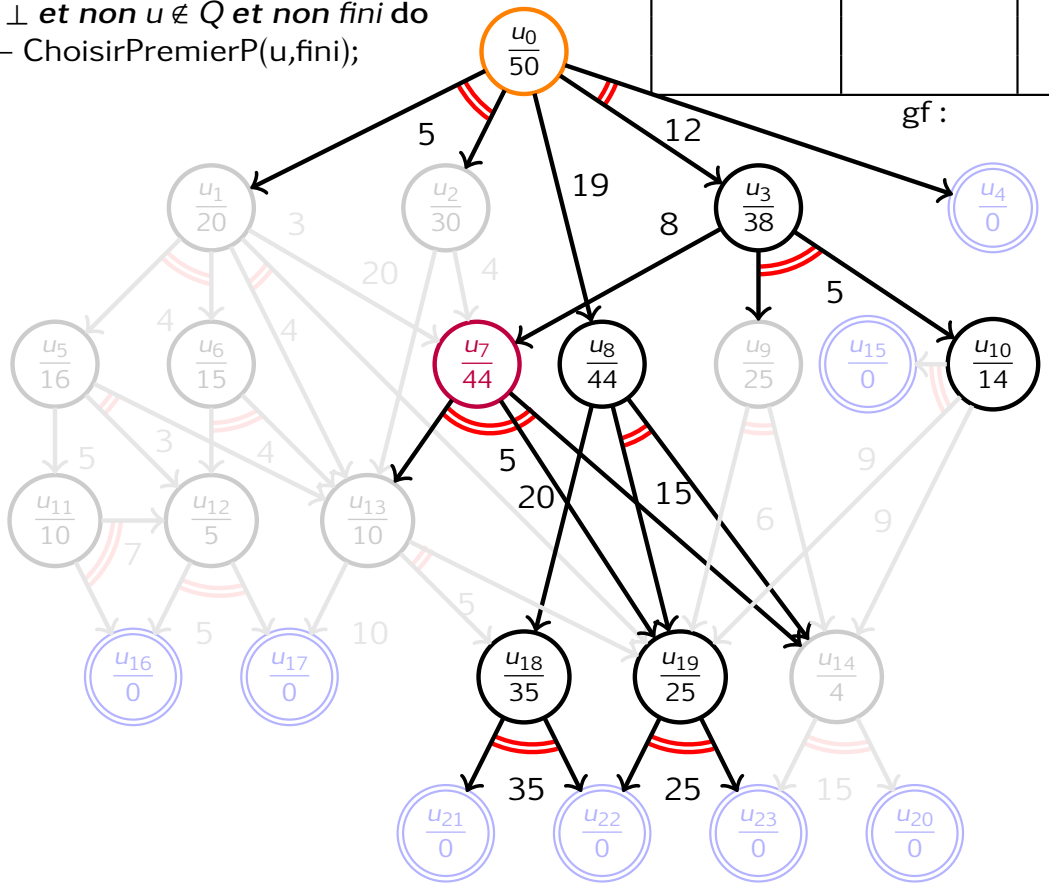

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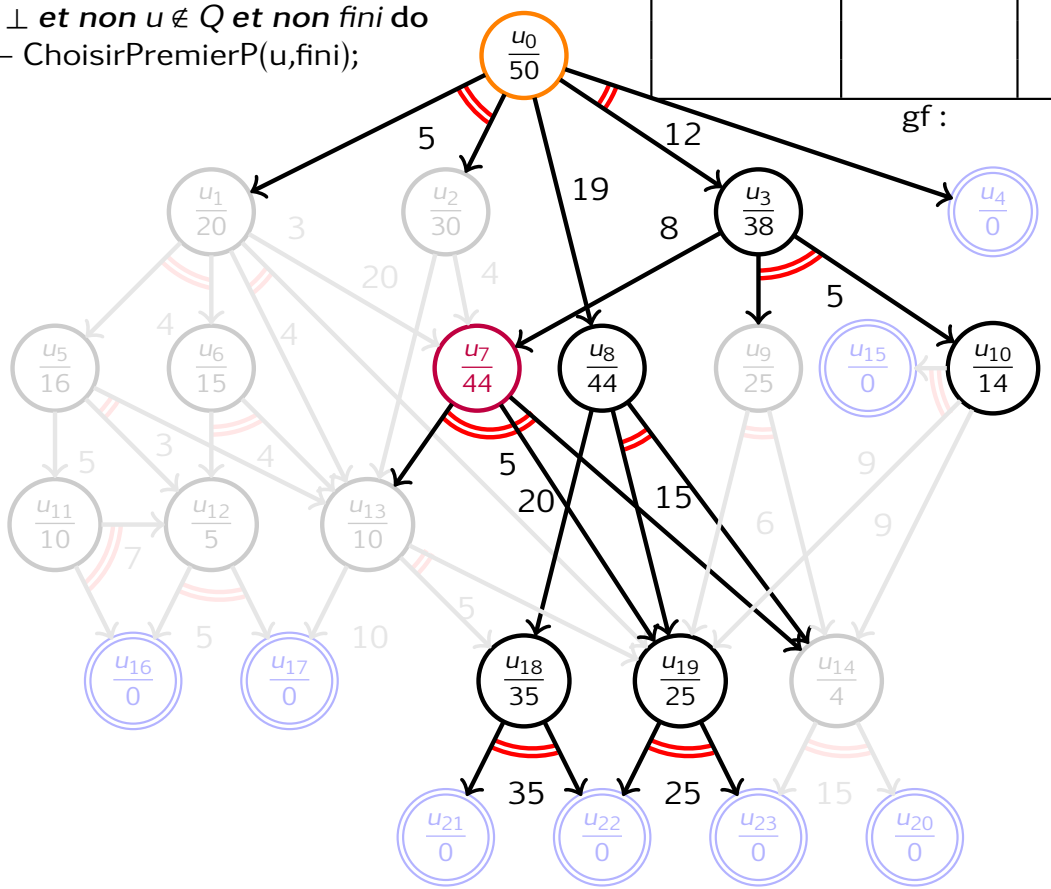

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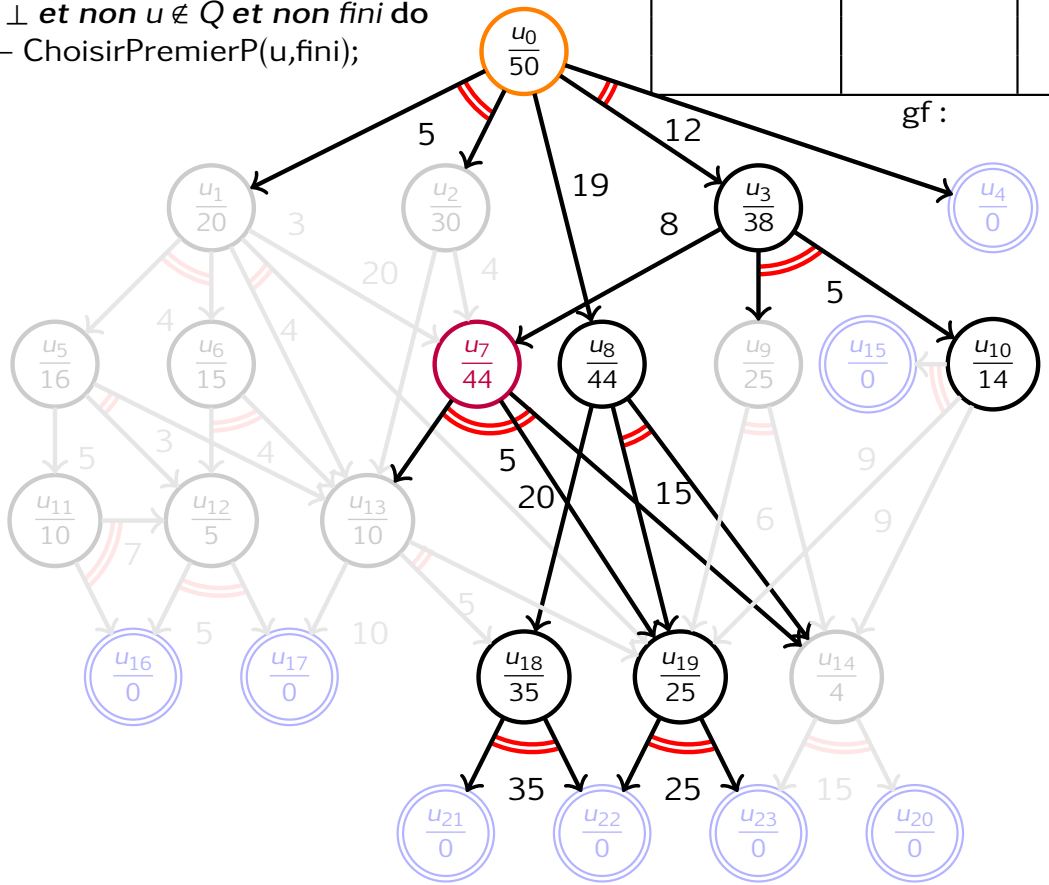

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u_3(12,44)	(52,44)	{u_0}	u_9(17,25) u_{10}(17,14)u_4(12,0)
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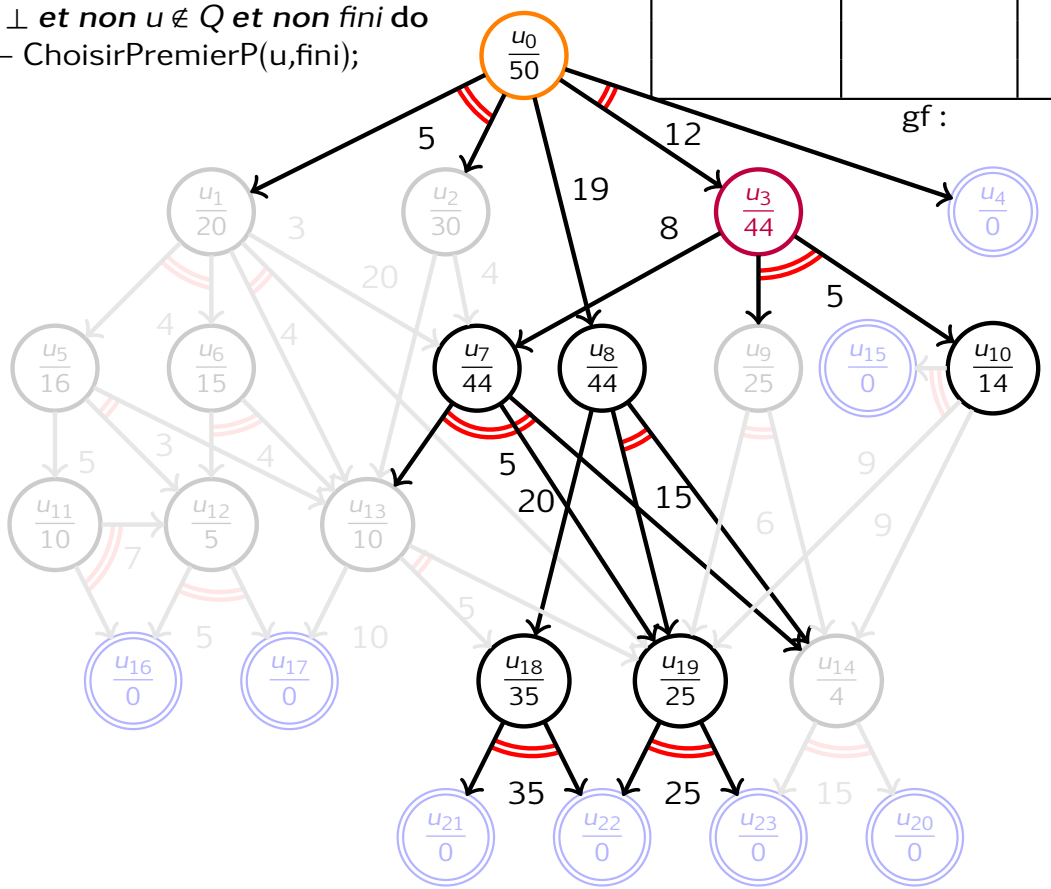

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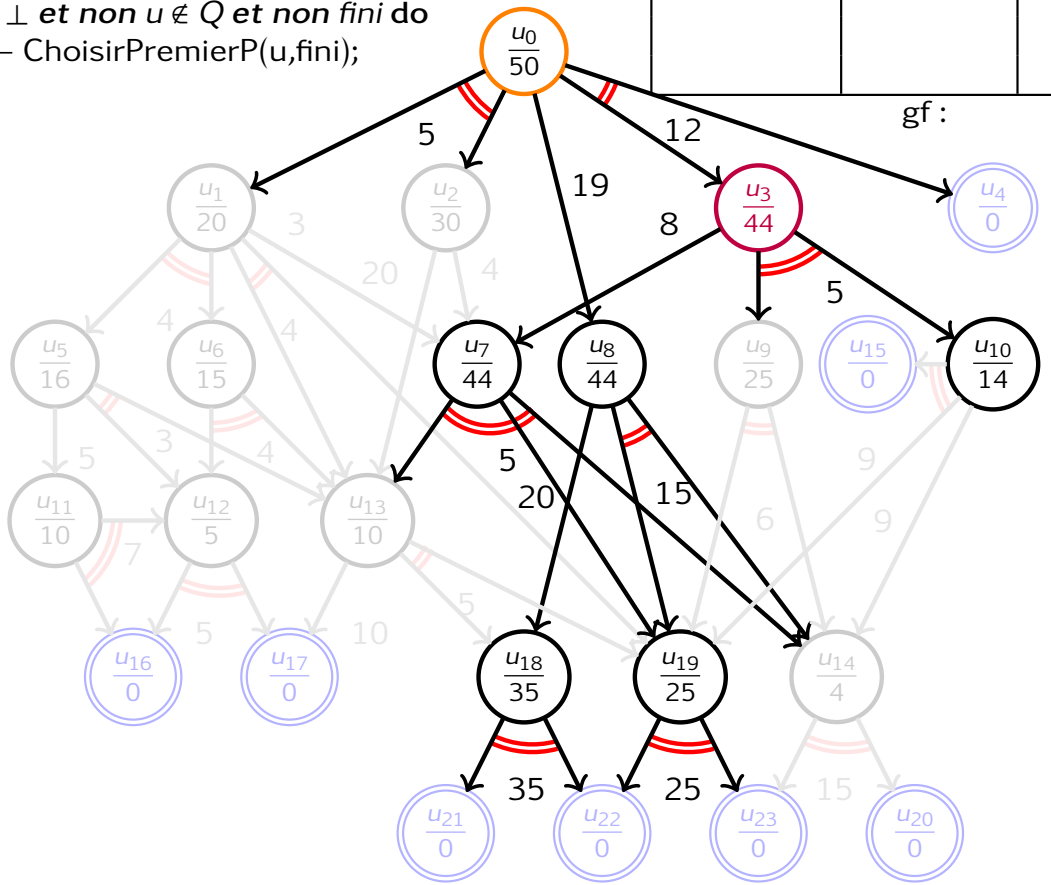

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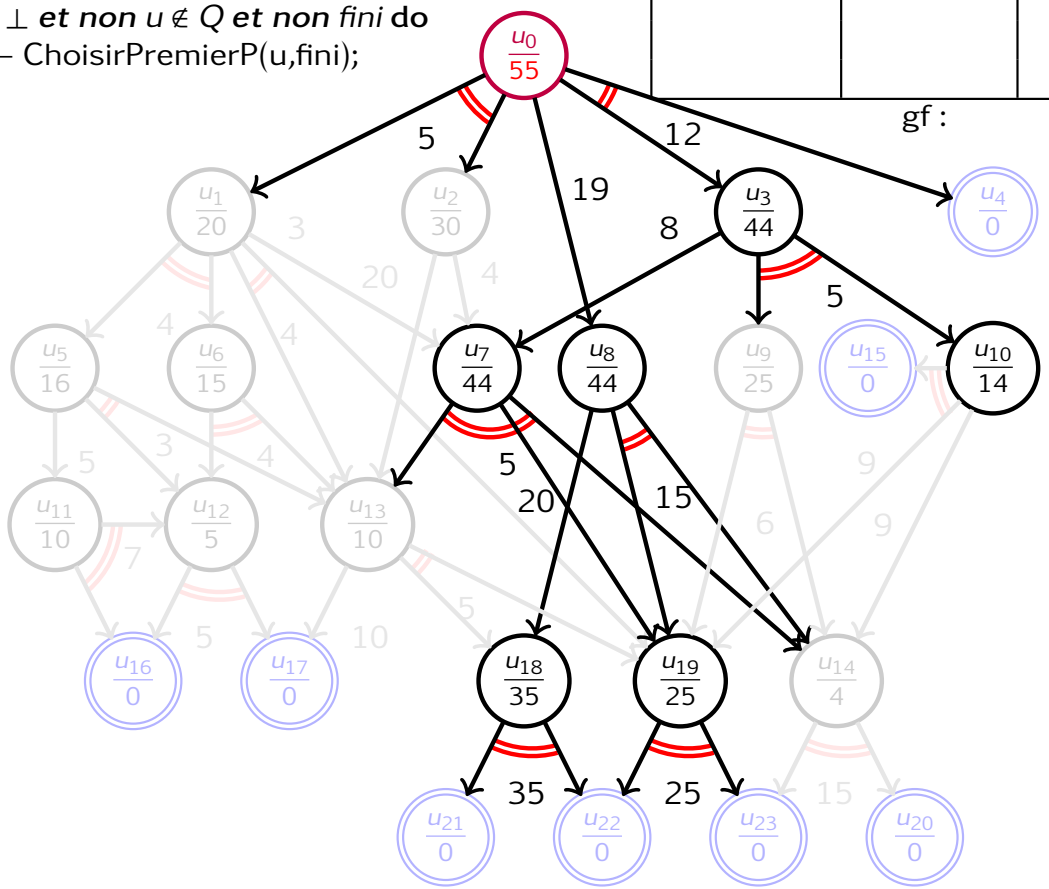
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
$u_0(0,55)$	(55,63,56)	{}	$u_1(5,20)$ $u_2(5,30)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,44)$	(52,44)	{}	$u_9(17,25)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,44)$	(44)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$

gf :

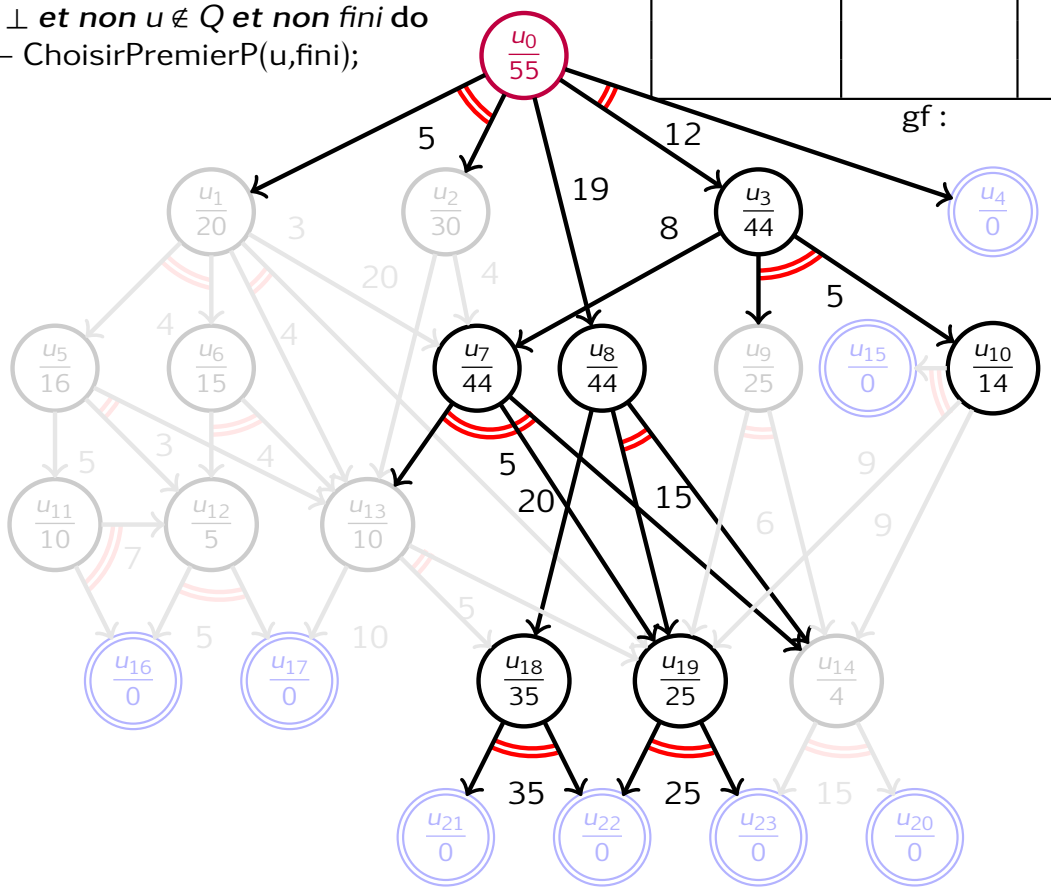


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,55)$	(55,63,56)	{}	$u_1(5,20)$ $u_2(5,30)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,44)$	(52,44)	{}	$u_9(17,25)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,44)$	(44)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$

gf :

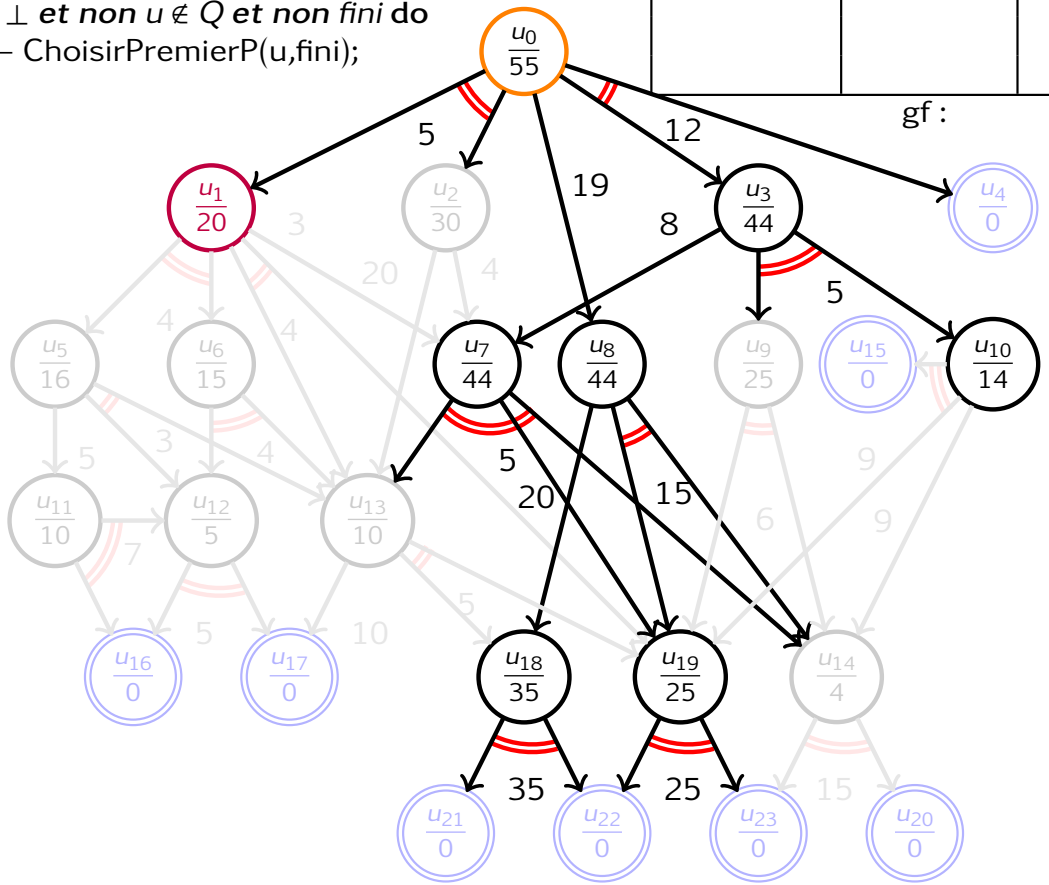


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,55)	(55,63,56)	{}	u_1(5,20) u_2(5,30)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,44)	(52,44)	{}	u_9(17,25) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,20)	(-,-,-)	{}	u_2(5,30)

gf :

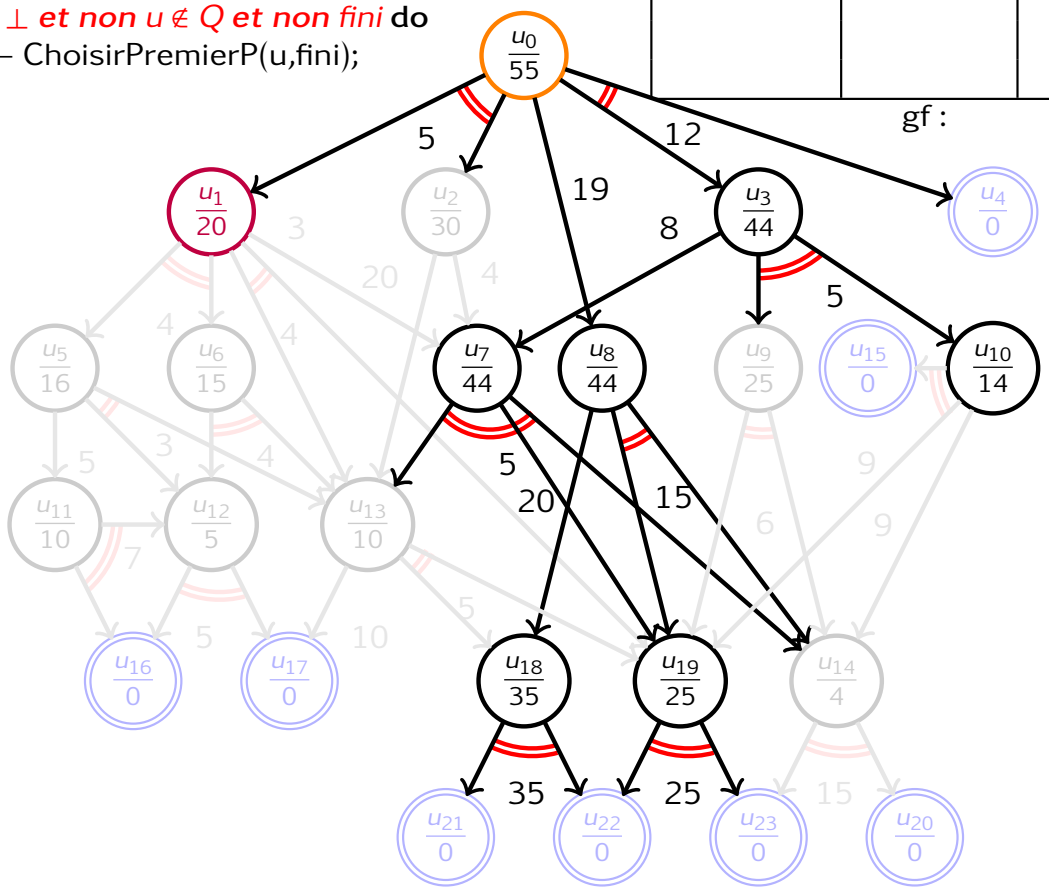


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,55)$	(55,63,56)	{}	$u_1(5,20)$ $u_2(5,30)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,44)$	(52,44)	{}	$u_9(17,25)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,44)$	(44)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,20)$	(-,-,-)	{}	$u_2(5,30)$

gf :

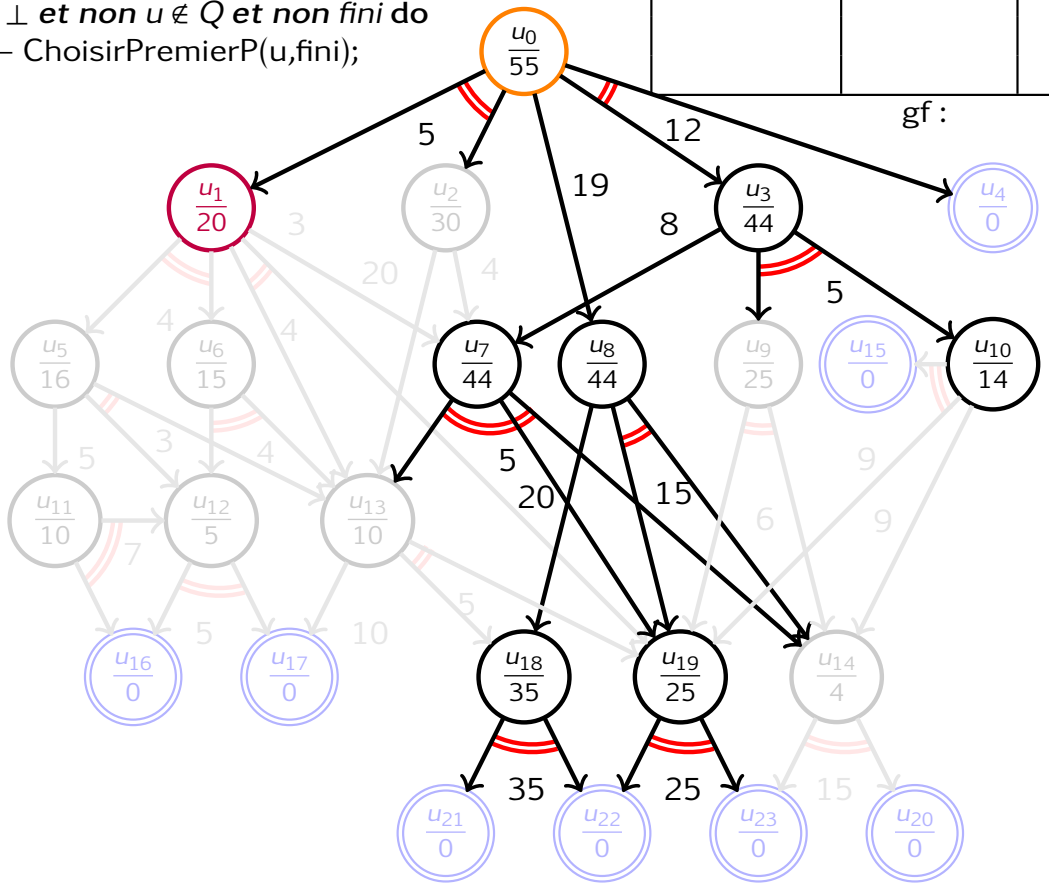


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,55)	(55,63,56)	{}	u_1(5,20) u_2(5,30)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,44)	(52,44)	{}	u_9(17,25) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,20)	(-,-,-)	{}	u_2(5,30)

gf :

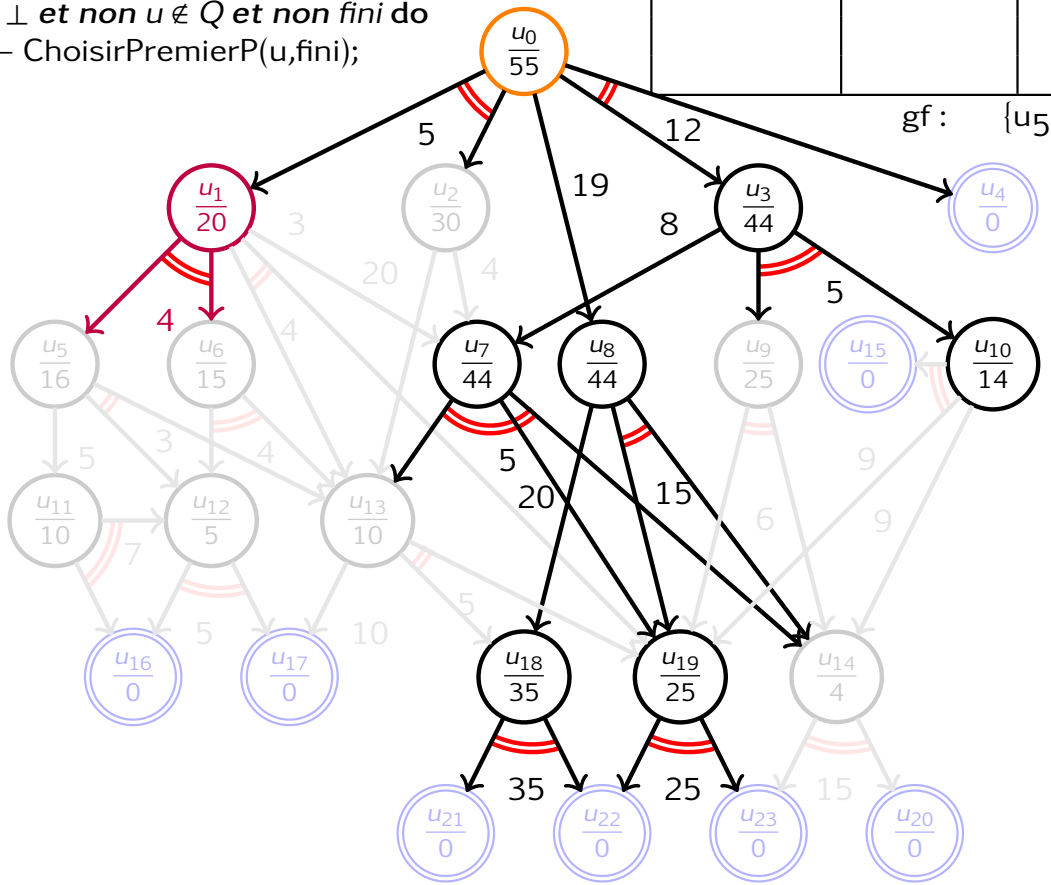


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
$u_0(0,55)$	(55,63,56)	{}	$u_1(5,20)$ $u_2(5,30)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,44)$	(52,44)	{}	$u_9(17,25)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,44)$	(44)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,20)$	(-,-,-)	{}	$u_2(5,30)$

gf : $\{u_5, u_6\}$

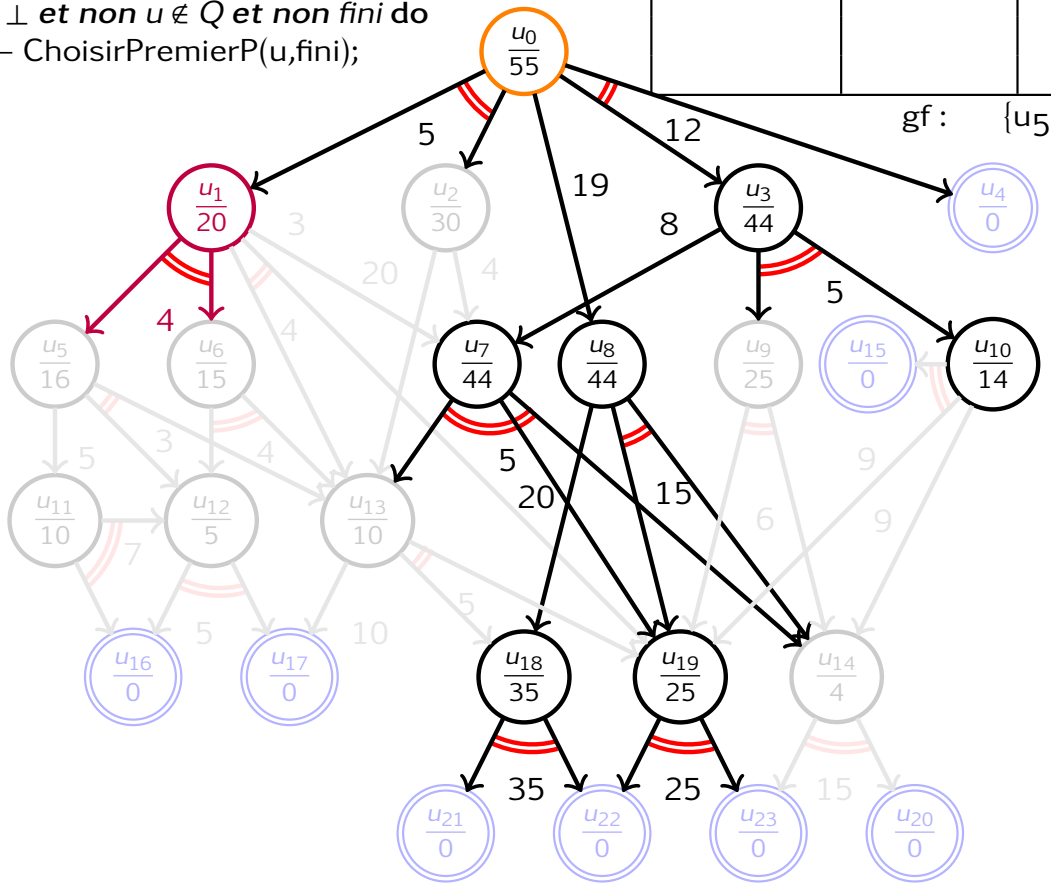


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,55)$	(55,63,56)	{}	$u_1(5,20)$ $u_2(5,30)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,44)$	(52,44)	{}	$u_9(17,25)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,44)$	(44)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,20)$	(-,-,-)	{}	$u_2(5,30)$

gf : $\{u_5, u_6\}$

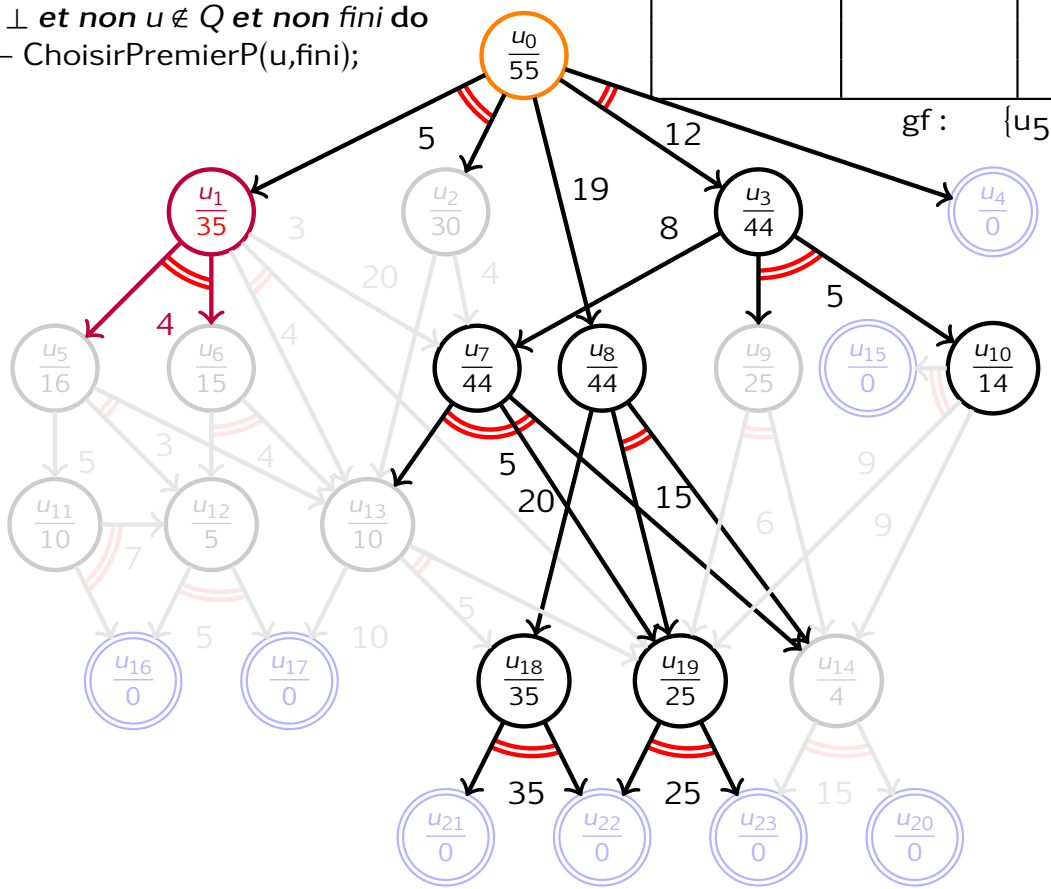


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,55)	(55,63,56)	{}	u_1(5,35) u_2(5,30)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,44)	(52,44)	{}	u_9(17,25) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,-,-)	{u_0}	u_5(9,16)u_6(9,15)u_2(5,30)

gf : $\{u_5, u_6\}$ = 4+16+15 = 35

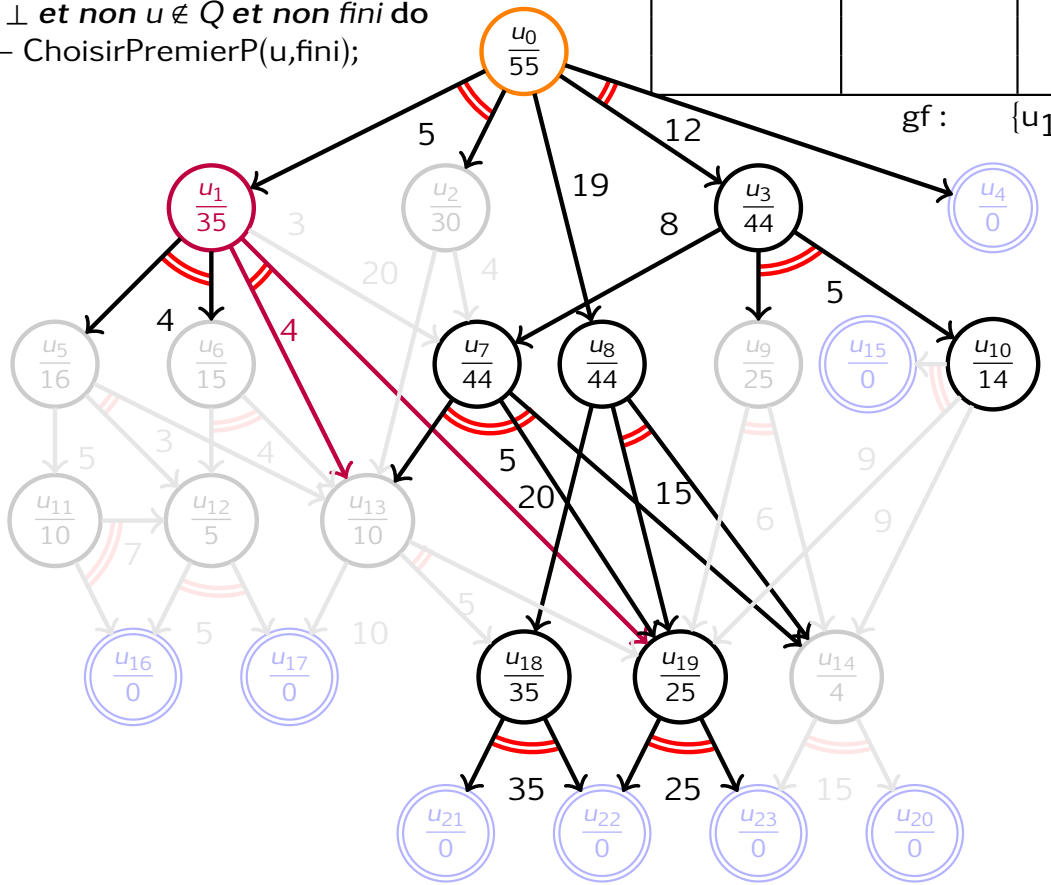


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,55)$	(55,63,56)	{}	$u_1(5,35)$ $u_2(5,30)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,44)$	(52,44)	{}	$u_9(17,25)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,44)$	(44)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,35)$	(35,-,-)	$\{u_0\}$	$u_5(9,16)u_6(9,15)u_2(5,30)$

gf : $\{u_{13}, u_{19}\}$

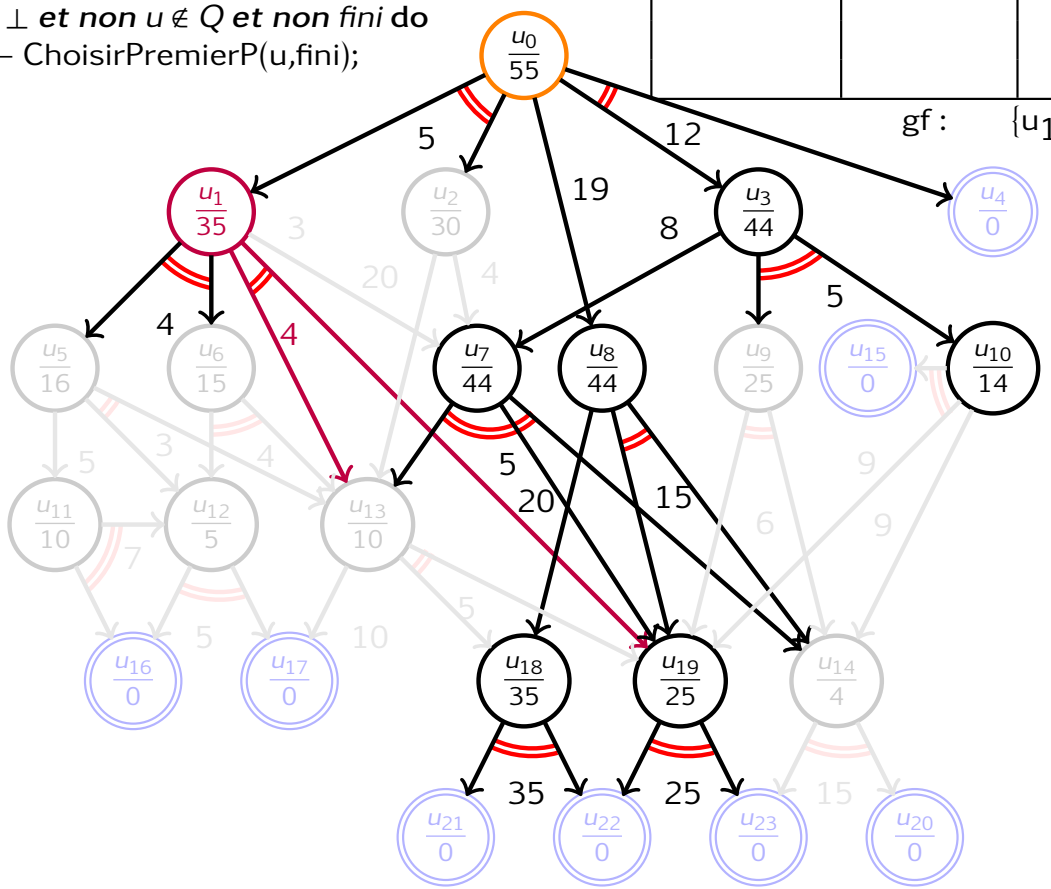


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while $u \neq \perp$ ***et non*** *fini* **do**
  gf ← groupeFils(u) ;
  while $gf \neq \emptyset$ **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while $A \neq \emptyset$ **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while $u \neq \perp$ ***et non*** $u \notin Q$ ***et non*** *fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,55)	(55,63,56)	{}	u_1(5,35) u_2(5,30)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,44)	(52,44)	{}	u_9(17,25) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,-,-)	$\{u_0\}$	u_5(9,16)u_6(9,15)u_2(5,30)

gf : $\{u_{13}, u_{19}\}$

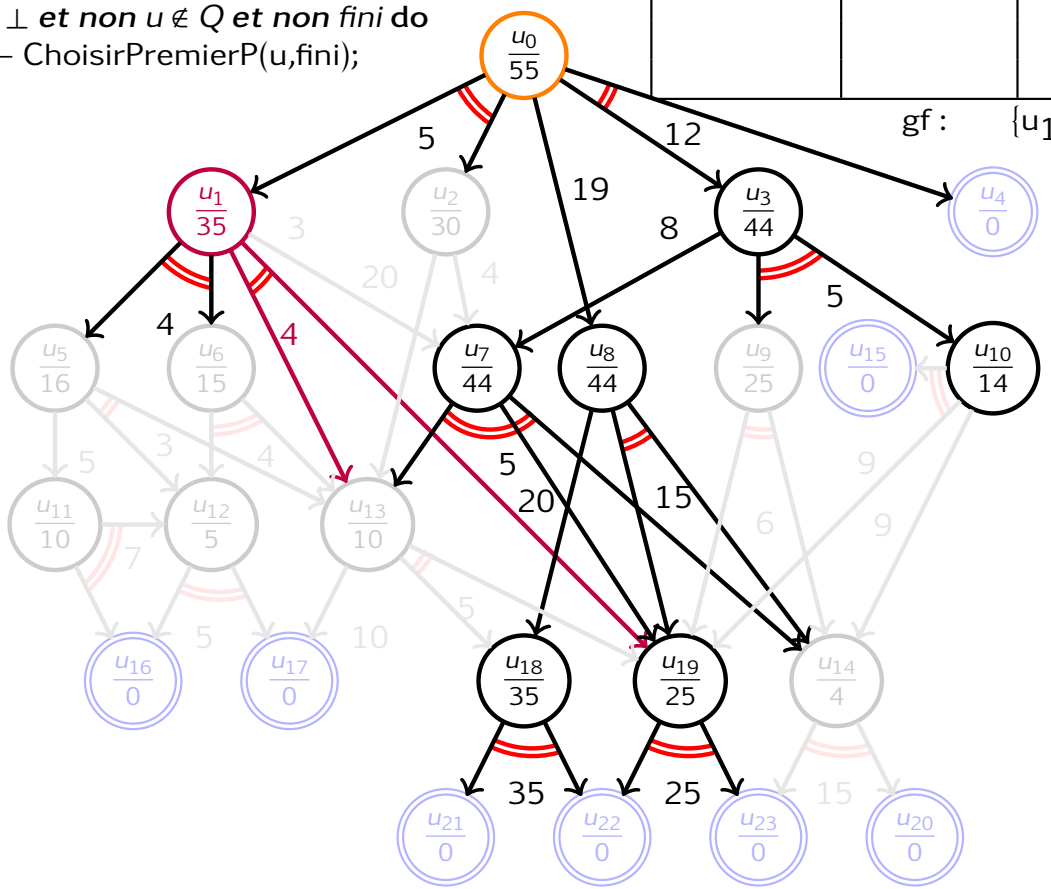


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,55)$	(55,63,56)	{}	$u_1(5,35)$ $u_2(5,30)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,44)$	(52,44)	{}	$u_9(17,25)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,44)$	(44)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,35)$	(35,39,-)	$\{u_0\}$	$u_5(9,16)u_6(9,15)u_2(5,30)$

gf : $\{u_{13}, u_{19}\}$ = 4+10+25 = 39

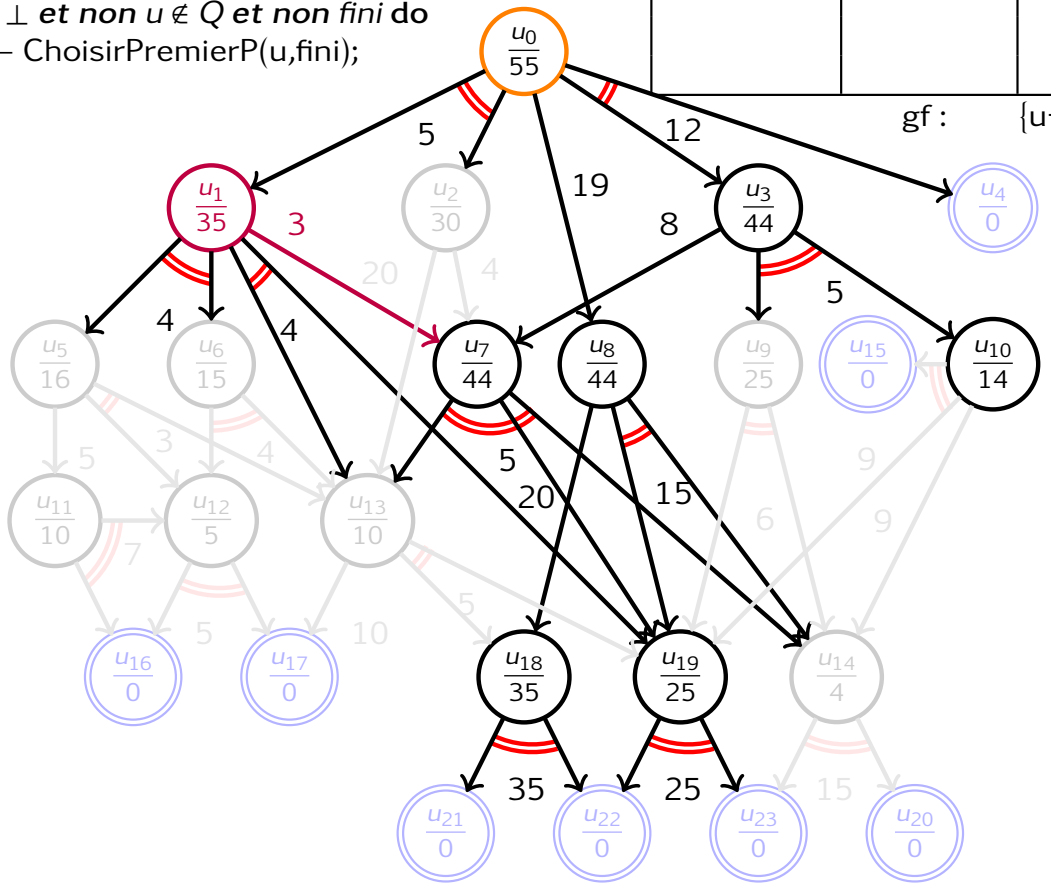


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,55)	(55,63,56)	{}	u_1(5,35) u_2(5,30)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,44)	(52,44)	{}	u_9(17,25) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,-)	{u_0}	u_5(9,16)u_6(9,15)u_2(5,30)

gf : {u_7}

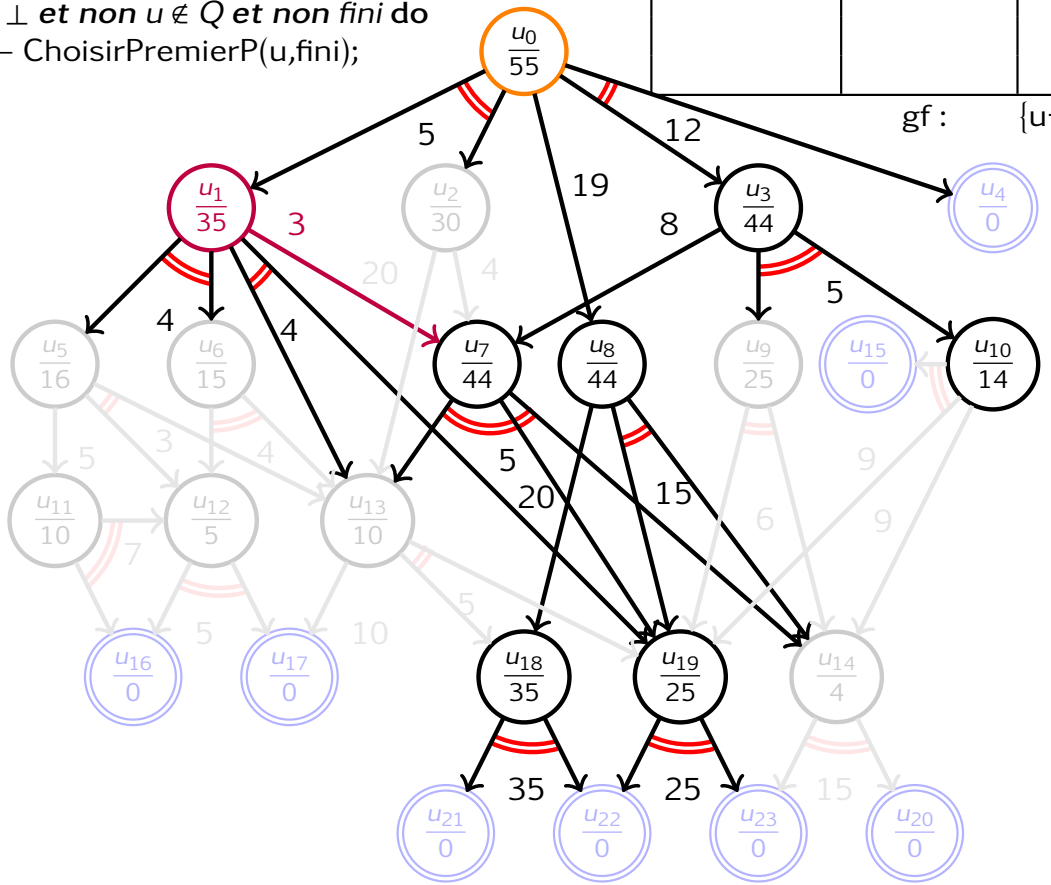


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,55)	(55,63,56)	{}	u_1(5,35) u_2(5,30)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,44)	(52,44)	{}	u_9(17,25) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,-)	{u_0}	u_5(9,16)u_6(9,15)u_2(5,30)

gf : {u_7}

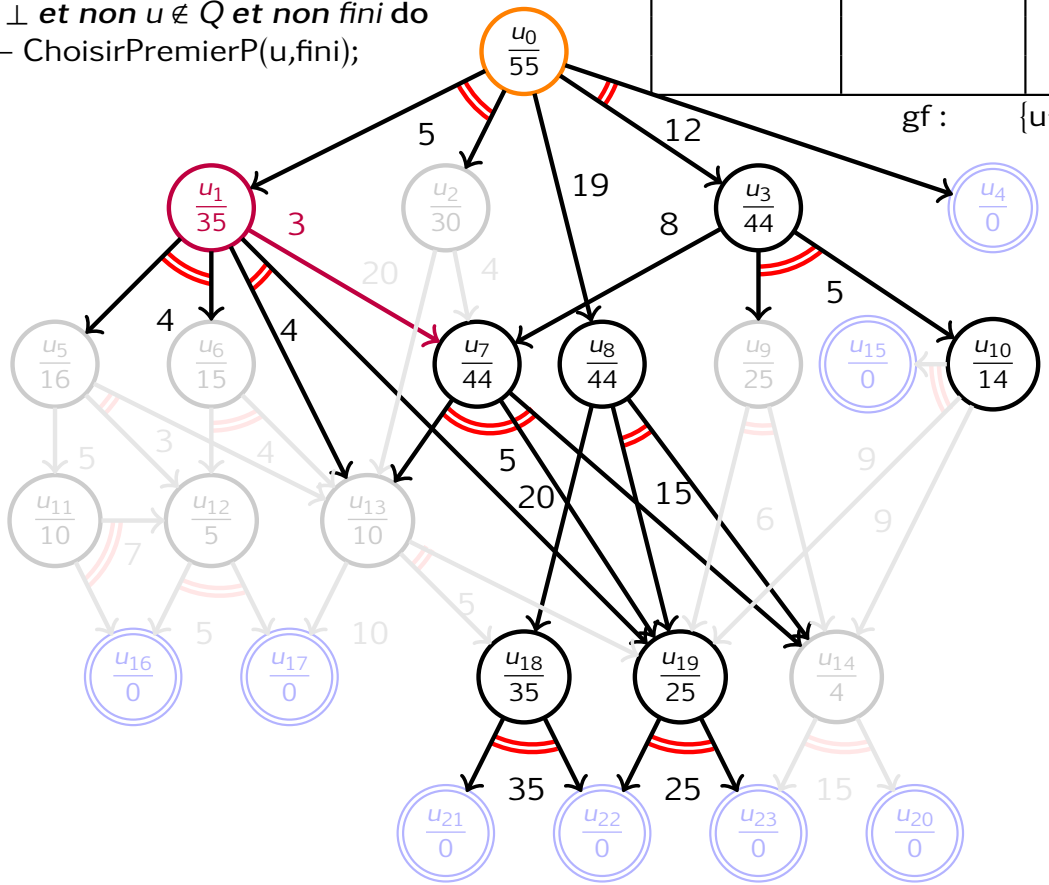


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,55)	(55,63,56)	{}	u_1(5,35) u_2(5,30)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,44)	(52,44)	{}	u_9(17,25) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{u_0}	u_5(9,16)u_6(9,15)u_2(5,30)

gf : {u_7} = 3+44 = 47

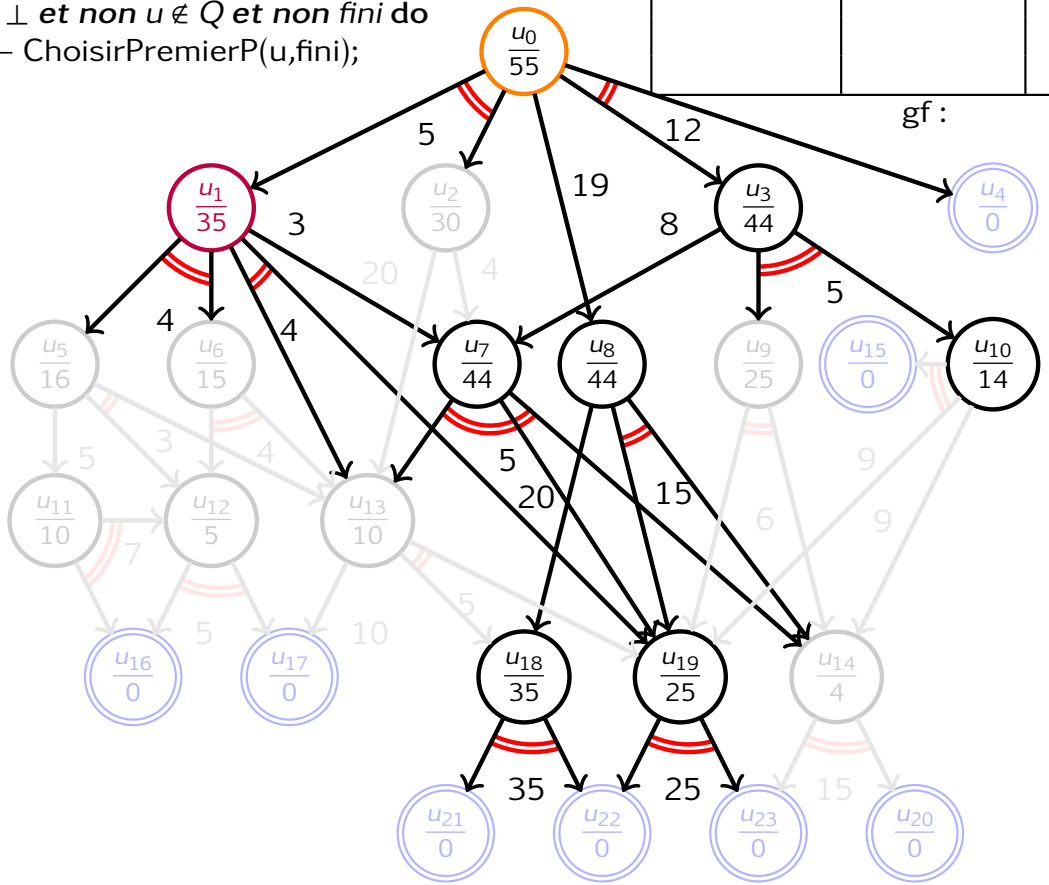


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,55)	(55,63,56)	{}	u_1(5,35) u_2(5,30)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,44)	(52,44)	{}	u_9(17,25) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{u_0}	u_5(9,16)u_6(9,15)u_2(5,30)

gf :

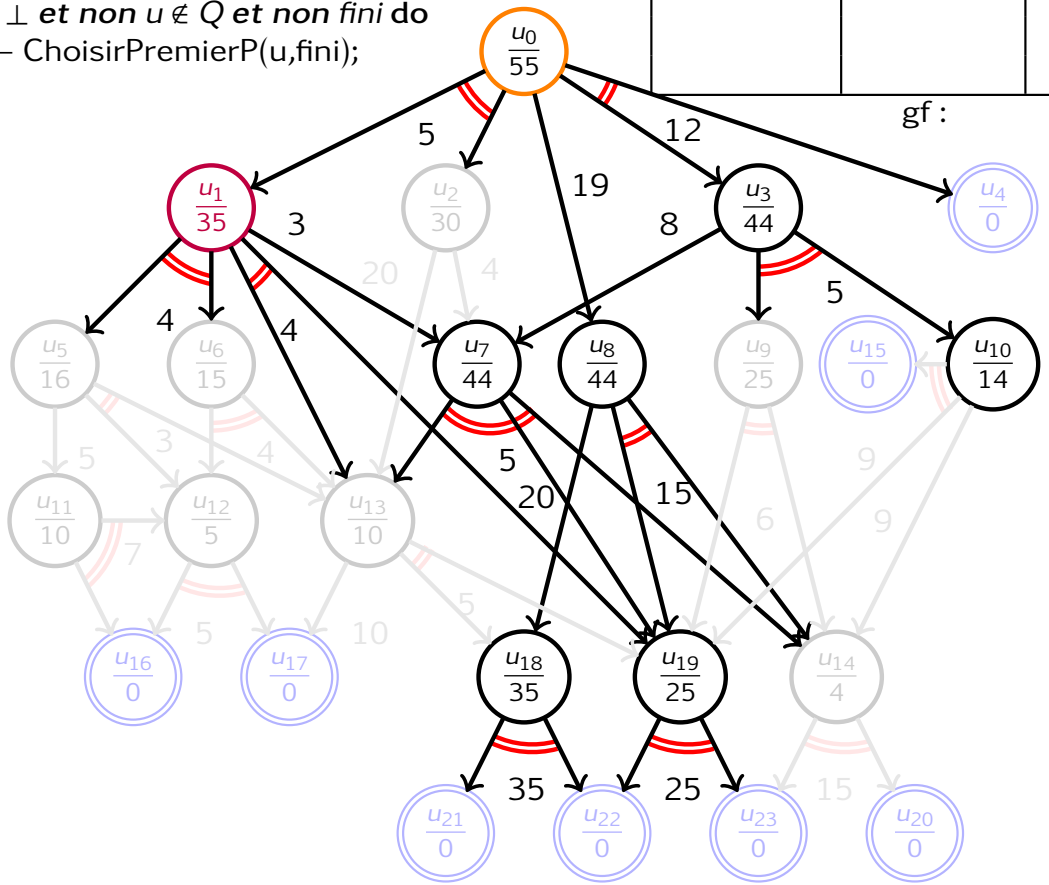


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while $u \neq \perp$ ***et non*** *fini* **do**
  gf ← groupeFils(u) ;
  while $gf \neq \emptyset$ **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while $A \neq \emptyset$ **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while $u \neq \perp$ ***et non*** $u \notin Q$ ***et non*** *fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,55)	(55,63,56)	{}	u_1(5,35) u_2(5,30)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,44)	(52,44)	{}	u_9(17,25) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{u_0}	u_5(9,16)u_6(9,15)u_2(5,30)

gf :

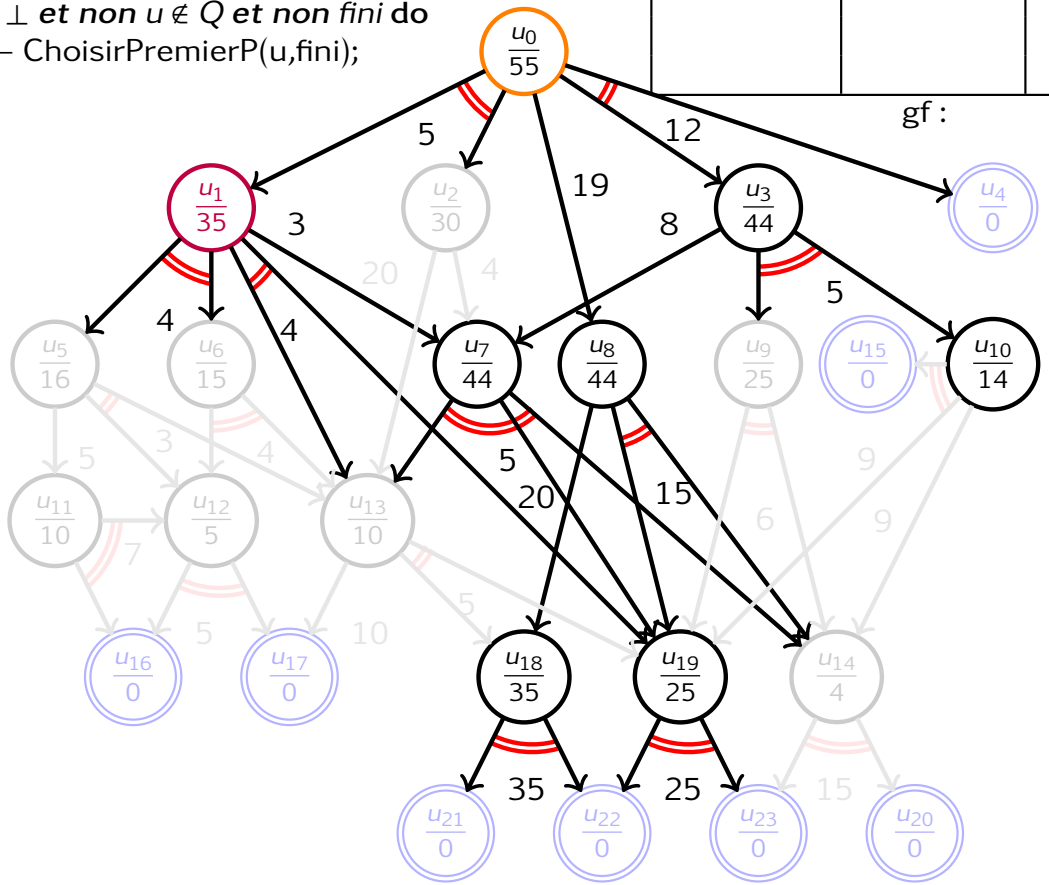


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,55)	(55,63,56)	{}	u_1(5,35) u_2(5,30)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,44)	(52,44)	{}	u_9(17,25) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{u_0}	u_5(9,16)u_6(9,15)u_2(5,30)

gf :

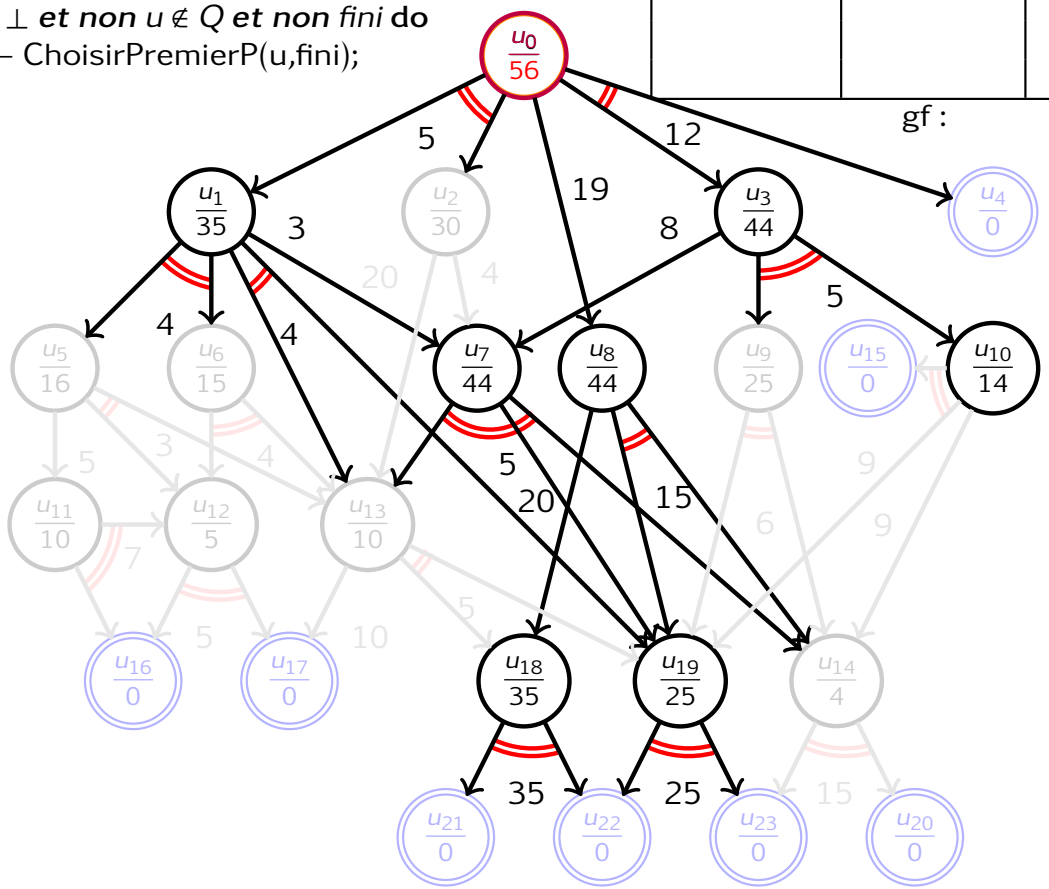


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,56)	(70,63,56)	{}	u_3(12,44) u_4(12,0)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,44)	(52,44)	{}	u_9(17,25) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{}	u_5(9,16)u_6(9,15)u_2(5,30)

gf :

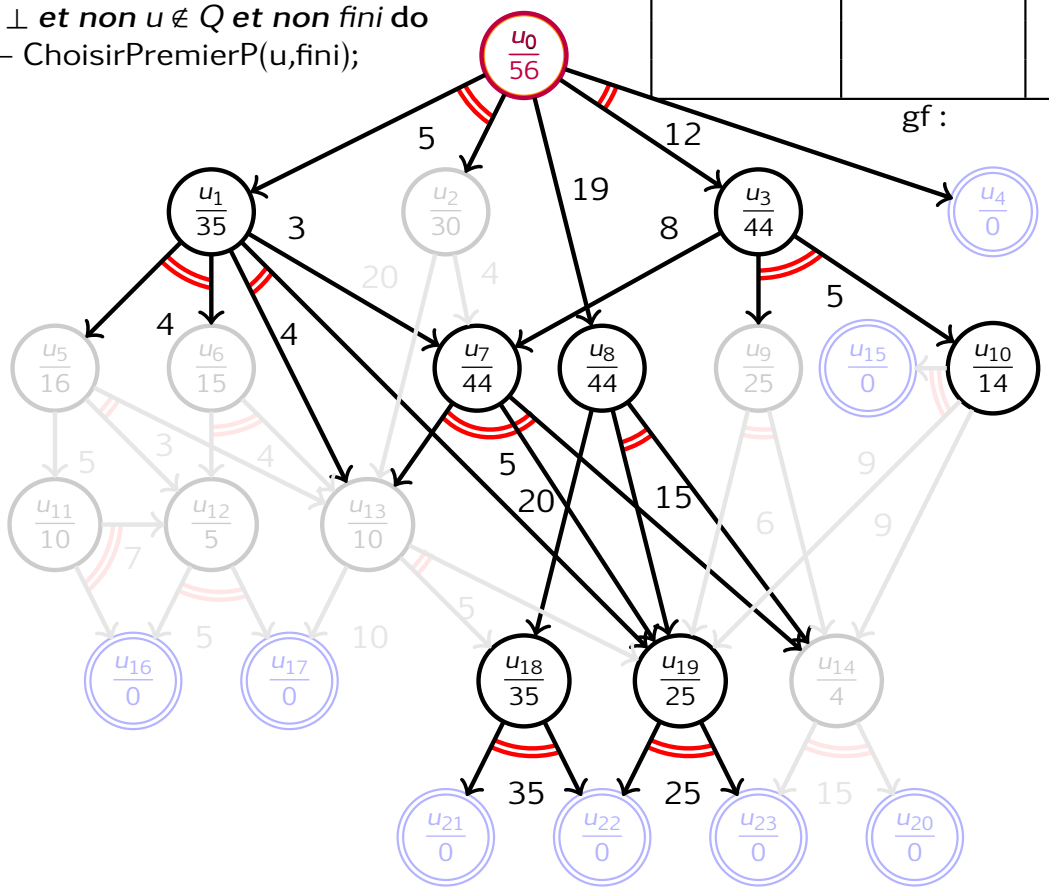


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,56)	(70,63,56)	{}	u_3(12,44) u_4(12,0)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,44)	(52,44)	{}	u_9(17,25) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{}	u_5(9,16)u_6(9,15)u_2(5,30)

gf :

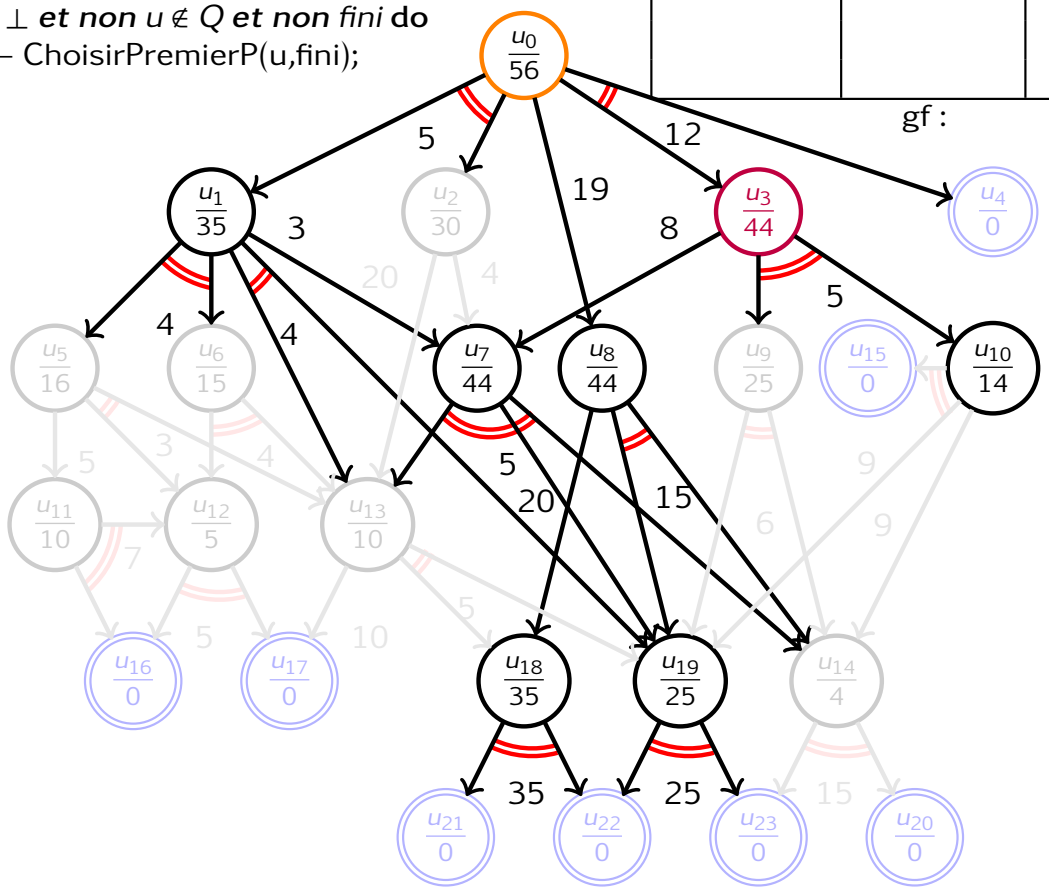


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,56)	(70,63,56)	{}	u_3(12,44) u_4(12,0)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,44)	(52,44)	{}	u_9(17,25) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{}	u_5(9,16)u_6(9,15)u_2(5,30)

gf :

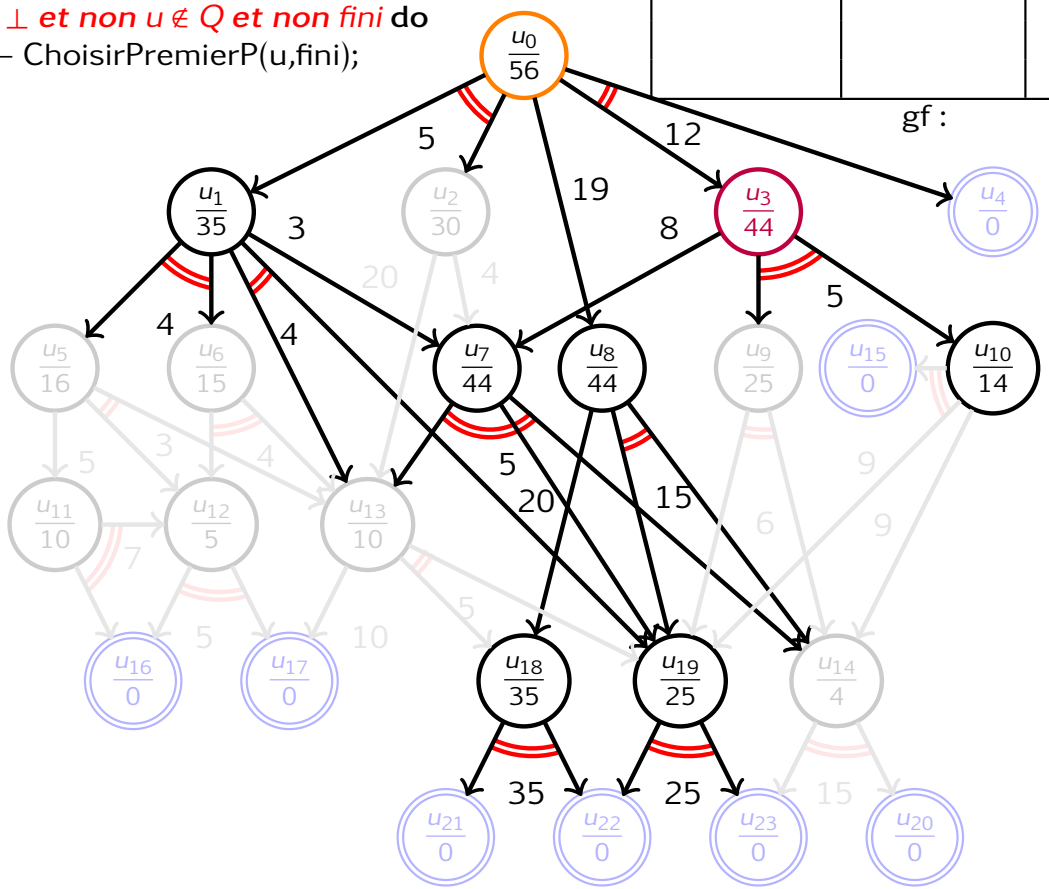


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,56)	(70,63,56)	{}	u_3(12,44) u_4(12,0)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,44)	(52,44)	{}	u_9(17,25) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{}	u_5(9,16)u_6(9,15)u_2(5,30)

gf :

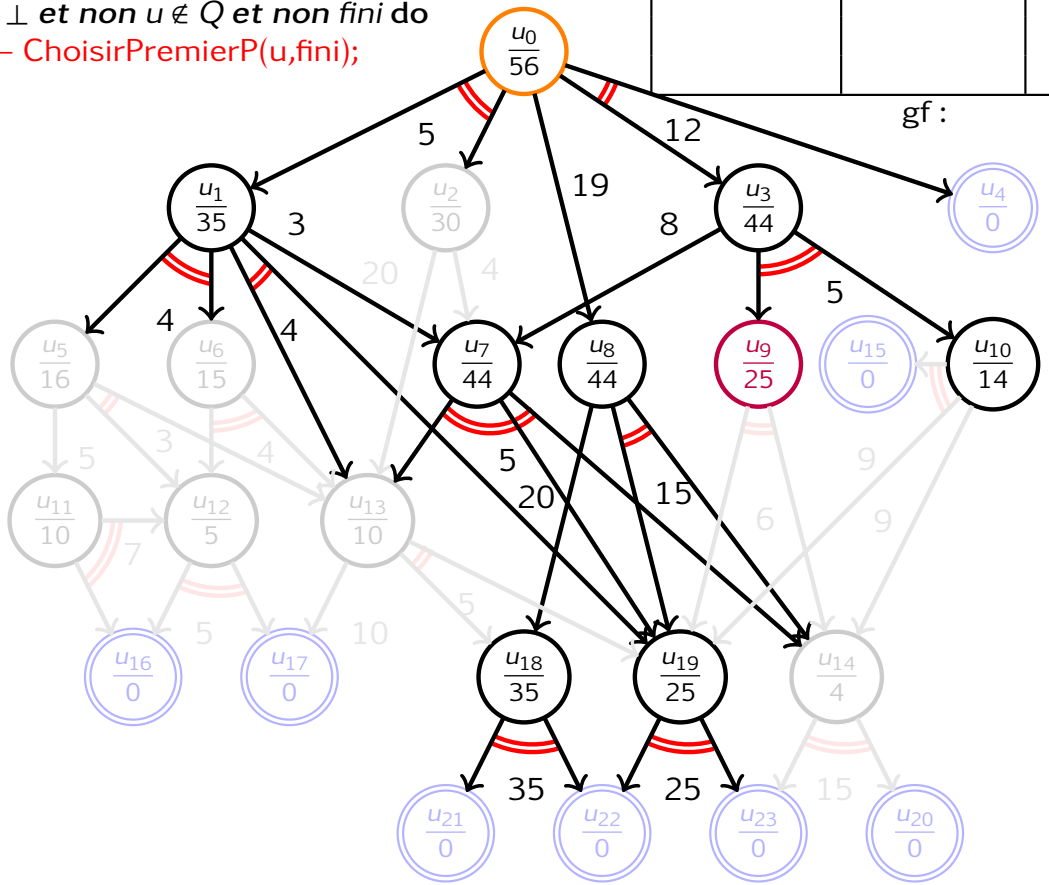


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,56)	(70,63,56)	{}	u_3(12,44) u_4(12,0)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,44)	(52,44)	{}	u_9(17,25) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,25)	(-)	{}	u_{10}(17,31)u_{14}(34,38)

gf :

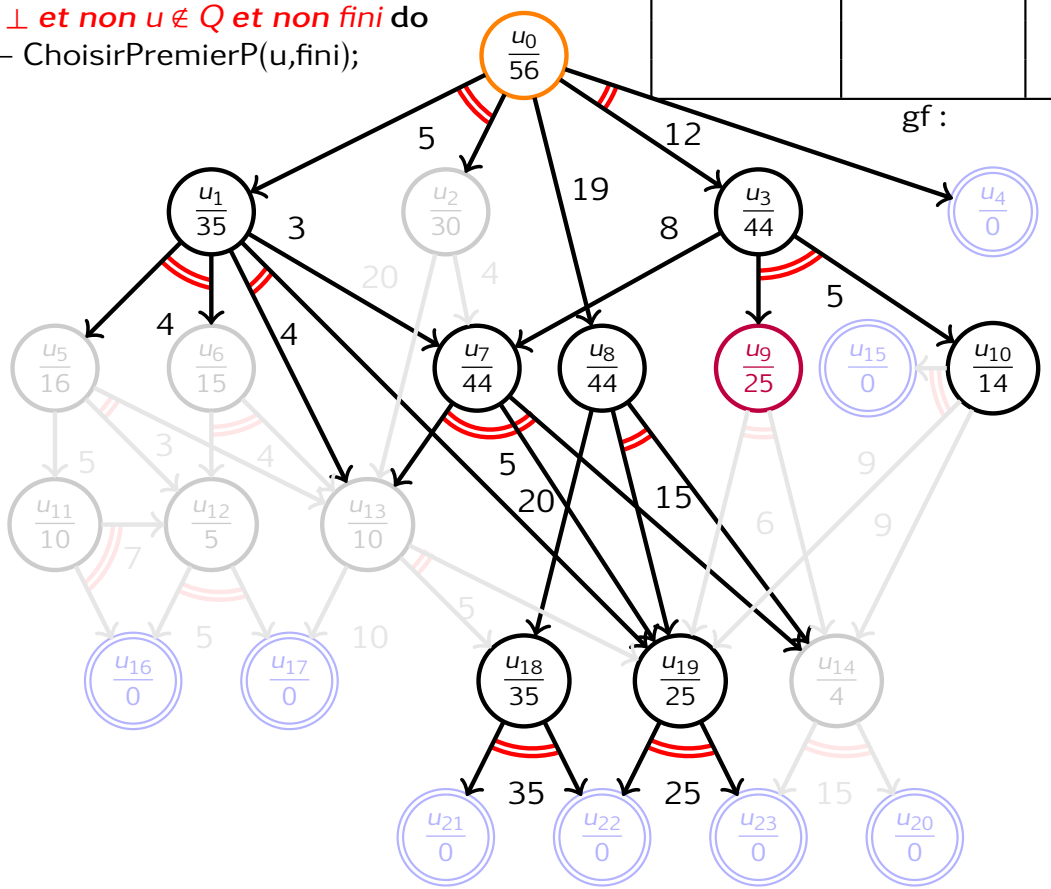


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,56)	(70,63,56)	{}	u_3(12,44) u_4(12,0)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,44)	(52,44)	{}	u_9(17,25) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,25)	(-)	{}	u_{10}(17,31)u_{14}(34,38)

gf :

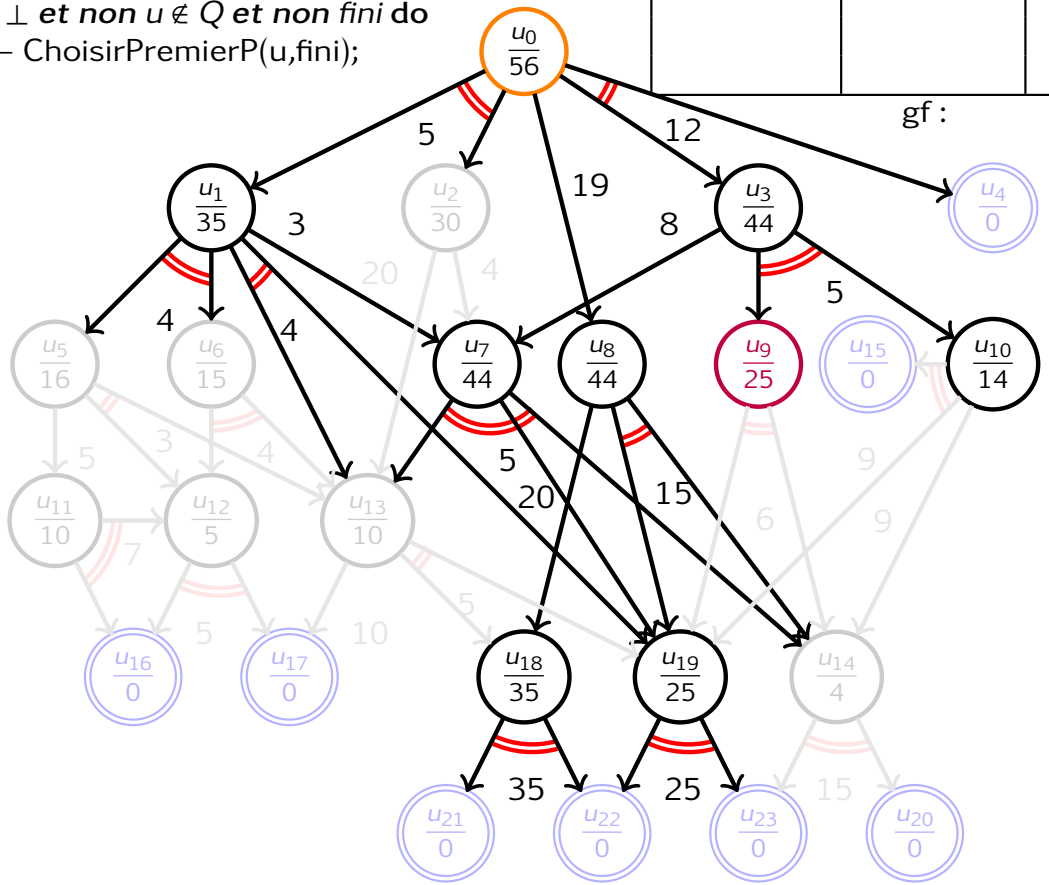


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,56)	(70,63,56)	{}	u_3(12,44) u_4(12,0)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,44)	(52,44)	{}	u_9(17,25) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,25)	(-)	{}	u_{10}(17,31)u_{14}(34,38)

gf :

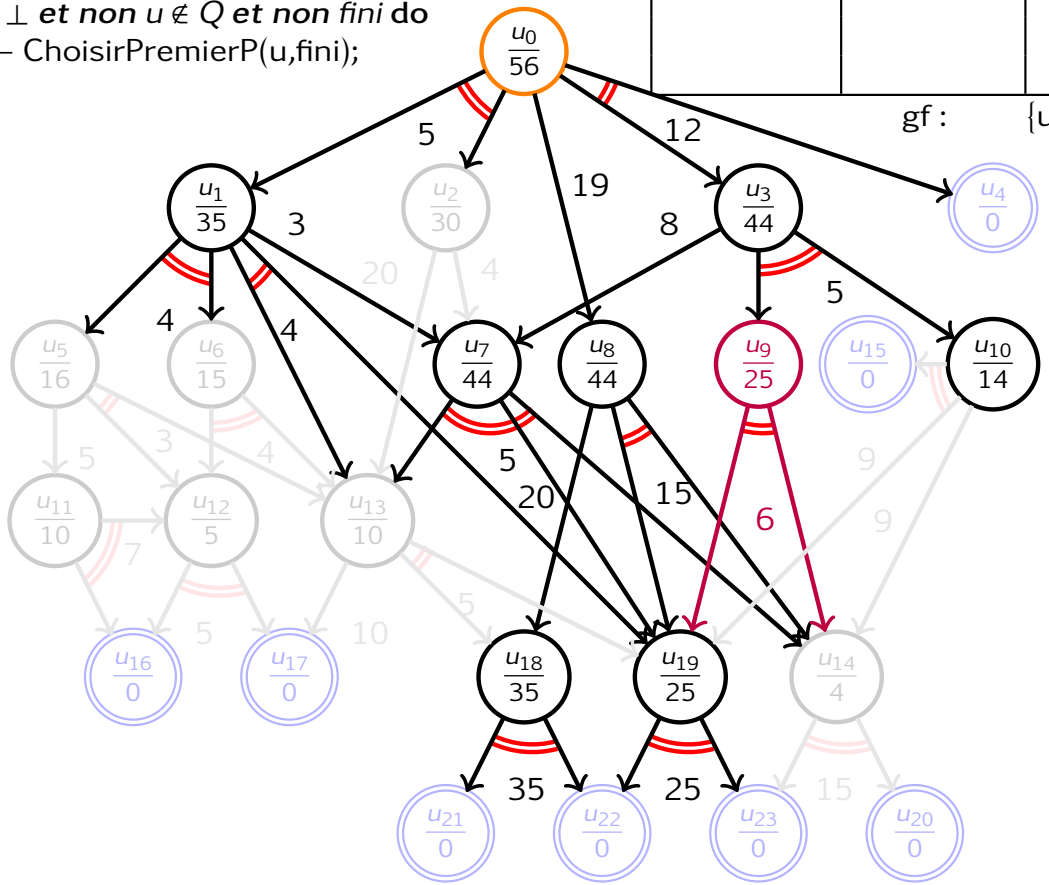


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,56)	(70,63,56)	{}	u_3(12,44) u_4(12,0)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,44)	(52,44)	{}	u_9(17,25) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,25)	(-)	{}	u_{10}(17,31)u_{14}(34,38)

gf : $\{u_{19}, u_{14}\}$

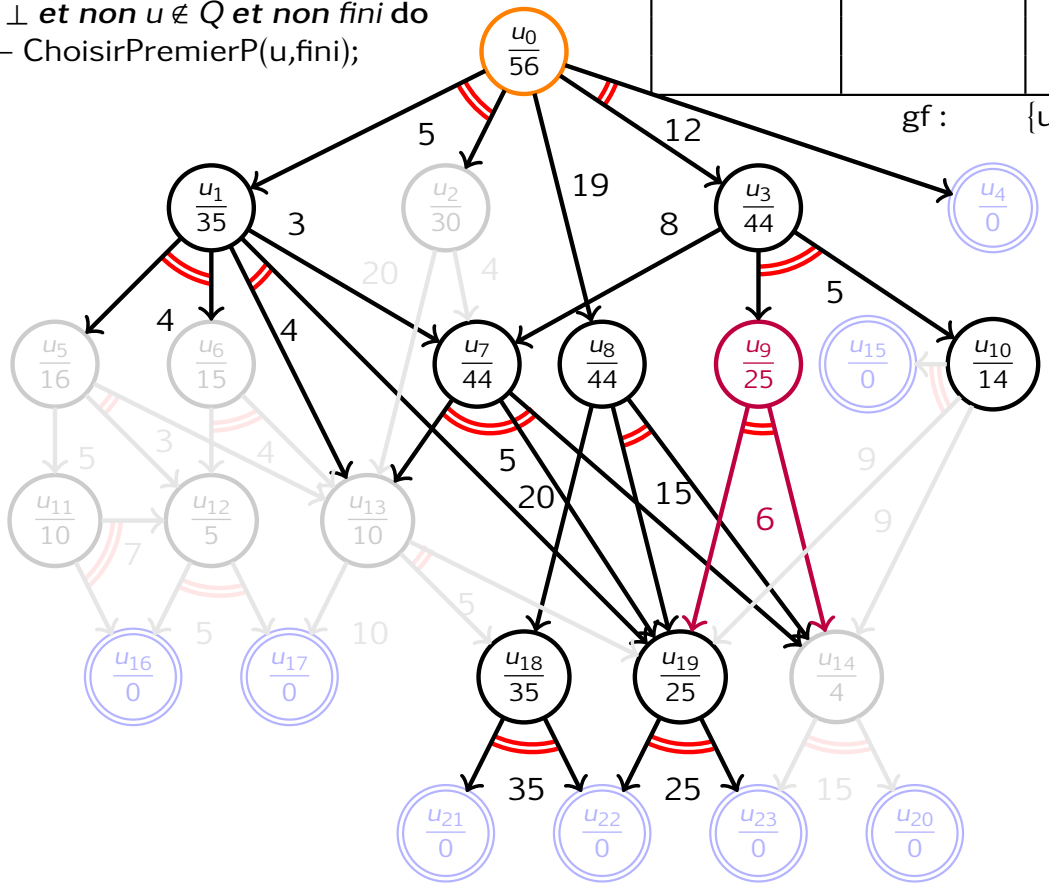


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
$u_0(0,56)$	(70,63,56)	{}	$u_3(12,44)$ $u_4(12,0)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,44)$	(52,44)	{}	$u_9(17,25)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,44)$	(44)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,35)$	(35,39,47)	{}	$u_5(9,16)u_6(9,15)u_2(5,30)$
$u_9(17,25)$	(-)	{}	$u_{10}(17,31)u_{14}(34,38)$

gf : $\{u_{19},u_{14}\}$

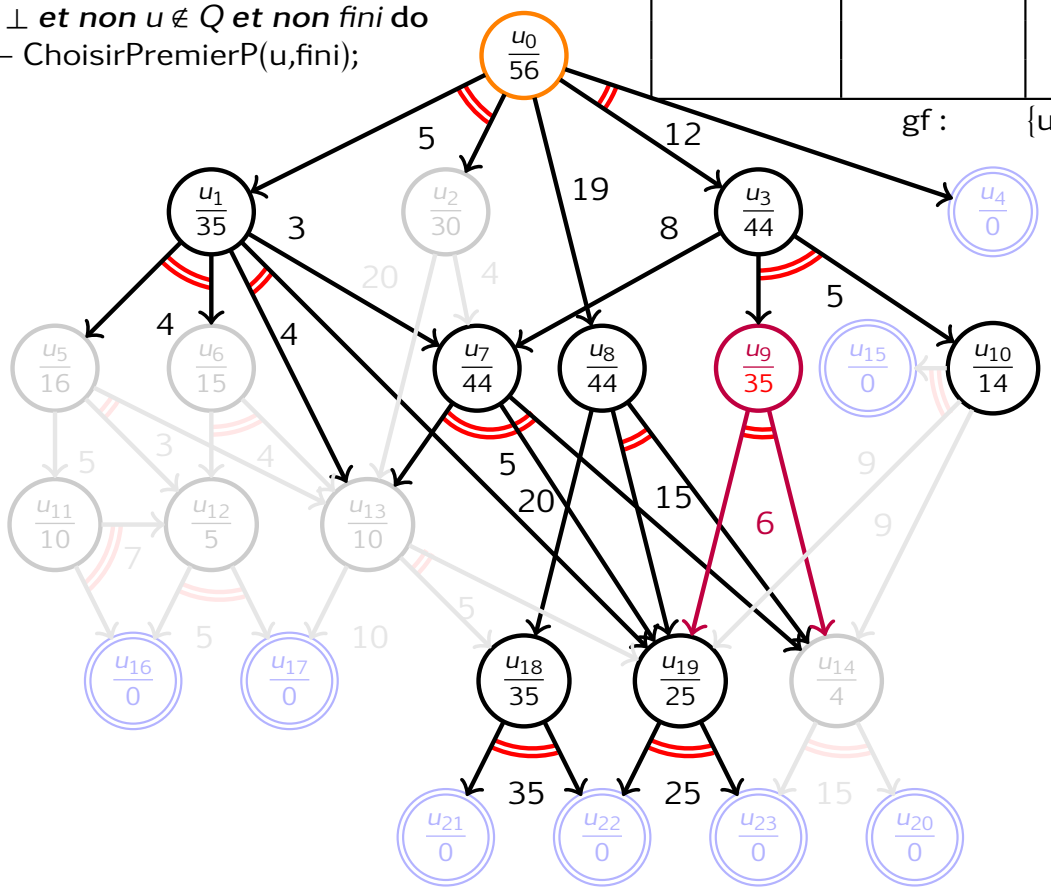


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,56)	(70,63,56)	{}	u_3(12,44) u_4(12,0)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,44)	(52,44)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,35)	(35)	{u_3}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,

gf : {u_{19},u_{14}} = 6+25+4 = 35

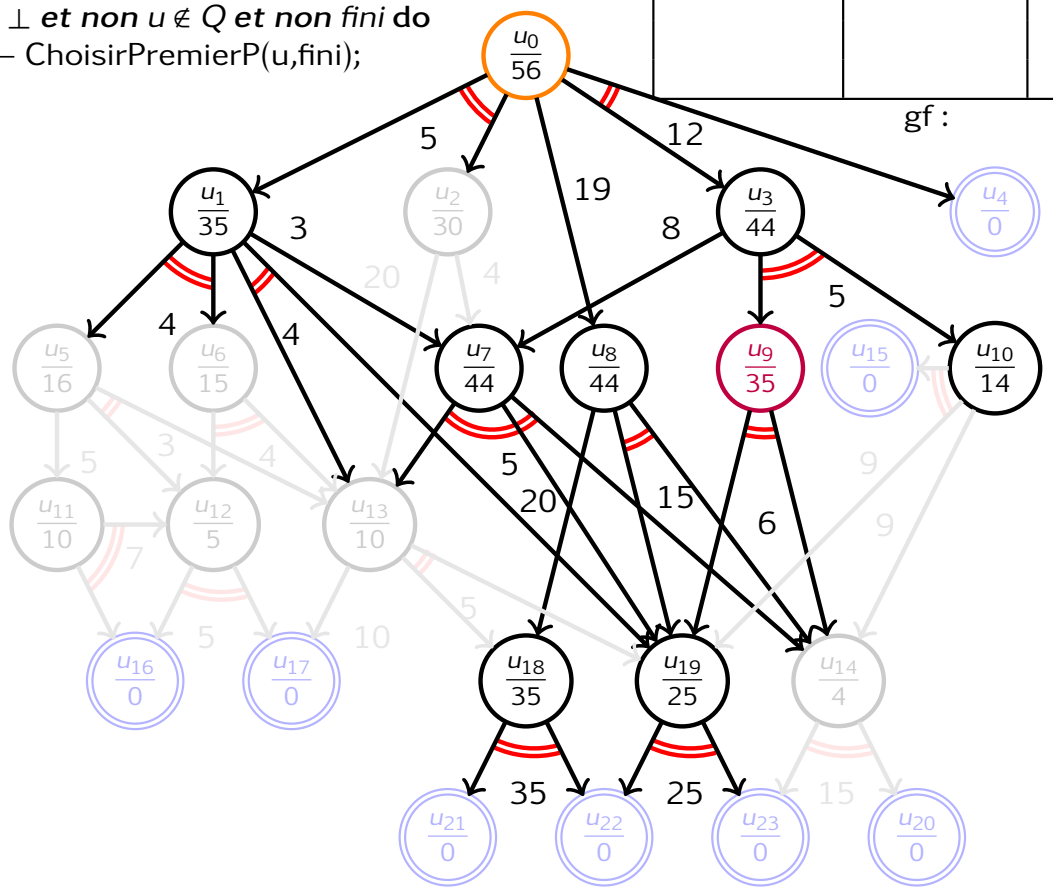


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,56)	(70,63,56)	{}	u_3(12,44) u_4(12,0)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,44)	(52,44)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,35)	(35)	{u_3}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,

gf :

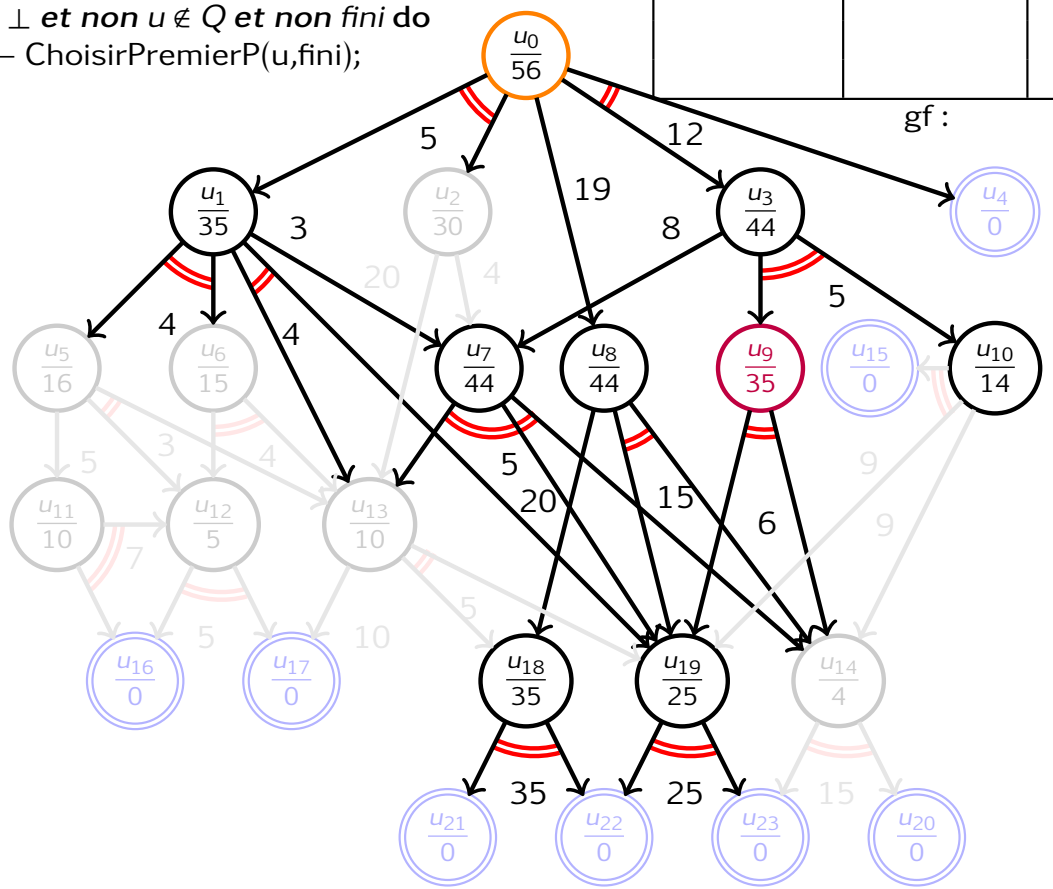


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while $u \neq \perp$ ***et non*** *fini* **do**
  gf ← groupeFils(u) ;
  while $gf \neq \emptyset$ **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while $A \neq \emptyset$ **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while $u \neq \perp$ ***et non*** $u \notin Q$ ***et non*** *fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,56)	(70,63,56)	{}	u_3(12,44) u_4(12,0)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,44)	(52,44)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,35)	(35)	{u_3}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,

gf :

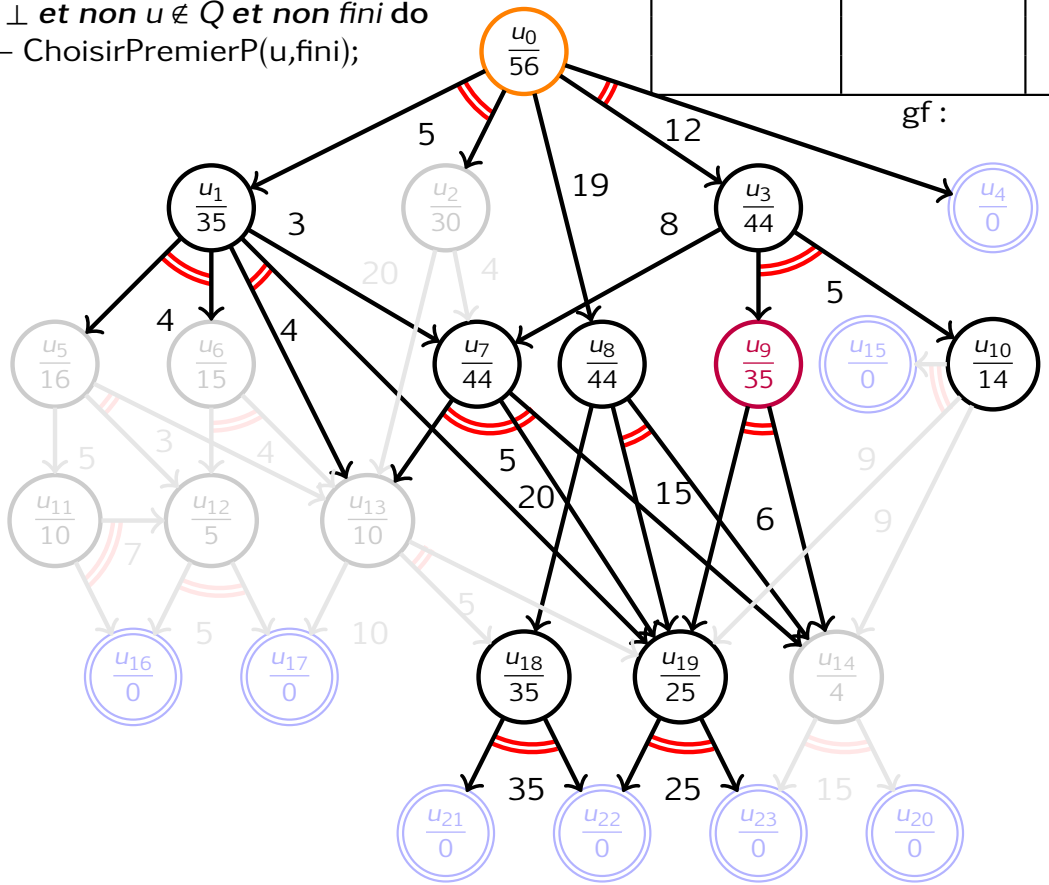


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
$u_0(0,56)$	(70,63,56)	{}	$u_3(12,44)$ $u_4(12,0)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,44)$	(52,44)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,44)$	(44)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,35)$	(35,39,47)	{}	$u_5(9,16)u_6(9,15)u_2(5,30)$
$u_9(17,35)$	(35)	$\{u_3\}$	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,$

gf :

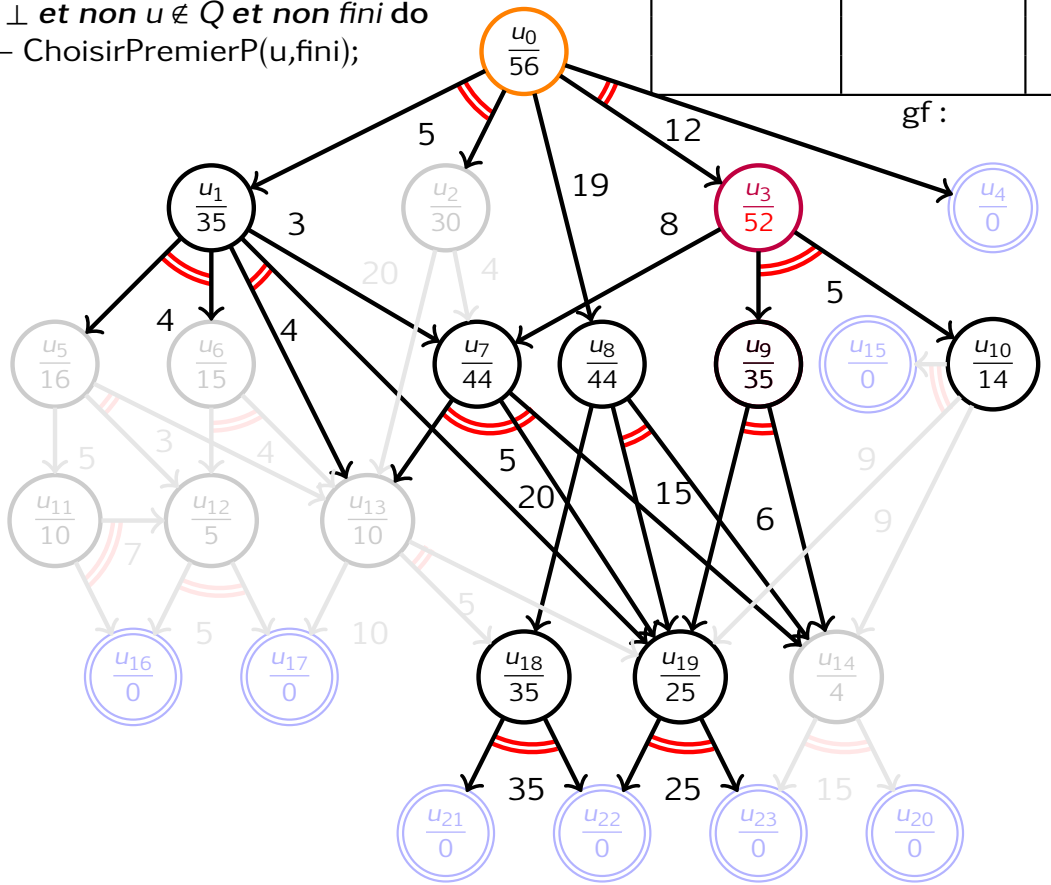


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,56)	(70,63,56)	{}	u_3(12,52) u_4(12,0)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,52)	(52,54)	{u_0}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,35)	(35)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,

gf :

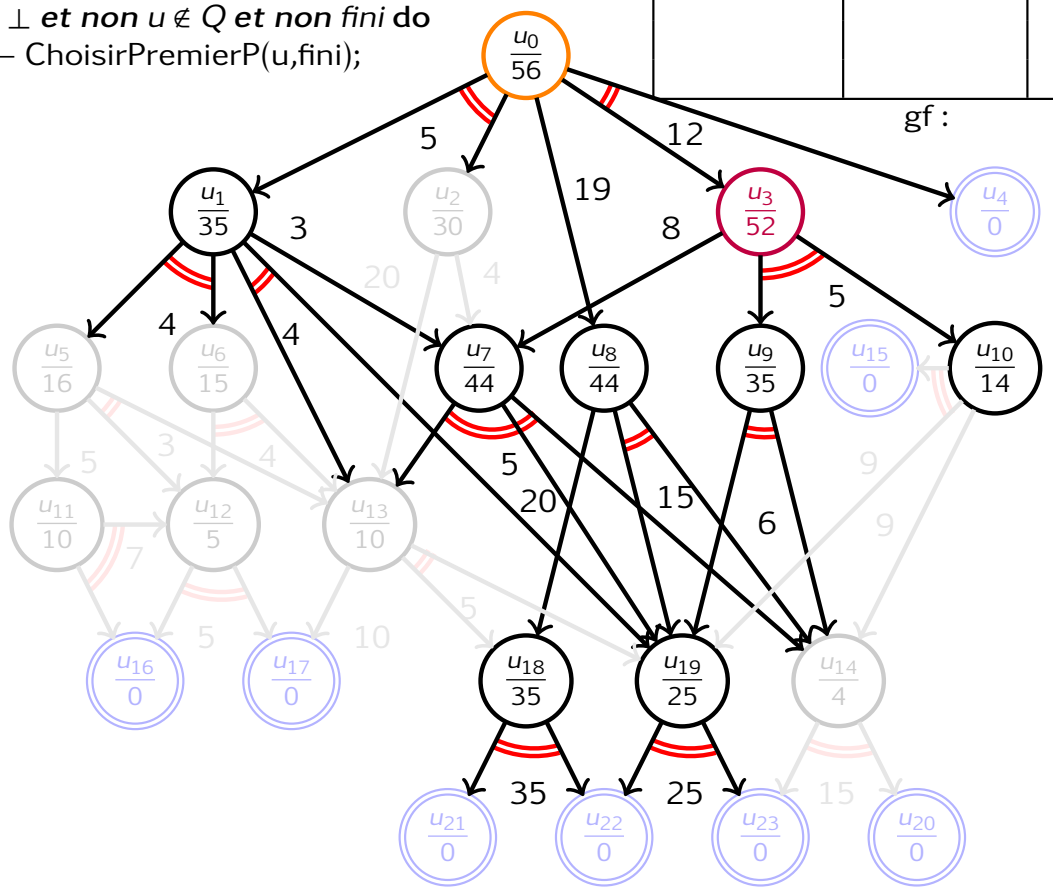


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,56)$	(70,63,56)	{}	$u_3(12,52)$ $u_4(12,0)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,52)$	(52,54)	$\{u_0\}$	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,44)$	(44)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,35)$	(35,39,47)	{}	$u_5(9,16)u_6(9,15)u_2(5,30)$
$u_9(17,35)$	(35)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,$

gf :

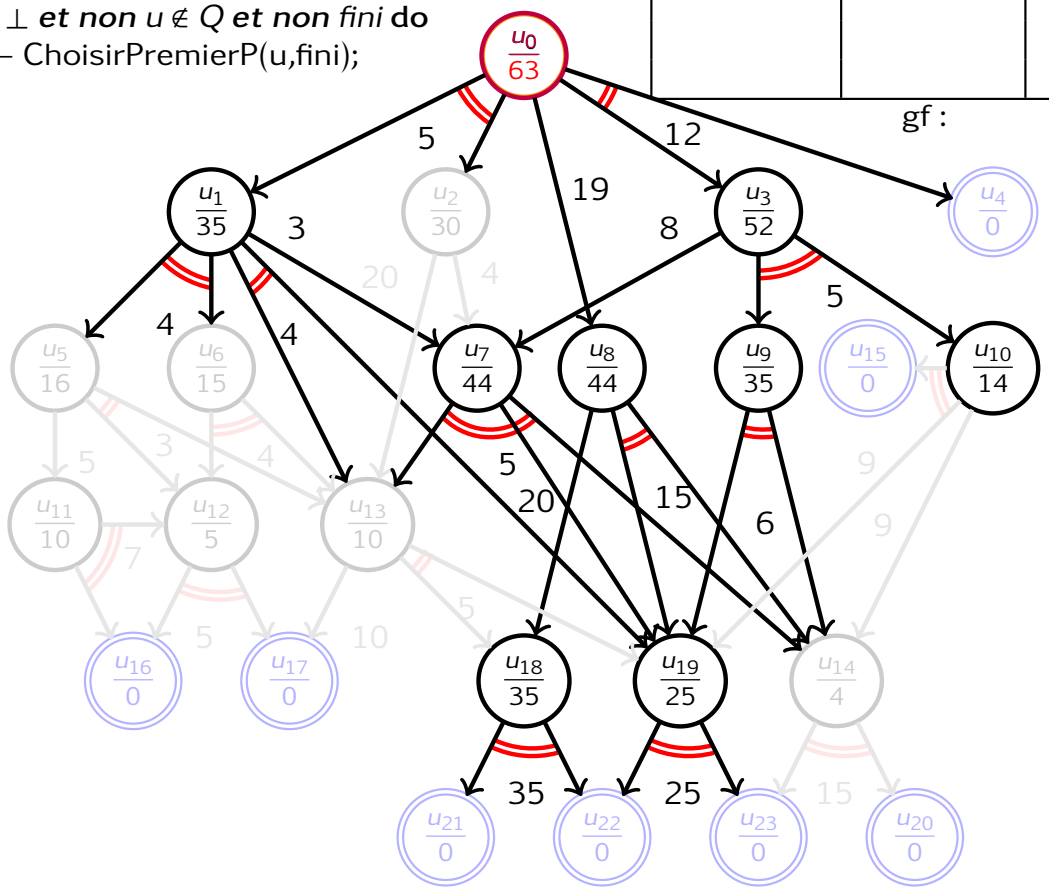


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
$u_0(0,63)$	(70,63,64)	{}	$u_8(19,44)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,52)$	(52,54)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,44)$	(44)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,35)$	(35,39,47)	{}	$u_5(9,16)u_6(9,15)u_2(5,30)$
$u_9(17,35)$	(35)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)$

gf :

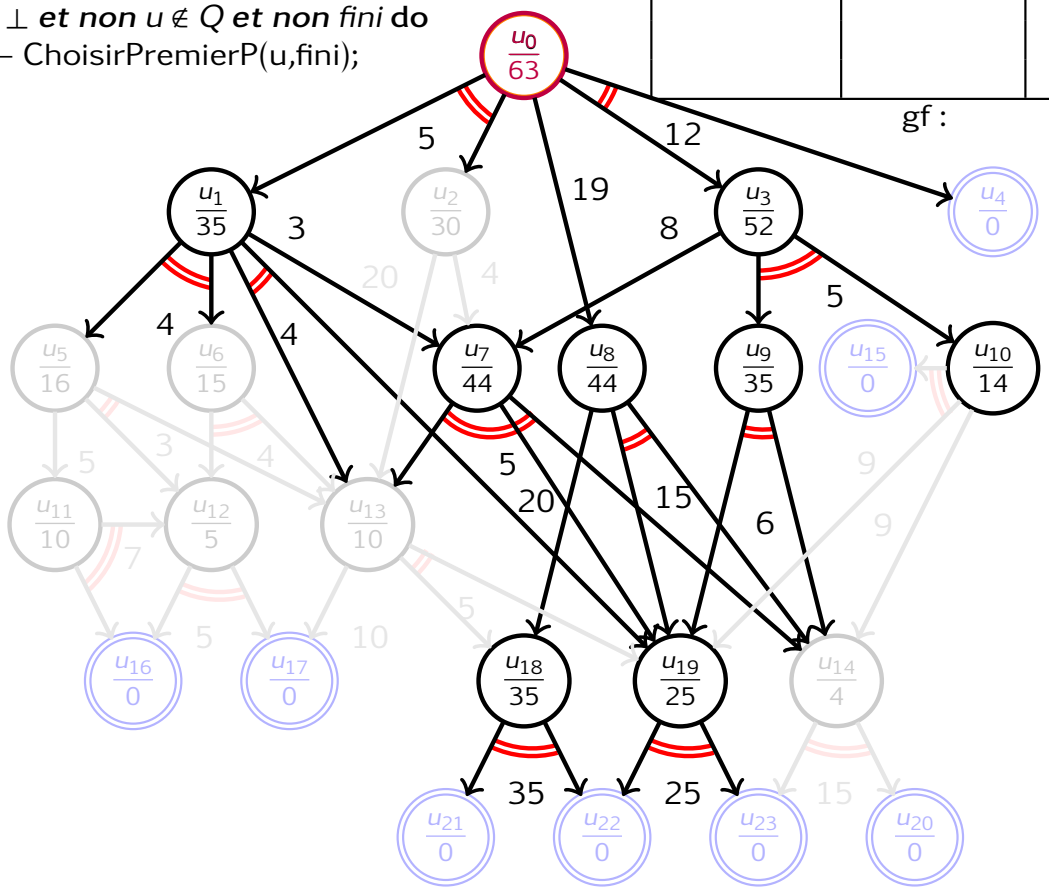


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,63)	(70,63,64)	{}	u_8(19,44)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,52)	(52,54)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,35)	(35)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)

gf :

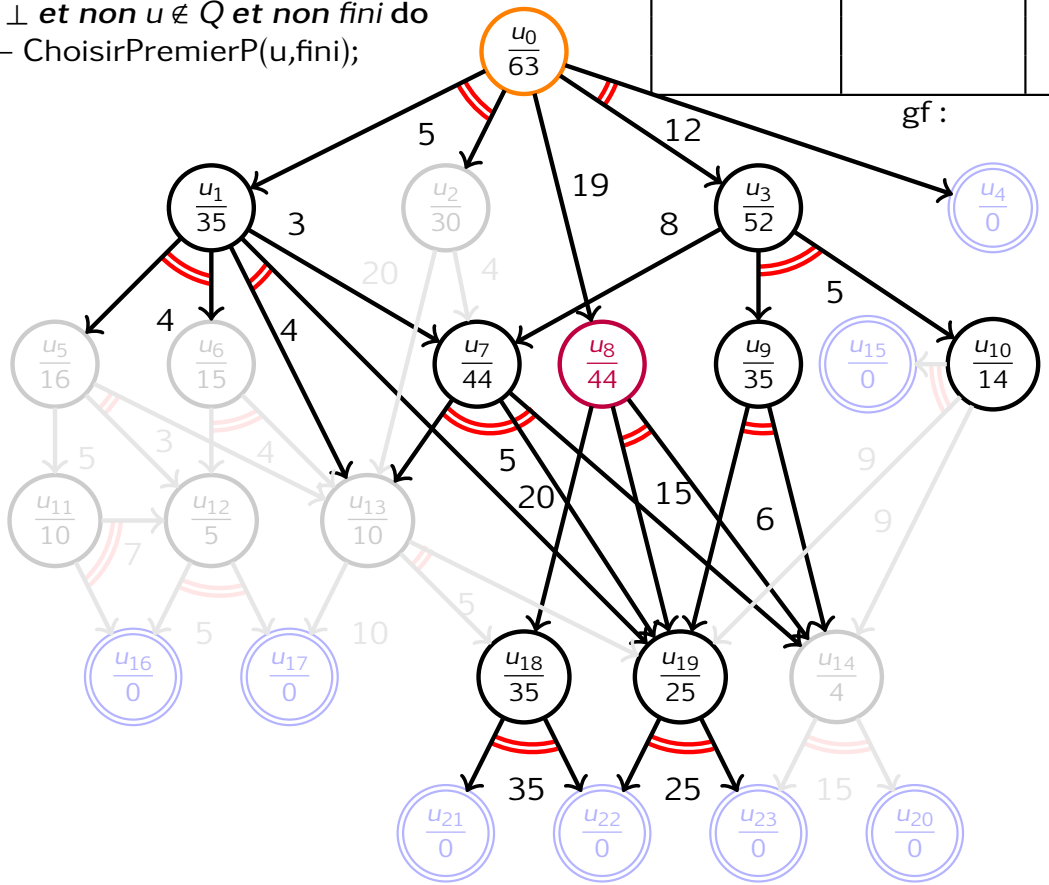


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,63)	(70,63,64)	{}	u_8(19,44)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,52)	(52,54)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,35)	(35)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)

gf :

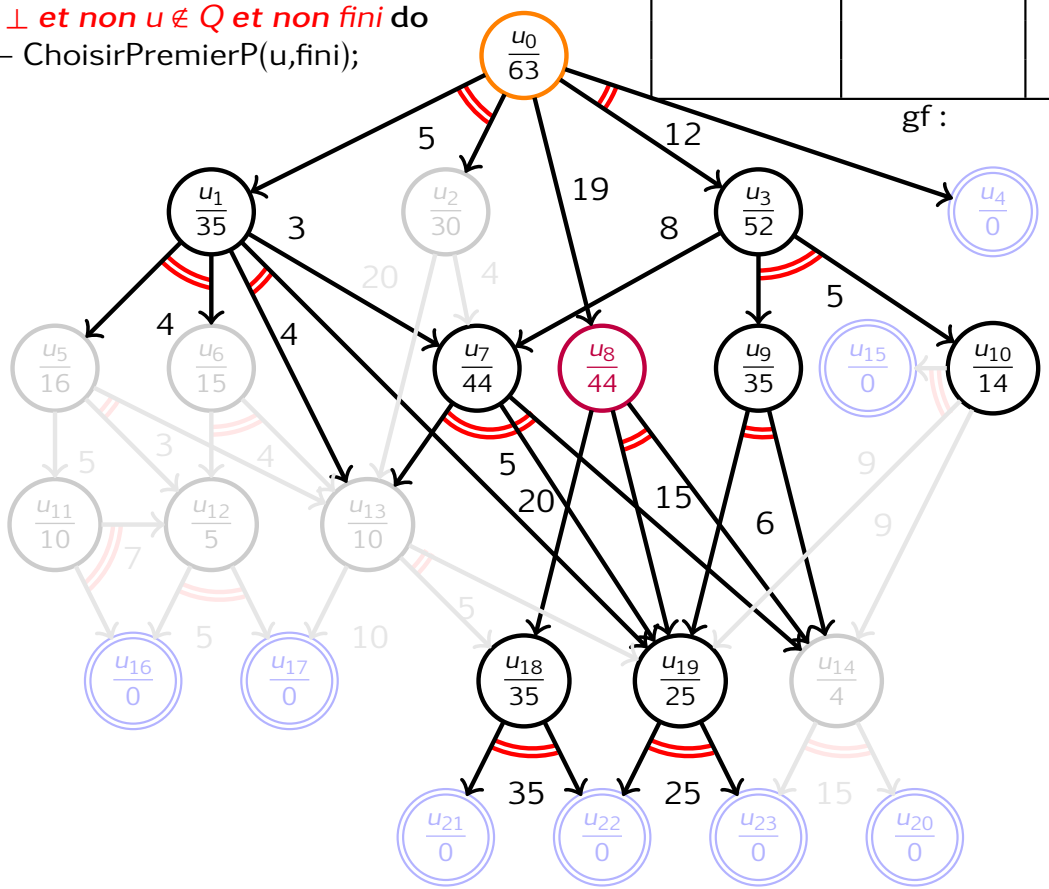


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,63)	(70,63,64)	{}	u_8(19,44)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,52)	(52,54)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,35)	(35)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)

gf :

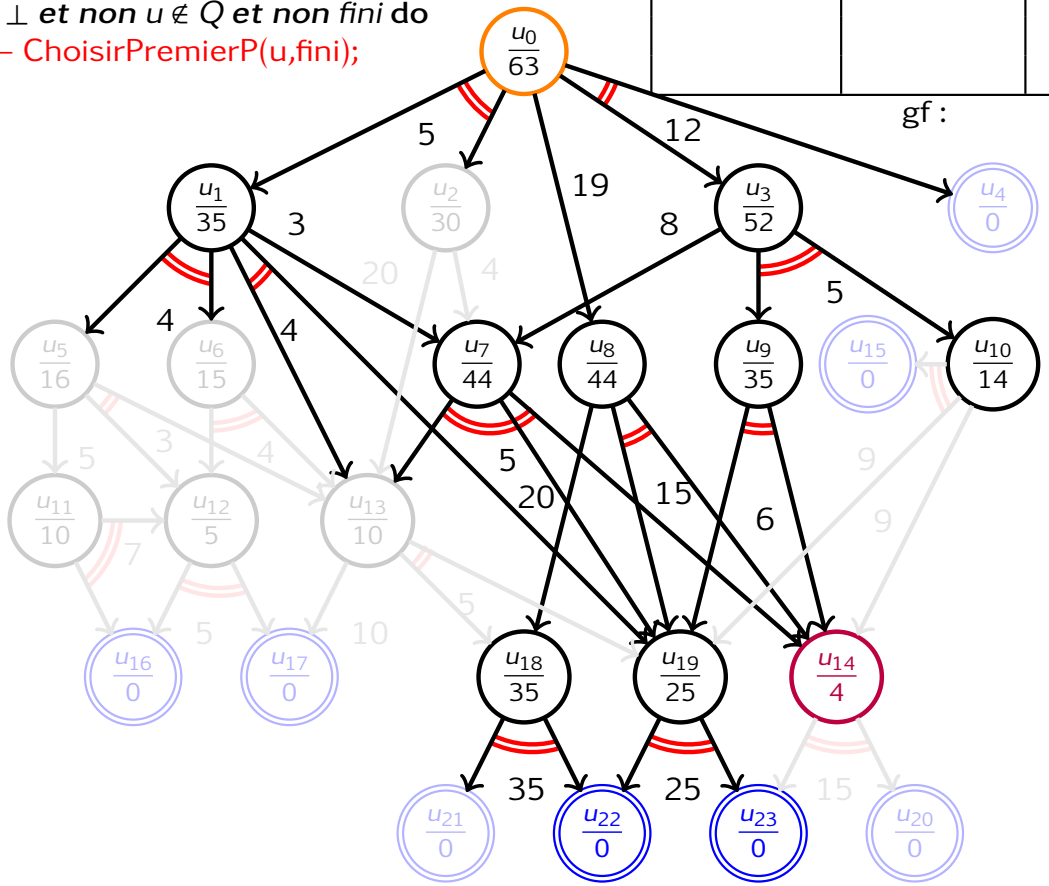


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,63)	(70,63,64)	{}	u_8(19,44)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,52)	(52,54)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,35)	(35)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,4)	(-)	{}	

gf :

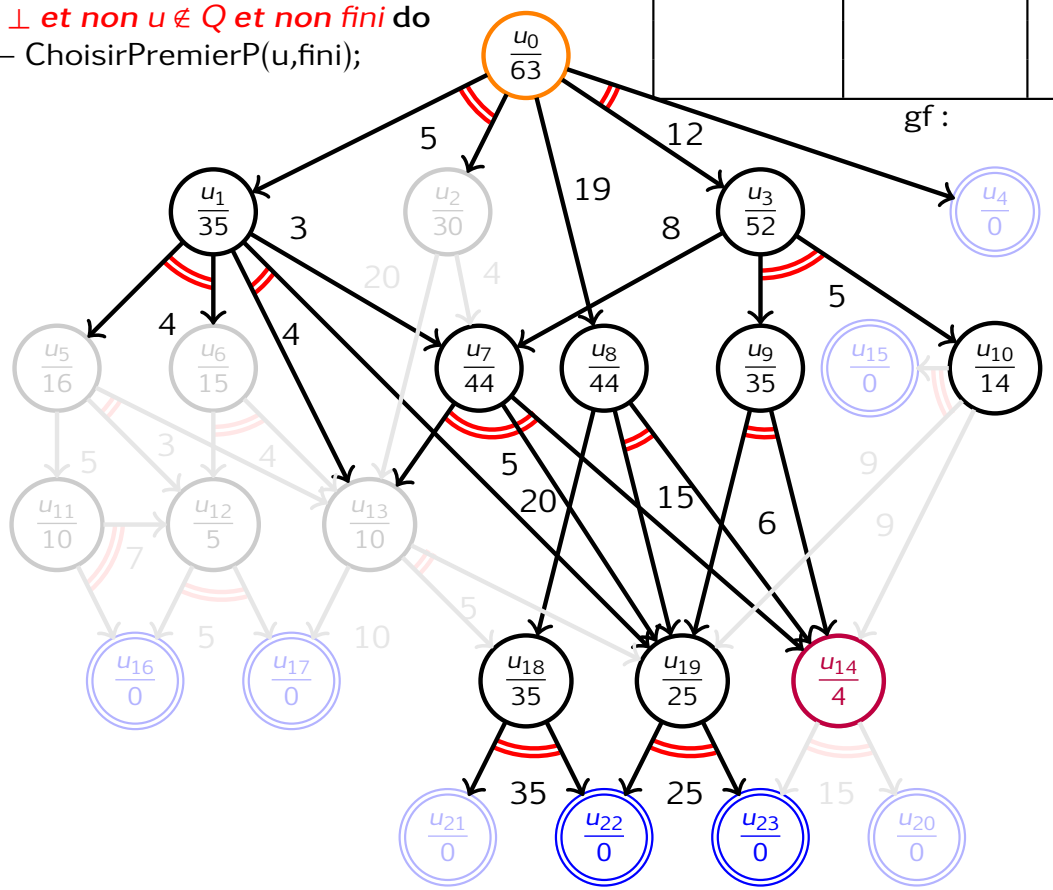


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,63)	(70,63,64)	{}	u_8(19,44)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,52)	(52,54)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,35)	(35)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,4)	(-)	{}	

gf :

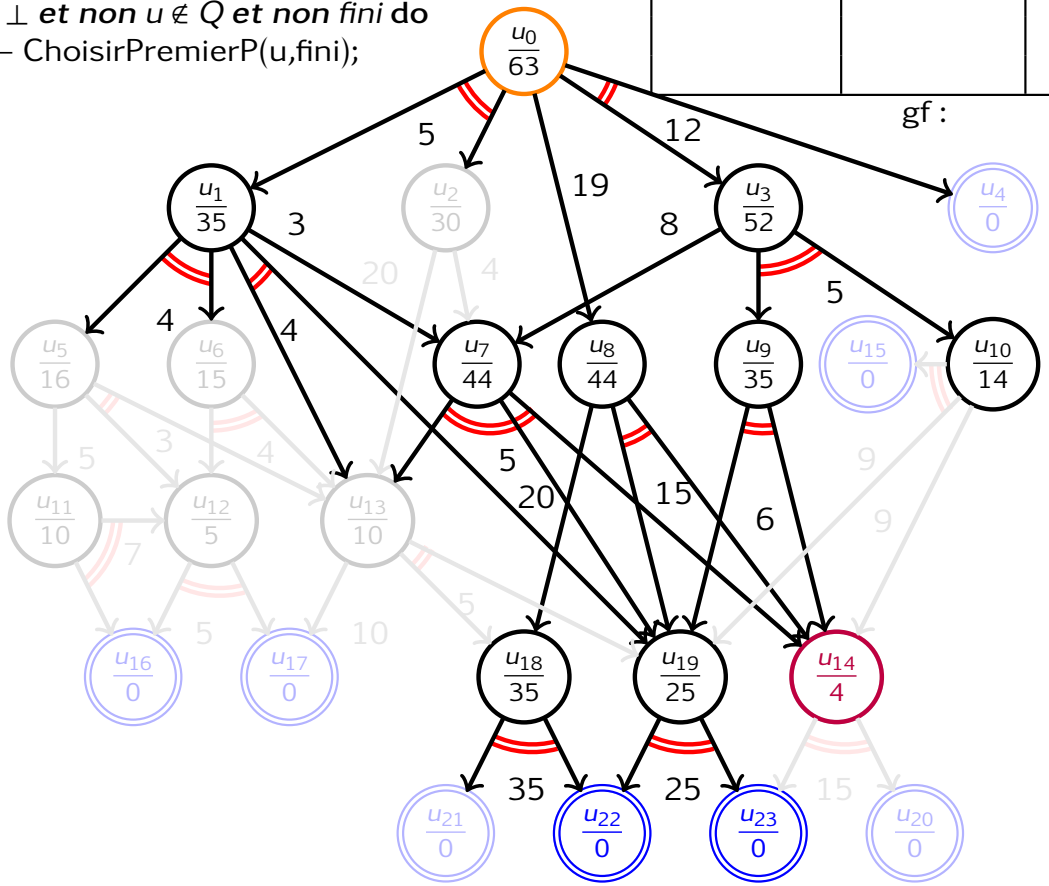


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,63)	(70,63,64)	{}	u_8(19,44)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,52)	(52,54)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,35)	(35)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,4)	(-)	{}	

gf :

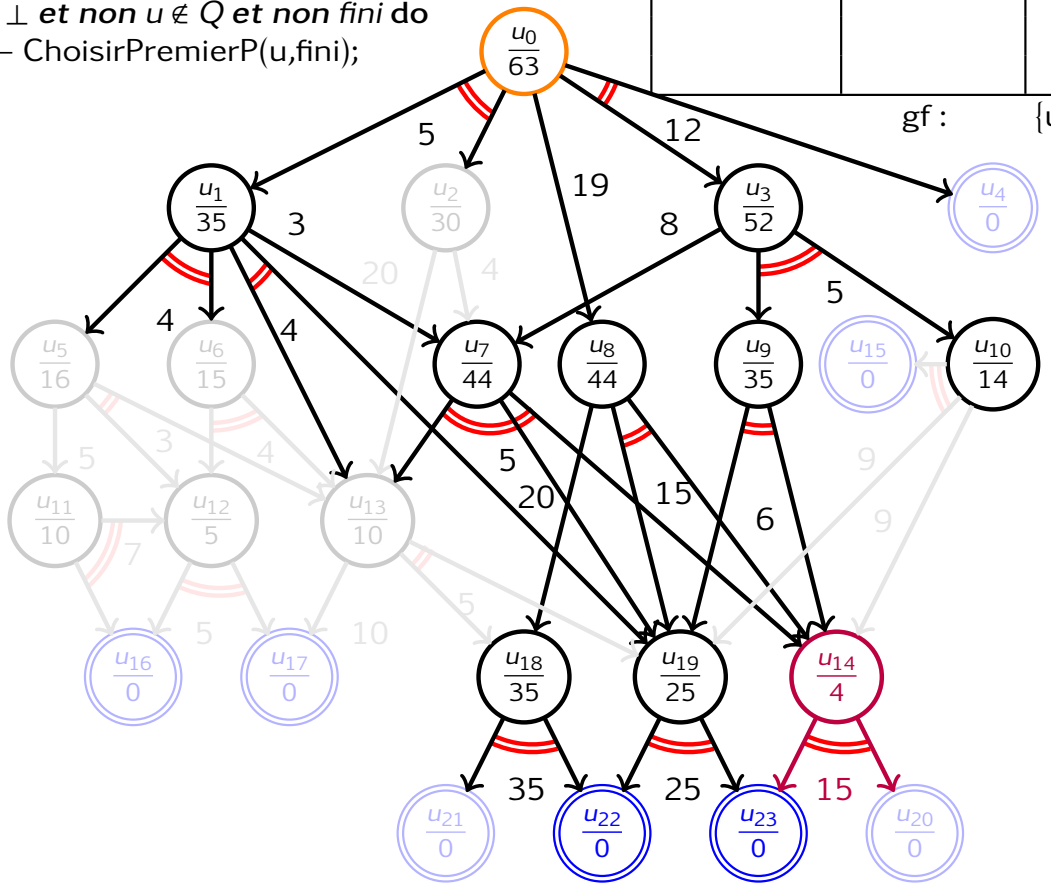


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,63)	(70,63,64)	{}	u_8(19,44)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,52)	(52,54)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,35)	(35)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,4)	(-)	{}	

gf : $\{u_{23}, u_{20}\}$

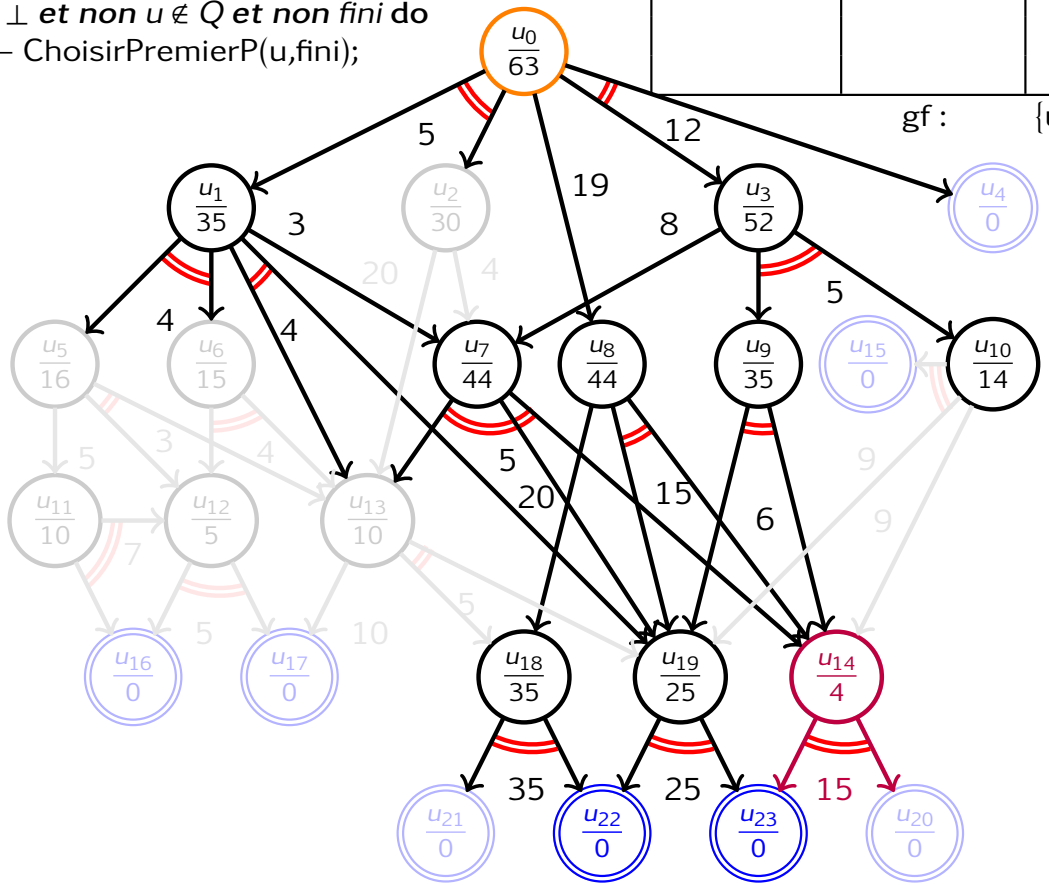


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,63)	(70,63,64)	{}	u_8(19,44)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,52)	(52,54)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,35)	(35)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,4)	(-)	{}	

gf : $\{u_{23}, u_{20}\}$



Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
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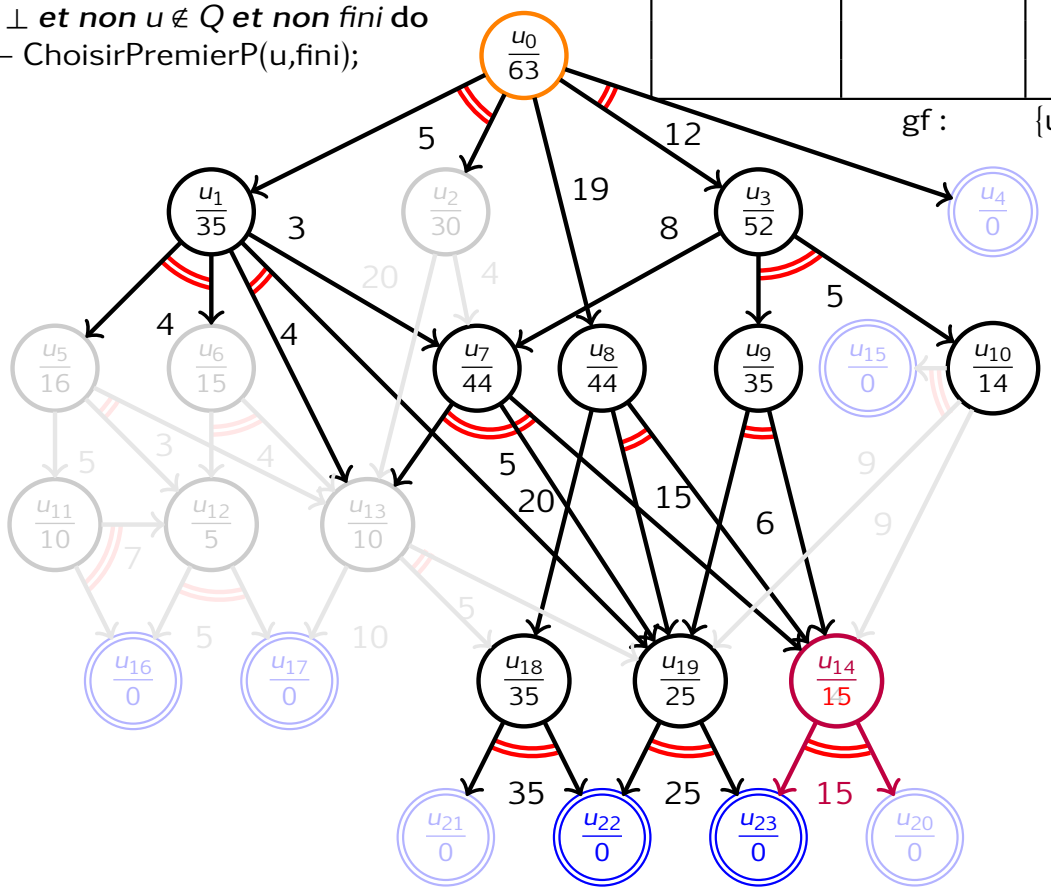
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,63)$	(70,63,64)	{}	$u_8(19,44)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,52)$	(52,54)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,44)$	(44)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,35)$	(35,39,47)	{}	$u_5(9,16)u_6(9,15)u_2(5,30)$
$u_9(17,35)$	(35)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_1$
$u_{14}(34,15)$	(15)	$\{u_7\ u_8\ u_9\}$	$u_{23}(49,0)$ $u_{20}(49,0)$

gf : $\{u_{23},u_{20}\}$ = 15+0+0 = 15



Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
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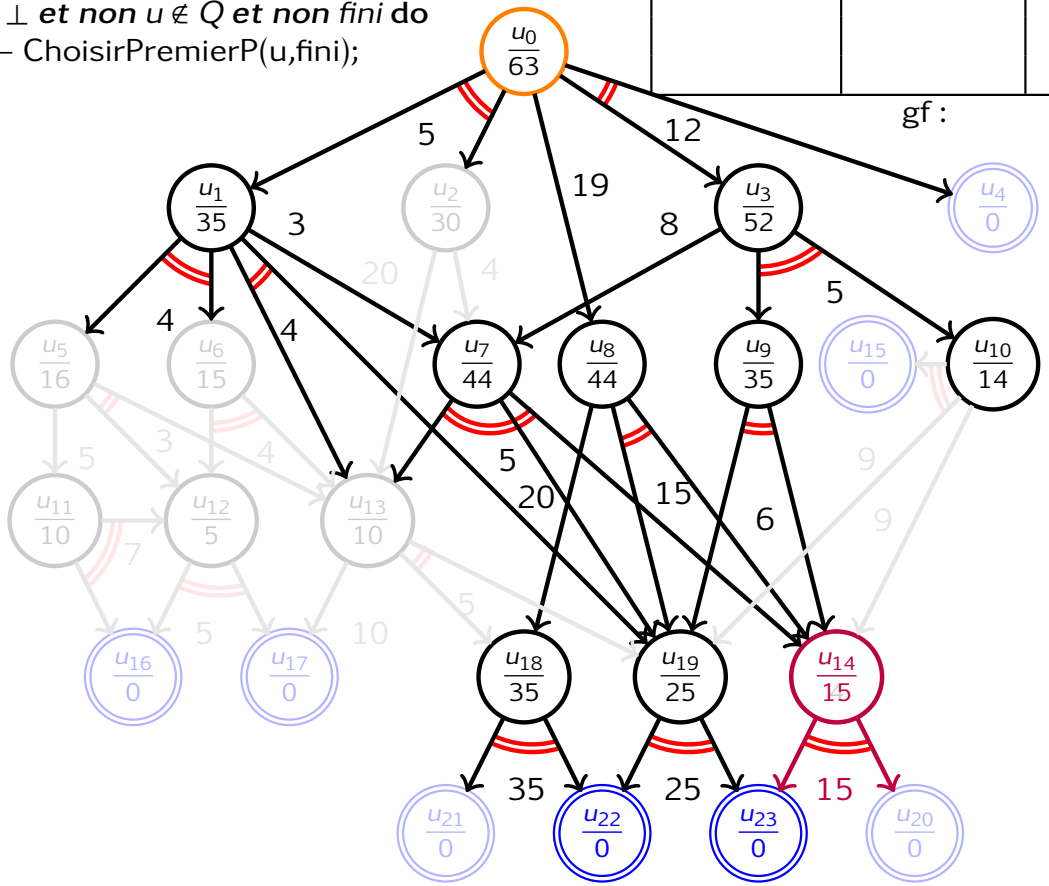
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,63)$	(70,63,64)	{}	$u_8(19,44)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,52)$	(52,54)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,44)$	(44)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,35)$	(35,39,47)	{}	$u_5(9,16)u_6(9,15)u_2(5,30)$
$u_9(17,35)$	(35)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_1$
$u_{14}(34,15)$	(15)	$\{u_7\ u_8\ u_9\}$	$u_{23}(49,0)$ $u_{20}(49,0)$

gf :



Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
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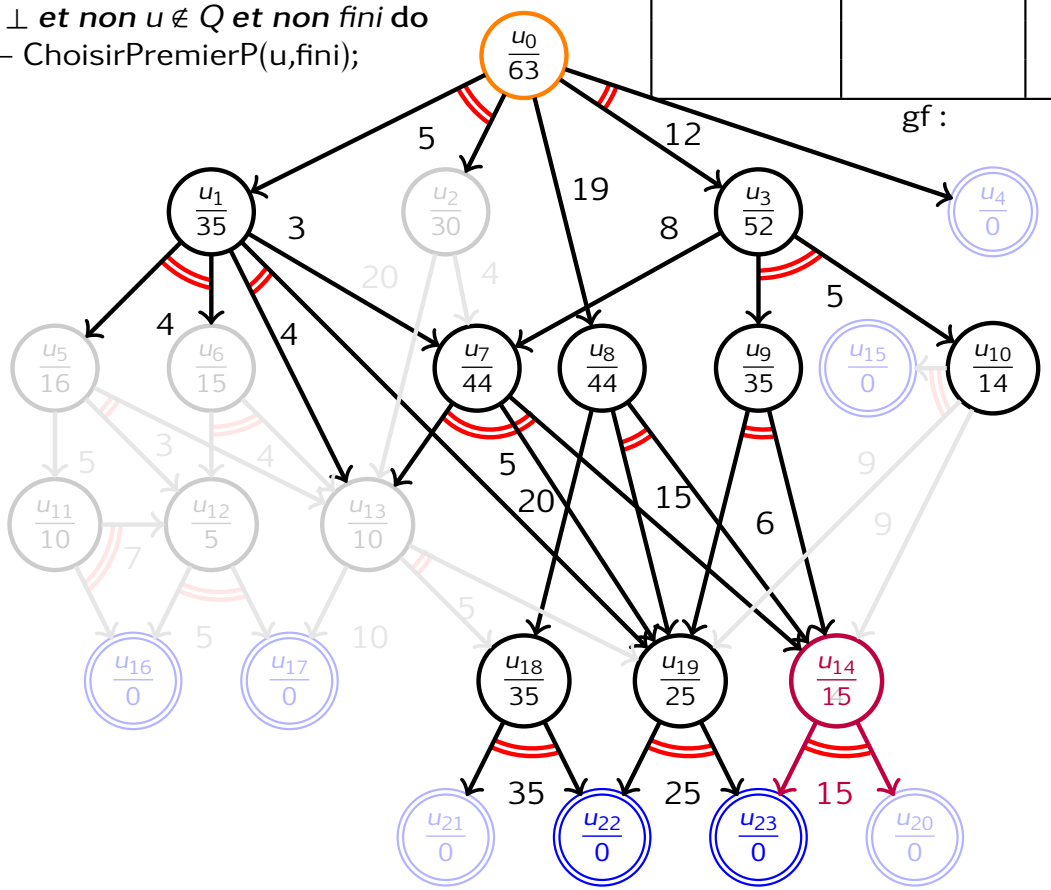
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,63)$	(70,63,64)	{}	$u_8(19,44)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,52)$	(52,54)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,44)$	(44)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,35)$	(35,39,47)	{}	$u_5(9,16)u_6(9,15)u_2(5,30)$
$u_9(17,35)$	(35)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_1$
$u_{14}(34,15)$	(15)	$\{u_7\ u_8\ u_9\}$	$u_{23}(49,0)$ $u_{20}(49,0)$

gf :



Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
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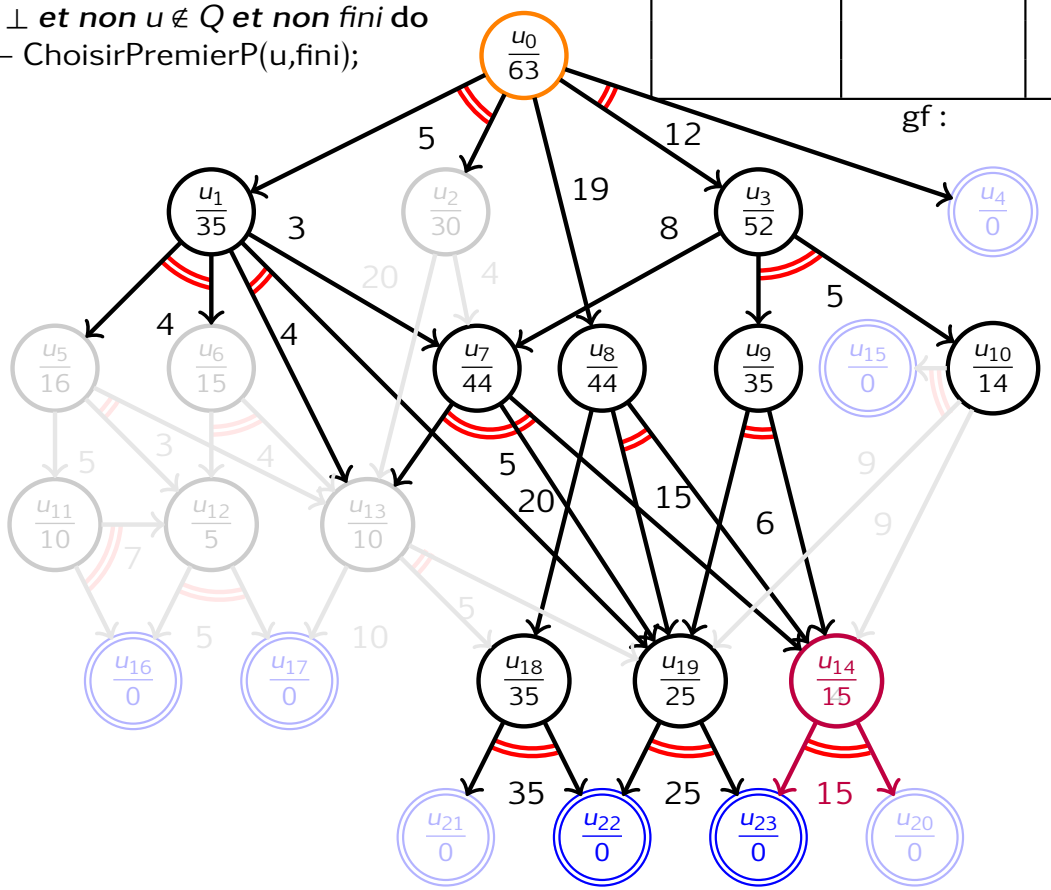
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,63)	(70,63,64)	{}	u_8(19,44)
u_8(19,44)	(55,44)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,52)	(52,54)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,44)	(44)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,47)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,35)	(35)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_1
u_{14}(34,15)	(15)	{u_7 u_8 u_9}	u_{23}(49,0) u_{20}(49,0)

gf :



Algorithme OA*(Liste,Sol)

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    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
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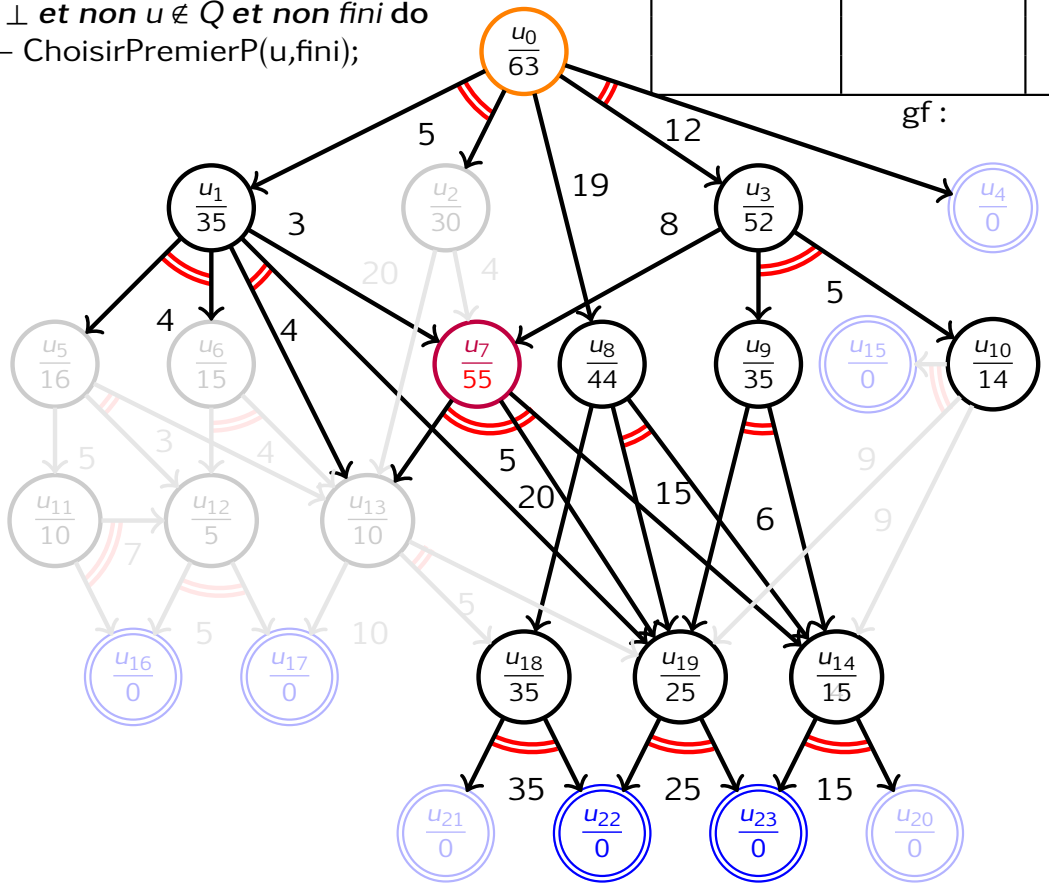
u(g(u),h(u))	fils(u)	A	P
$u_0(0,63)$	(70,63,64)	{}	$u_8(19,44)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,52)$	(52,54)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	$\{u_8 u_9 u_1 u_3\}$	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,35)$	(35,39,47)	{}	$u_5(9,16)u_6(9,15)u_2(5,30)$
$u_9(17,35)$	(35)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$

gf :



Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
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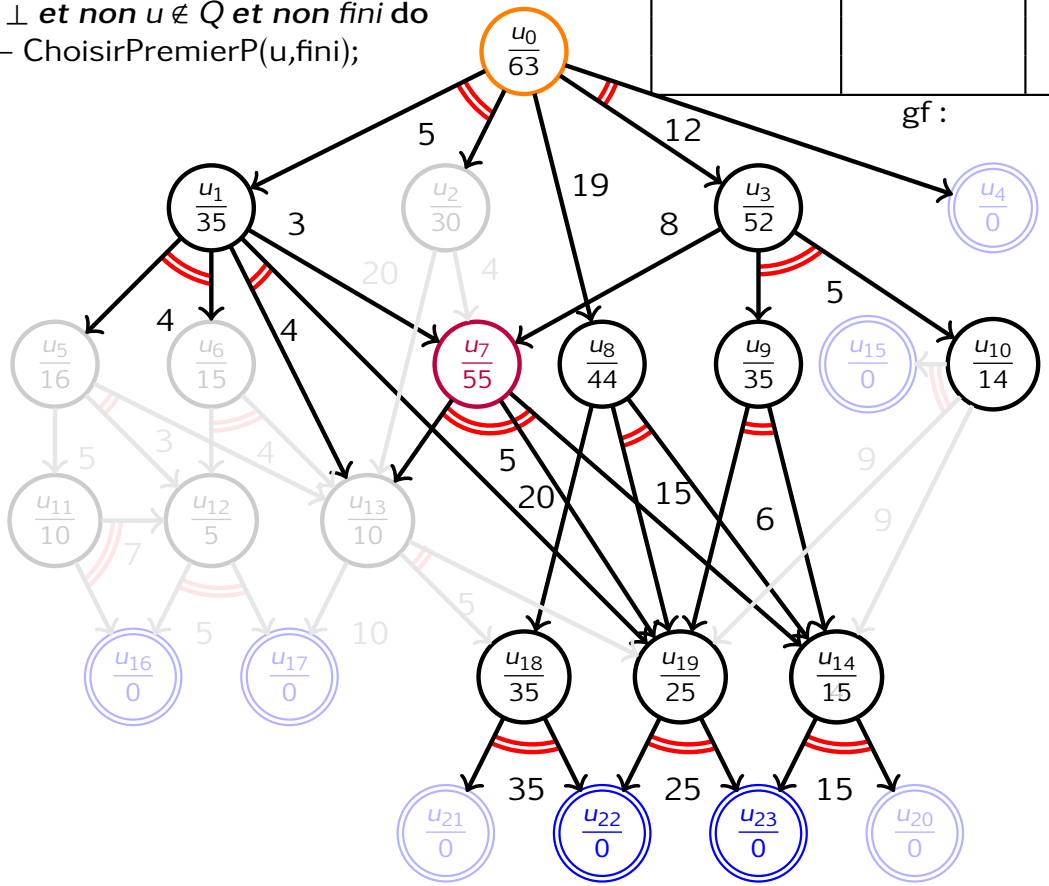
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,63)$	(70,63,64)	{}	$u_8(19,44)$
$u_8(19,44)$	(55,44)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,52)$	(52,54)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	$\{u_8 u_9 u_1 u_3\}$	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,35)$	(35,39,47)	{}	$u_5(9,16)u_6(9,15)u_2(5,30)$
$u_9(17,35)$	(35)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$

gf :



Algorithme OA*(Liste,Sol)

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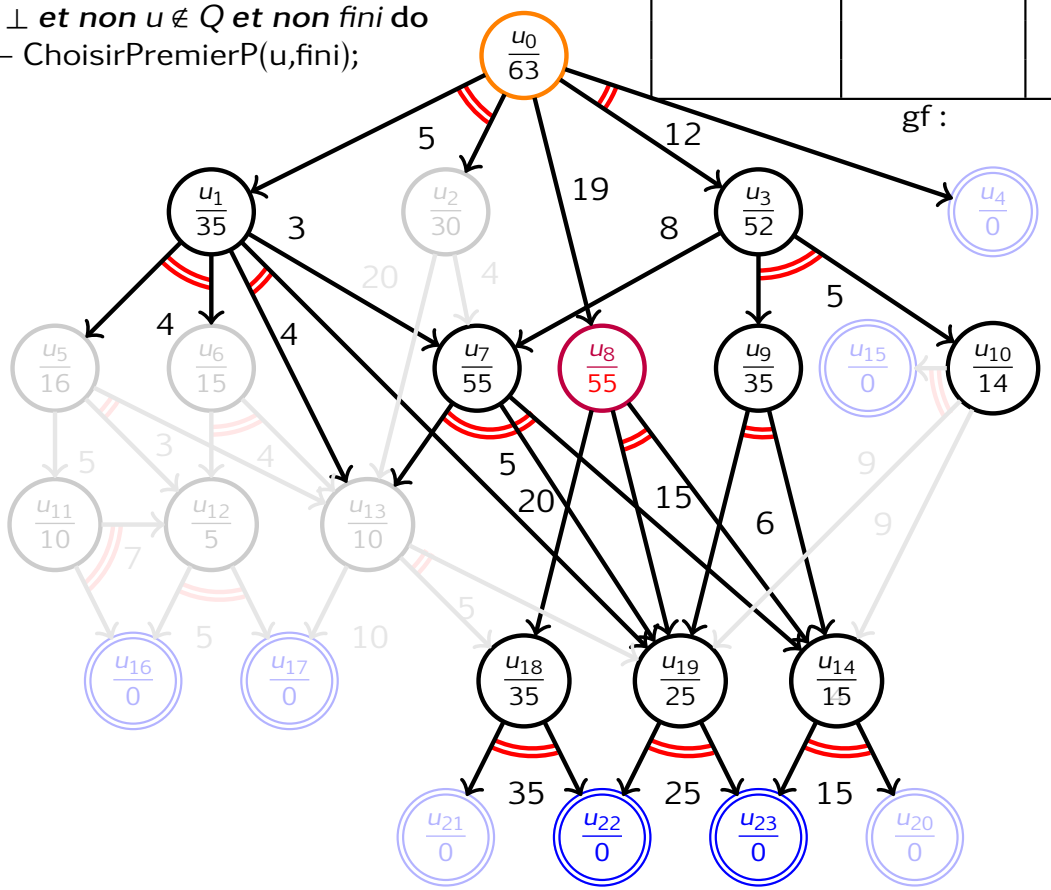
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,63)$	(70,63,64)	{}	$u_8(19,55)$
$u_8(19,55)$	(55,55)	$\{u_9 u_1 u_3 u_0\}$	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,52)$	(52,54)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,35)$	(35,39,47)	{}	$u_5(9,16)u_6(9,15)u_2(5,30)$
$u_9(17,35)$	(35)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$

gf :

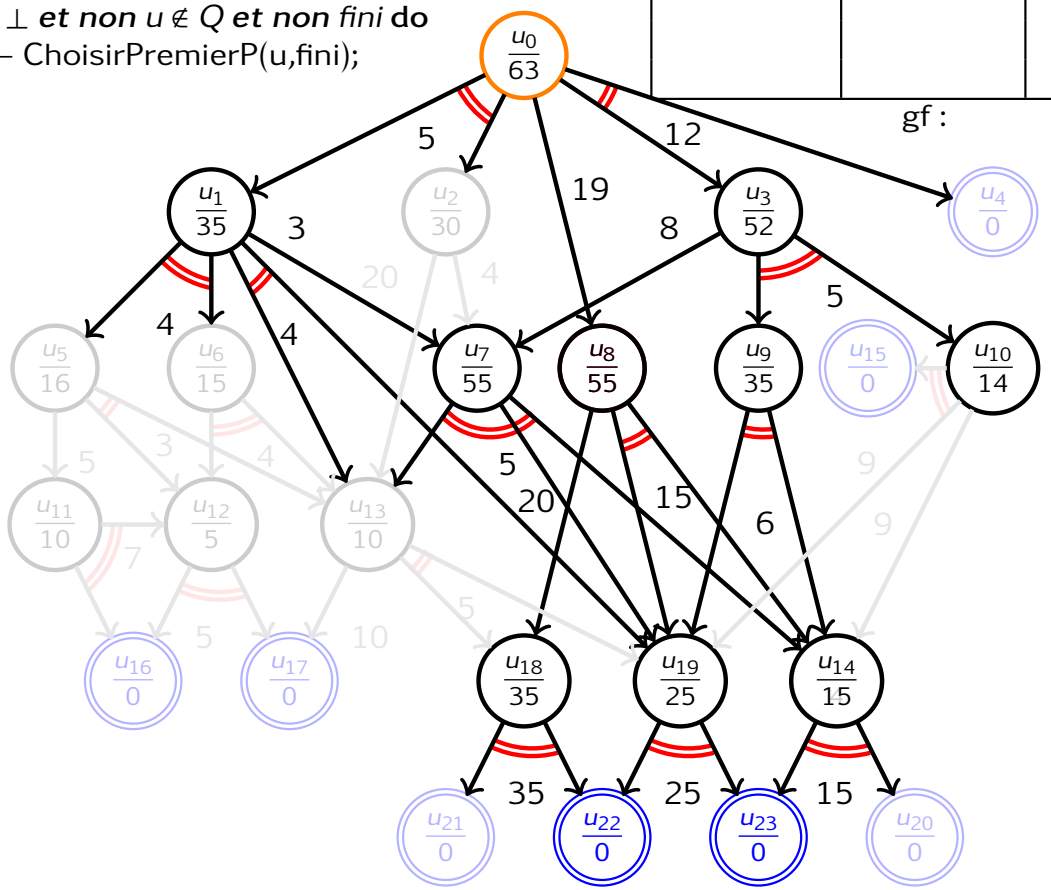


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,63)$	(70,63,64)	{}	$u_8(19,55)$
$u_8(19,55)$	(55,55)	$\{u_9 u_1 u_3 u_0\}$	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,52)$	(52,54)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,35)$	(35,39,47)	{}	$u_5(9,16)u_6(9,15)u_2(5,30)$
$u_9(17,35)$	(35)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$

gf :

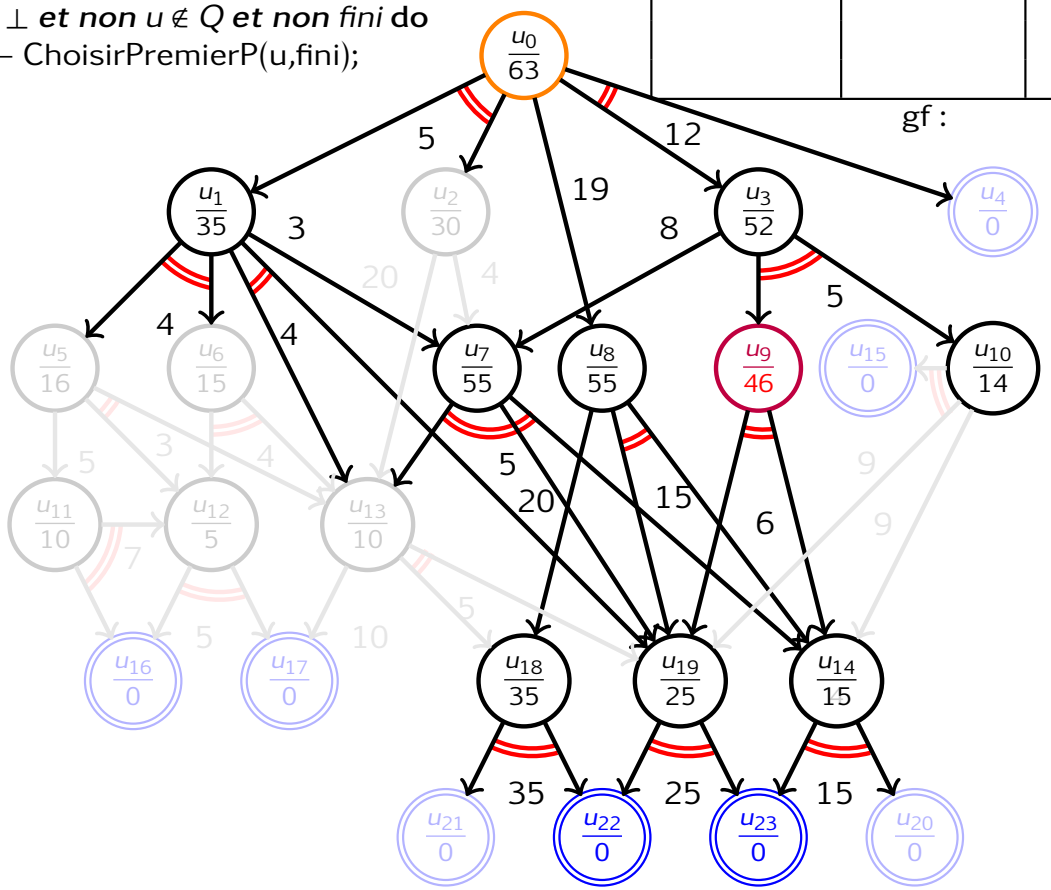


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
$u_0(0,63)$	(70,63,64)	{}	$u_8(19,55)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,52)$	(52,54)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,35)$	(35,39,47)	{}	$u_5(9,16)u_6(9,15)u_2(5,30)$
$u_9(17,46)$	(46)	$\{u_1 u_3 u_0\}$	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$

gf :



Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
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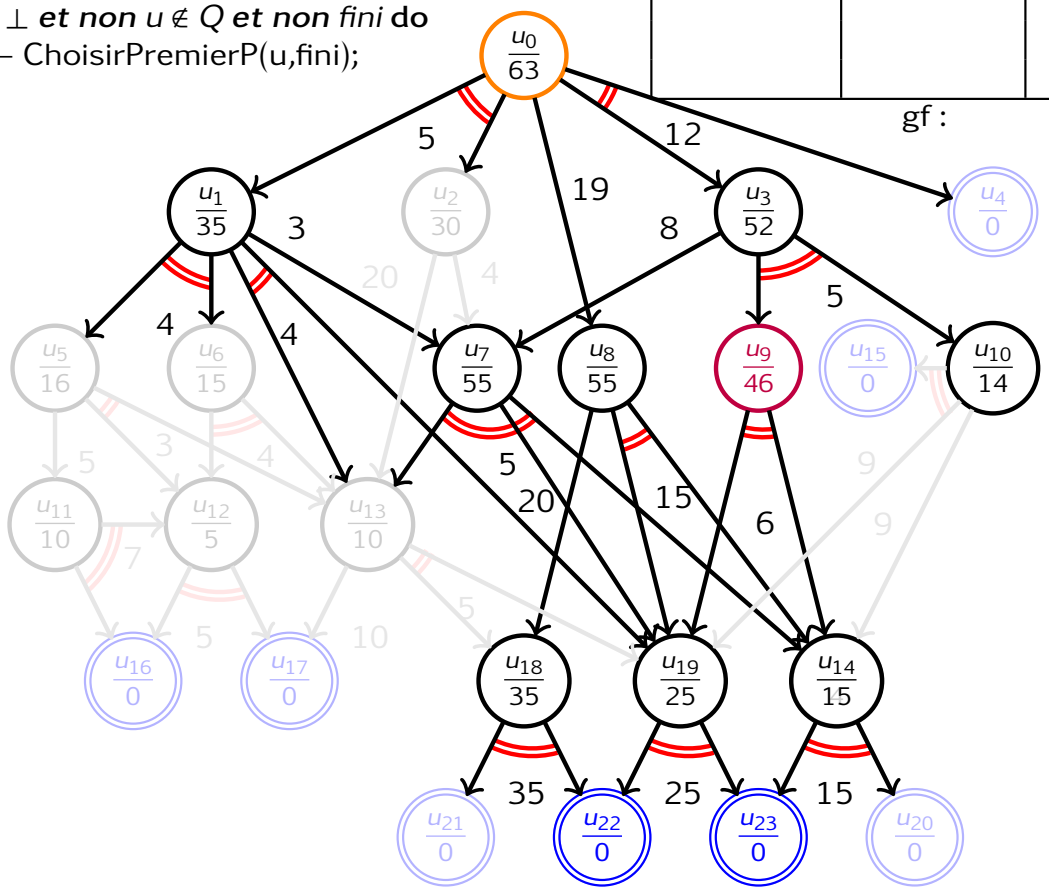
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,63)$	(70,63,64)	{}	$u_8(19,55)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,52)$	(52,54)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,35)$	(35,39,47)	{}	$u_5(9,16)u_6(9,15)u_2(5,30)$
$u_9(17,46)$	(46)	$\{u_1 u_3 u_0\}$	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$

gf :



Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
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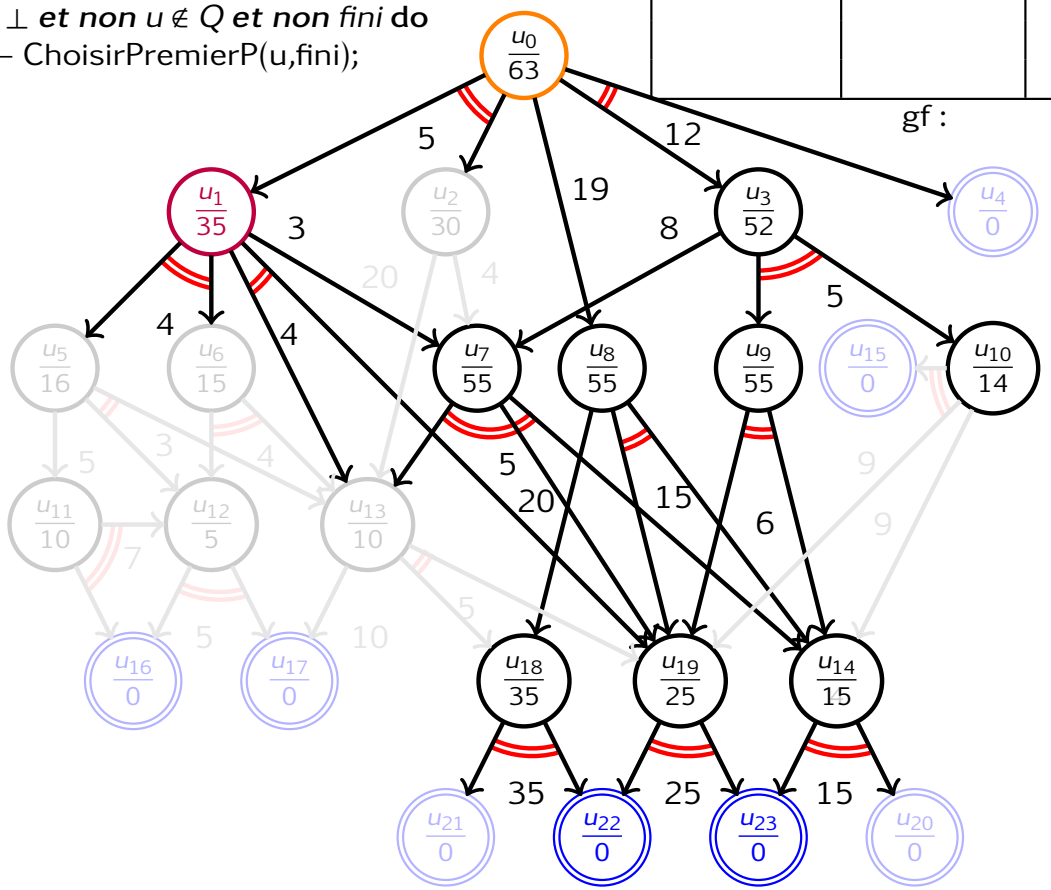
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,63)$	(70,63,64)	{}	$u_8(19,55)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,52)$	(52,54)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,35)$	(35,39,58)	$\{u_3\ u_0\}$	$u_5(9,16)u_6(9,15)u_2(5,30)$
$u_9(17,46)$	(46)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}($
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$

gf :



Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
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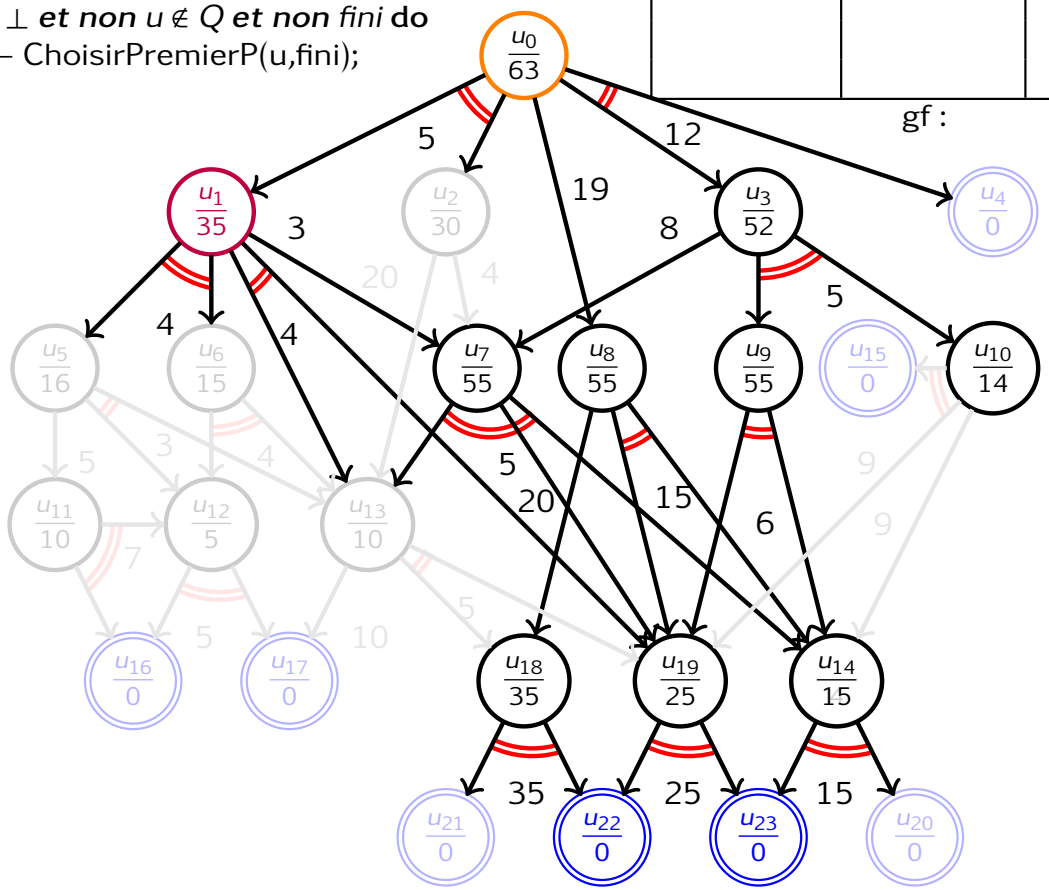
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,63)$	(70,63,64)	{}	$u_8(19,55)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,52)$	(52,54)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,35)$	(35,39,58)	$\{u_3\ u_0\}$	$u_5(9,16)u_6(9,15)u_2(5,30)$
$u_9(17,46)$	(46)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}($
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$

gf :



Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
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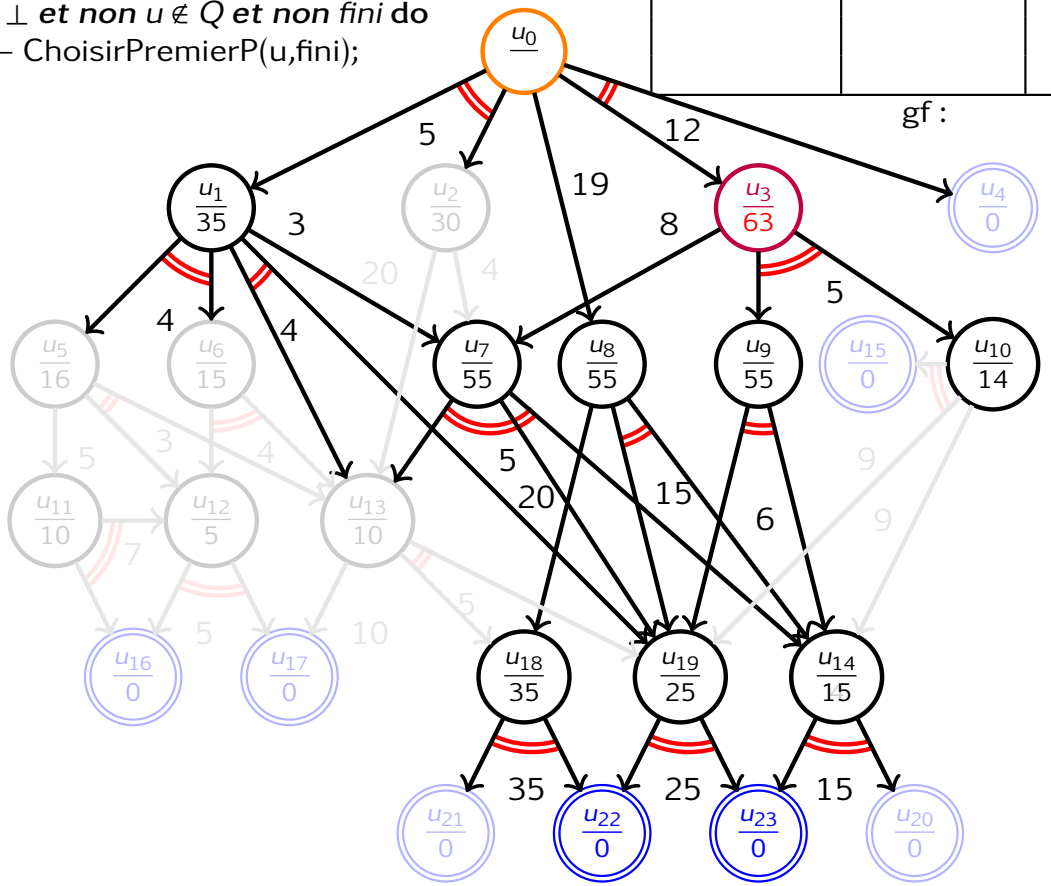
u(g(u),h(u))	fils(u)	A	P
$u_0(0,63)$	(70,63,64)	{}	$u_8(19,55)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,63)$	(63,74)	$\{u_0\}$	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,35)$	(35,39,58)	{}	$u_5(9,16)u_6(9,15)u_2(5,30)$
$u_9(17,46)$	(46)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$

gf :



Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
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  while *gf ≠ ∅* **do**
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    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
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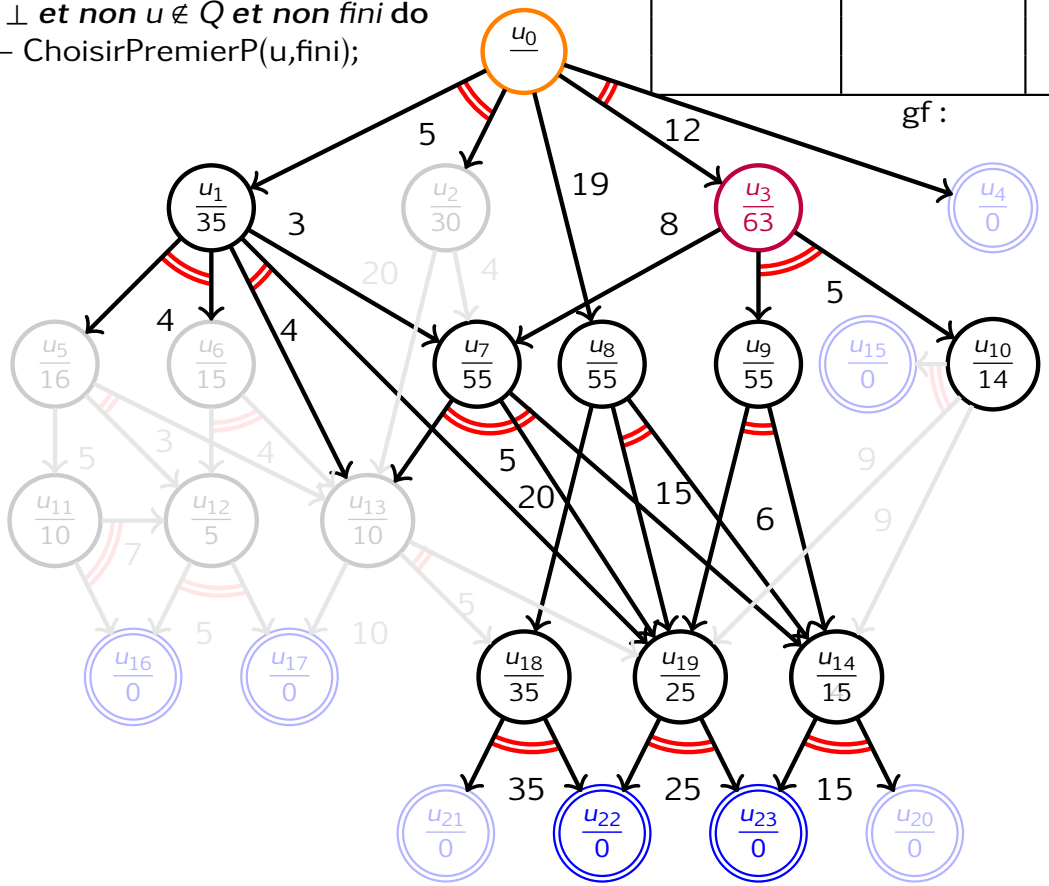
u(g(u),h(u))	fils(u)	A	P
$u_0(0,63)$	(70,63,64)	{}	$u_8(19,55)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,63)$	(63,74)	$\{u_0\}$	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,35)$	(35,39,58)	{}	$u_5(9,16)u_6(9,15)u_2(5,30)$
$u_9(17,46)$	(46)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$

gf :



Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
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    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
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  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
$u_0(0,70)$	(70,74,75)	{}	$u_1(5,35)$ $u_2(5,30)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,63)$	(63,74)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,35)$	(35,39,58)	{}	$u_5(9,16)u_6(9,15)u_2(5,30)$
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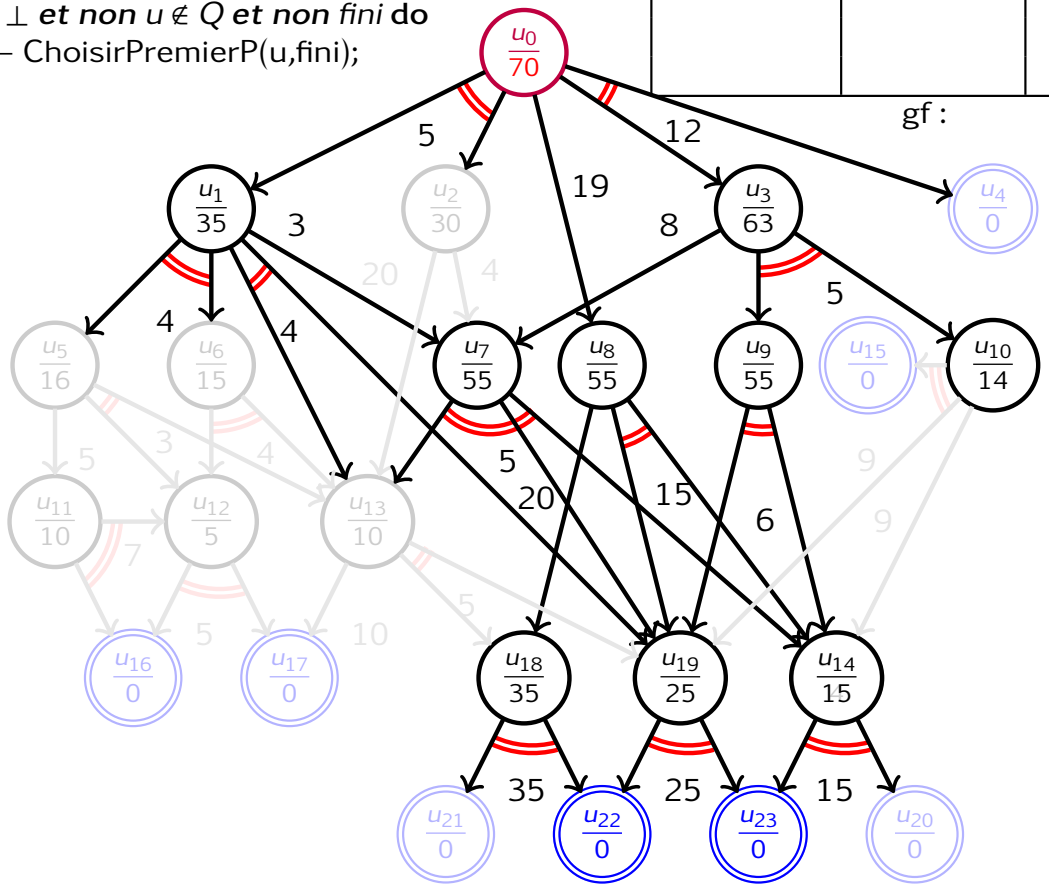
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$

gf :



Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
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  while *gf ≠ ∅* **do**
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    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
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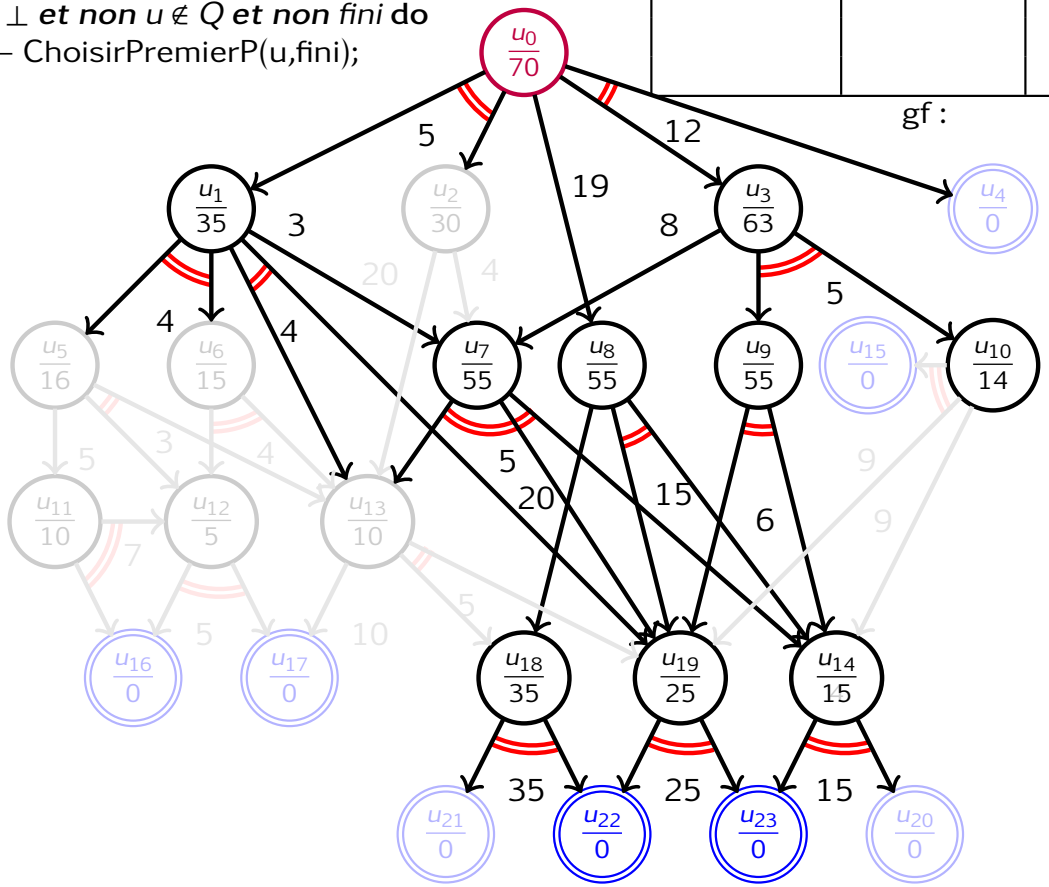
u(g(u),h(u))	fils(u)	A	P
u_0(0,70)	(70,74,75)	{}	u_1(5,35) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,58)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)

gf :



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    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
$u_0(0,70)$	(70,74,75)	{}	$u_1(5,35)$ $u_2(5,30)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
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$u_3(12,63)$	(63,74)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,35)$	(35,39,58)	{}	$u_5(9,16)u_6(9,15)u_2(5,30)$
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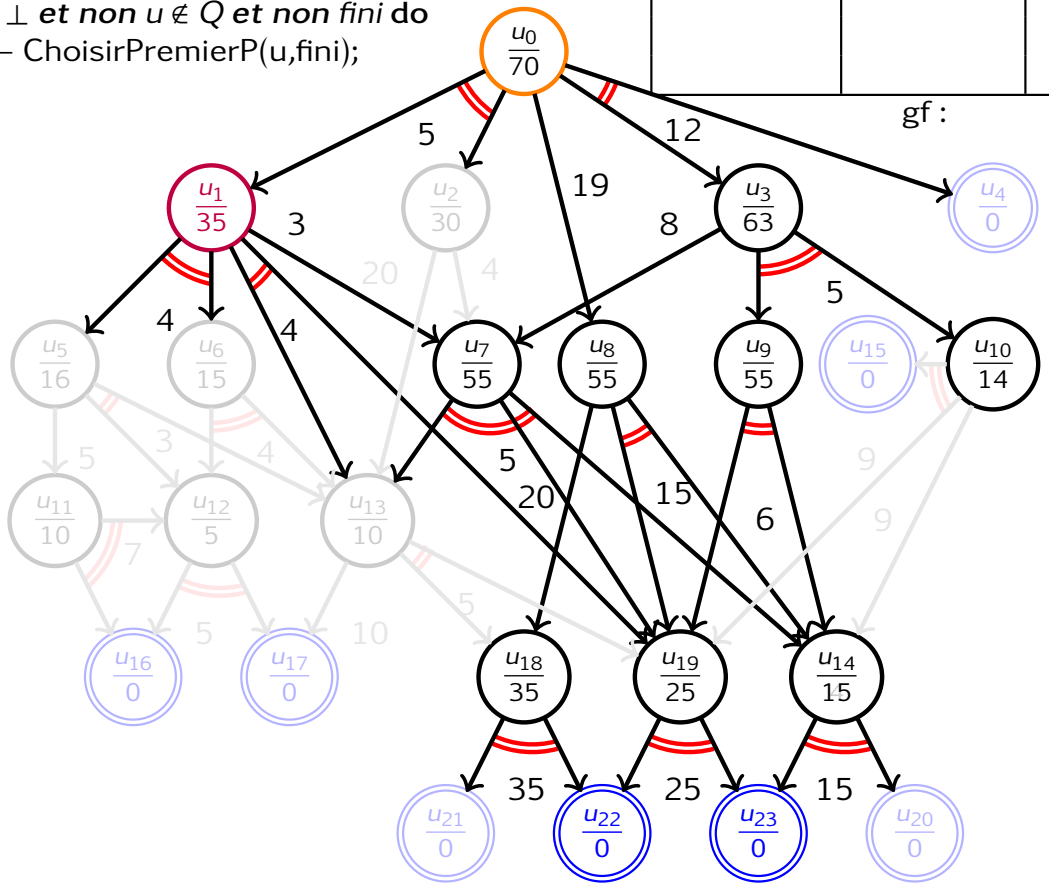
gf :



Algorithme OA*(Liste,Sol)

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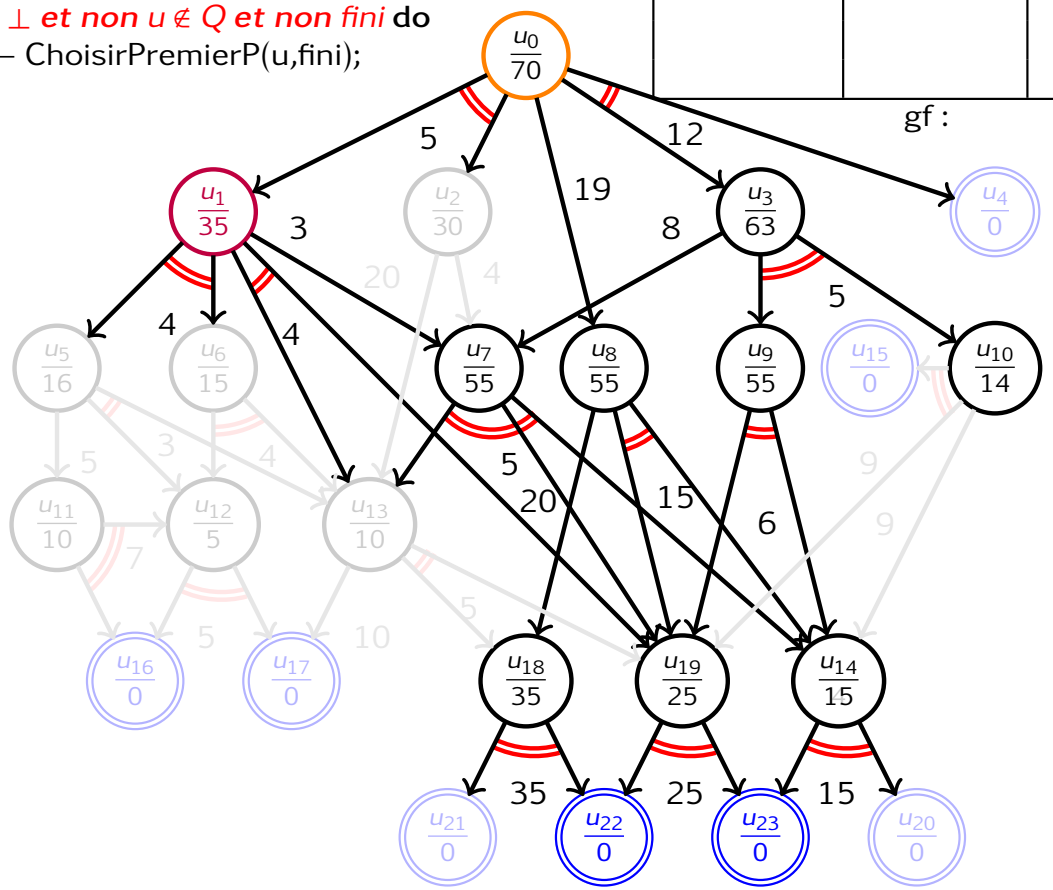
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,70)	(70,74,75)	{}	u_1(5,35) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,58)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,16)	(-,-)	{}	u_6(9,15) u_2(5,30)

gf :

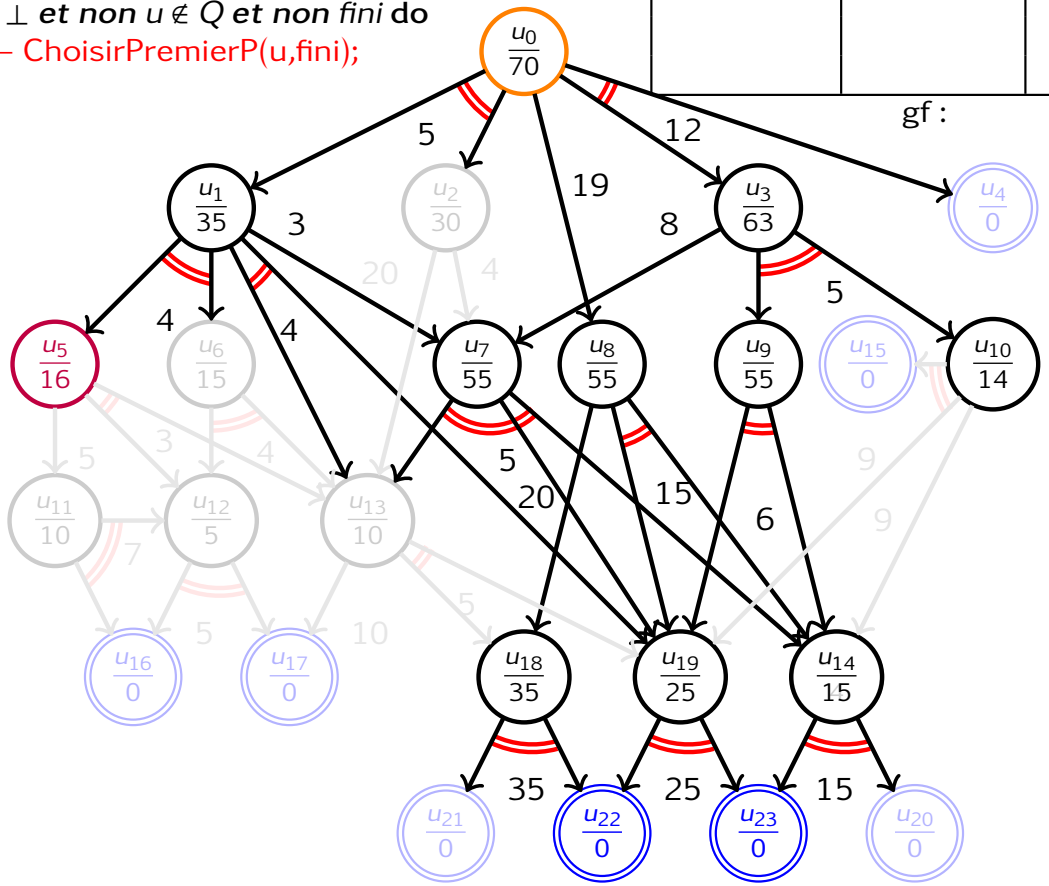


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,70)	(70,74,75)	{}	u_1(5,35) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,58)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,16)	(-,-)	{}	u_6(9,15) u_2(5,30)

gf :

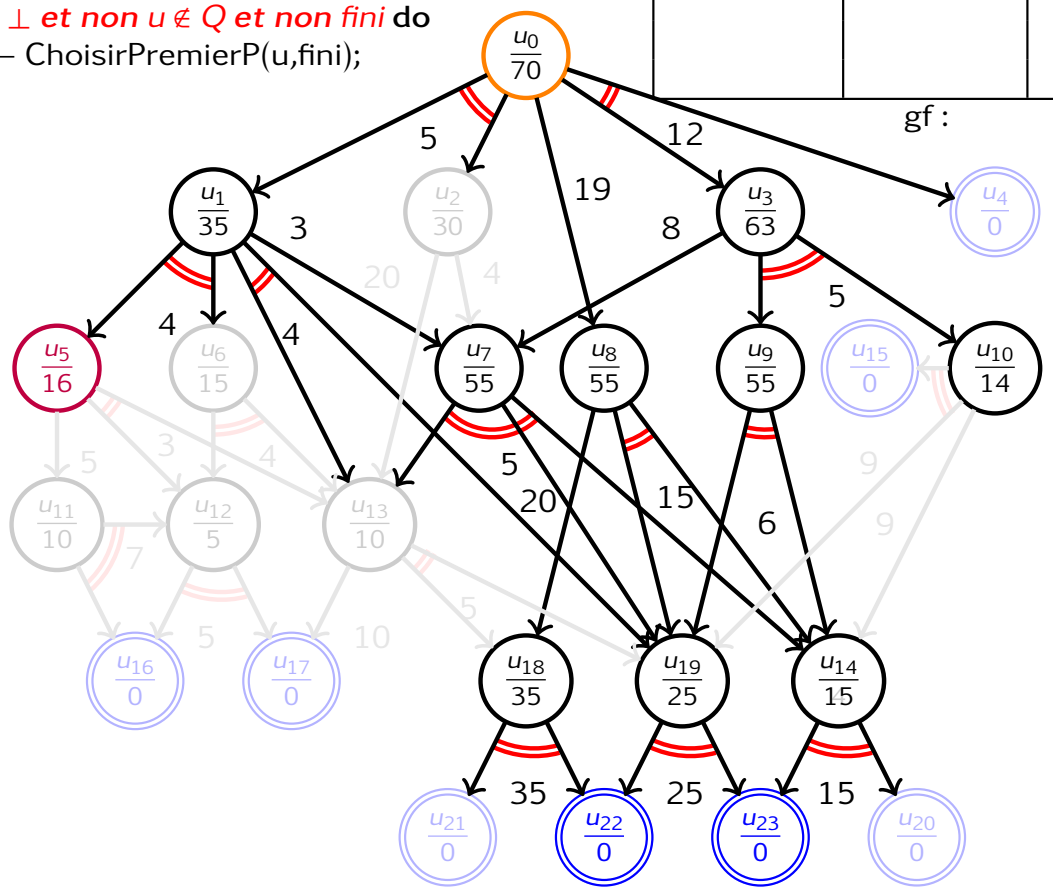


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,70)	(70,74,75)	{}	u_1(5,35) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,58)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,16)	(-,-)	{}	u_6(9,15) u_2(5,30)

gf :

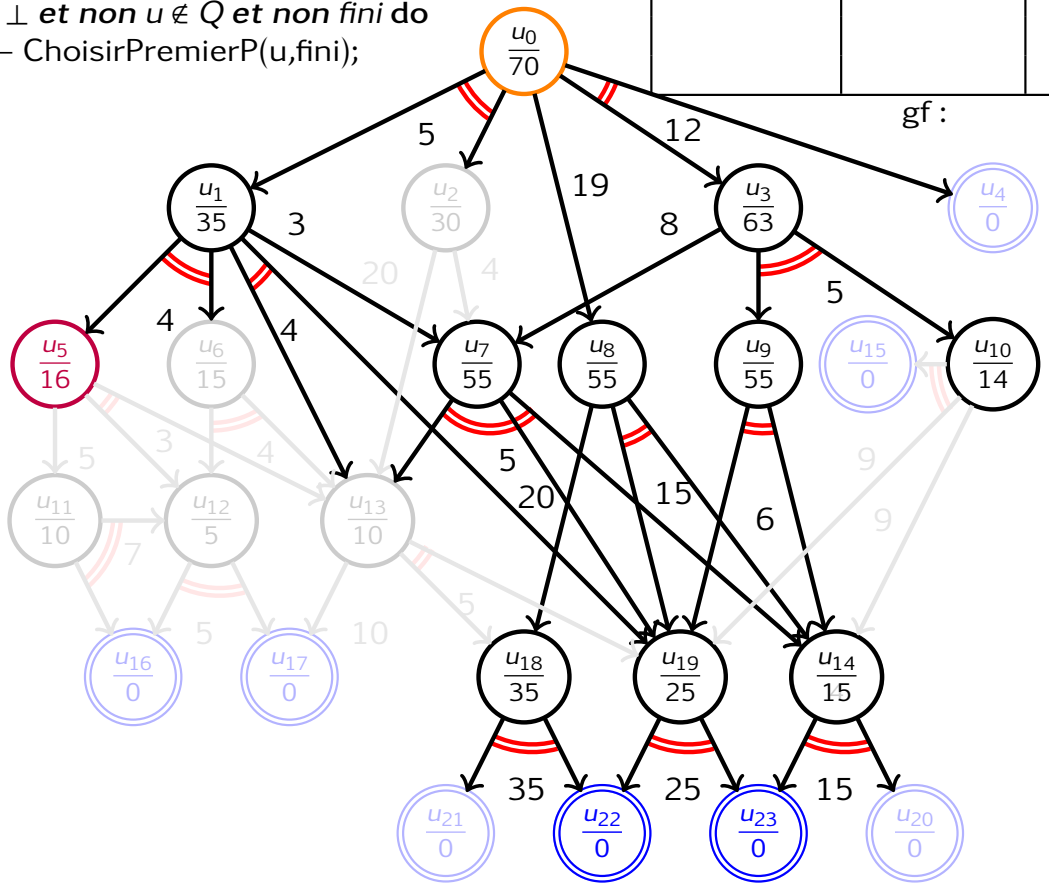


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,70)	(70,74,75)	{}	u_1(5,35) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,58)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,16)	(-,-)	{}	u_6(9,15) u_2(5,30)

gf :

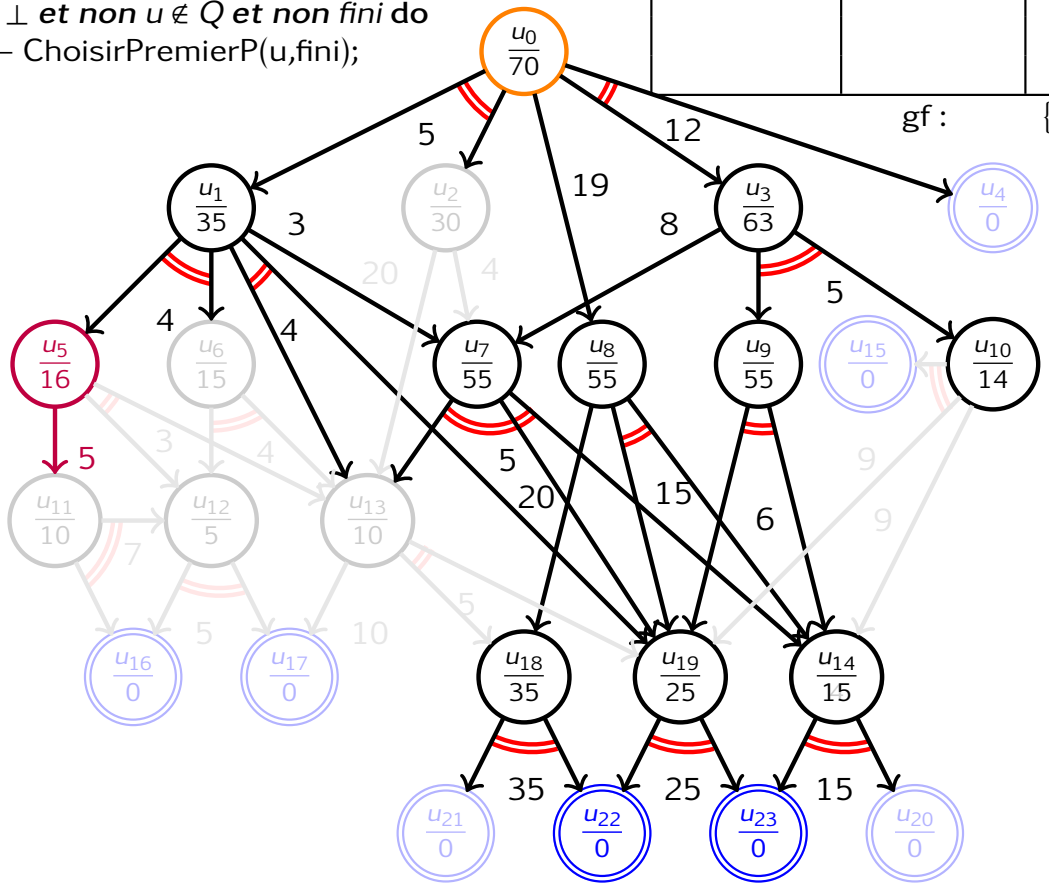


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,70)	(70,74,75)	{}	u_1(5,35) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,58)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,16)	(-,-)	{}	u_6(9,15) u_2(5,30)

gf : $\{u_{11}\}$

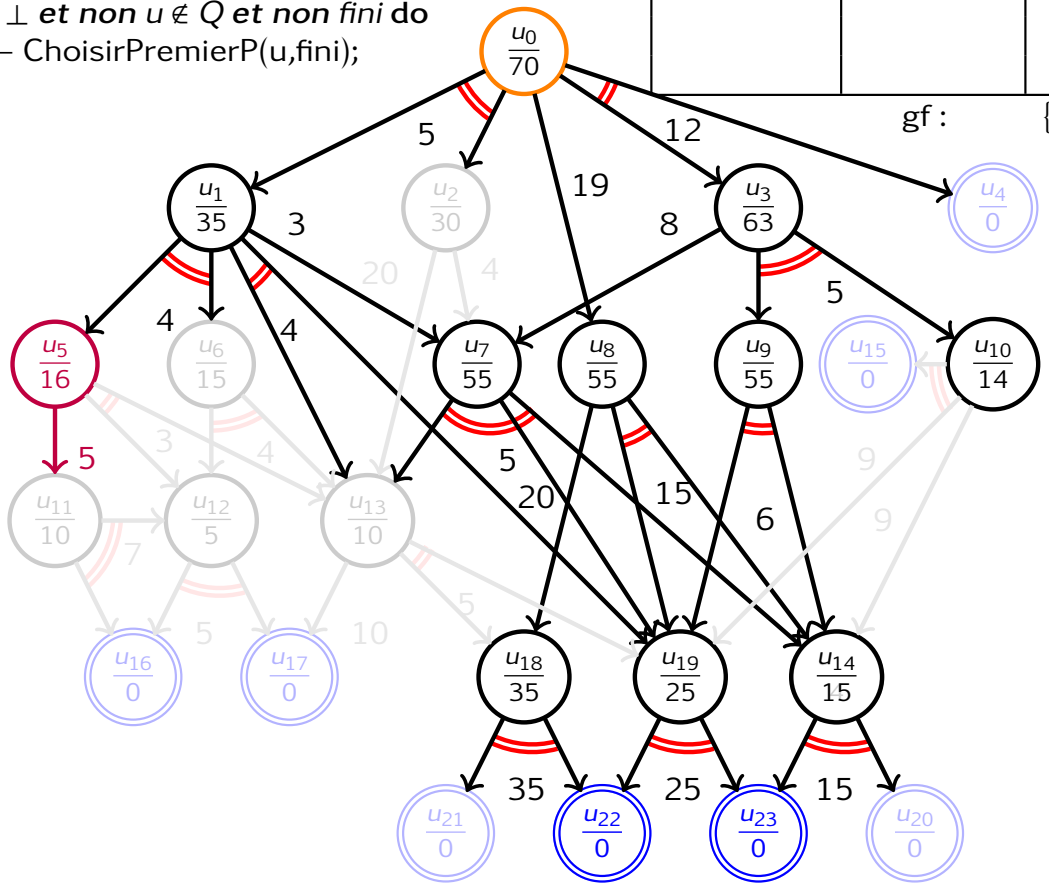


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,70)	(70,74,75)	{}	u_1(5,35) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,58)	{}	u_5(9,16)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,16)	(-,-)	{}	u_6(9,15) u_2(5,30)

gf : $\{u_{11}\}$

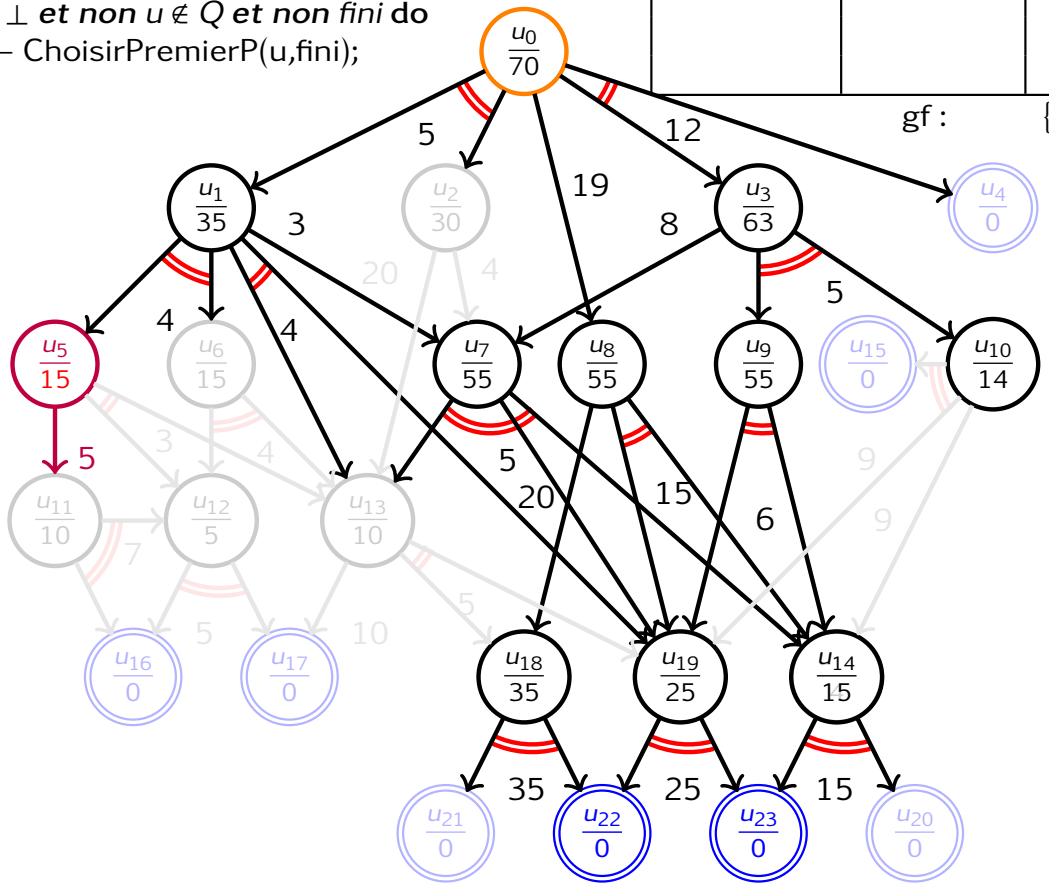


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,70)	(70,74,75)	{}	u_1(5,35) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,58)	{}	u_5(9,15)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,15)	(15,-)	{u_1}	u_{11}(14,15)u_6(9,15) u_2(5,30)

gf : {u_{11}} = 5+10 = 15

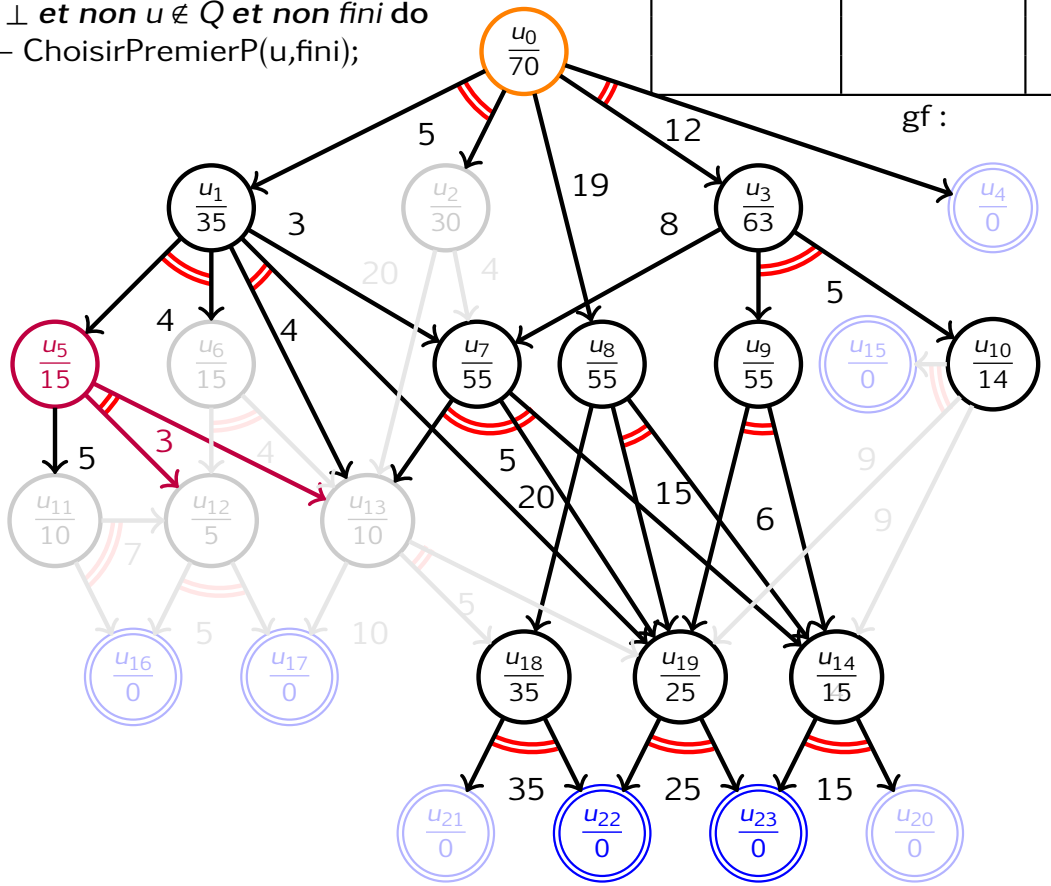


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,70)	(70,74,75)	{}	u_1(5,35) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,58)	{}	u_5(9,15)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,15)	(15,-)	{u_1}	u_{11}(14,15)u_6(9,15) u_2(5,30)

gf : {u_{12},u_{13}}

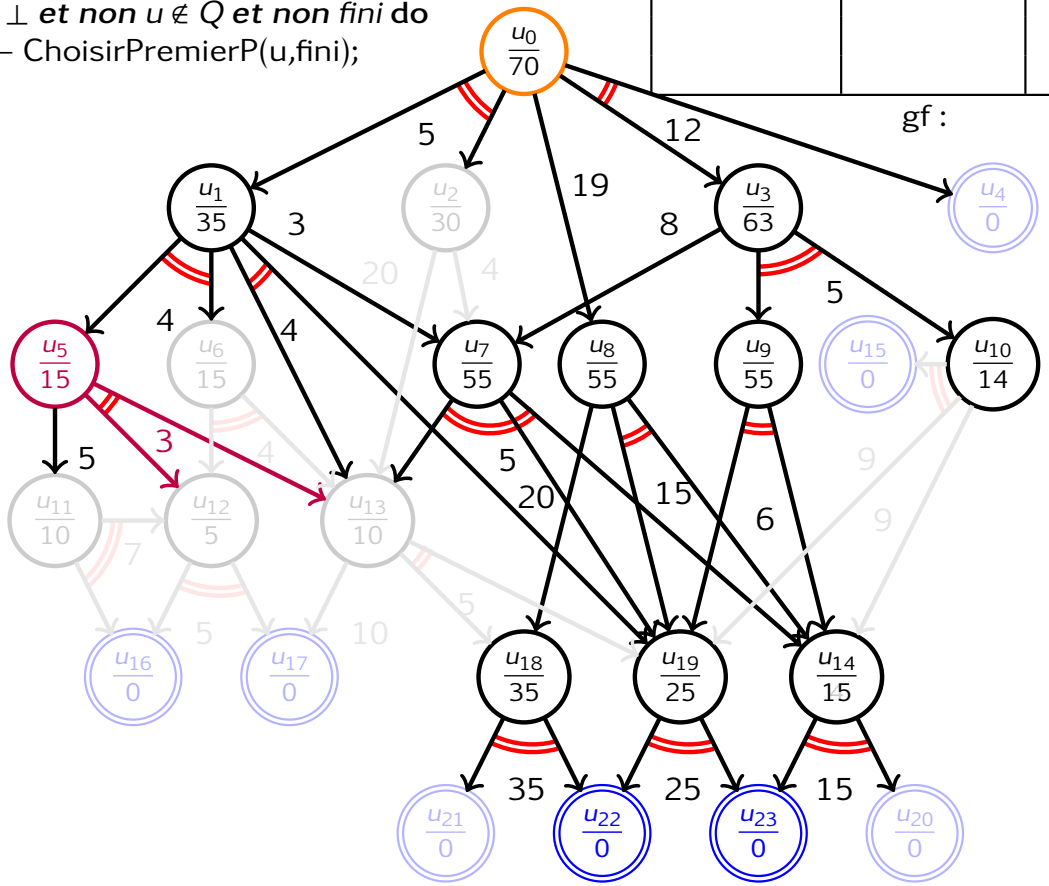


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,70)	(70,74,75)	{}	u_1(5,35) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,58)	{}	u_5(9,15)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,15)	(15,-)	{u_1}	u_{11}(14,15)u_6(9,15) u_2(5,30)

gf : {u_{12},u_{13}}

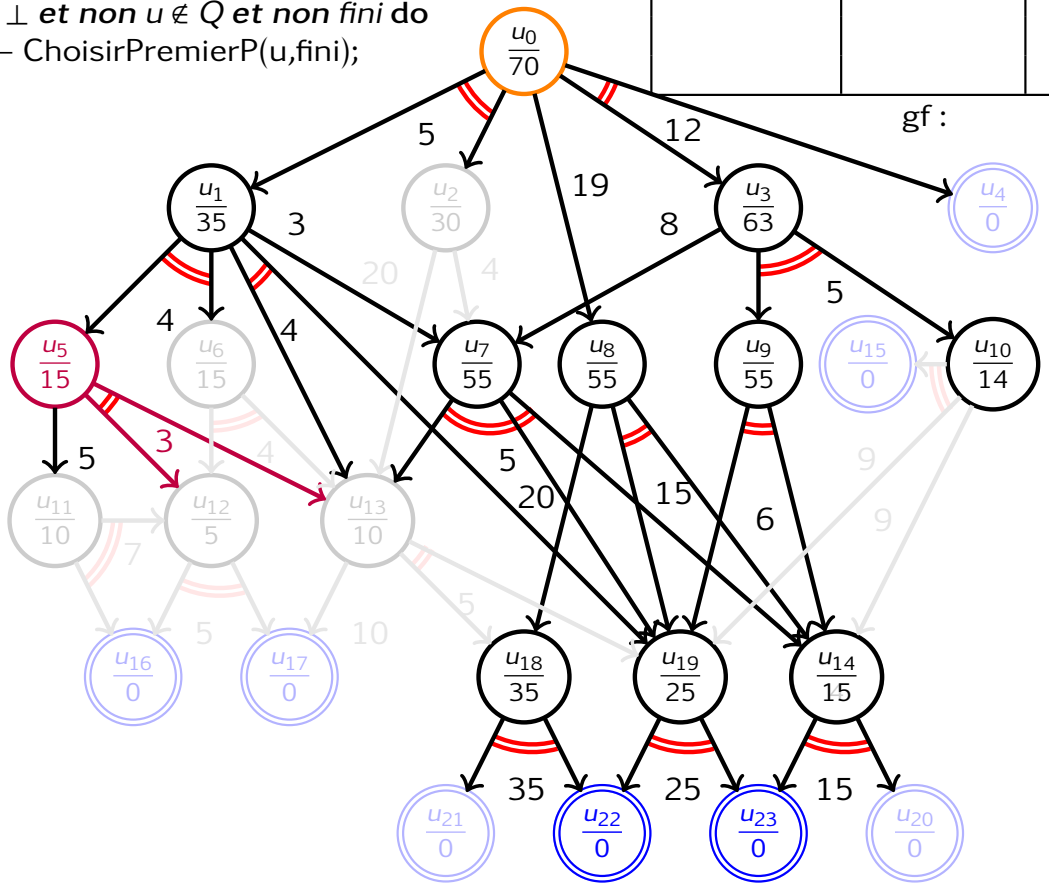


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,70)	(70,74,75)	{}	u_1(5,35) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,58)	{}	u_5(9,15)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,15)	(15,18)	{u_1}	u_{11}(14,15)u_6(9,15) u_2(5,30)

gf : {u_{12},u_{13}} = 3+5+10 = 18

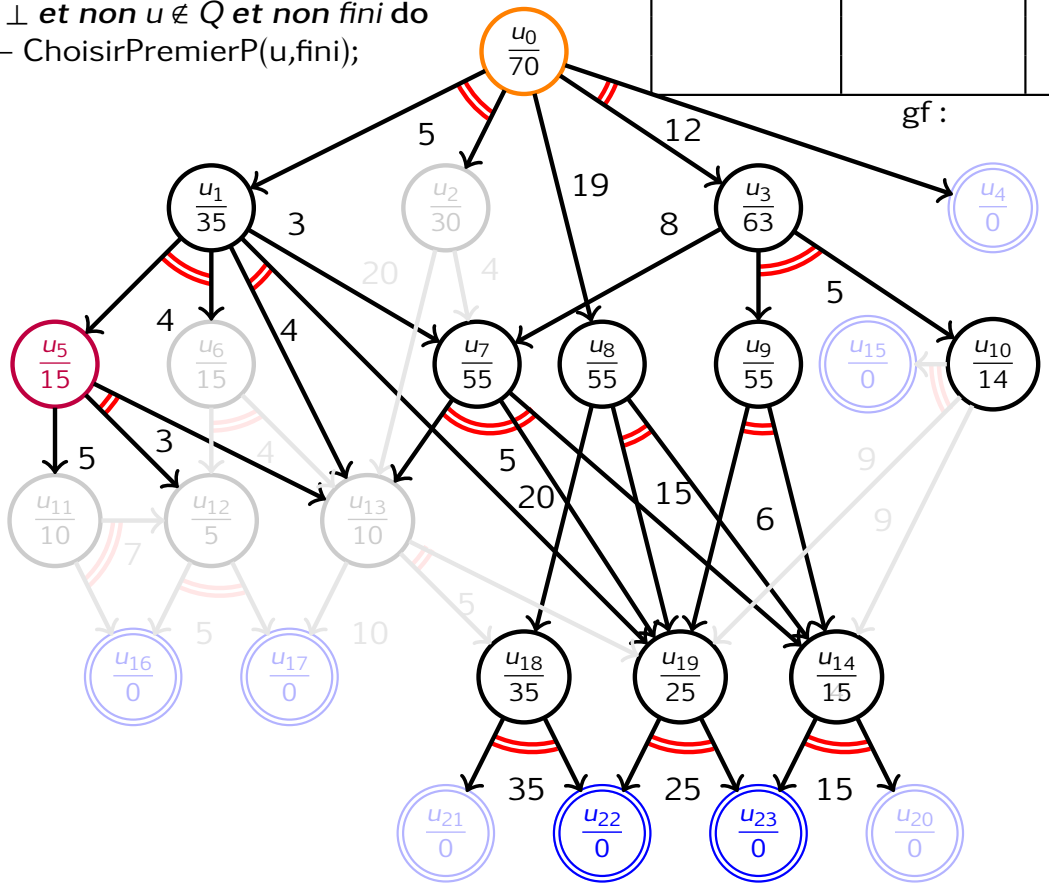


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,70)	(70,74,75)	{}	u_1(5,35) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,58)	{}	u_5(9,15)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,15)	(15,18)	{u_1}	u_{11}(14,15)u_6(9,15) u_2(5,30)

gf :

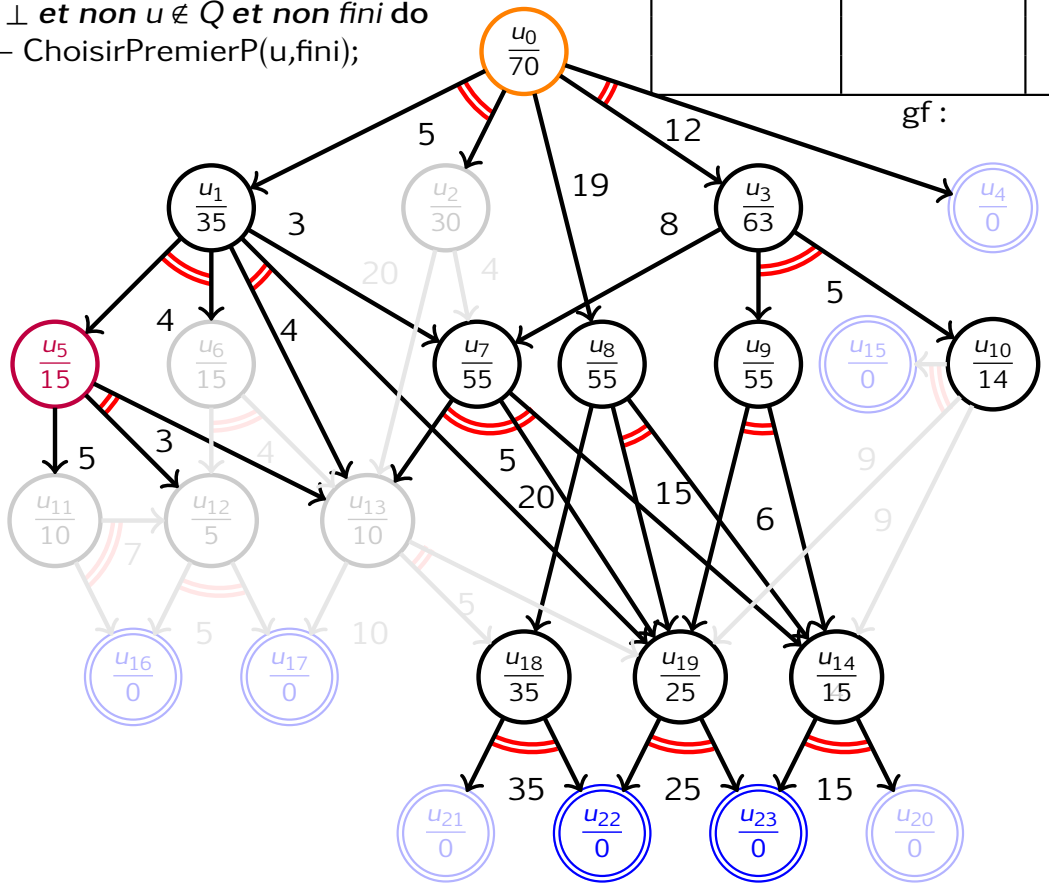


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,70)	(70,74,75)	{}	u_1(5,35) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,58)	{}	u_5(9,15)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,15)	(15,18)	{u_1}	u_{11}(14,15)u_6(9,15) u_2(5,30)

gf :



Algorithme OA*(Liste,Sol)

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u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
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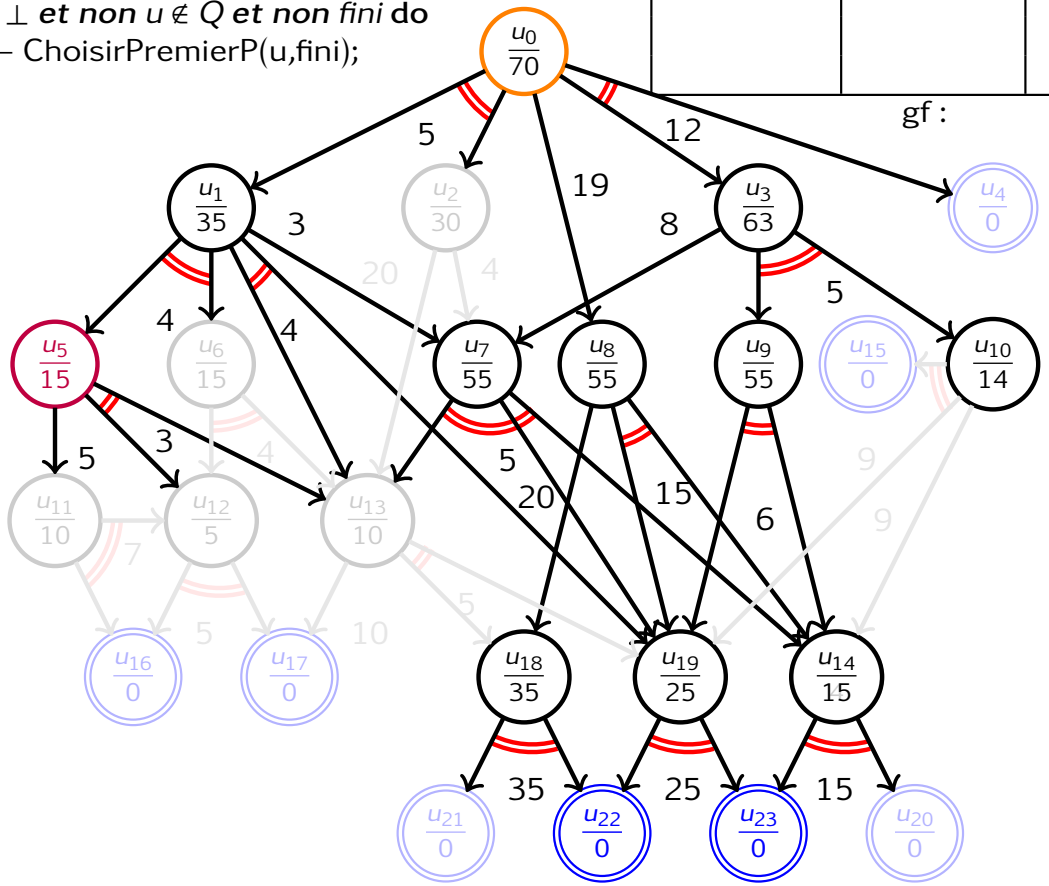
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,70)	(70,74,75)	{}	u_1(5,35) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,35)	(35,39,58)	{}	u_5(9,15)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,15)	(15,18)	{u_1}	u_{11}(14,15)u_6(9,15) u_2(5,30)

gf :

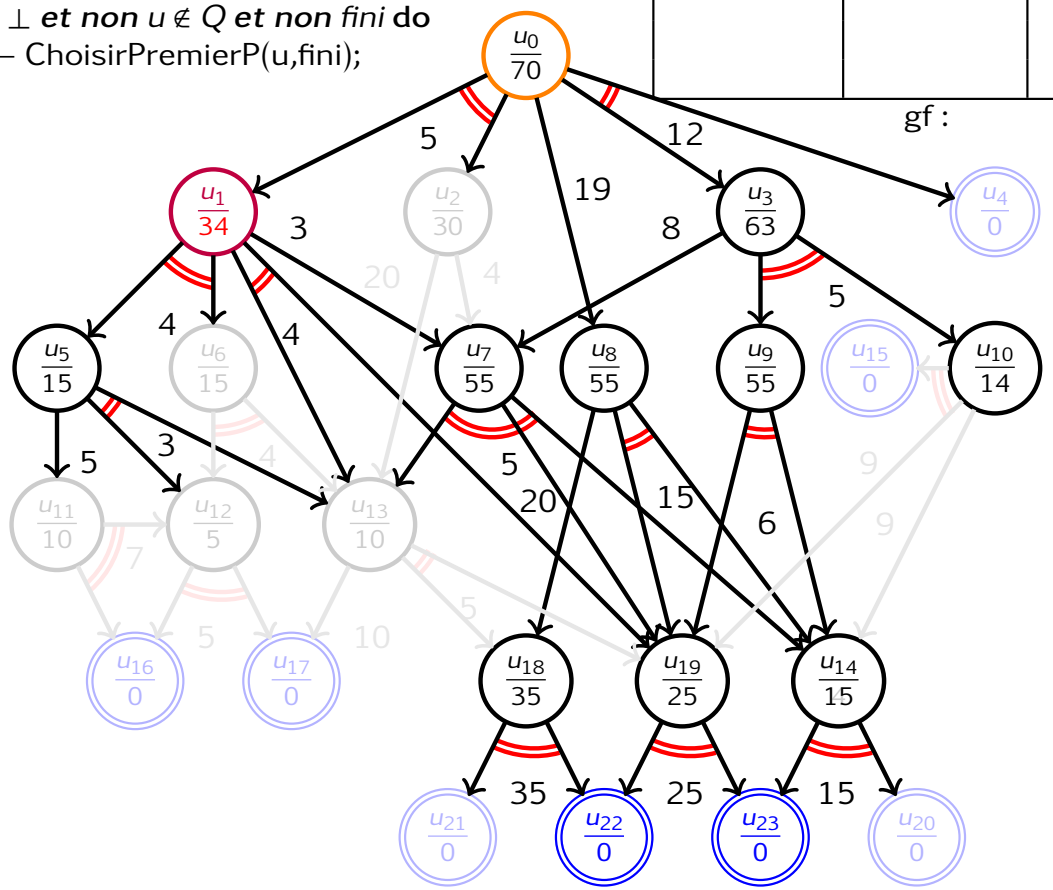


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,70)	(70,74,75)	{}	u_1(5,34) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,34)	(34,39,58)	{u_0}	u_5(9,15)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,15)	(15,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)

gf :



Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
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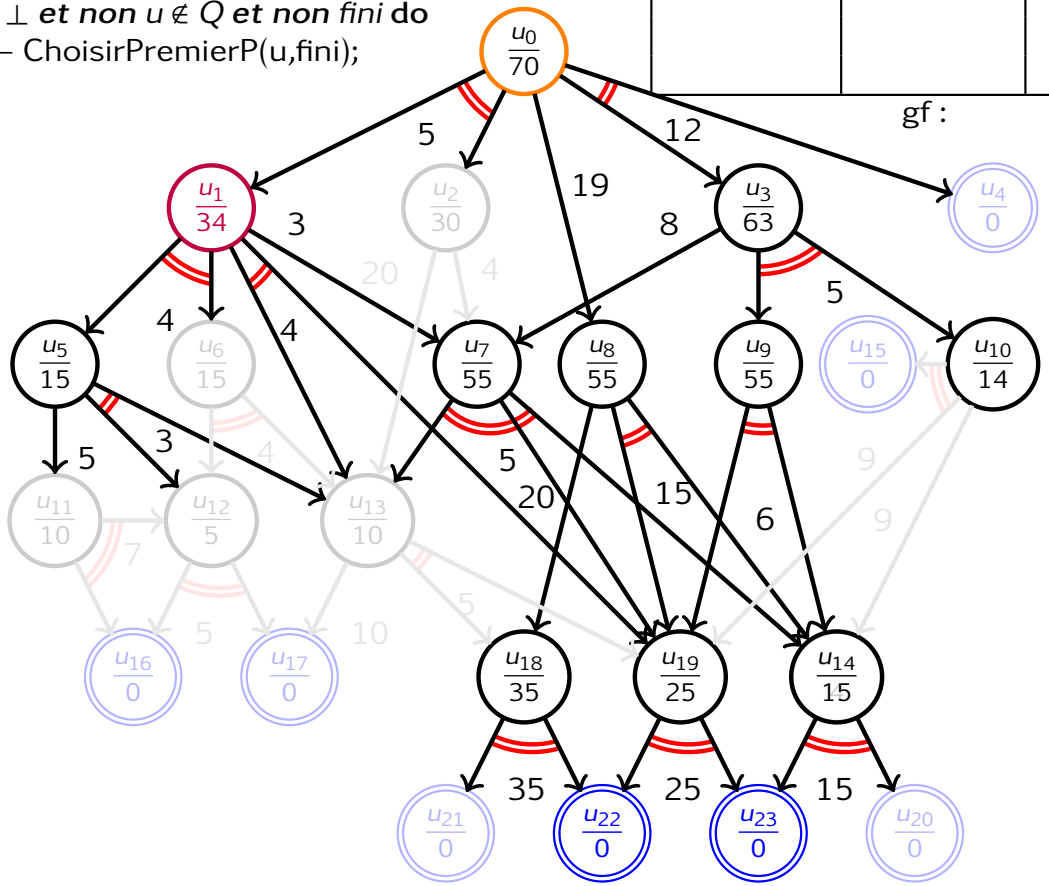
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,70)	(70,74,75)	{}	u_1(5,34) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,34)	(34,39,58)	{u_0}	u_5(9,15)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,15)	(15,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)

gf :

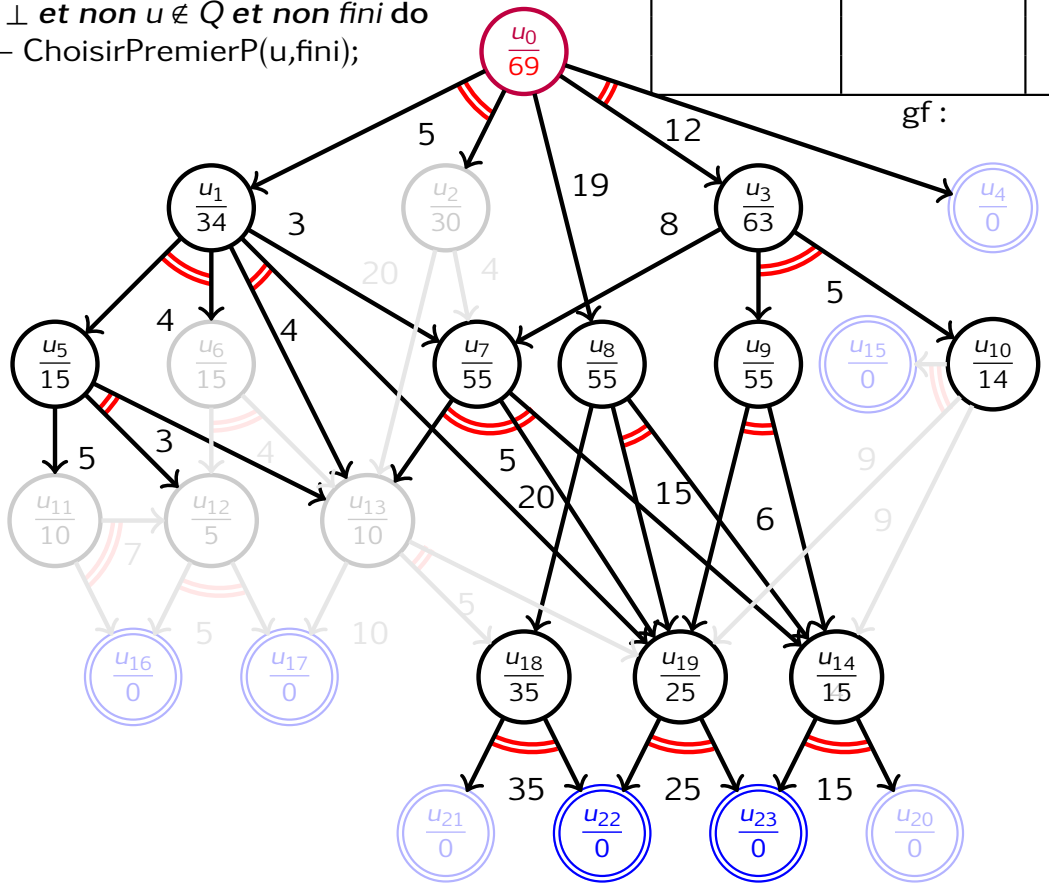


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,69)	(69,74,75)	{}	u_1(5,34) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,34)	(34,39,58)	{}	u_5(9,15)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,15)	(15,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)

gf :



Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
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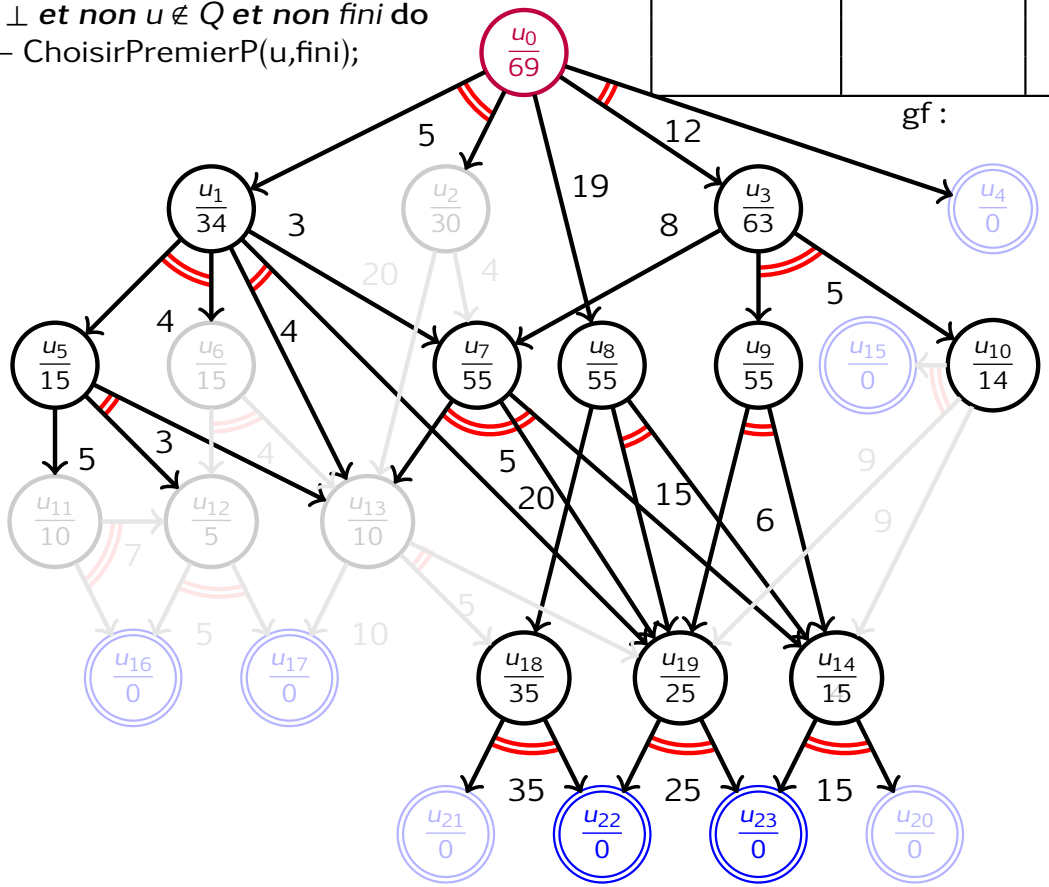
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,69)	(69,74,75)	{}	u_1(5,34) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,34)	(34,39,58)	{}	u_5(9,15)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,15)	(15,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)

gf :



Algorithme OA*(Liste,Sol)

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while *u ≠ ⊥ et non fini* **do**
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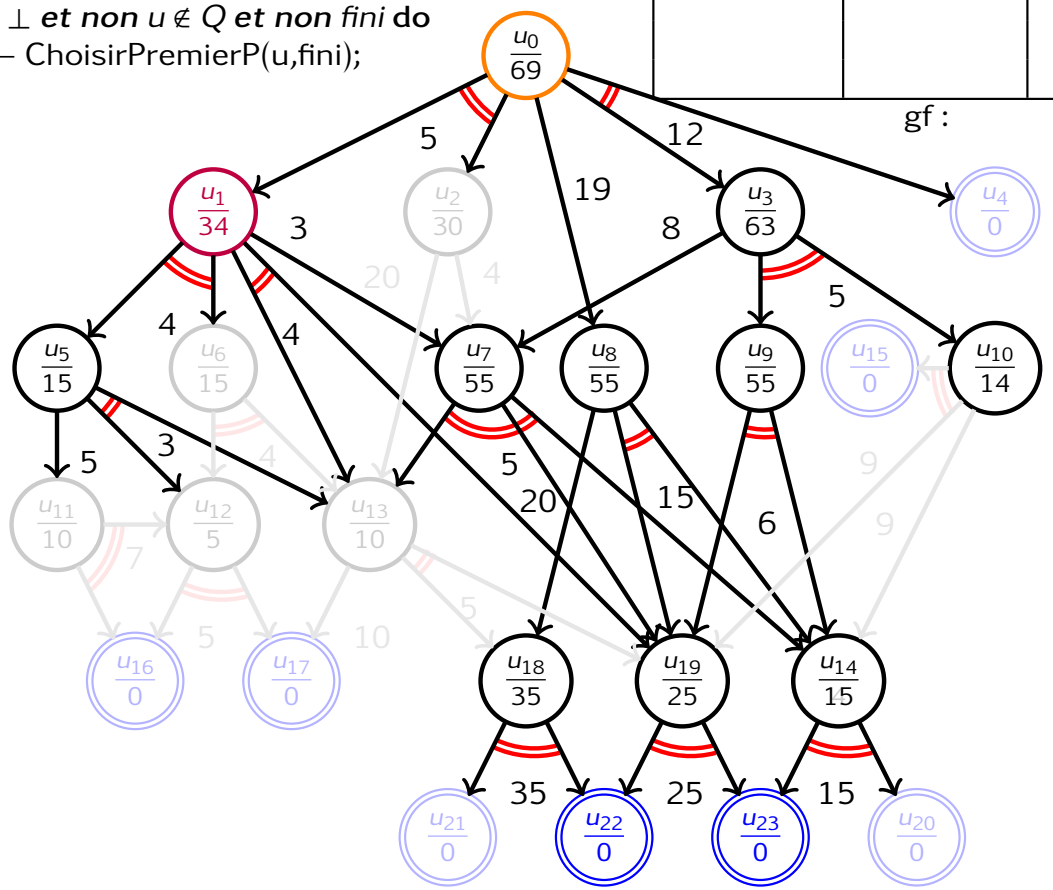
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,69)	(69,74,75)	{}	u_1(5,34) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,34)	(34,39,58)	{}	u_5(9,15)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,15)	(15,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)

gf :



Algorithme OA*(Liste,Sol)

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        gf ← gfSuivant(u);
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    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,69)	(69,74,75)	{}	u_1(5,34) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,34)	(34,39,58)	{}	u_5(9,15)u_6(9,15)u_2(5,30)
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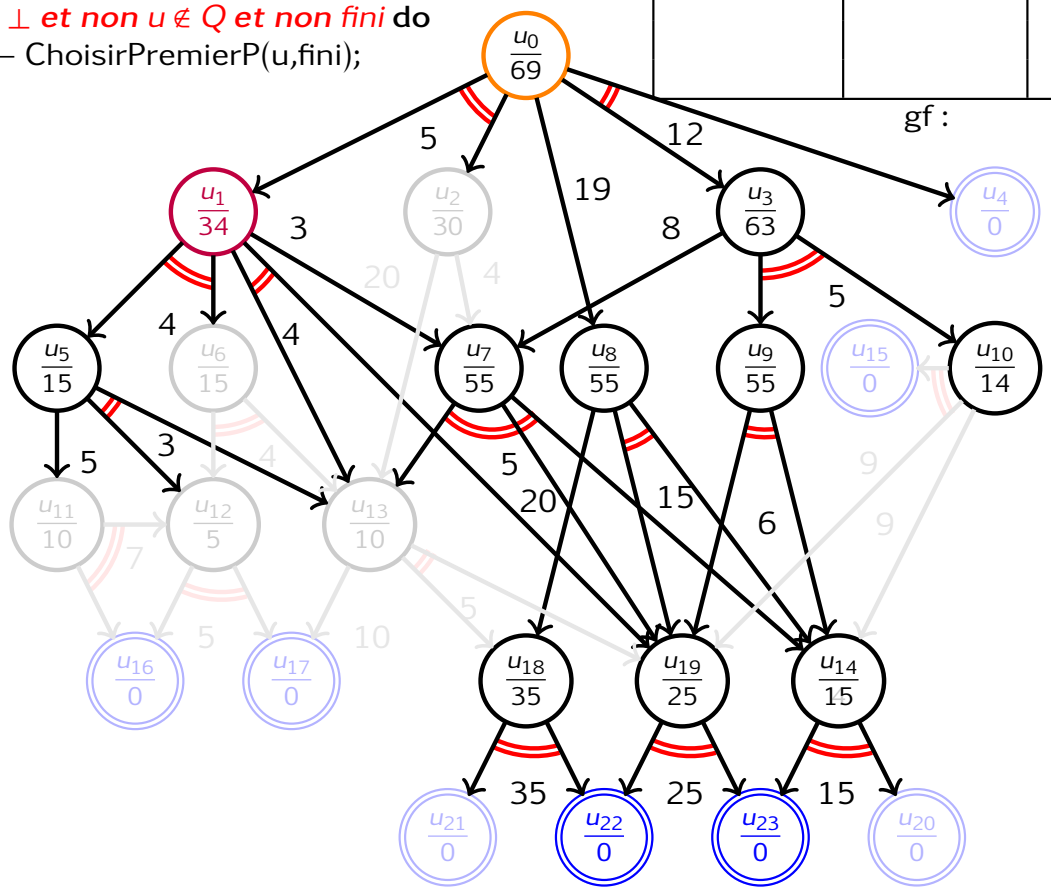
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,15)	(15,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)

gf :



Algorithme OA*(Liste,Sol)

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    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
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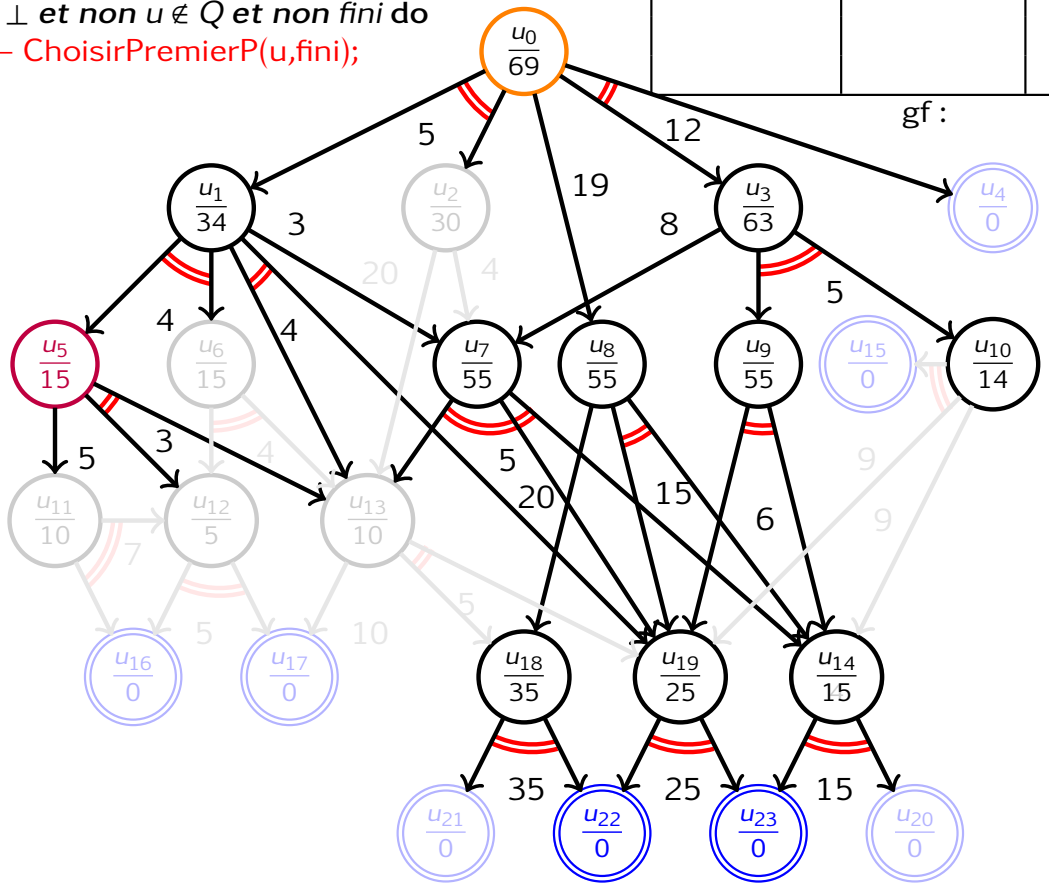
u(g(u),h(u))	fils(u)	A	P
u_0(0,69)	(69,74,75)	{}	u_1(5,34) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,34)	(34,39,58)	{}	u_5(9,15)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,15)	(15,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)

gf :



Algorithme OA*(Liste,Sol)

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        gf ← gfSuivant(u);
    while A ≠ ∅ do
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    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,69)	(69,74,75)	{}	u_1(5,34) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
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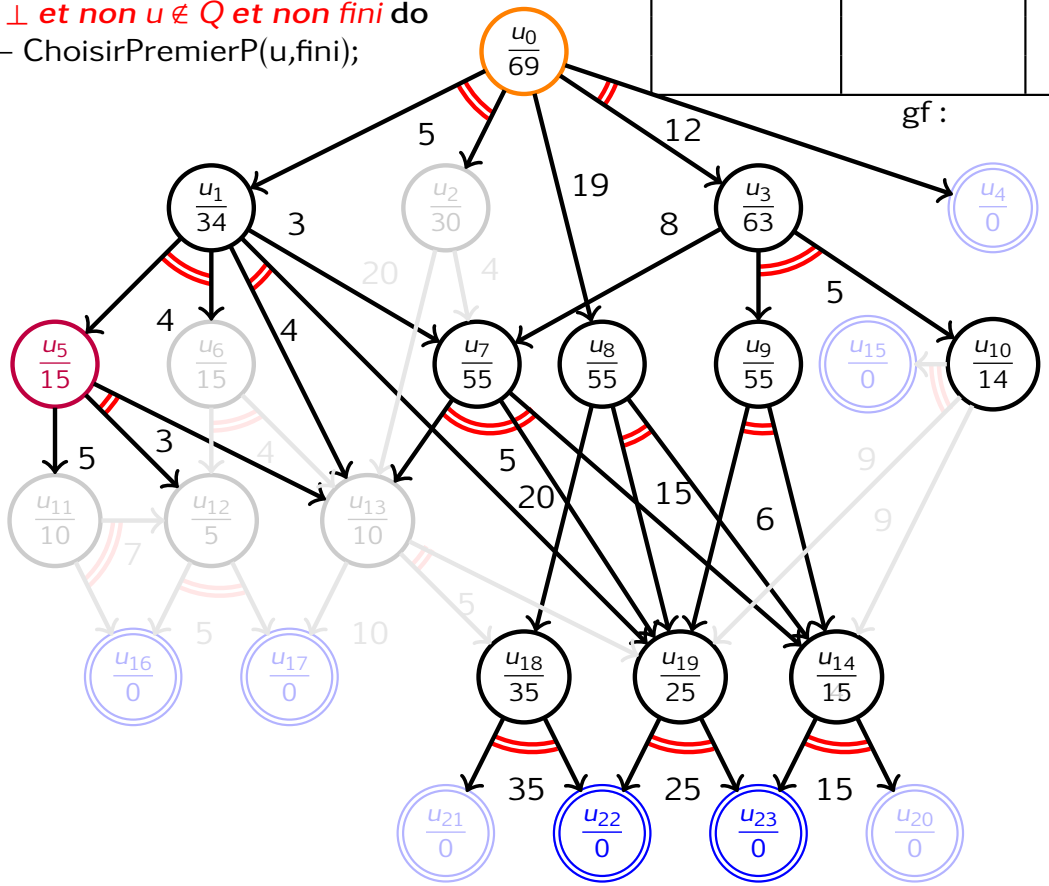
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,34)	(34,39,58)	{}	u_5(9,15)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,15)	(15,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)

gf :

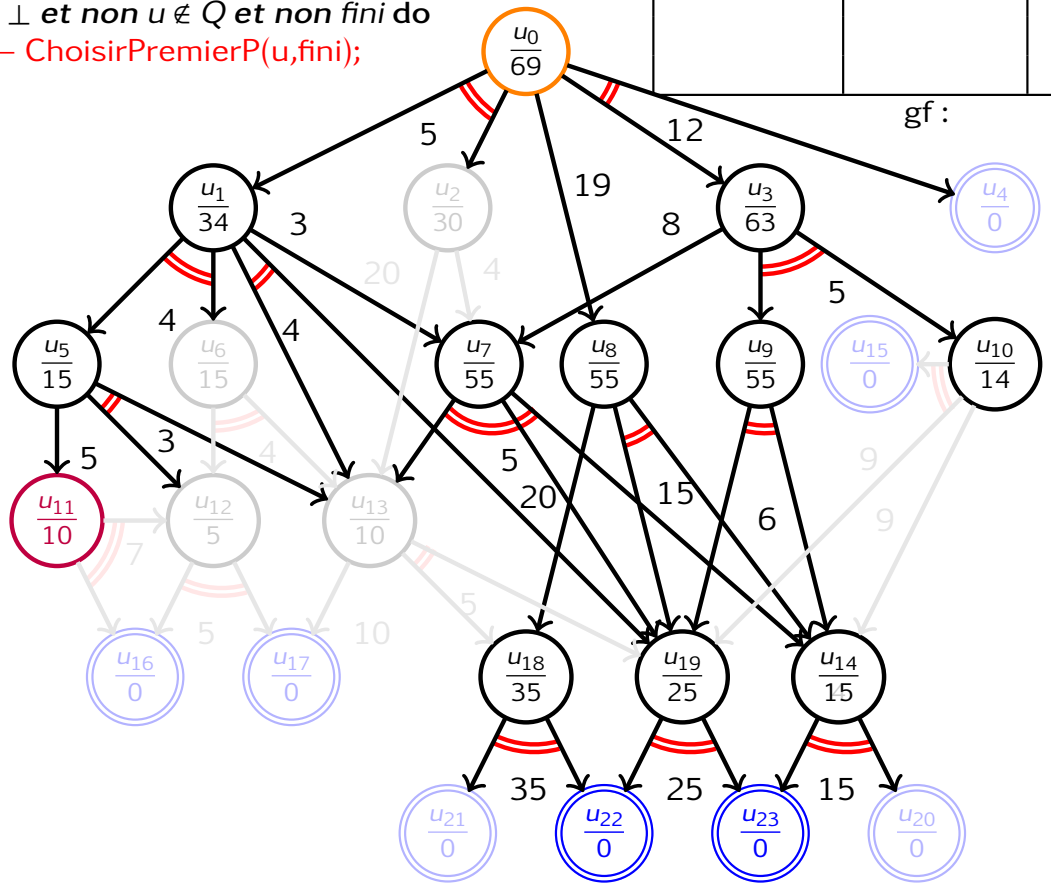


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,69)	(69,74,75)	{}	u_1(5,34) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,34)	(34,39,58)	{}	u_5(9,15)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,15)	(15,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)
u_{11}(14,10)	(-)	{}	u_6(9,15)u_2(5,30)

gf :

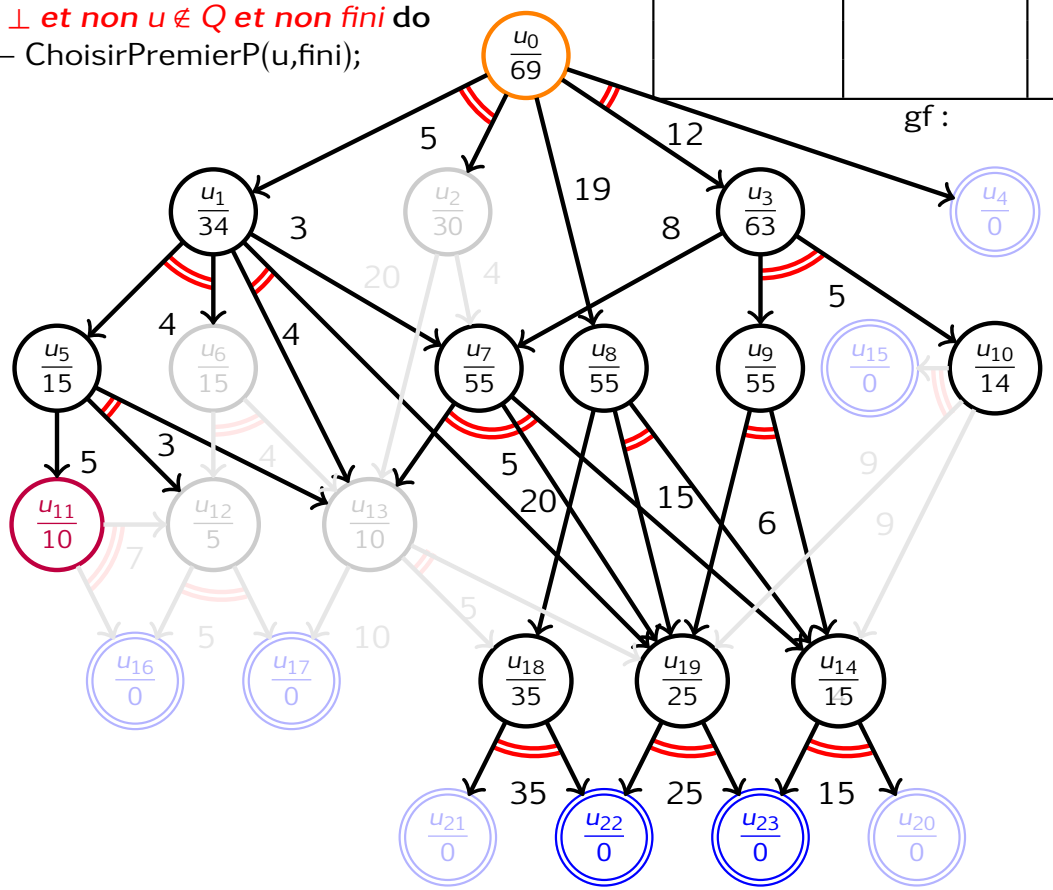


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,69)	(69,74,75)	{}	u_1(5,34) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,34)	(34,39,58)	{}	u_5(9,15)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,15)	(15,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)
u_{11}(14,10)	(-)	{}	u_6(9,15)u_2(5,30)

gf :

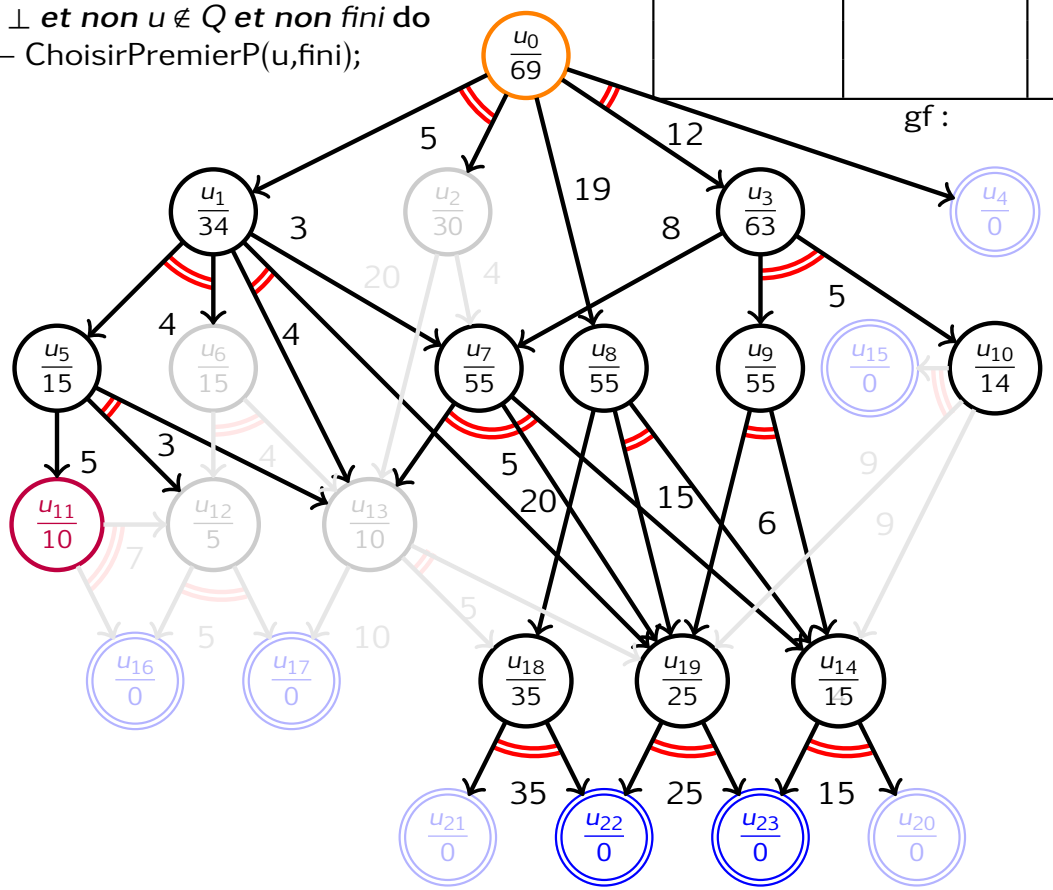


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,69)	(69,74,75)	{}	u_1(5,34) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,34)	(34,39,58)	{}	u_5(9,15)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,15)	(15,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)
u_{11}(14,10)	(-)	{}	u_6(9,15)u_2(5,30)

gf :

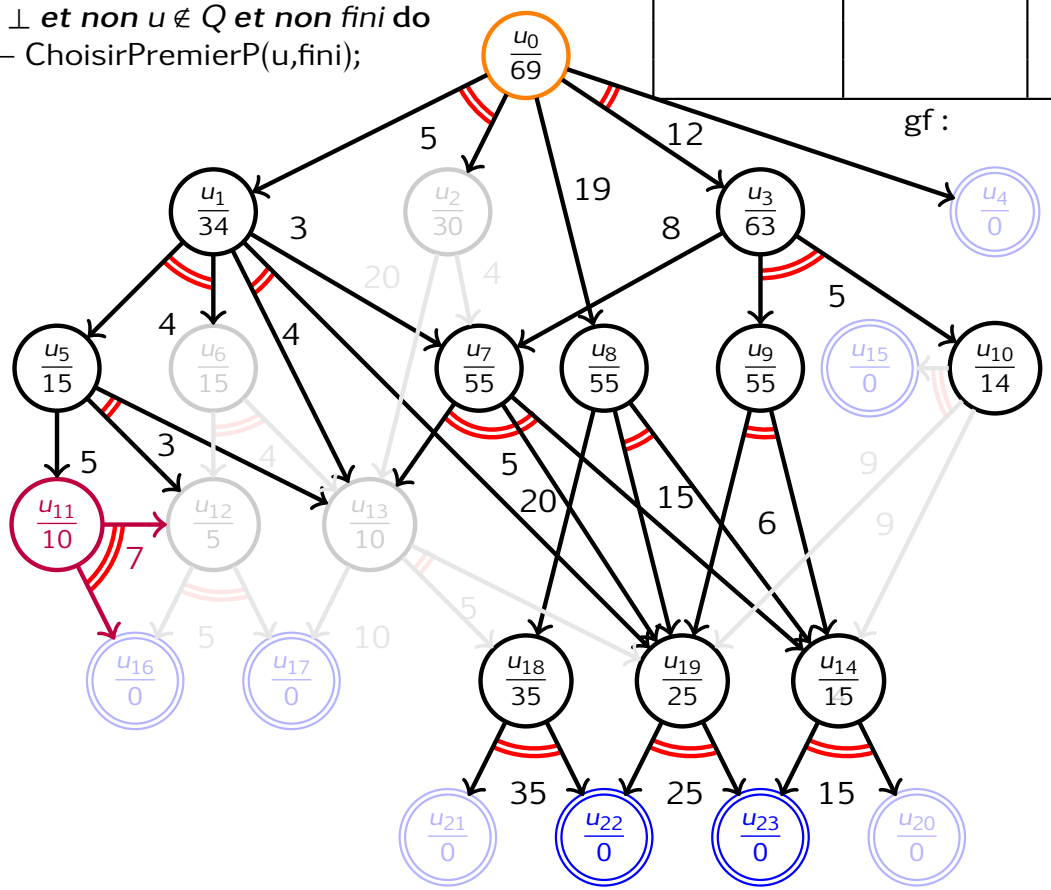


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,69)	(69,74,75)	{}	u_1(5,34) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,34)	(34,39,58)	{}	u_5(9,15)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,15)	(15,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)
u_{11}(14,10)	(-)	{}	u_6(9,15)u_2(5,30)

gf : $\{u_{12}, u_{16}\}$

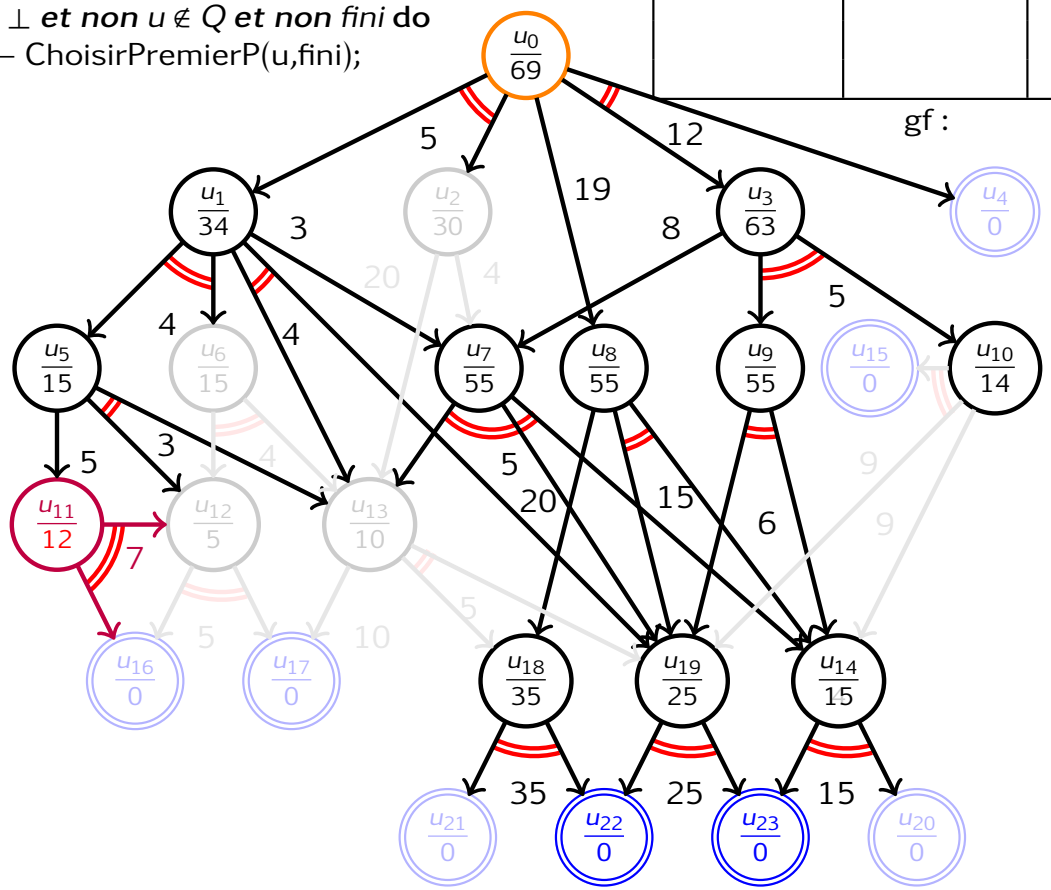


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,69)	(69,74,75)	{}	u_1(5,34) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,34)	(34,39,58)	{}	u_5(9,15)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,15)	(15,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)
u_{11}(14,10)	(-)	{}	u_6(9,15)u_2(5,30)

gf : $\{u_{12},u_{16}\}$

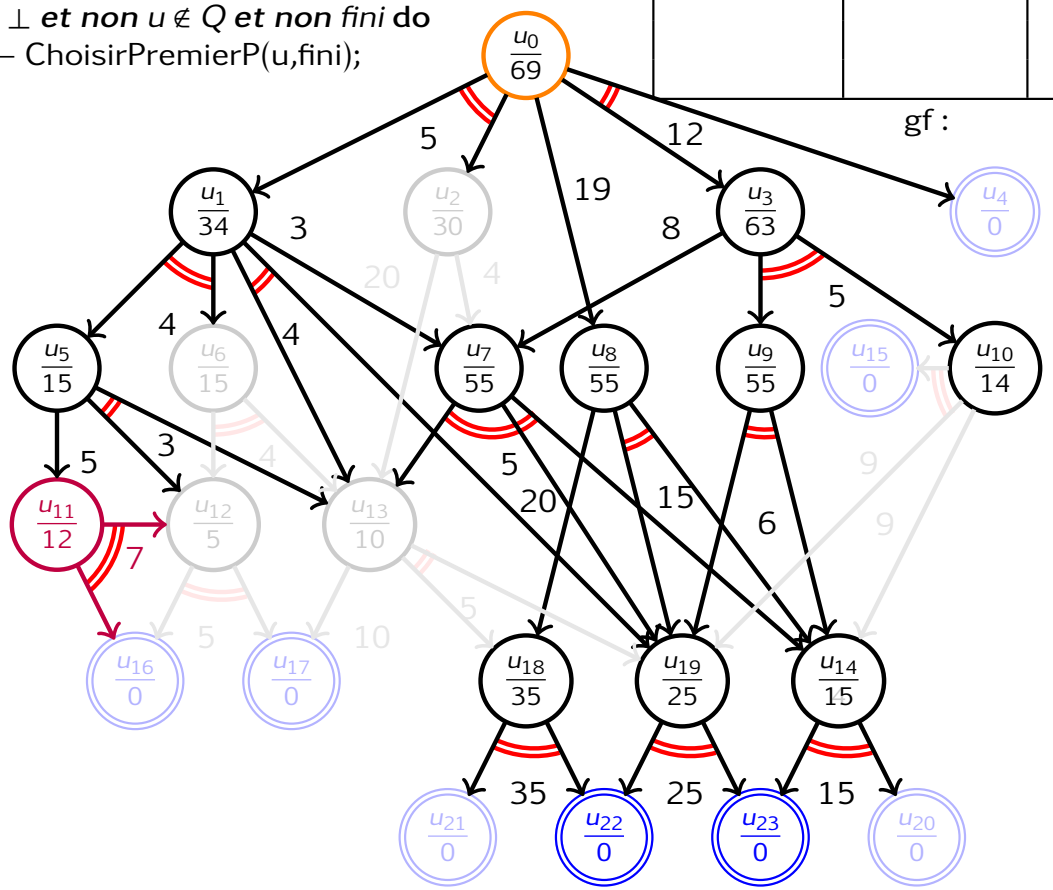


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,69)$	(69,74,75)	{}	$u_1(5,34)$ $u_2(5,30)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,63)$	(63,74)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,34)$	(34,39,58)	{}	$u_5(9,15)u_6(9,15)u_2(5,30)$
$u_9(17,46)$	(46)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$
$u_5(9,15)$	(15,18)	{}	$u_{11}(14,15)u_6(9,15)$ $u_2(5,30)$
$u_{11}(14,12)$	(12)	$\{u_5\}$	$u_{16}(7,0)u_{12}(7,5)u_6(9,15)u_2(5,30)$

gf : $\{u_{12},u_{16}\}$ = 7+0+5 = 12

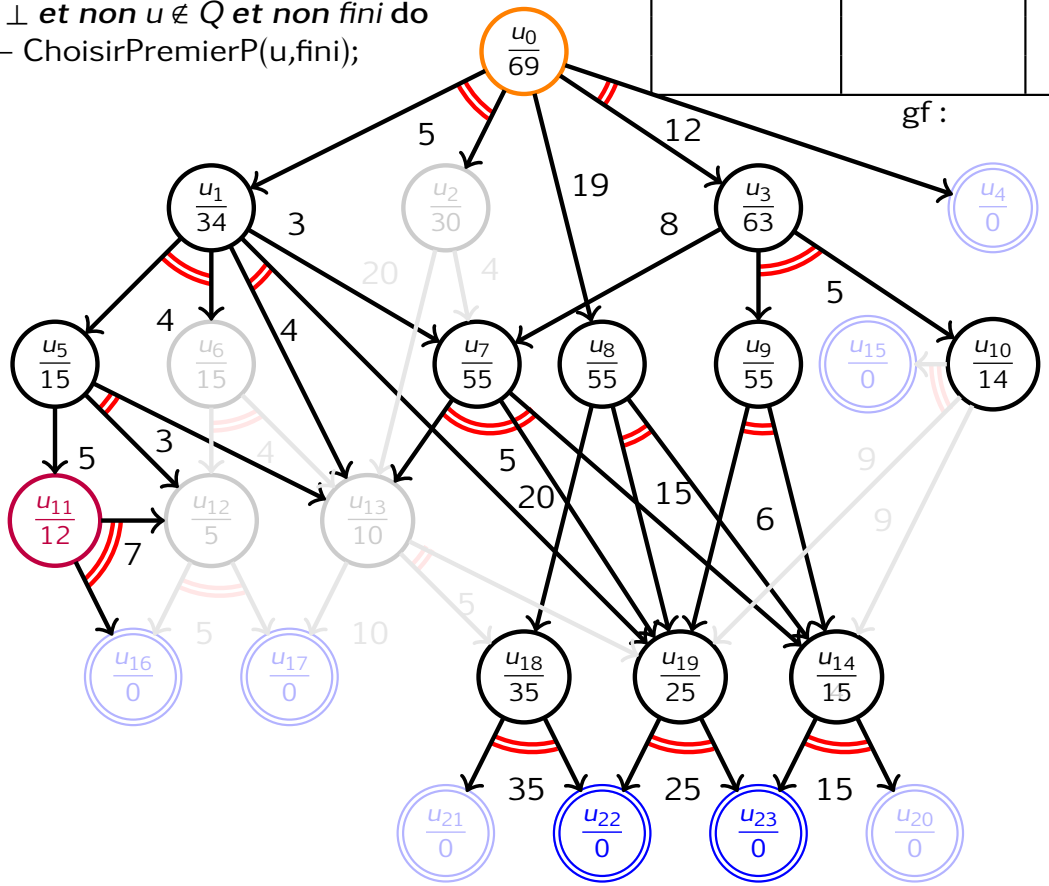


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
$u_0(0,69)$	(69,74,75)	{}	$u_1(5,34)$ $u_2(5,30)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,63)$	(63,74)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,34)$	(34,39,58)	{}	$u_5(9,15)u_6(9,15)u_2(5,30)$
$u_9(17,46)$	(46)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$
$u_5(9,15)$	(15,18)	{}	$u_{11}(14,15)u_6(9,15)$ $u_2(5,30)$
$u_{11}(14,12)$	(12)	$\{u_5\}$	$u_{16}(7,0)u_{12}(7,5)u_6(9,15)u_2(5,30)$

gf :

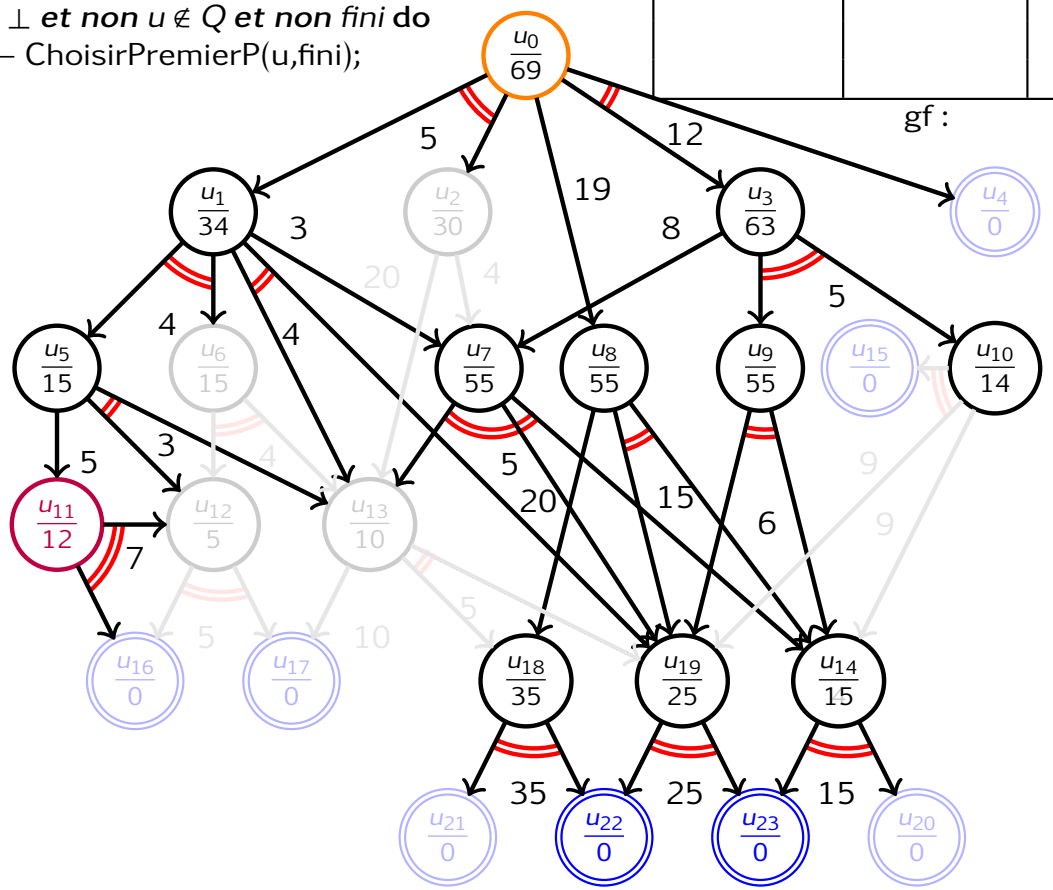


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while $u \neq \perp$ ***et non*** *fini* **do**
  gf ← groupeFils(u) ;
  while $gf \neq \emptyset$ **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while $A \neq \emptyset$ **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while $u \neq \perp$ ***et non*** $u \notin Q$ ***et non*** *fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,69)	(69,74,75)	{}	u_1(5,34) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,34)	(34,39,58)	{}	u_5(9,15)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,15)	(15,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)
u_{11}(14,12)	(12)	{u_5}	u_{16}(7,0)u_{12}(7,5)u_6(9,15)u_2(5,30)

gf :



Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
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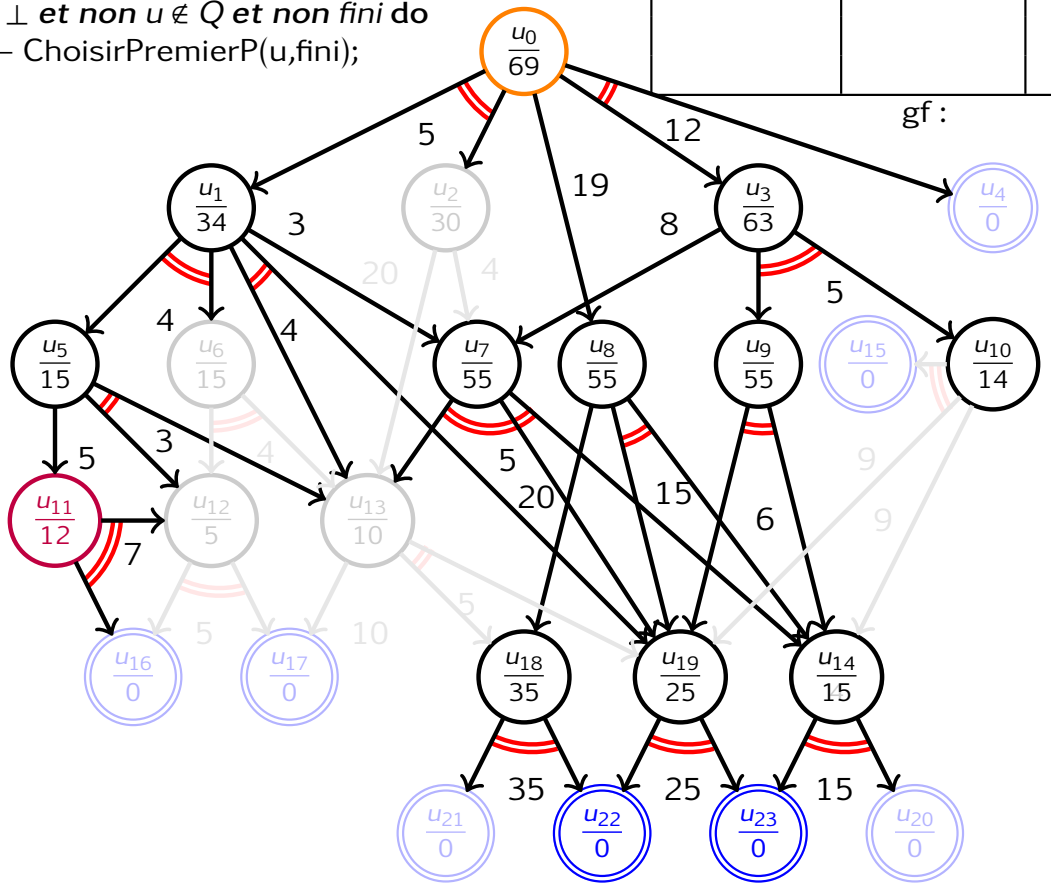
$u_0(0,69)$	(69,74,75)	{}	$u_1(5,34)$ $u_2(5,30)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,63)$	(63,74)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,34)$	(34,39,58)	{}	$u_5(9,15)u_6(9,15)u_2(5,30)$
$u_9(17,46)$	(46)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$
$u_5(9,15)$	(15,18)	{}	$u_{11}(14,15)u_6(9,15)$ $u_2(5,30)$
$u_{11}(14,12)$	(12)	$\{u_5\}$	$u_{16}(7,0)u_{12}(7,5)u_6(9,15)u_2(5,30)$

gf :



Algorithme OA*(Liste,Sol)

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  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
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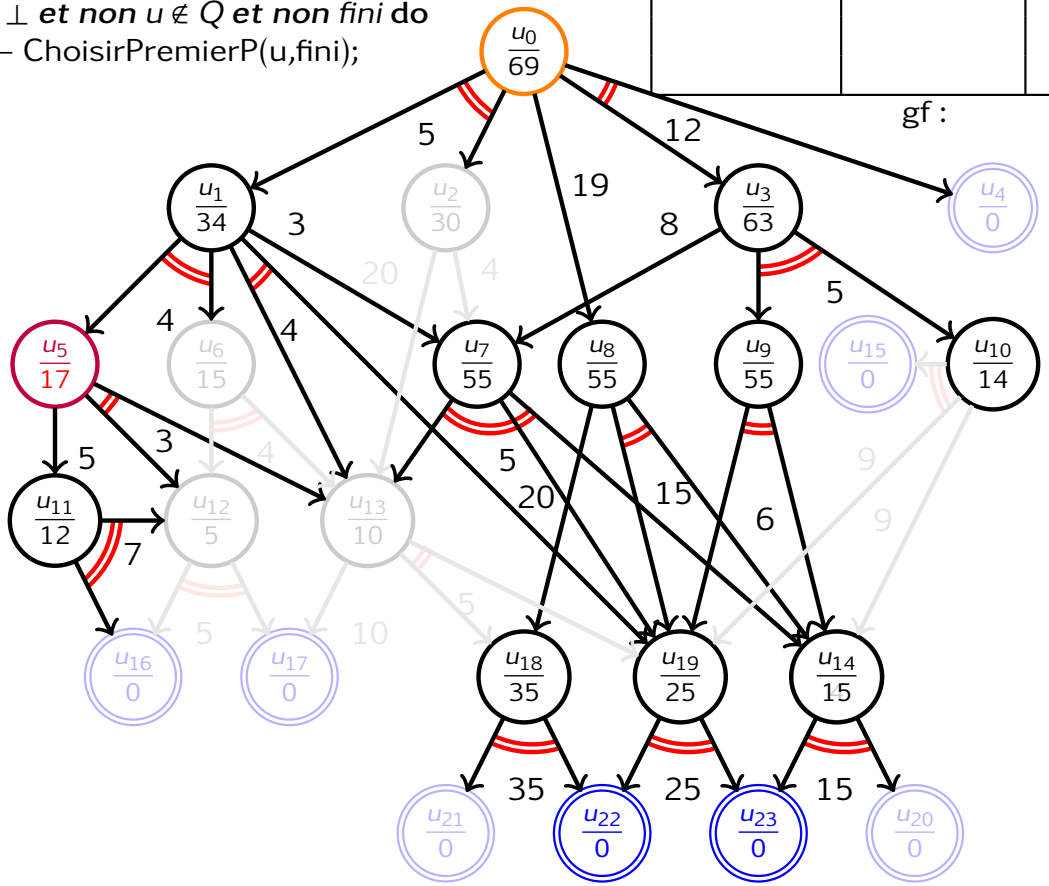
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,69)	(69,74,75)	{}	u_1(5,34) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,34)	(34,39,58)	{}	u_5(9,17)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{u_1}	u_{11}(14,15)u_6(9,15) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{16}(7,0)u_{12}(7,5)u_6(9,15)u_2(5,30)

gf :

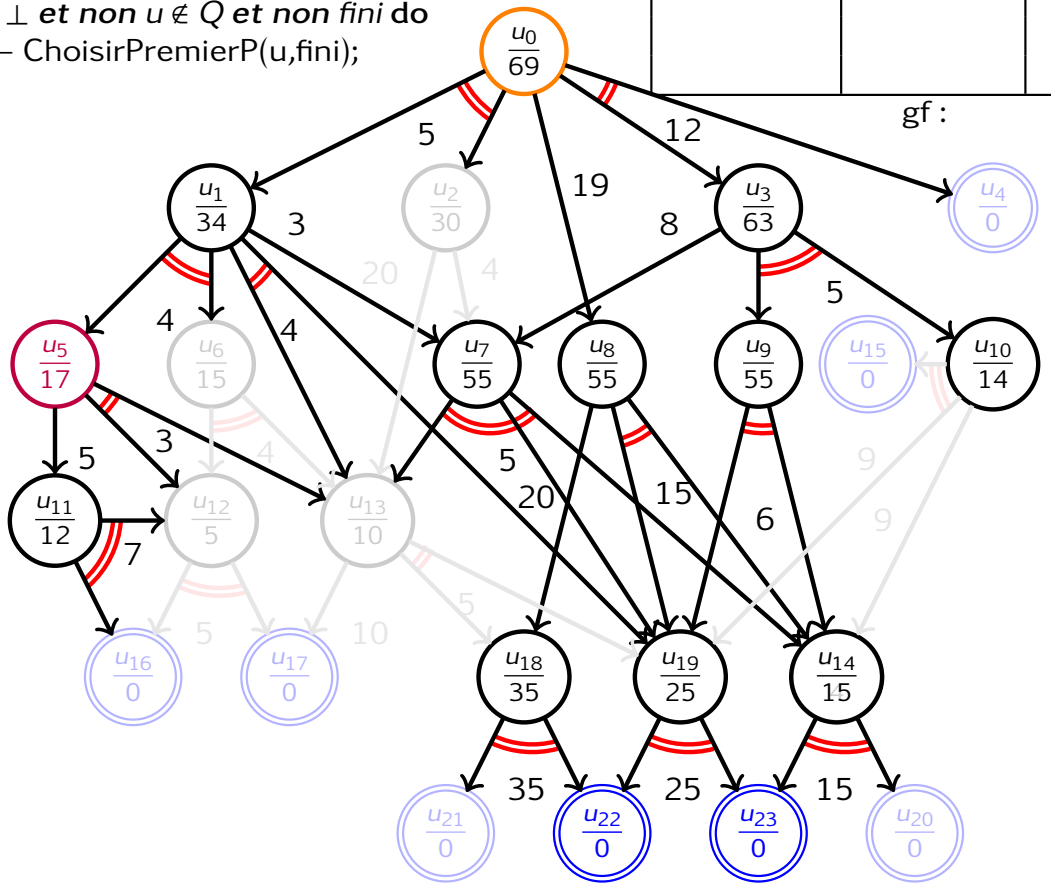


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
$u_0(0,69)$	(69,74,75)	{}	$u_1(5,34)$ $u_2(5,30)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,63)$	(63,74)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,34)$	(34,39,58)	{}	$u_5(9,17)u_6(9,15)u_2(5,30)$
$u_9(17,46)$	(46)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$
$u_5(9,17)$	(17,18)	$\{u_1\}$	$u_{11}(14,15)u_6(9,15)$ $u_2(5,30)$
$u_{11}(14,12)$	(12)	{}	$u_{16}(7,0)u_{12}(7,5)u_6(9,15)u_2(5,30)$

gf :



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u(g(u),h(u))	fils(u)	A	P
u_0(0,69)	(69,74,75)	{}	u_1(5,36) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,36)	(36,39,58)	{u_0}	u_5(9,17)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{16}(7,0)u_{12}(7,5)u_6(9,15)u_2(5,30)

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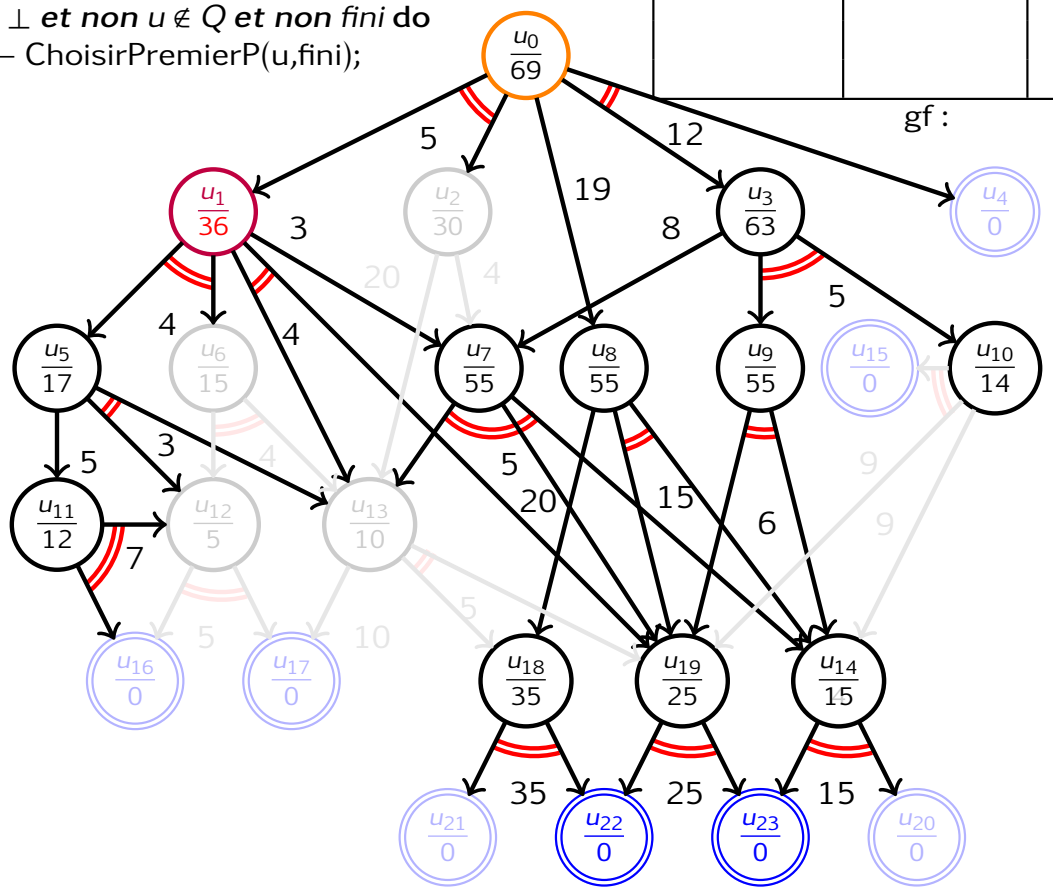

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$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
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$u_3(12,63)$	(63,74)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,36)$	(36,39,58)	$\{u_0\}$	$u_5(9,17)u_6(9,15)u_2(5,30)$
$u_9(17,46)$	(46)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,$
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$u_5(9,17)$	(17,18)	{}	$u_{11}(14,15)u_6(9,15)$ $u_2(5,30)$
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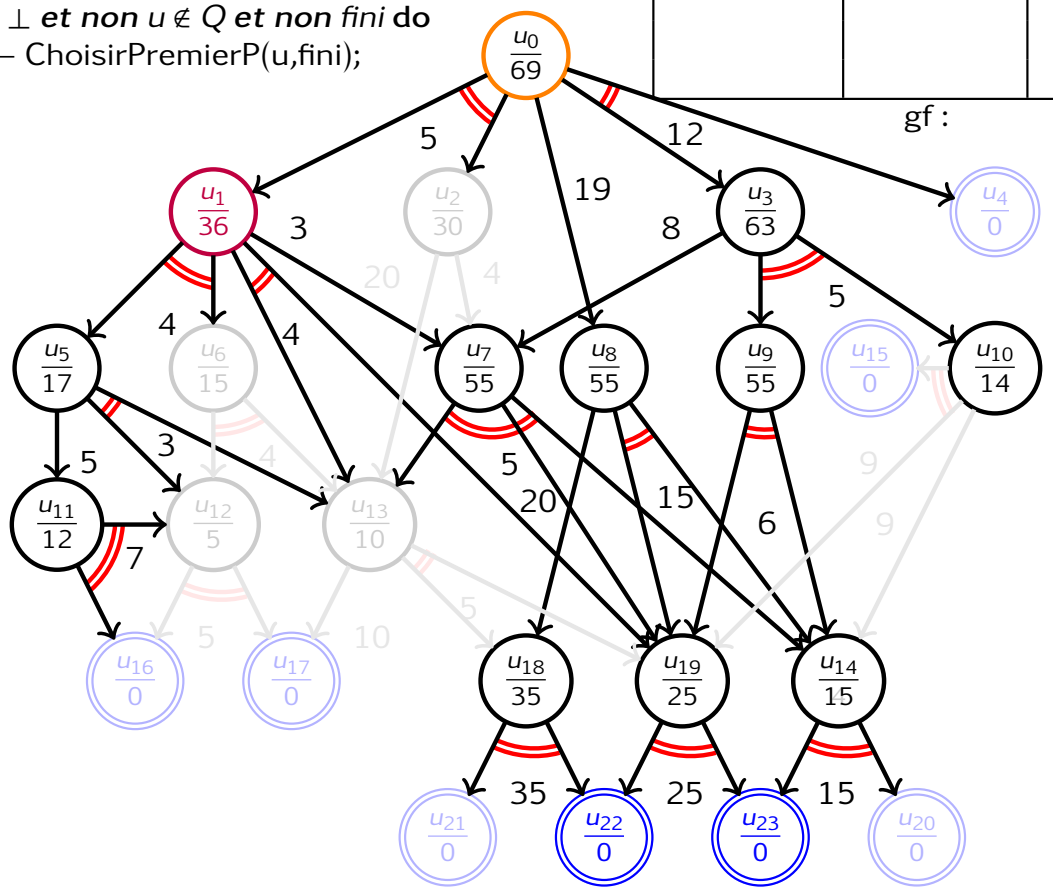

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$u_0(0,71)$	(71,74,75)	{}	$u_1(5,36)$ $u_2(5,30)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
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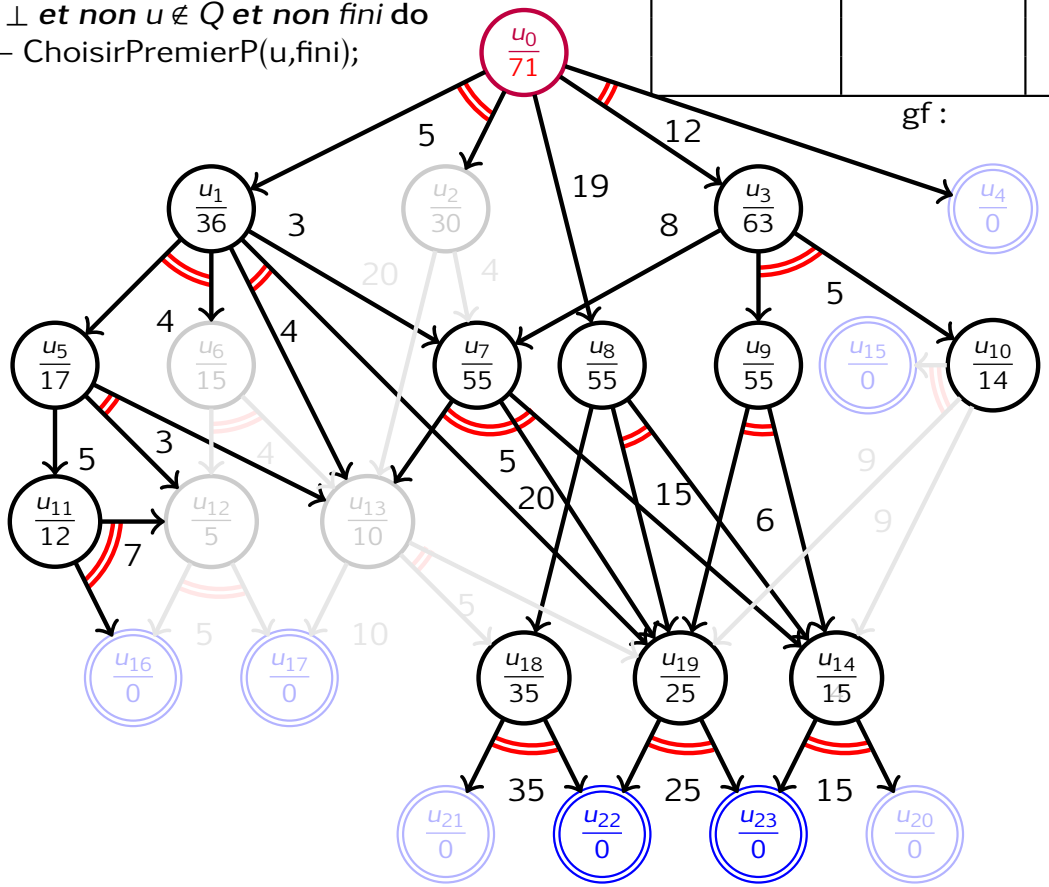

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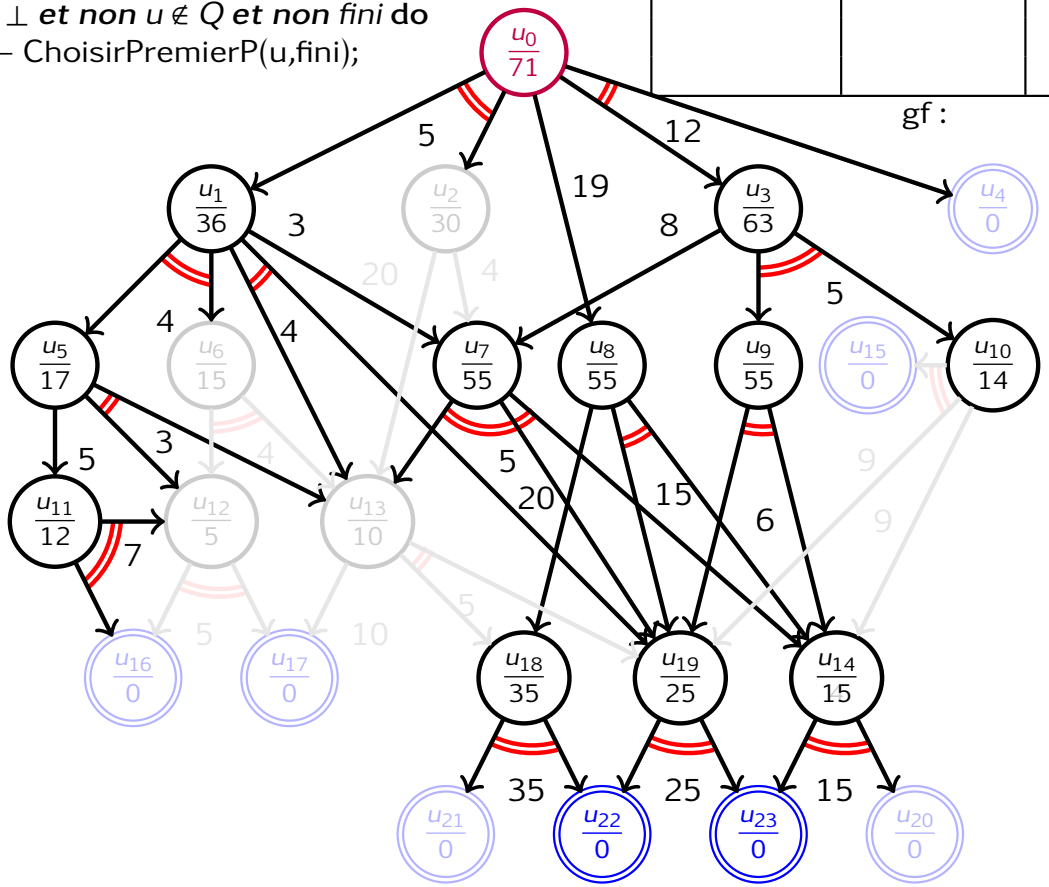

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u_1(5,36)	(36,39,58)	{}	u_5(9,17)u_6(9,15)u_2(5,30)
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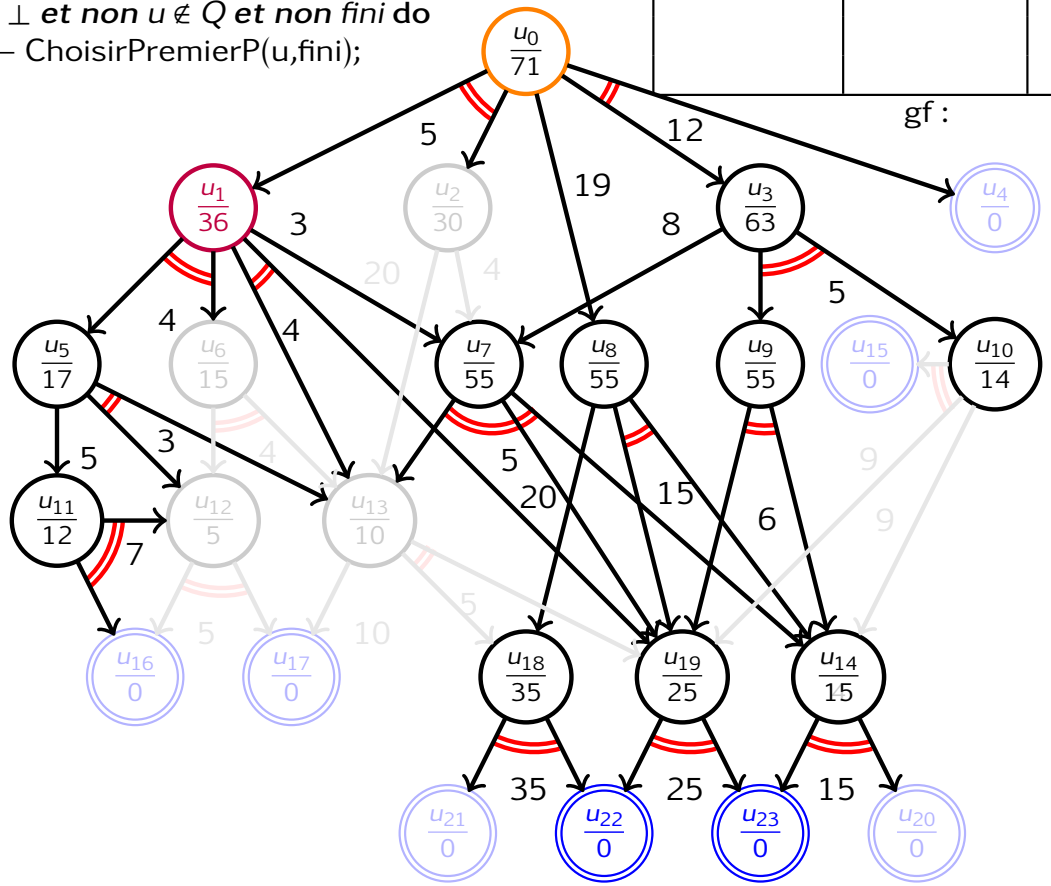

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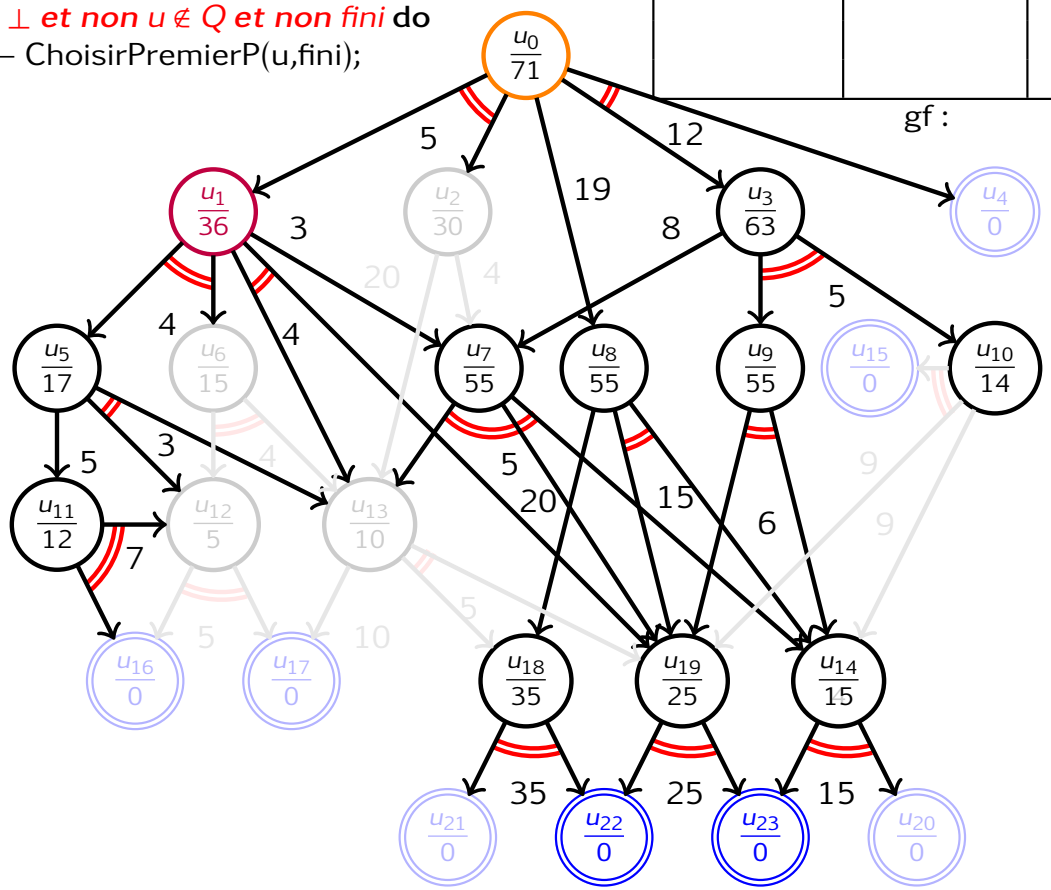

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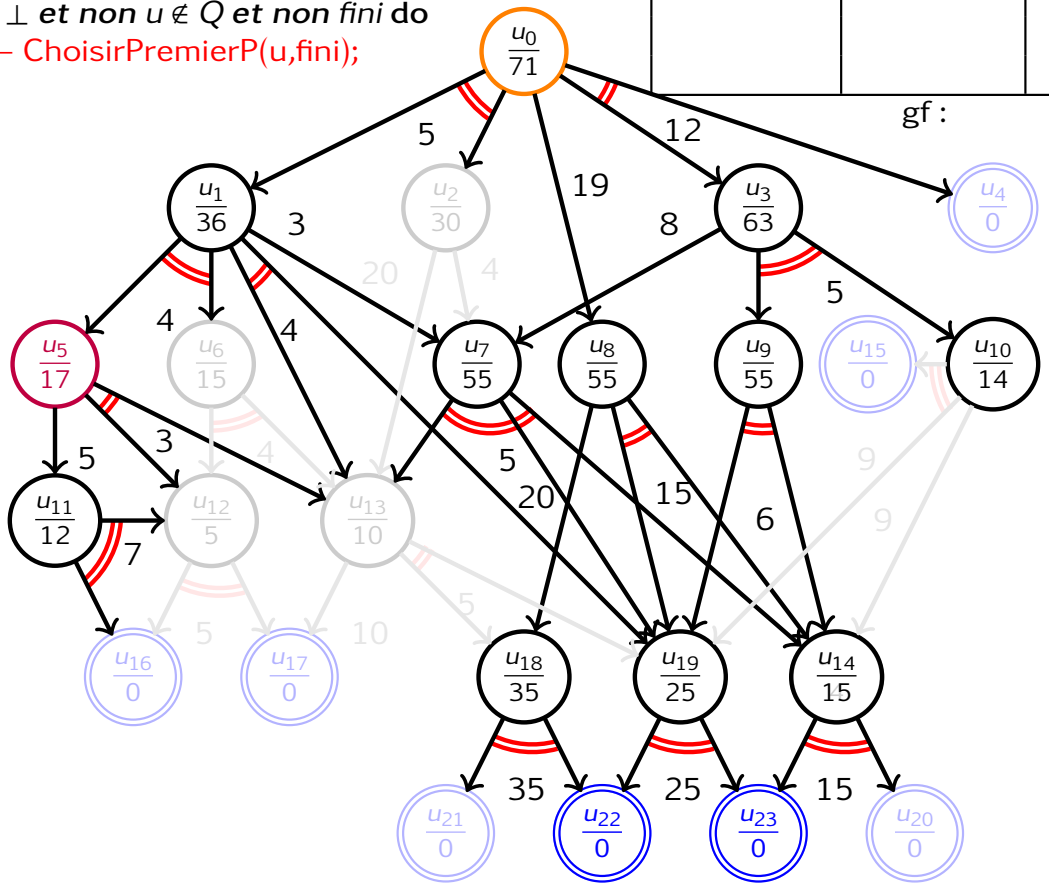

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u_1(5,36)	(36,39,58)	{}	u_5(9,17)u_6(9,15)u_2(5,30)
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u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)
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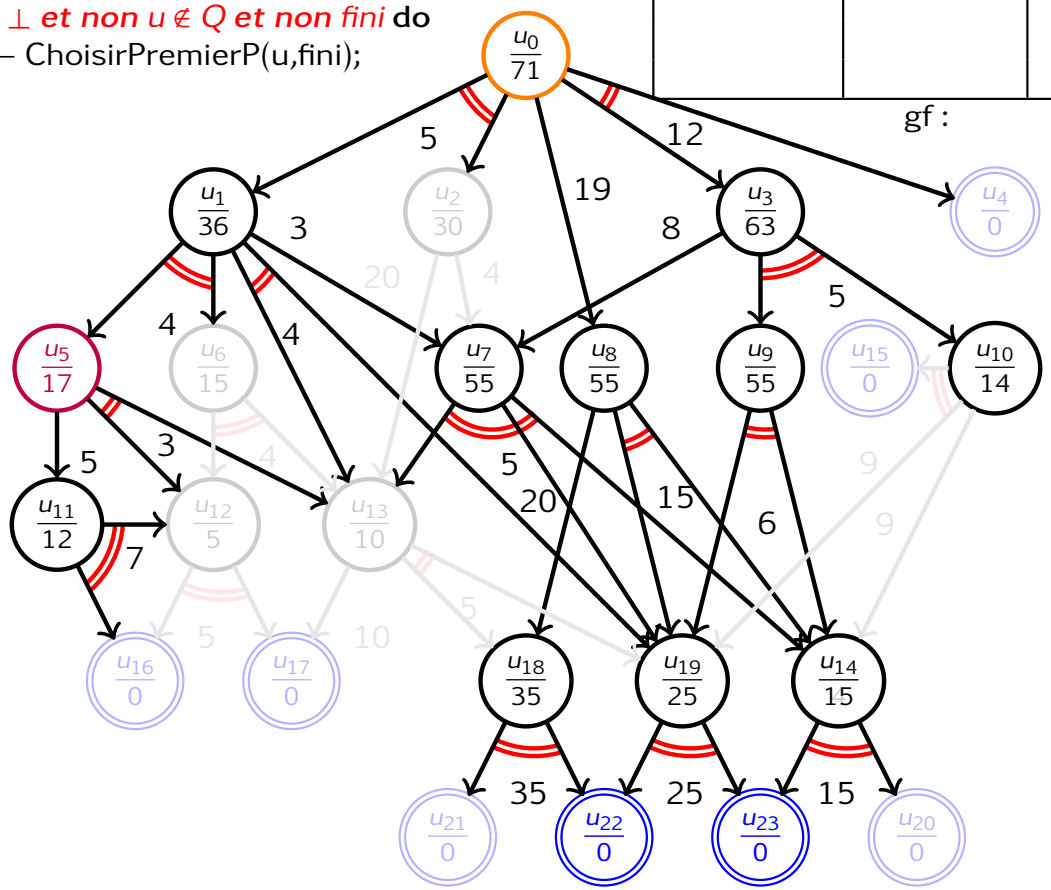

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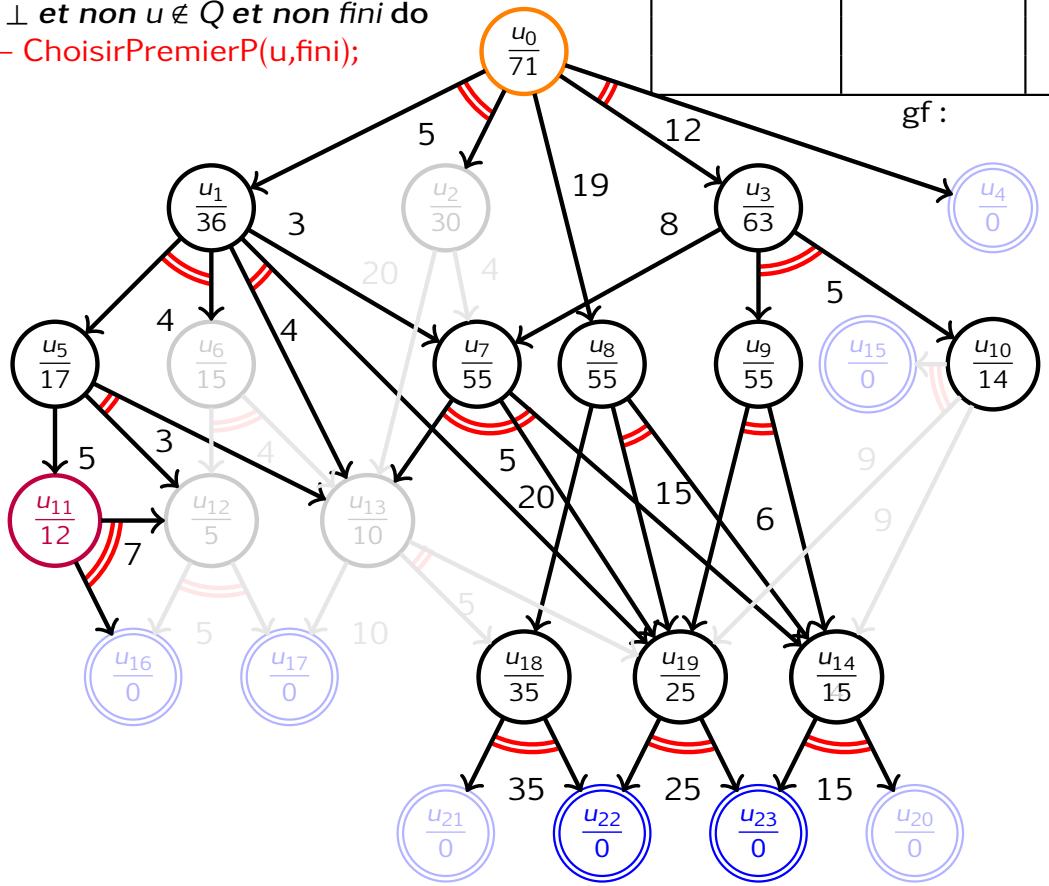
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,63)$	(63,74)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,36)$	(36,39,58)	{}	$u_5(9,17)u_6(9,15)u_2(5,30)$
$u_9(17,46)$	(46)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$
$u_5(9,17)$	(17,18)	{}	$u_{11}(14,15)u_6(9,15)$ $u_2(5,30)$
$u_{11}(14,12)$	(12)	{}	$u_{16}(7,0)u_{12}(7,5)u_6(9,15)u_2(5,30)$

gf :

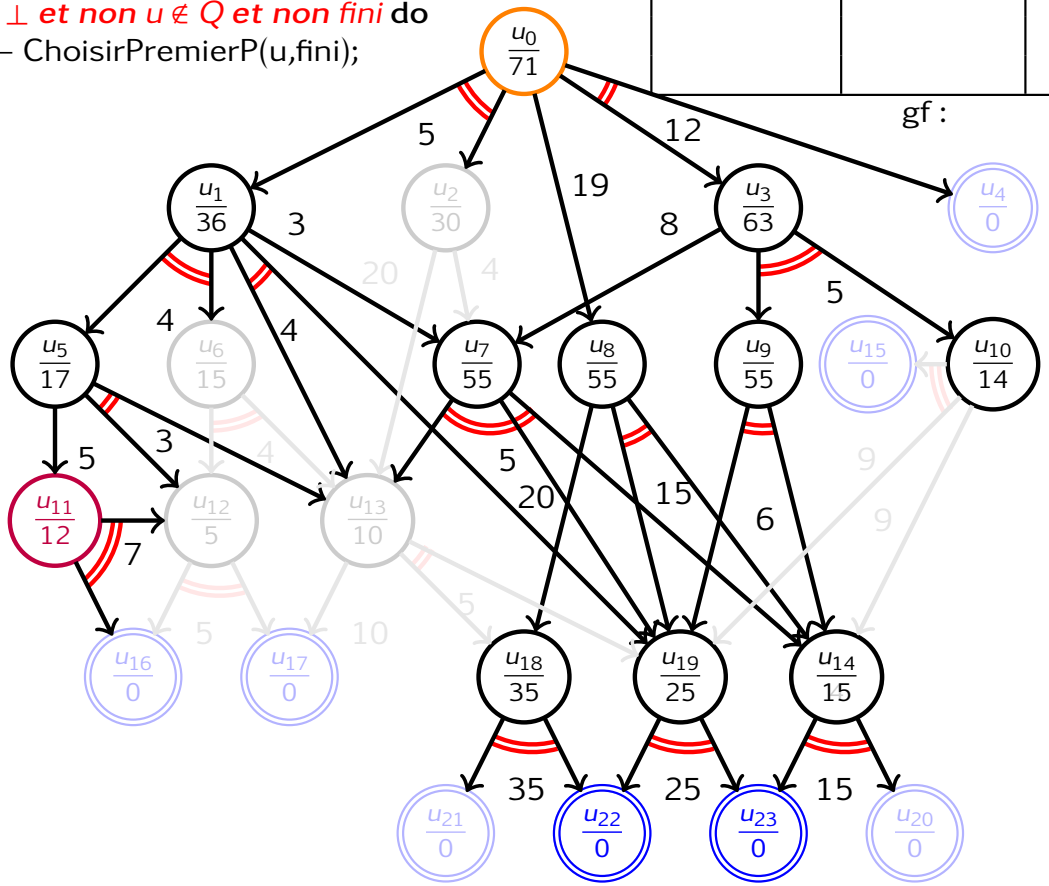


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,71)$	(71,74,75)	{}	$u_1(5,36)$ $u_2(5,30)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,63)$	(63,74)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,36)$	(36,39,58)	{}	$u_5(9,17)u_6(9,15)u_2(5,30)$
$u_9(17,46)$	(46)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$
$u_5(9,17)$	(17,18)	{}	$u_{11}(14,15)u_6(9,15)$ $u_2(5,30)$
$u_{11}(14,12)$	(12)	{}	$u_{16}(7,0)u_{12}(7,5)u_6(9,15)u_2(5,30)$

gf :

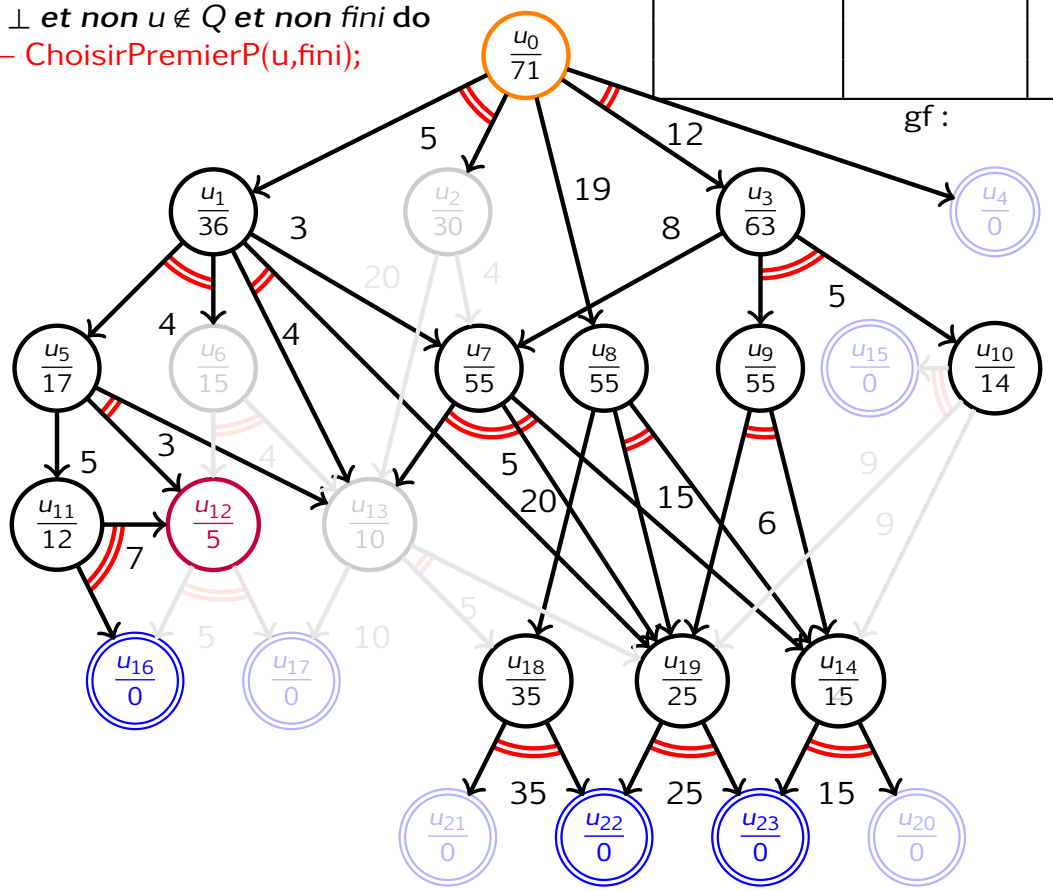


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
$u_0(0,71)$	(71,74,75)	{}	$u_1(5,36)$ $u_2(5,30)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,63)$	(63,74)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,36)$	(36,39,58)	{}	$u_5(9,17)u_6(9,15)u_2(5,30)$
$u_9(17,46)$	(46)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$
$u_5(9,17)$	(17,18)	{}	$u_{11}(14,15)u_6(9,15)$ $u_2(5,30)$
$u_{11}(14,12)$	(12)	{}	$u_{12}(7,5)u_6(9,15)u_2(5,30)$
$u_{12}(21,5)$	(-)	{}	$u_6(9,15)u_2(5,30)$

gf :

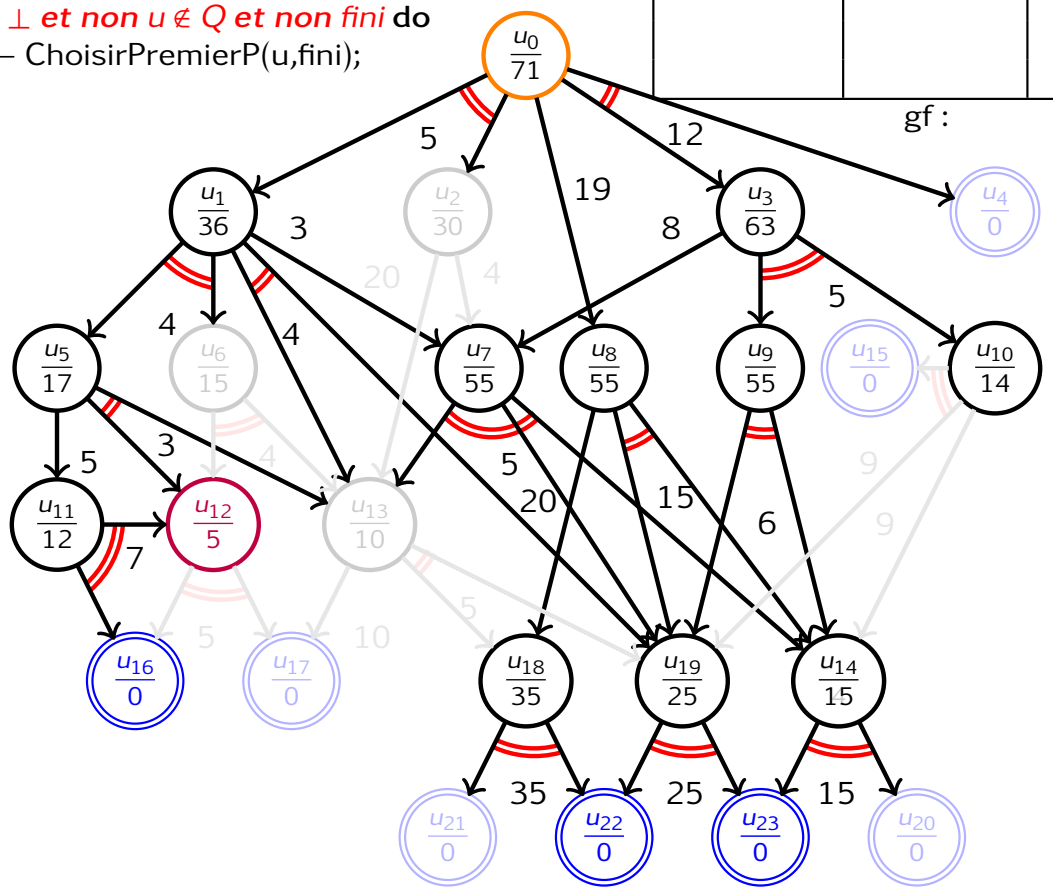


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
$u_0(0,71)$	(71,74,75)	{}	$u_1(5,36)$ $u_2(5,30)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,63)$	(63,74)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,36)$	(36,39,58)	{}	$u_5(9,17)u_6(9,15)u_2(5,30)$
$u_9(17,46)$	(46)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$
$u_5(9,17)$	(17,18)	{}	$u_{11}(14,15)u_6(9,15)$ $u_2(5,30)$
$u_{11}(14,12)$	(12)	{}	$u_{12}(7,5)u_6(9,15)u_2(5,30)$
$u_{12}(21,5)$	(-)	{}	$u_6(9,15)u_2(5,30)$

gf :

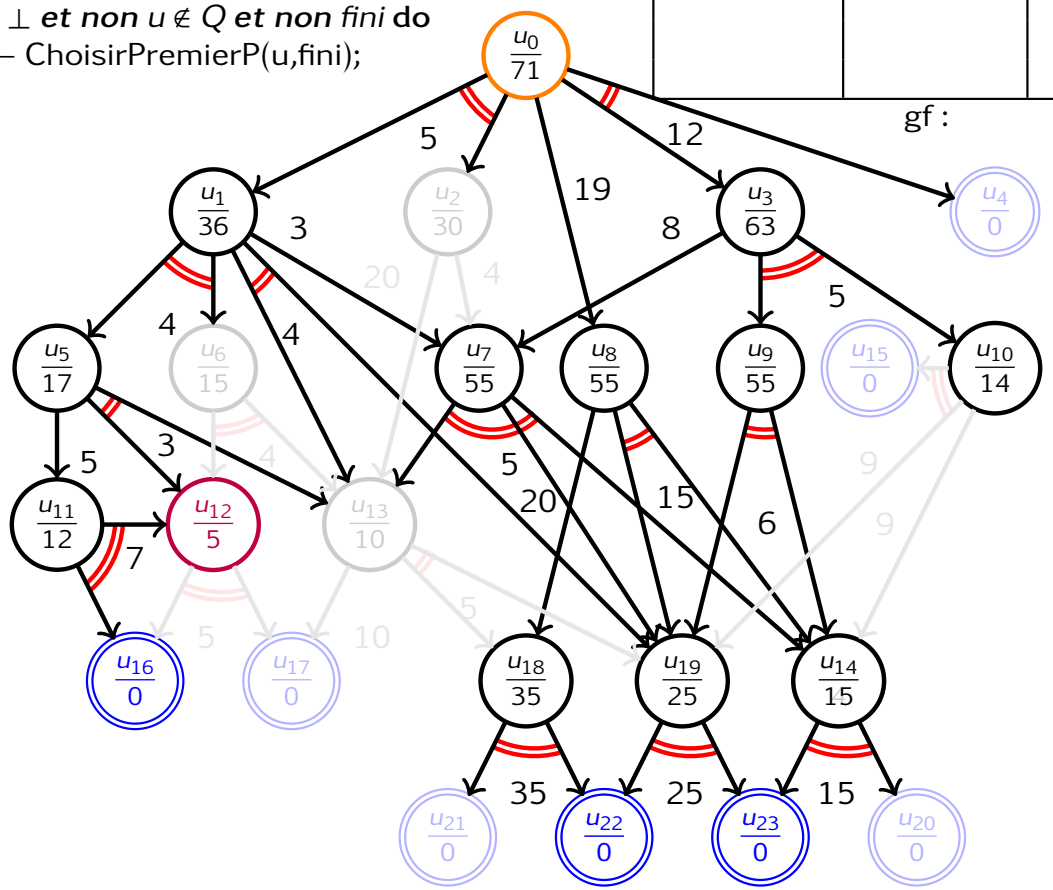


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
$u_0(0,71)$	(71,74,75)	{}	$u_1(5,36)$ $u_2(5,30)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,63)$	(63,74)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,36)$	(36,39,58)	{}	$u_5(9,17)u_6(9,15)u_2(5,30)$
$u_9(17,46)$	(46)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$
$u_5(9,17)$	(17,18)	{}	$u_{11}(14,15)u_6(9,15)$ $u_2(5,30)$
$u_{11}(14,12)$	(12)	{}	$u_{12}(7,5)u_6(9,15)u_2(5,30)$
$u_{12}(21,5)$	(-)	{}	$u_6(9,15)u_2(5,30)$

gf :

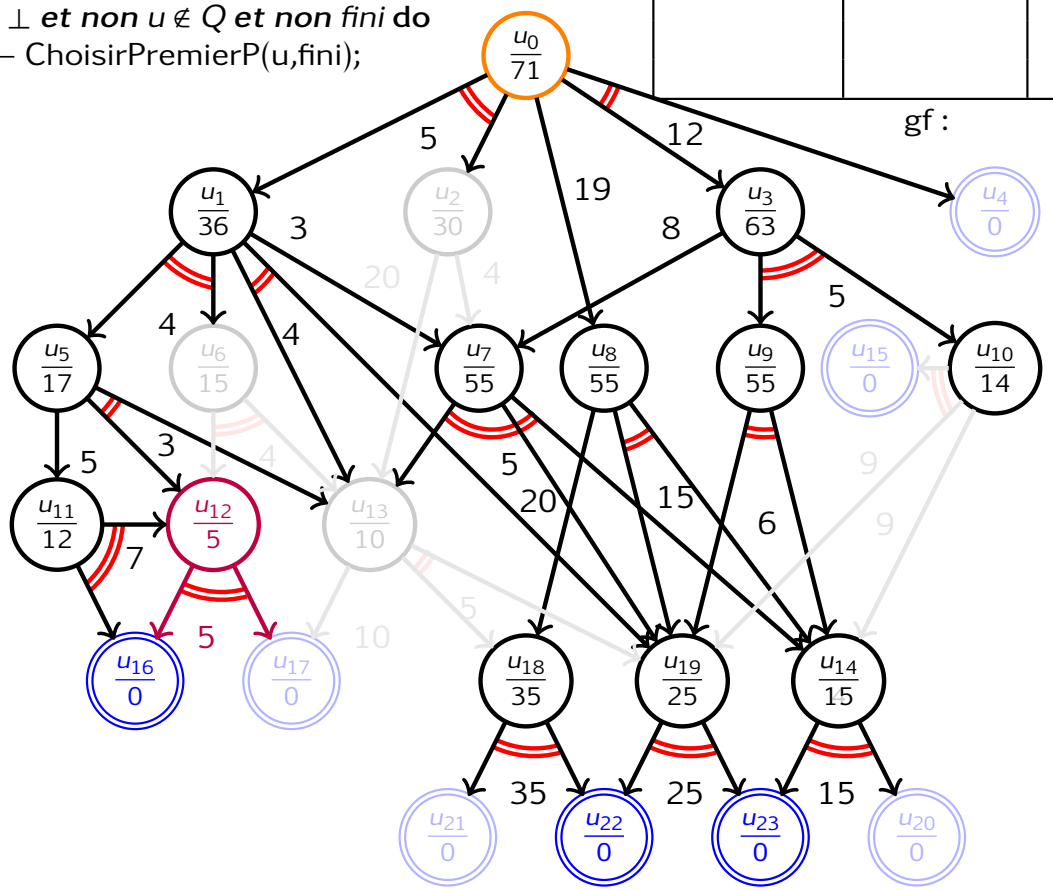


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
$u_0(0,71)$	(71,74,75)	{}	$u_1(5,36)$ $u_2(5,30)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,63)$	(63,74)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,36)$	(36,39,58)	{}	$u_5(9,17)u_6(9,15)u_2(5,30)$
$u_9(17,46)$	(46)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$
$u_5(9,17)$	(17,18)	{}	$u_{11}(14,15)u_6(9,15)$ $u_2(5,30)$
$u_{11}(14,12)$	(12)	{}	$u_{12}(7,5)u_6(9,15)u_2(5,30)$
$u_{12}(21,5)$	(-)	{}	$u_6(9,15)u_2(5,30)$

gf : $\{u_{16},u_{17}\}$

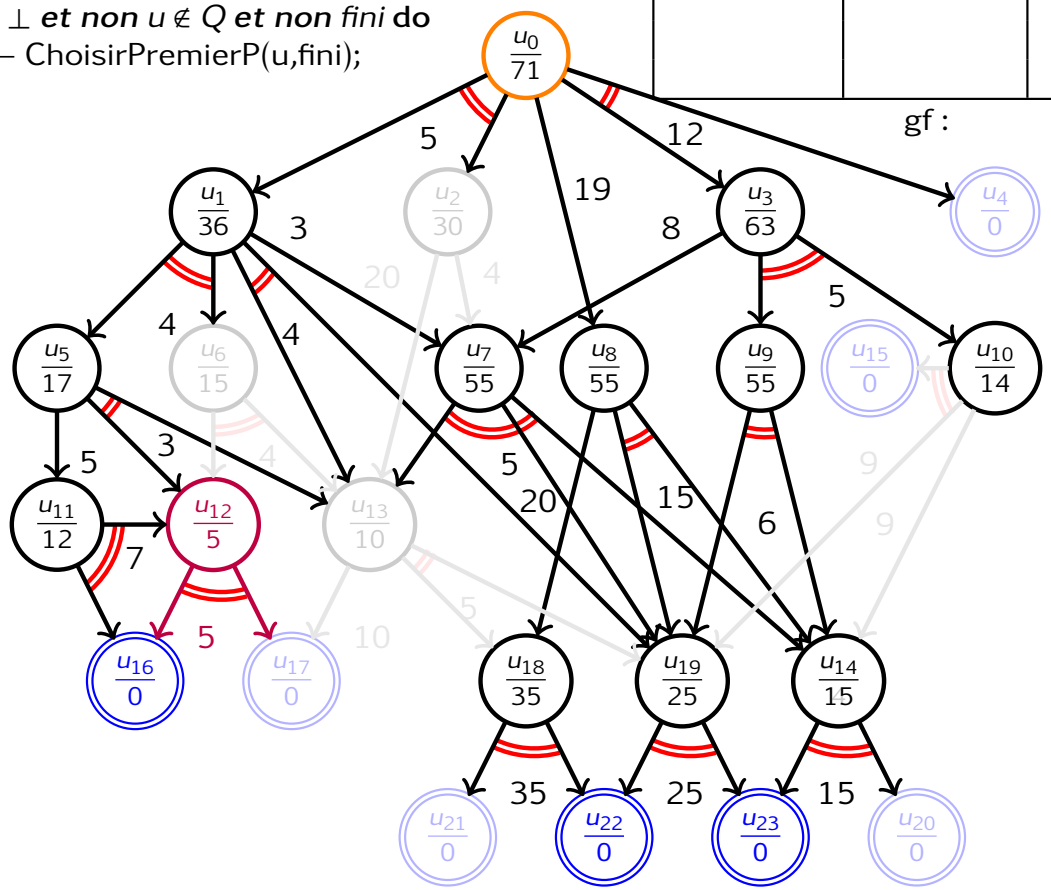


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
$u_0(0,71)$	(71,74,75)	{}	$u_1(5,36)$ $u_2(5,30)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,63)$	(63,74)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,36)$	(36,39,58)	{}	$u_5(9,17)u_6(9,15)u_2(5,30)$
$u_9(17,46)$	(46)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$
$u_5(9,17)$	(17,18)	{}	$u_{11}(14,15)u_6(9,15)$ $u_2(5,30)$
$u_{11}(14,12)$	(12)	{}	$u_{12}(7,5)u_6(9,15)u_2(5,30)$
$u_{12}(21,5)$	(-)	{}	$u_6(9,15)u_2(5,30)$

gf : $\{u_{16},u_{17}\}$

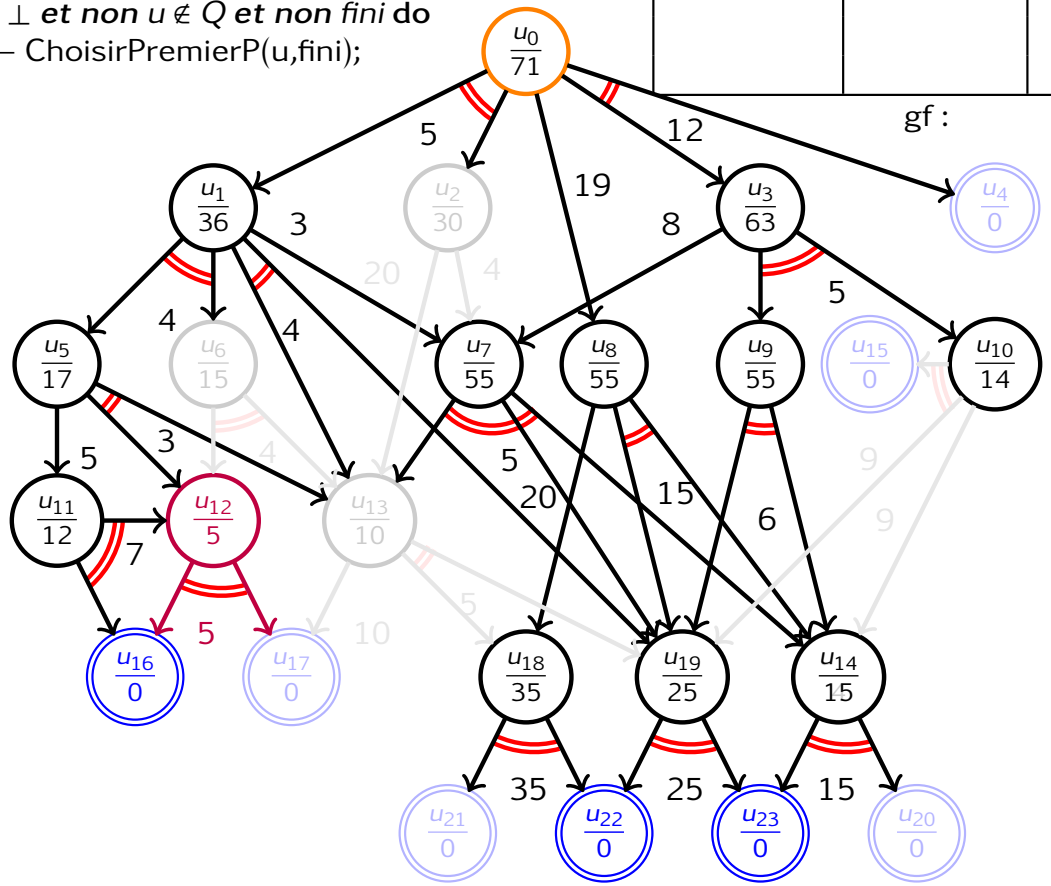


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,71)	(71,74,75)	{}	u_1(5,36) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,36)	(36,39,58)	{}	u_5(9,17)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,15)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,15)u_2(5,30)

gf : $\{u_{16},u_{17}\}$ = 5+0+0 = 5

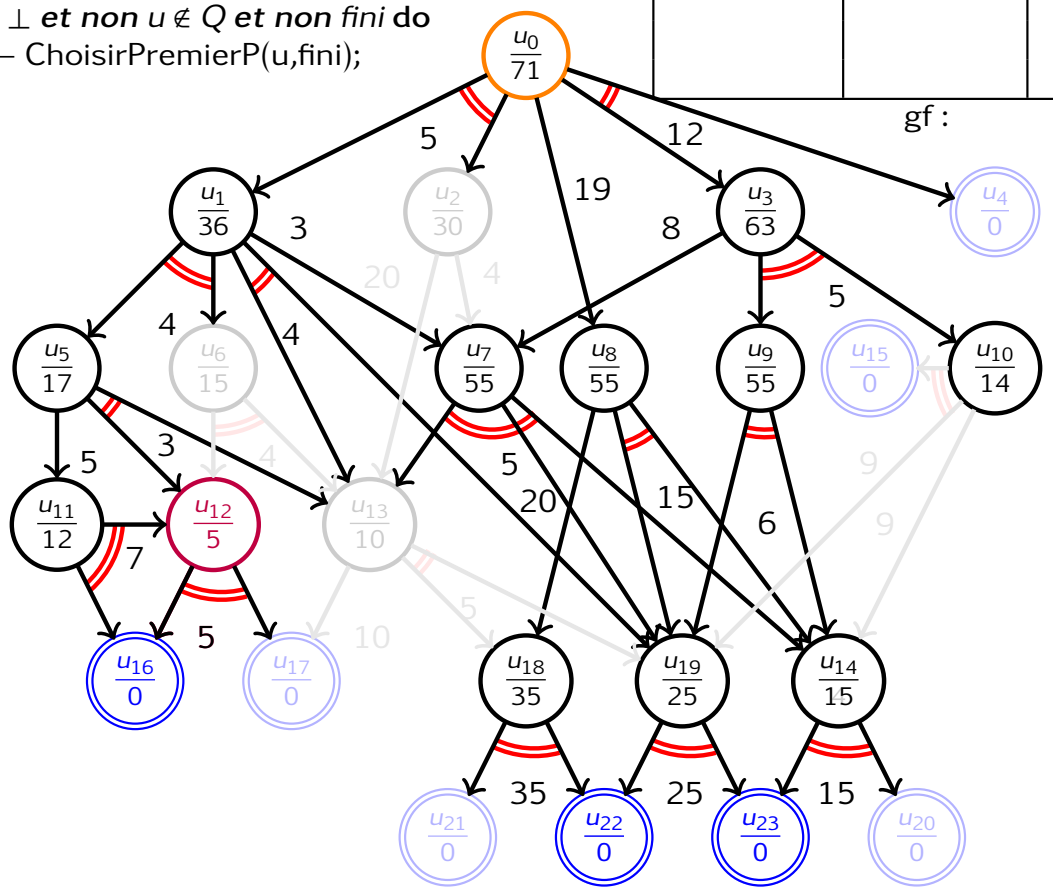


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,71)	(71,74,75)	{}	u_1(5,36) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,36)	(36,39,58)	{}	u_5(9,17)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,15)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,15)u_2(5,30)

gf :

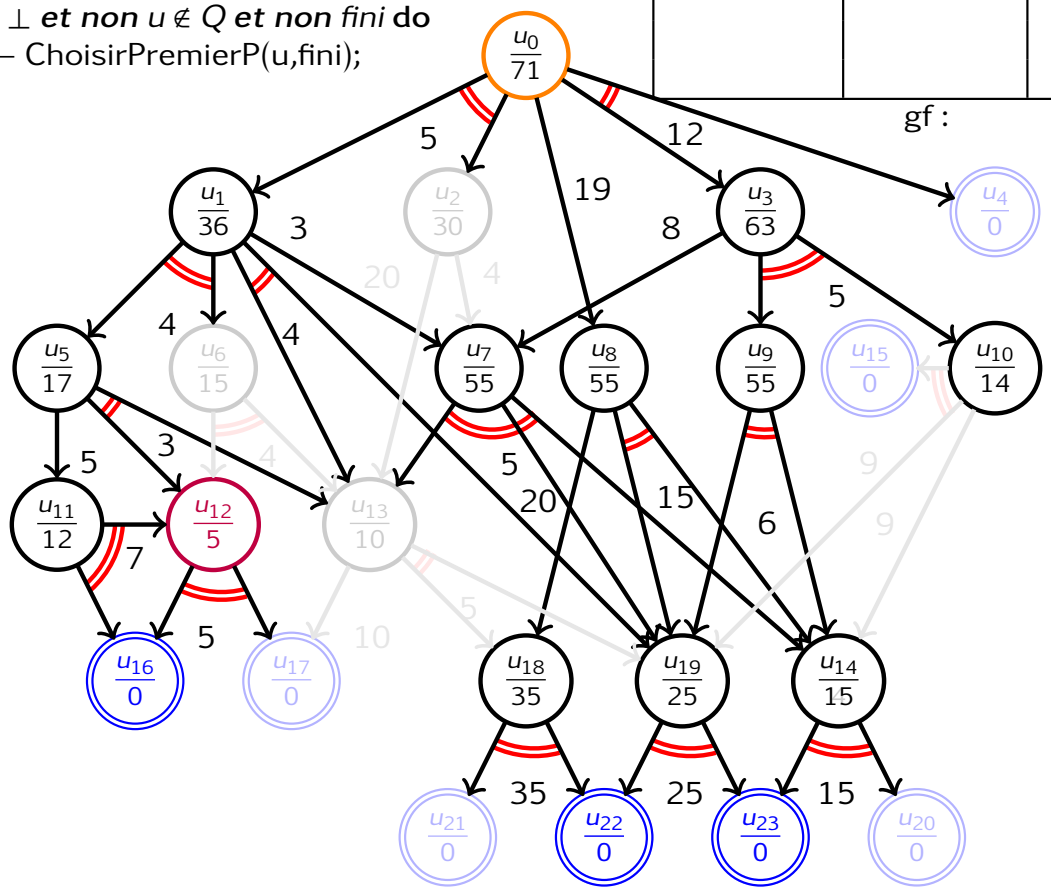


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while $u \neq \perp$ ***et non*** *fini* **do**
  gf ← groupeFils(u) ;
  while $gf \neq \emptyset$ **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while $A \neq \emptyset$ **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while $u \neq \perp$ ***et non*** $u \notin Q$ ***et non*** *fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,71)	(71,74,75)	{}	u_1(5,36) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,36)	(36,39,58)	{}	u_5(9,17)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,15)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,15)u_2(5,30)

gf :

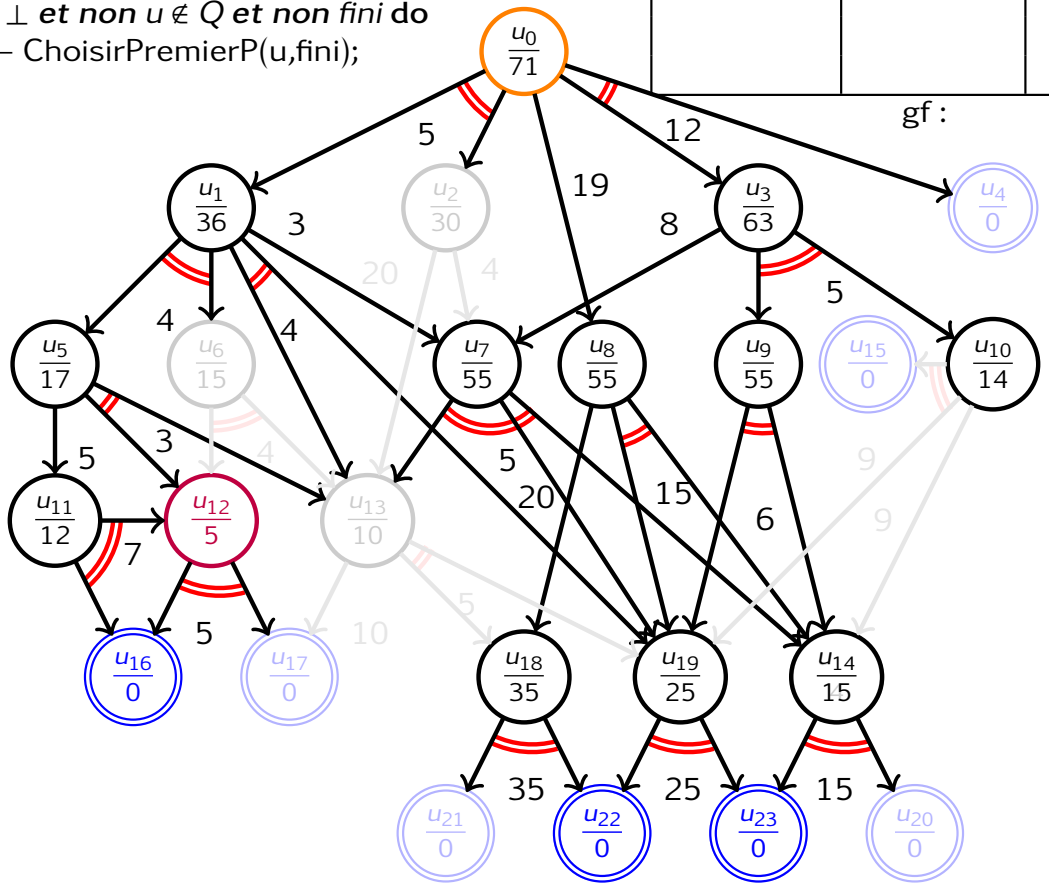


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,71)	(71,74,75)	{}	u_1(5,36) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,36)	(36,39,58)	{}	u_5(9,17)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,15)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,15)u_2(5,30)

gf :



Algorithme OA*(Liste,Sol)

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while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
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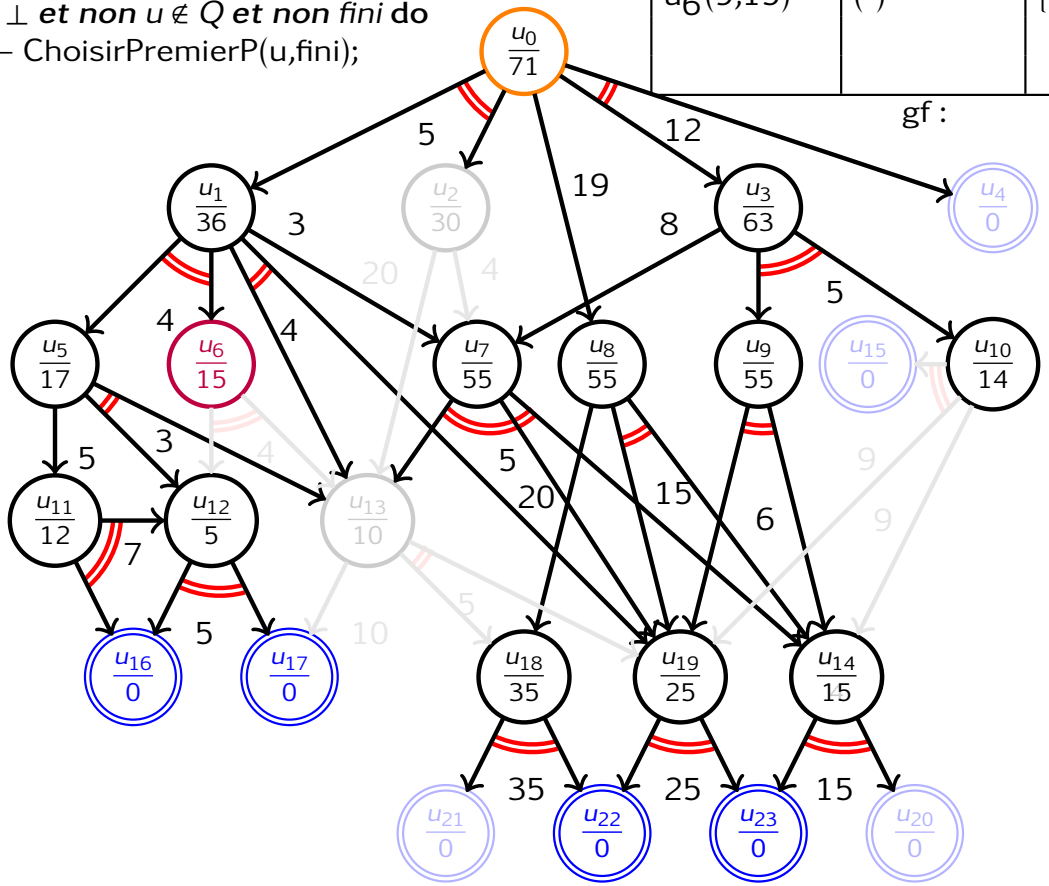
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,71)	(71,74,75)	{}	u_1(5,36) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,36)	(36,39,58)	{}	u_5(9,17)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,15)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,15)u_2(5,30)
u_6(9,15)	(-)	{}	u_2(5,30)

gf :



Algorithme OA*(Liste,Sol)

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while *u ≠ ⊥ et non fini* **do**
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    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
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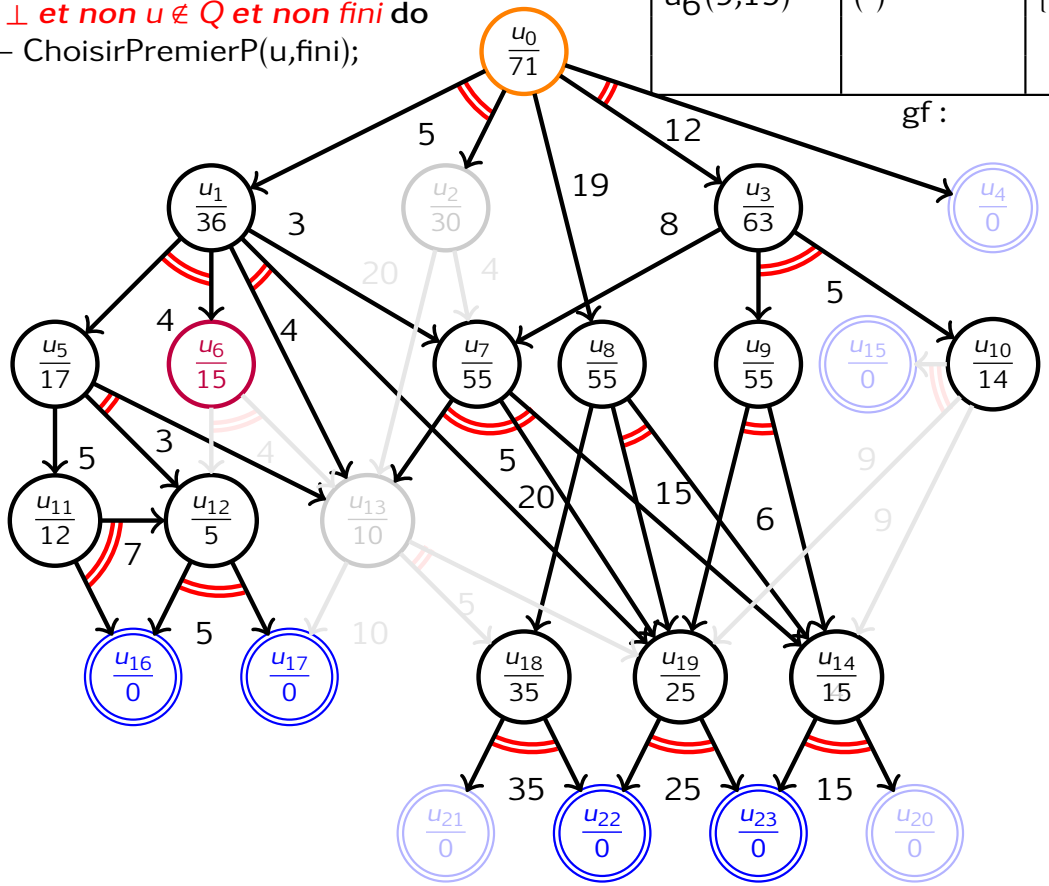
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,71)	(71,74,75)	{}	u_1(5,36) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,36)	(36,39,58)	{}	u_5(9,17)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,15)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,15)u_2(5,30)
u_6(9,15)	(-)	{}	u_2(5,30)

gf :

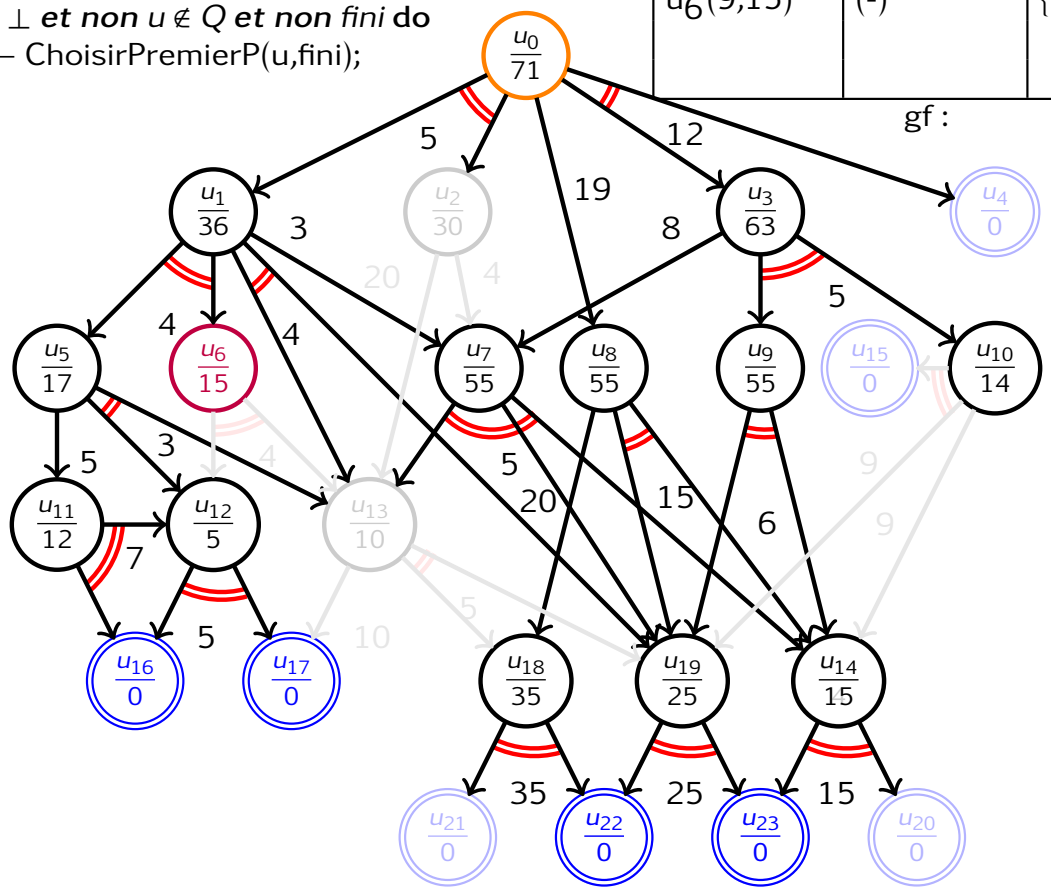


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
$u_0(0,71)$	(71,74,75)	{}	$u_1(5,36)$ $u_2(5,30)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,63)$	(63,74)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,36)$	(36,39,58)	{}	$u_5(9,17)u_6(9,15)u_2(5,30)$
$u_9(17,46)$	(46)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$
$u_5(9,17)$	(17,18)	{}	$u_{11}(14,15)u_6(9,15)$ $u_2(5,30)$
$u_{11}(14,12)$	(12)	{}	$u_{12}(7,5)u_6(9,15)u_2(5,30)$
$u_{12}(21,5)$	(5)	{}	$u_{17}(5,0)$ $u_6(9,15)u_2(5,30)$
$u_6(9,15)$	(-)	{}	$u_2(5,30)$

gf :

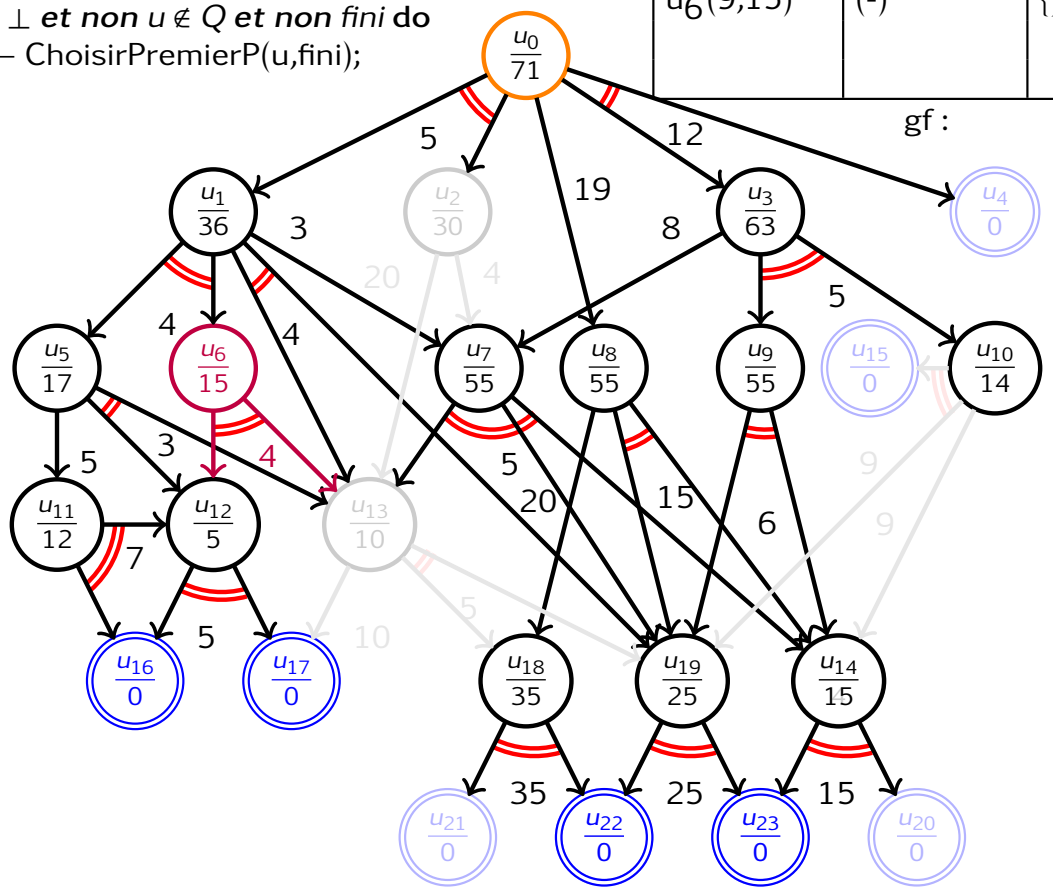


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,71)	(71,74,75)	{}	u_1(5,36) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,36)	(36,39,58)	{}	u_5(9,17)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,15)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,15)u_2(5,30)
u_6(9,15)	(-)	{}	u_2(5,30)

gf : $\{u_{12}, u_{13}\}$



Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
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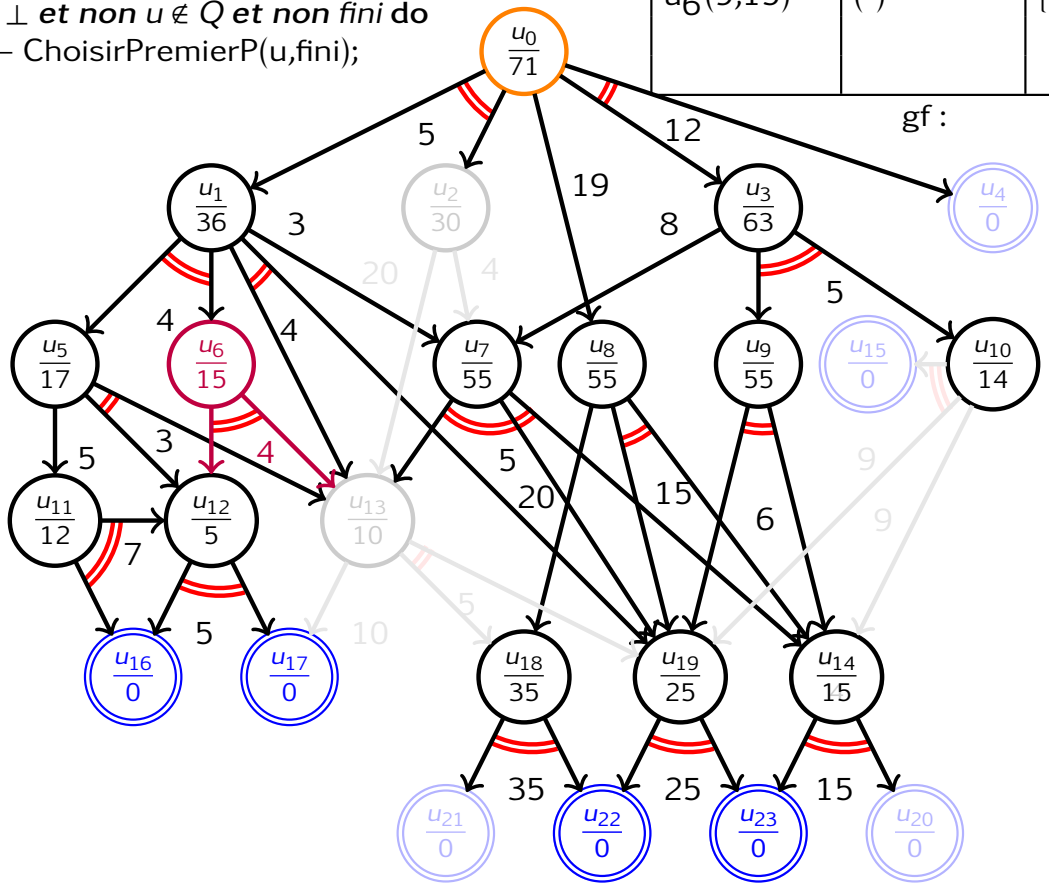
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,71)	(71,74,75)	{}	u_1(5,36) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,36)	(36,39,58)	{}	u_5(9,17)u_6(9,15)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,15) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,15)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,15)u_2(5,30)
u_6(9,15)	(-)	{}	u_2(5,30)

gf : $\{u_{12}, u_{13}\}$



Algorithme OA*(Liste,Sol)

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        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
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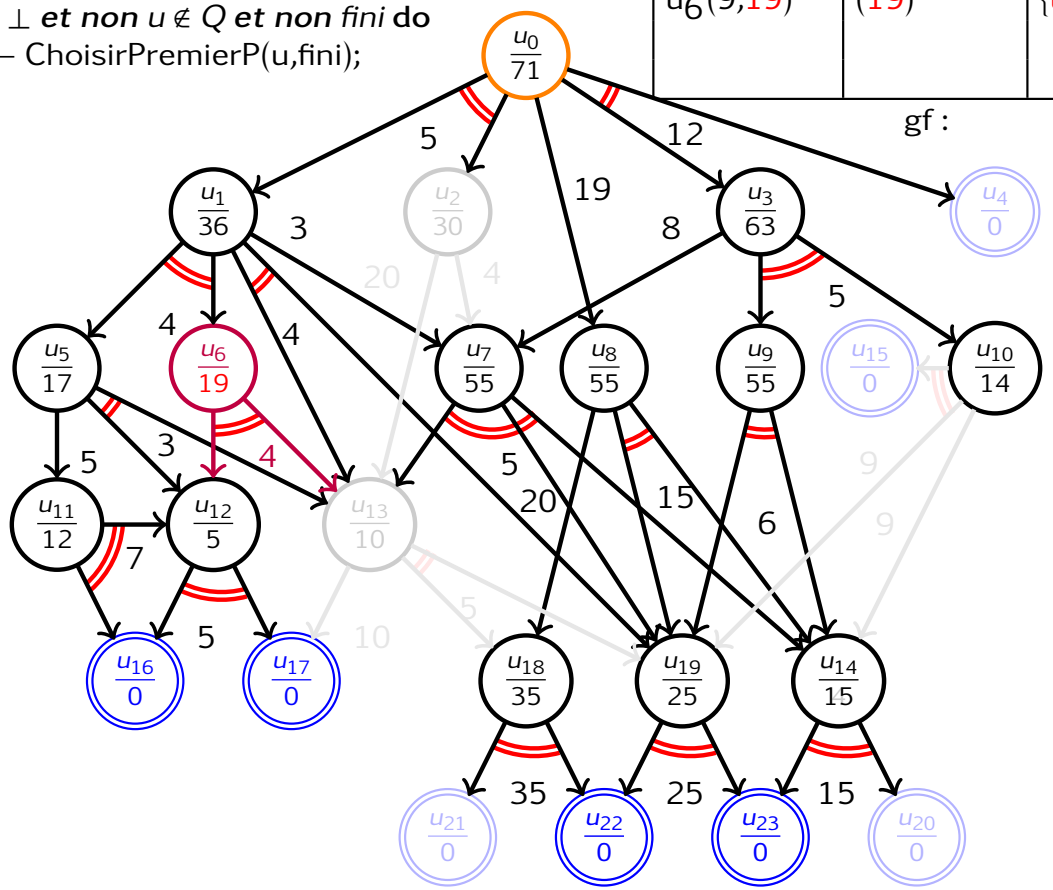
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,71)	(71,74,75)	{}	u_1(5,36) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,36)	(36,39,58)	{}	u_5(9,17)u_6(9,19)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{u_1}	u_{13}(4,10)u_2(5,30)

gf : {u_{12},u_{13}} = 4+5+10 = 19



Algorithme OA*(Liste,Sol)

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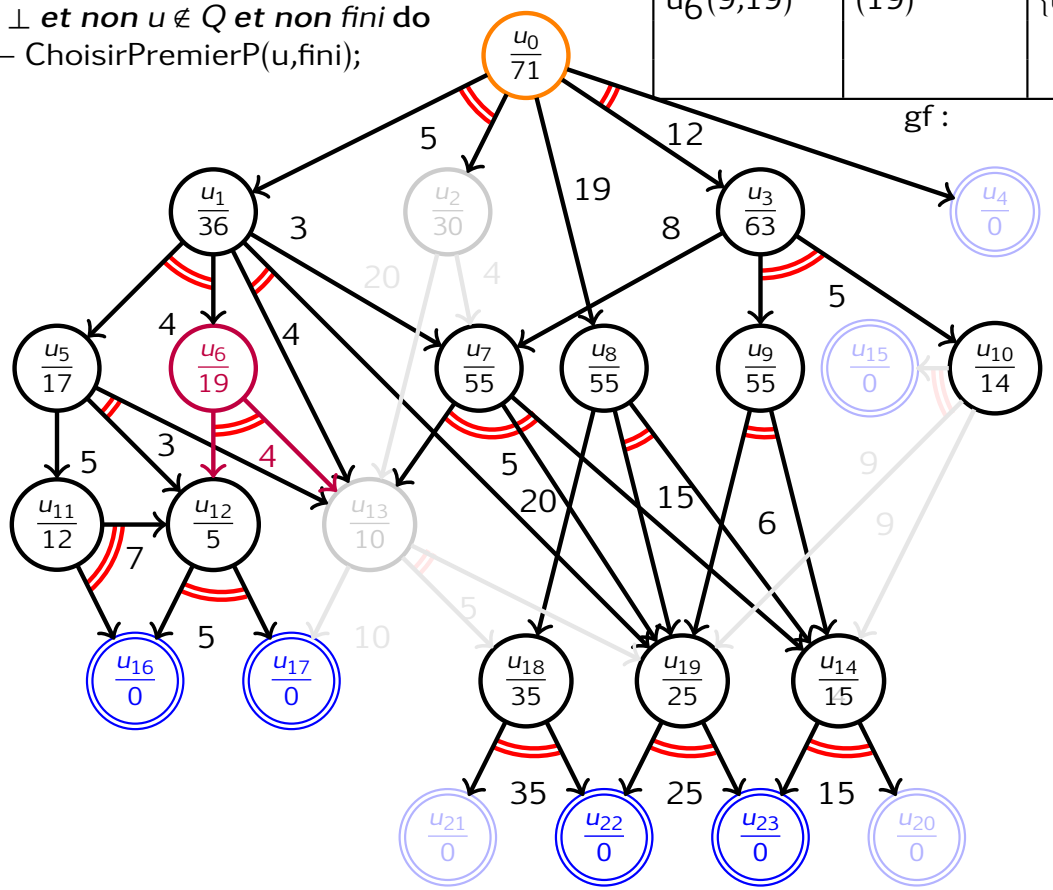
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,71)	(71,74,75)	{}	u_1(5,36) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,36)	(36,39,58)	{}	u_5(9,17)u_6(9,19)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{u_1}	u_{13}(4,10)u_2(5,30)

gf :



Algorithme OA*(Liste,Sol)

```
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    gf ← groupeFils(u) ;
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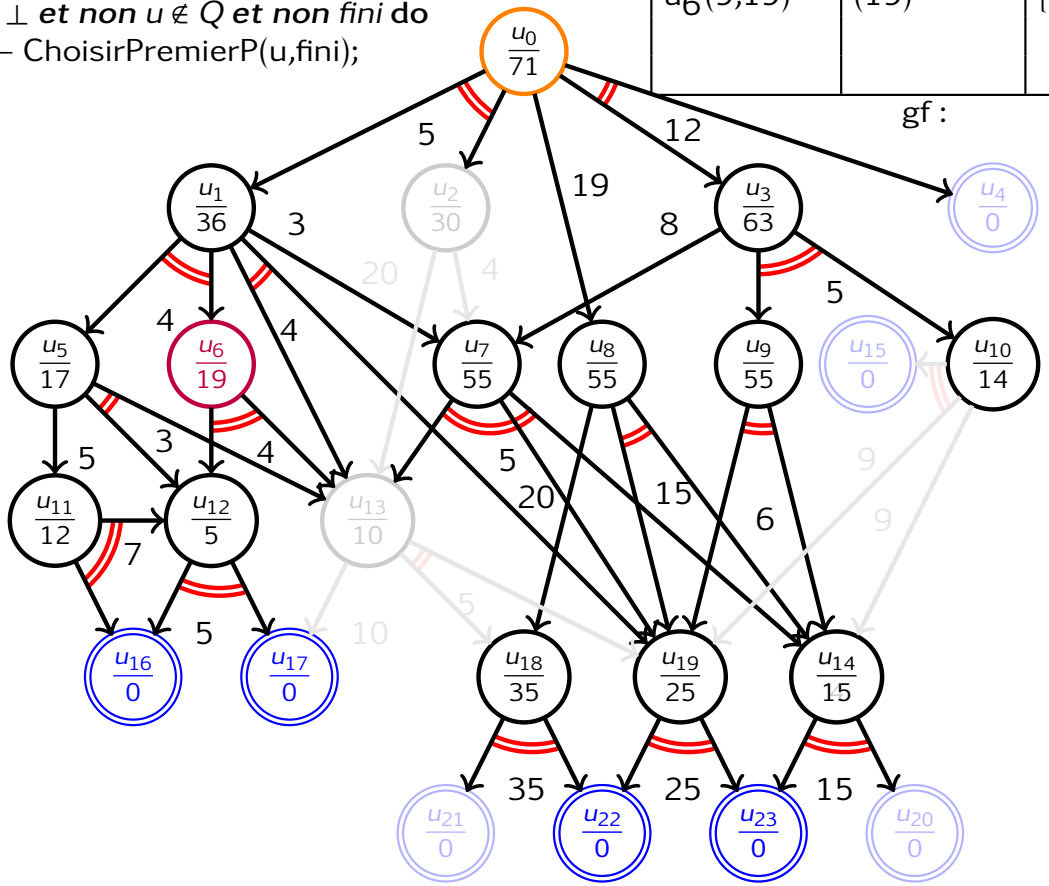
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,71)	(71,74,75)	{}	u_1(5,36) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,36)	(36,39,58)	{}	u_5(9,17)u_6(9,19)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{u_1}	u_{13}(4,10)u_2(5,30)

gf :

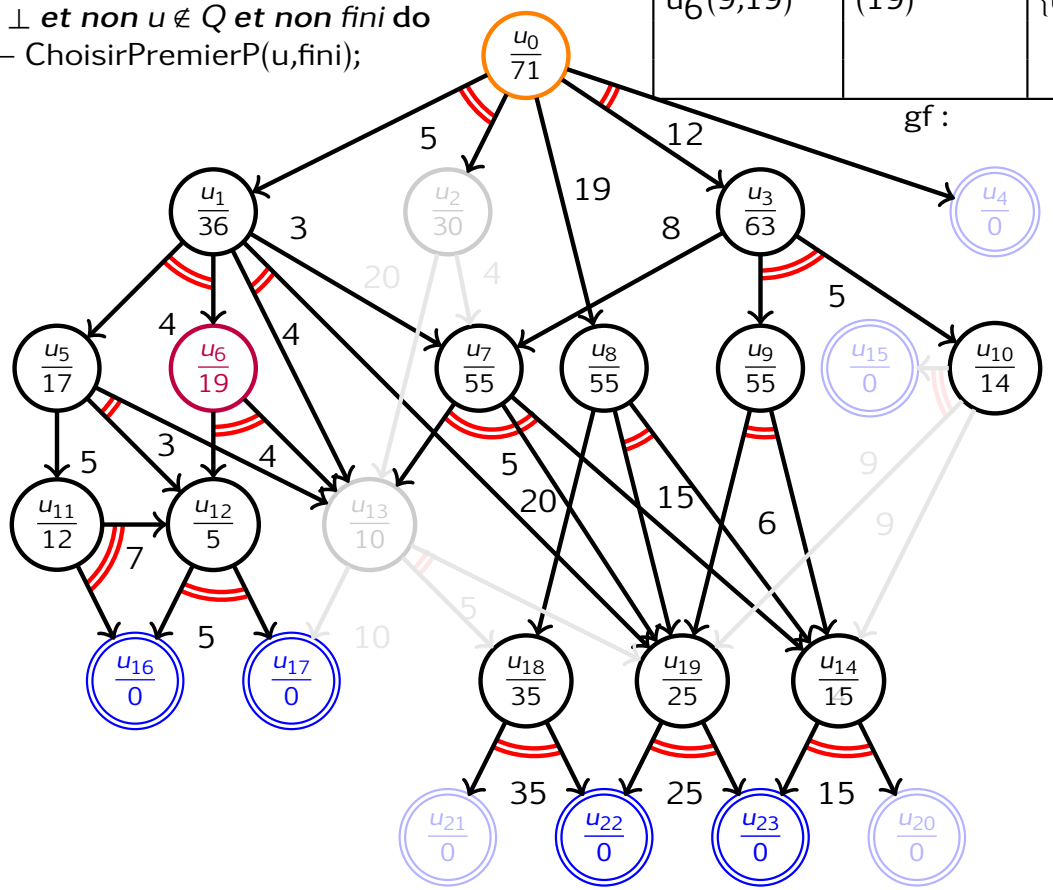


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,71)	(71,74,75)	{}	u_1(5,36) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,36)	(36,39,58)	{}	u_5(9,17)u_6(9,19)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{u_1}	u_{13}(4,10)u_2(5,30)

gf :

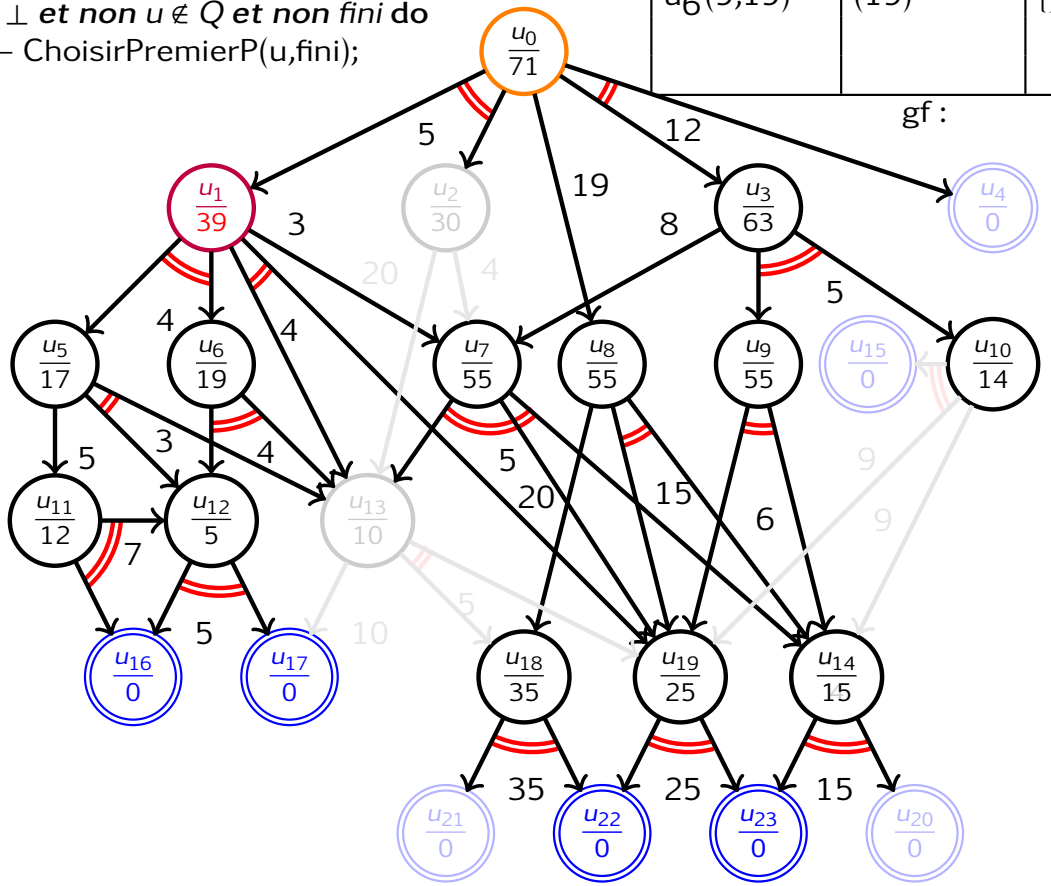


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,71)	(71,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{u_0}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)

gf :

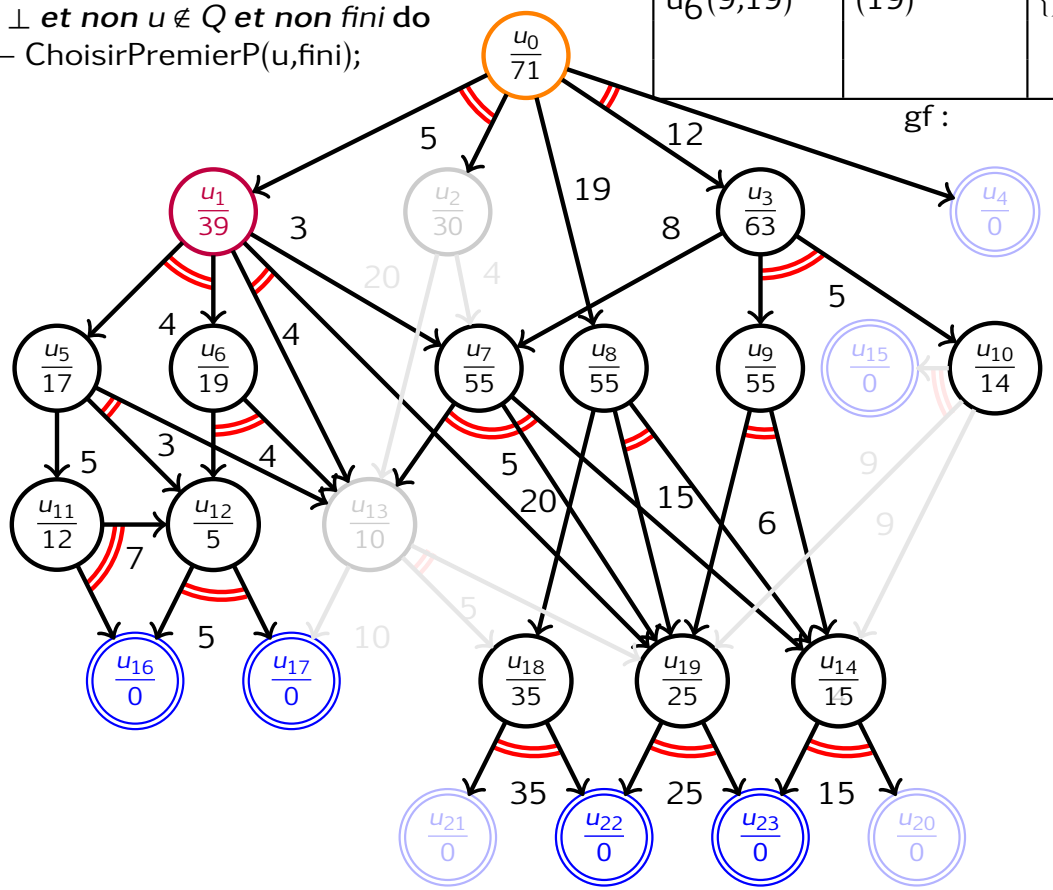


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,71)	(71,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{u_0}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)

gf :

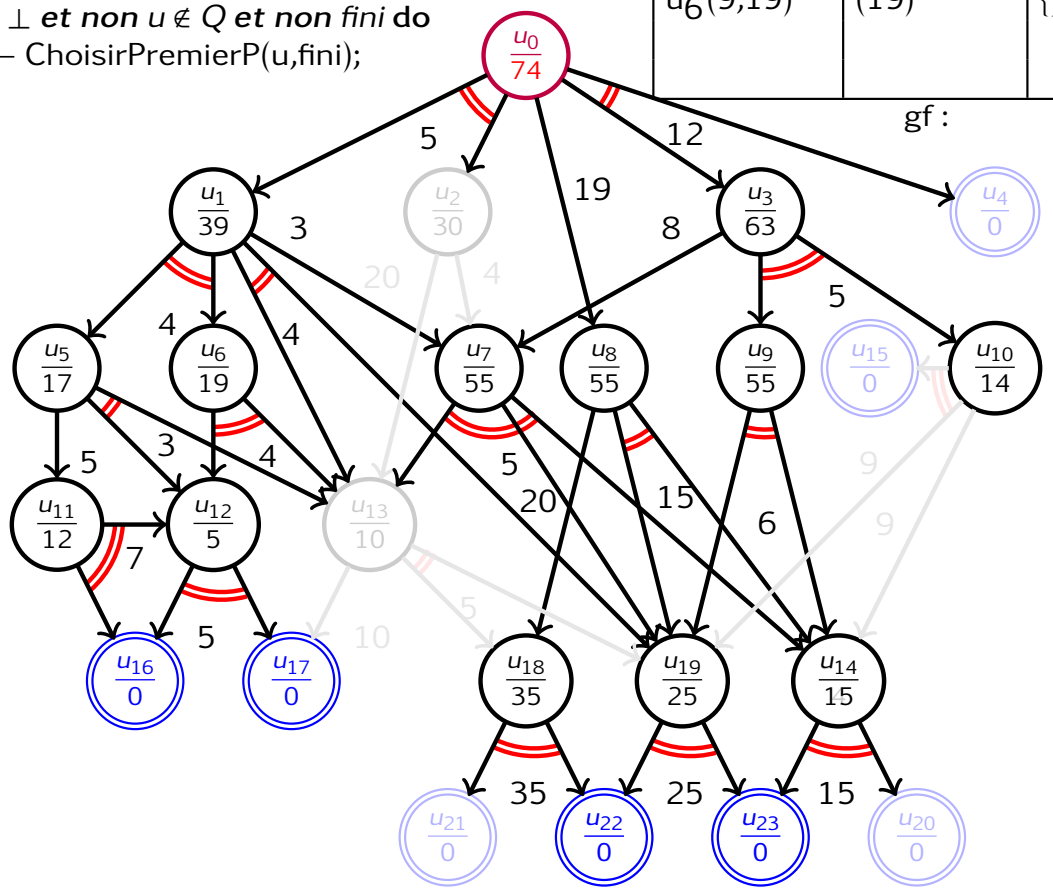


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
$u_0(0,74)$	(74,74,75)	{}	$u_1(5,39)$ $u_2(5,30)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,63)$	(63,74)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,39)$	(40,39,58)	{}	$u_{13}(4,10)u_{19}(4,25)u_2(5,30)$
$u_9(17,46)$	(46)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$
$u_5(9,17)$	(17,18)	{}	$u_{11}(14,15)u_6(9,19)$ $u_2(5,30)$
$u_{11}(14,12)$	(12)	{}	$u_{12}(7,5)u_6(9,19)u_2(5,30)$
$u_{12}(21,5)$	(5)	{}	$u_{17}(5,0)$ $u_6(9,19)u_2(5,30)$
$u_6(9,19)$	(19)	{}	$u_{13}(4,10)u_2(5,30)$

gf :

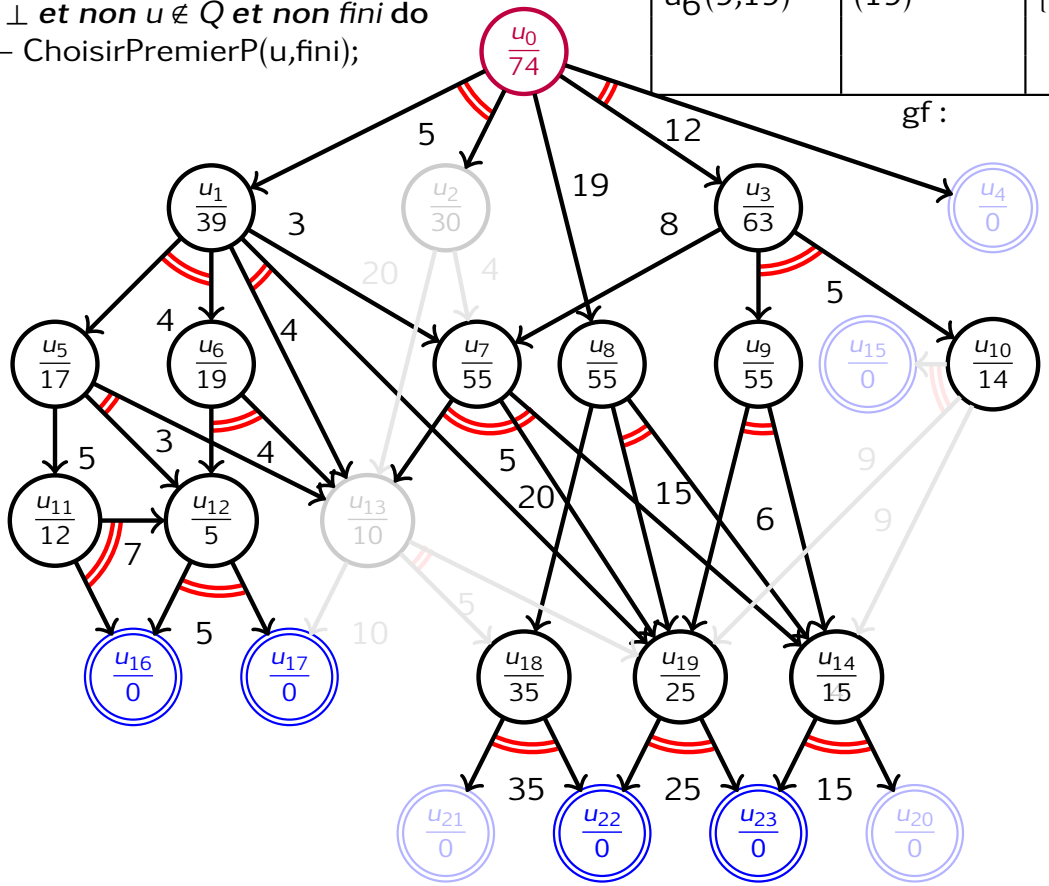


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)

gf :

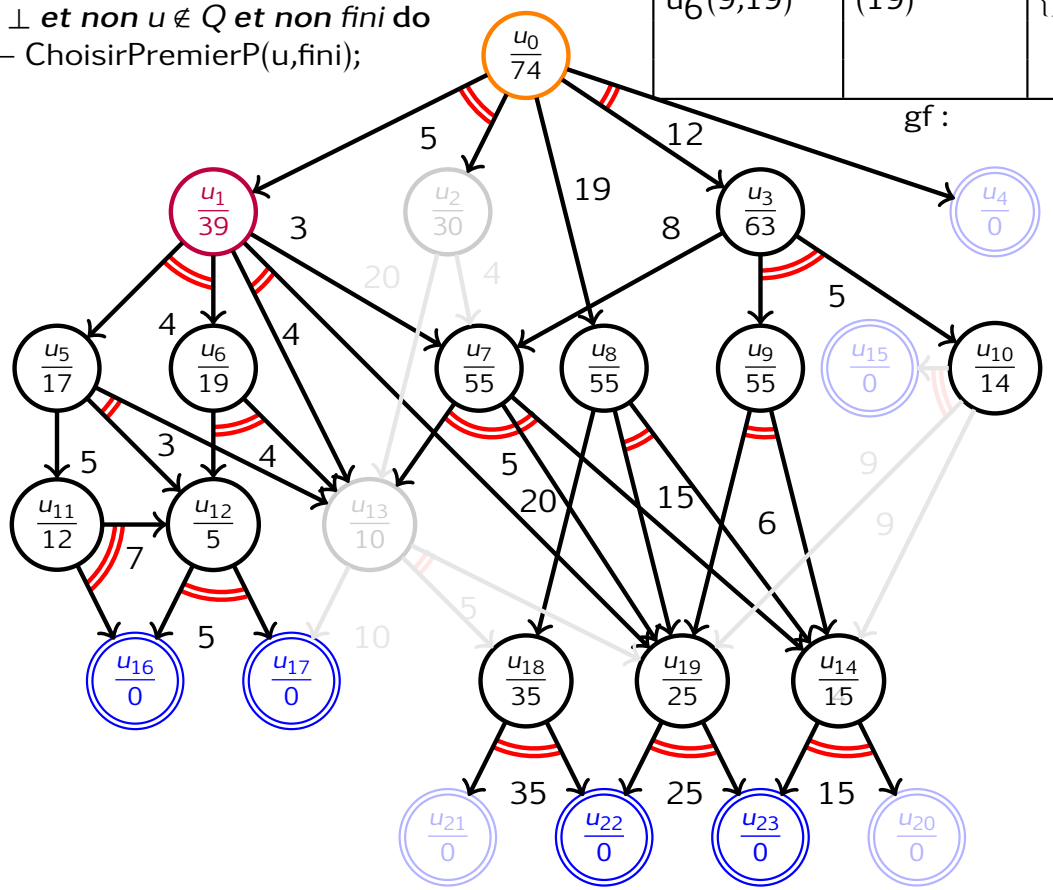


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)

gf :



Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
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  while *gf ≠ ∅* **do**
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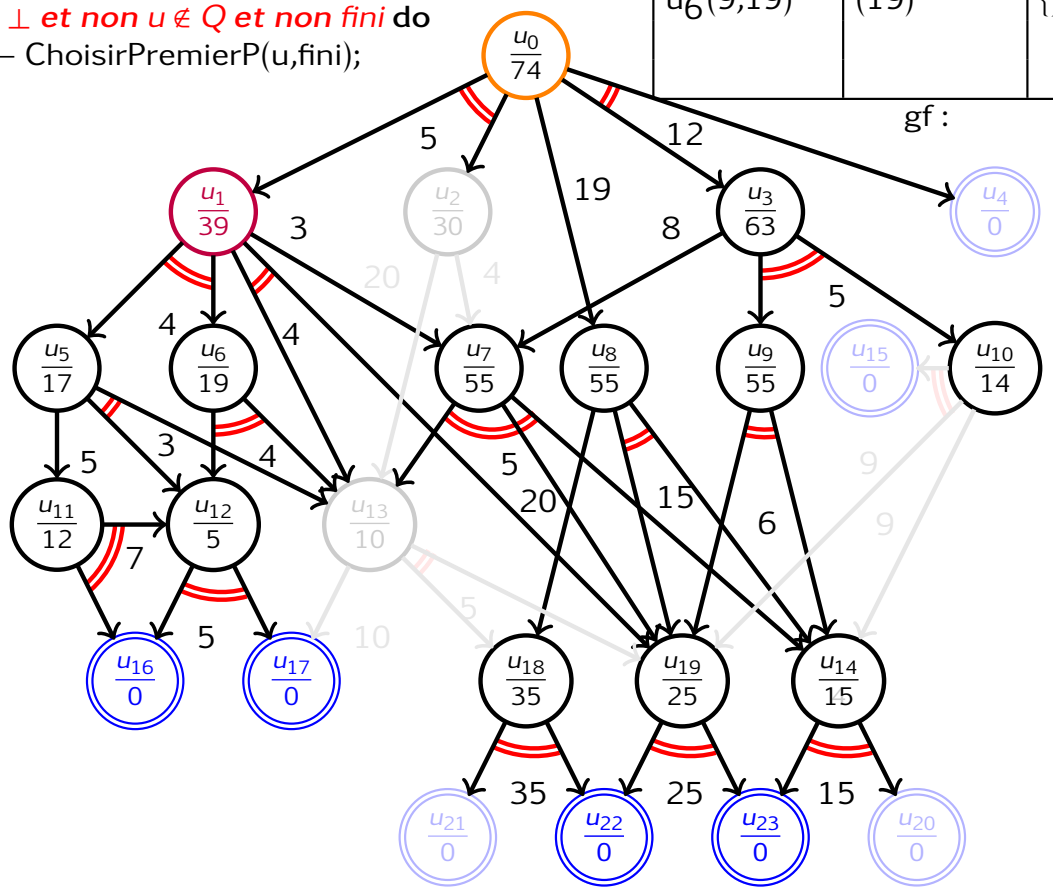
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)

gf :

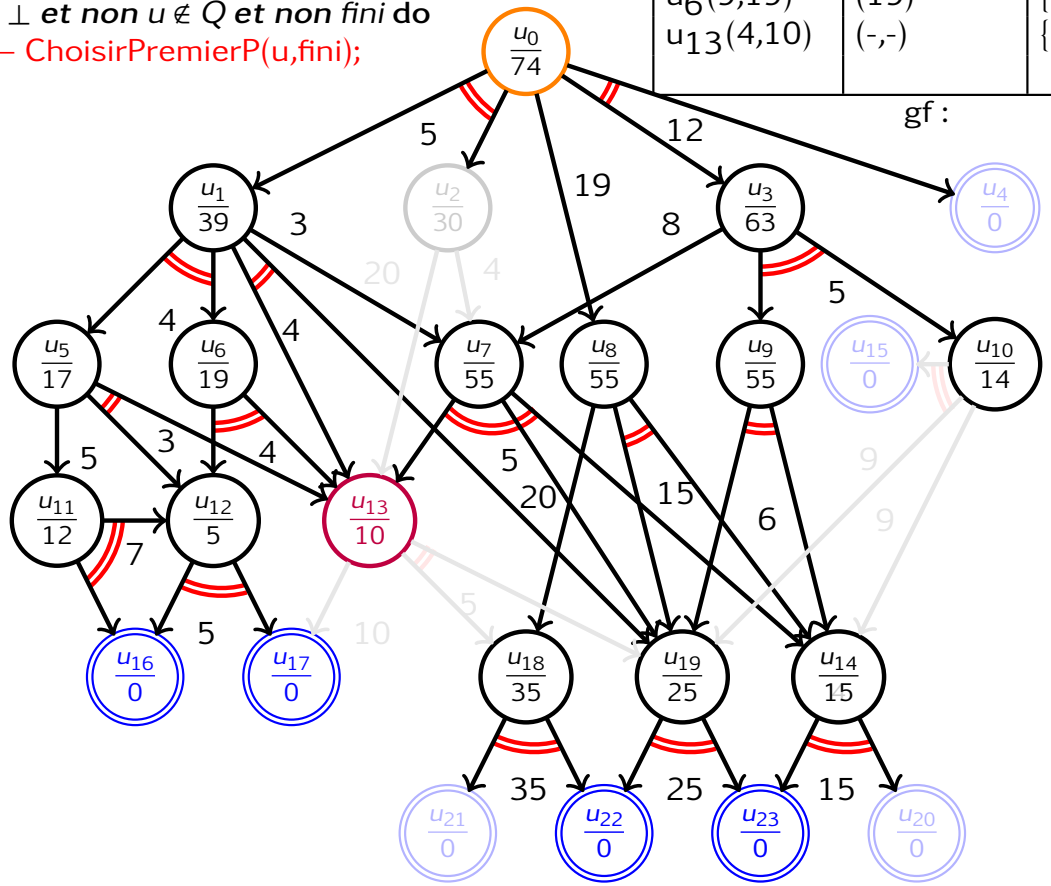


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(-,-)	{}	u_2(5,30)

gf :

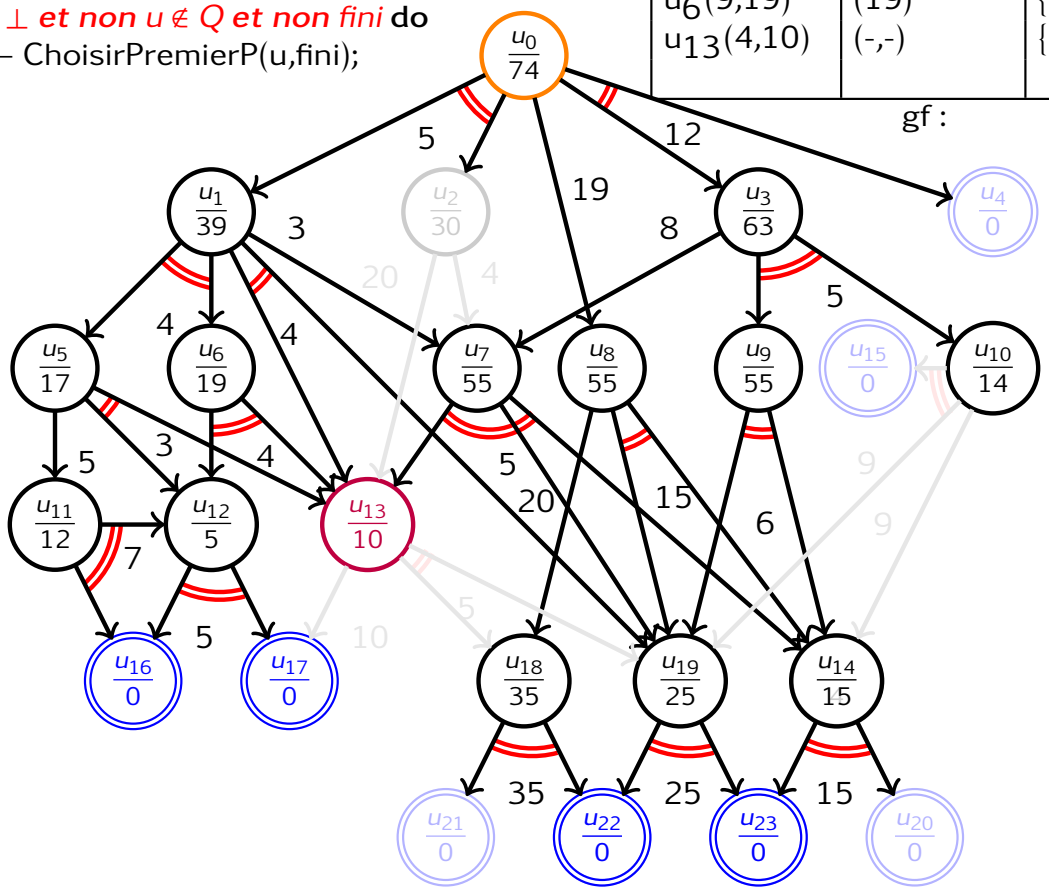


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(-,-)	{}	u_2(5,30)

gf :

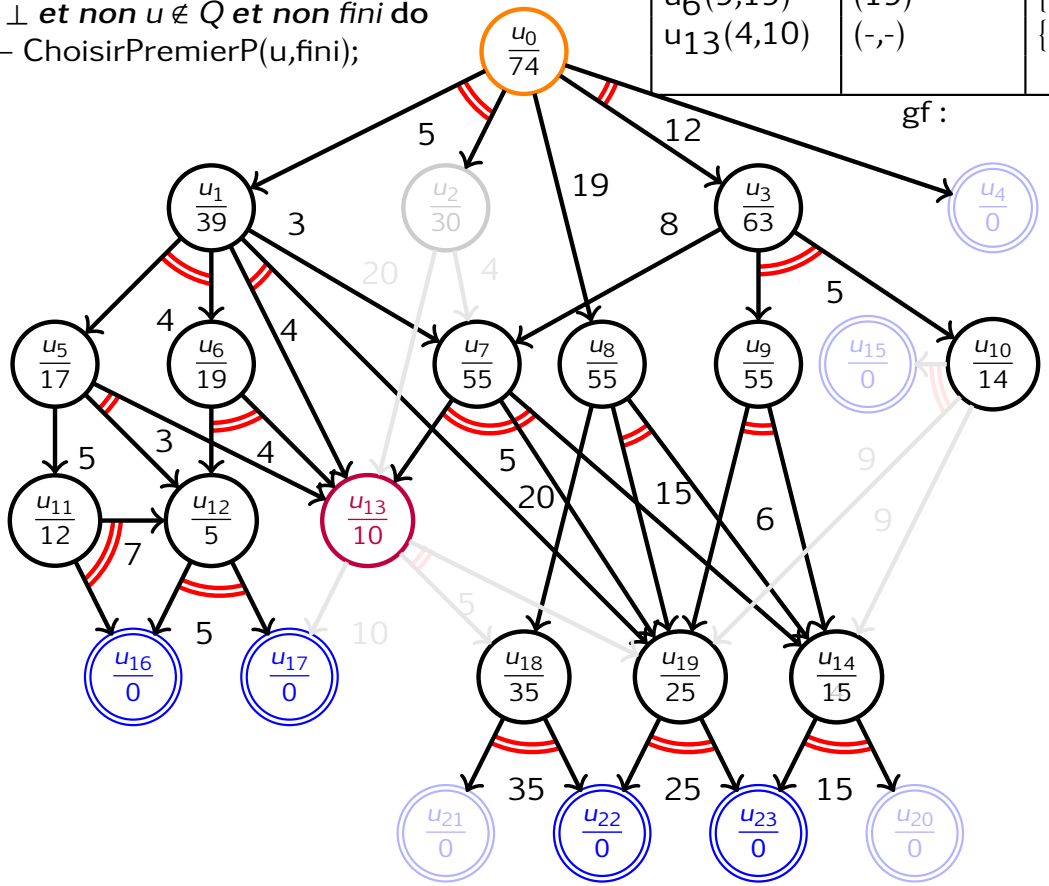


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(-,-)	{}	u_2(5,30)

gf :

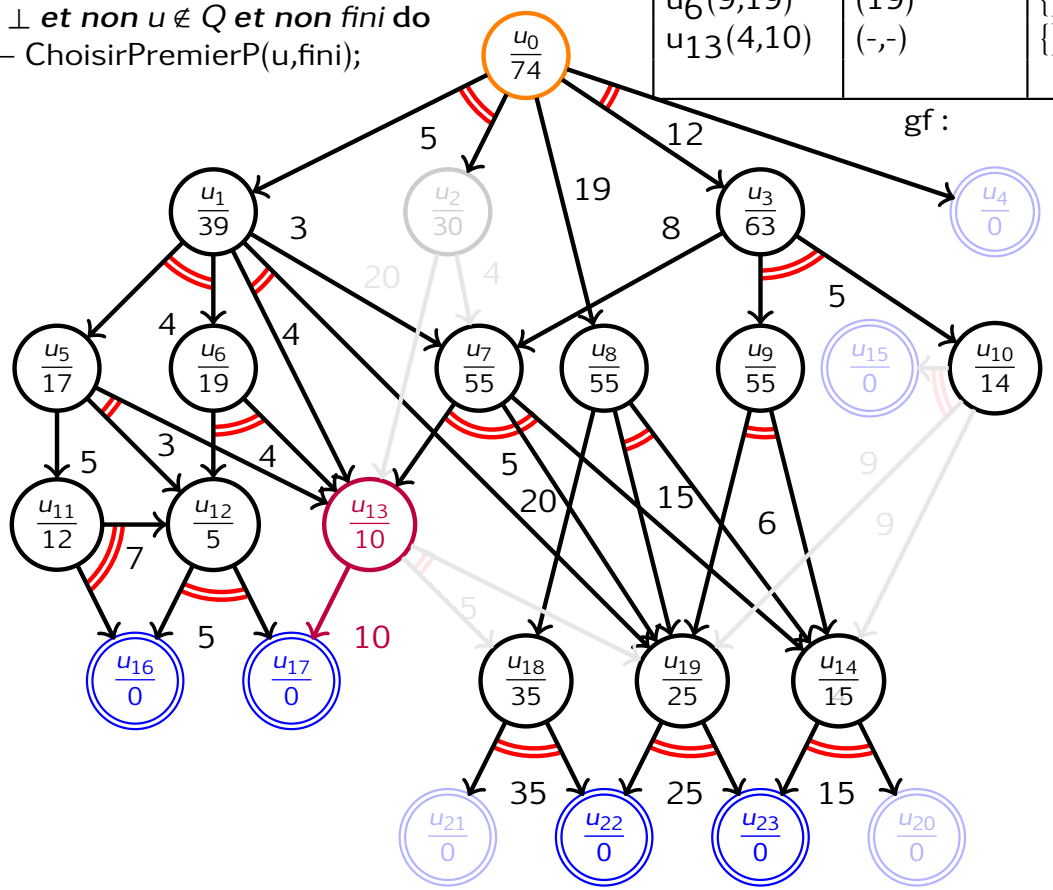


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(-,-)	{}	u_2(5,30)

gf : $\{u_{17}\}$

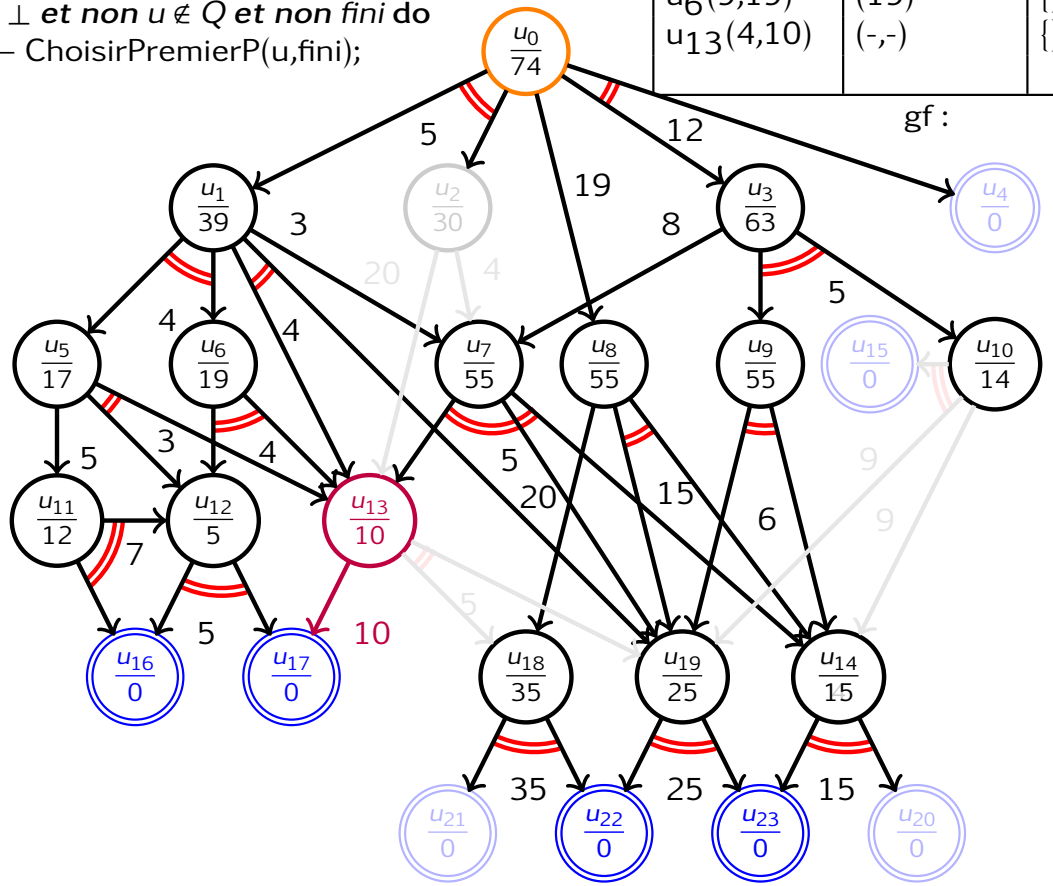


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
$u_0(0,74)$	(74,74,75)	{}	$u_1(5,39)$ $u_2(5,30)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,63)$	(63,74)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,39)$	(40,39,58)	{}	$u_{13}(4,10)u_{19}(4,25)u_2(5,30)$
$u_9(17,46)$	(46)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$
$u_5(9,17)$	(17,18)	{}	$u_{11}(14,15)u_6(9,19)$ $u_2(5,30)$
$u_{11}(14,12)$	(12)	{}	$u_{12}(7,5)u_6(9,19)u_2(5,30)$
$u_{12}(21,5)$	(5)	{}	$u_{17}(5,0)$ $u_6(9,19)u_2(5,30)$
$u_6(9,19)$	(19)	{}	$u_{13}(4,10)u_2(5,30)$
$u_{13}(4,10)$	(-,-)	{}	$u_2(5,30)$

gf : $\{u_{17}\}$

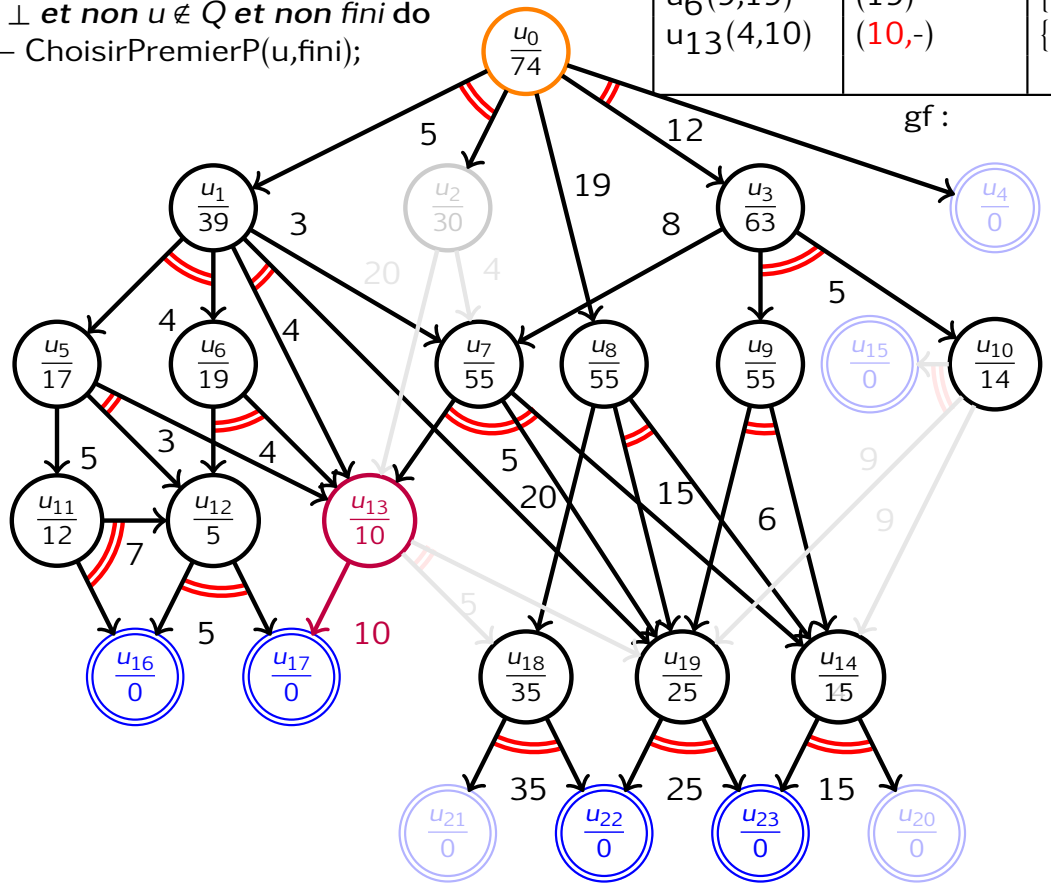


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(10,-)	{}	u_2(5,30)

gf : $\{u_{17}\}$ = 10+0 = 10

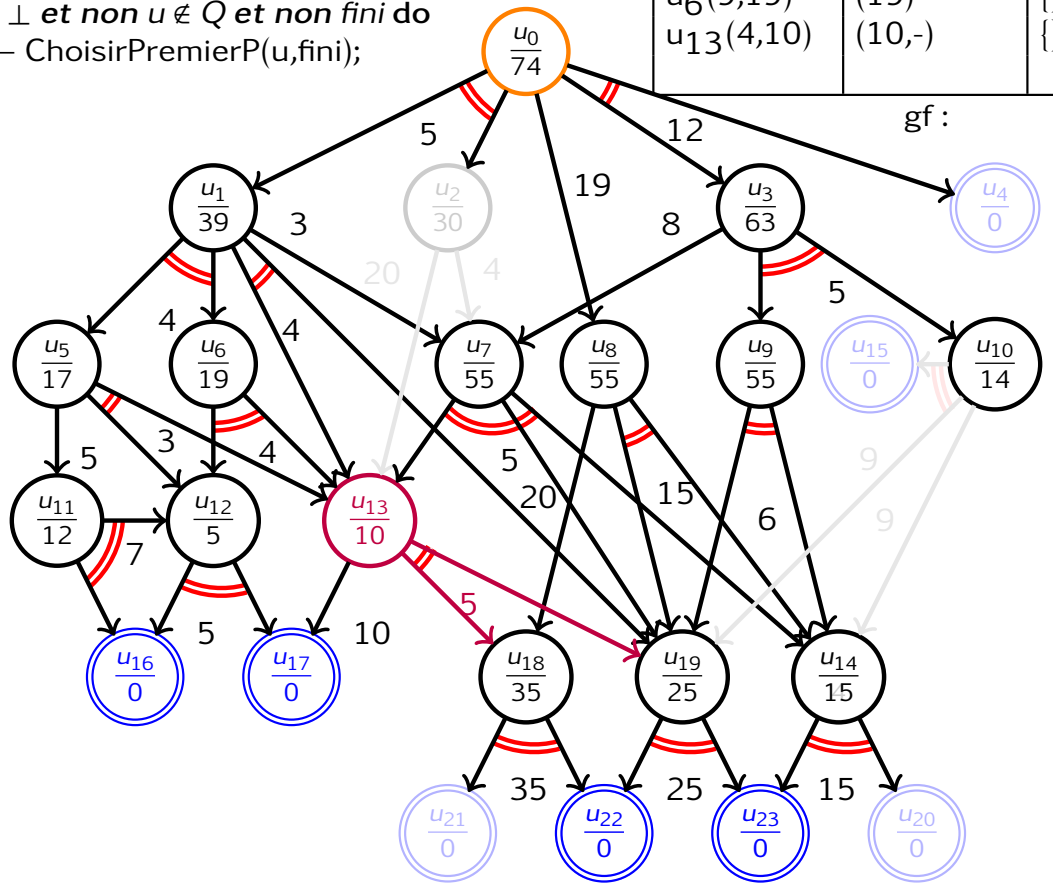


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(10,-)	{}	u_2(5,30)

gf : $\{u_{18}, u_{19}\}$

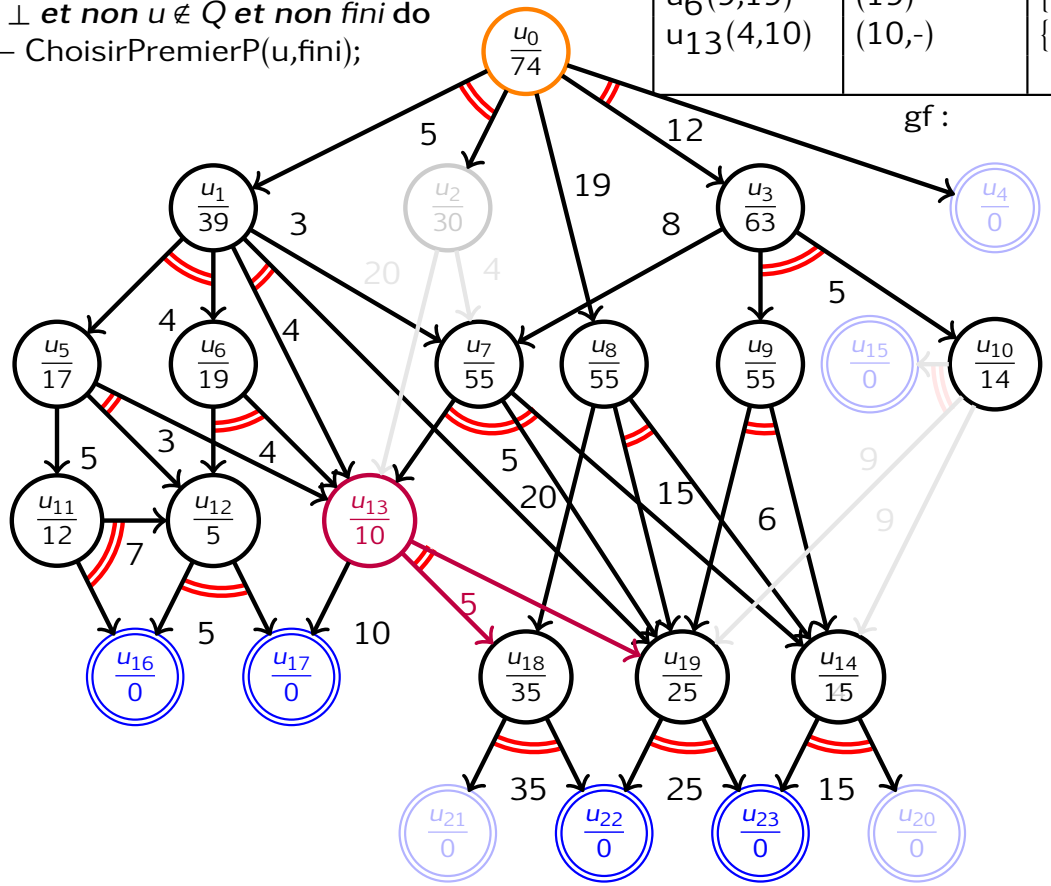


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(10,-)	{}	u_2(5,30)

gf : $\{u_{18}, u_{19}\}$

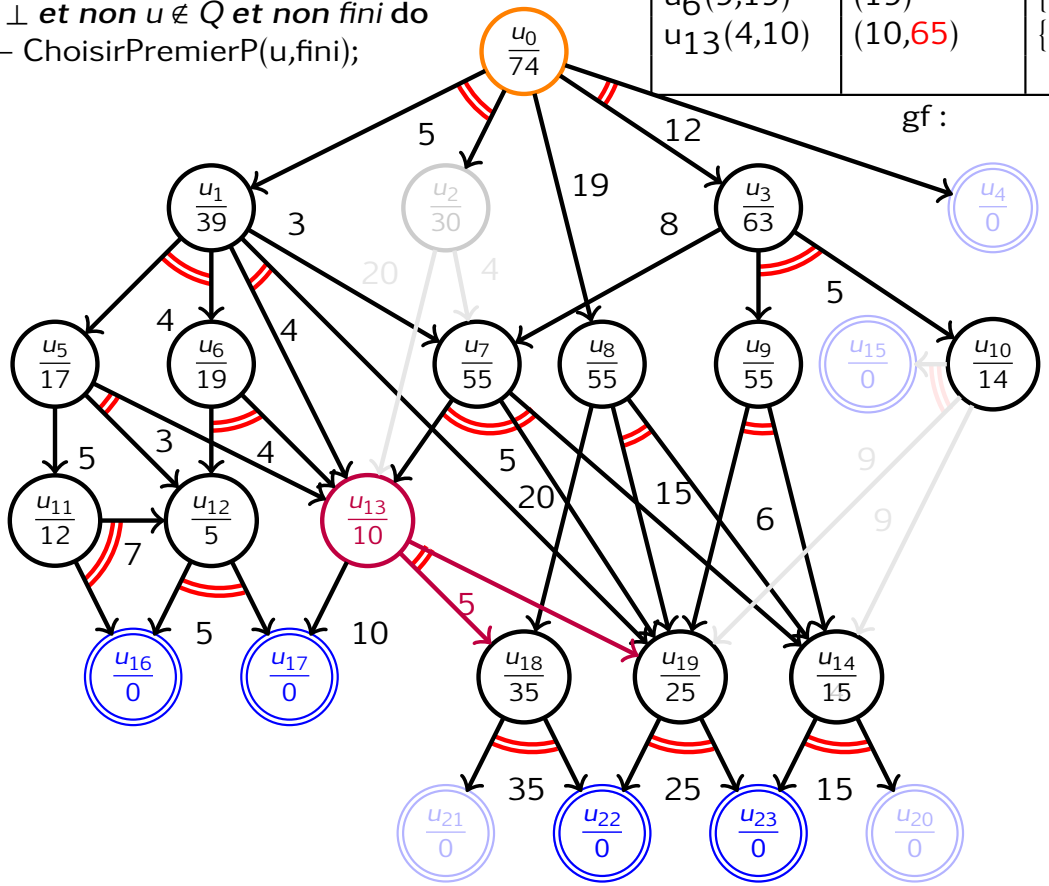


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(10,65)	{}	u_2(5,30)

gf : $\{u_{18},u_{19}\}$ = 5+35+25 = 65

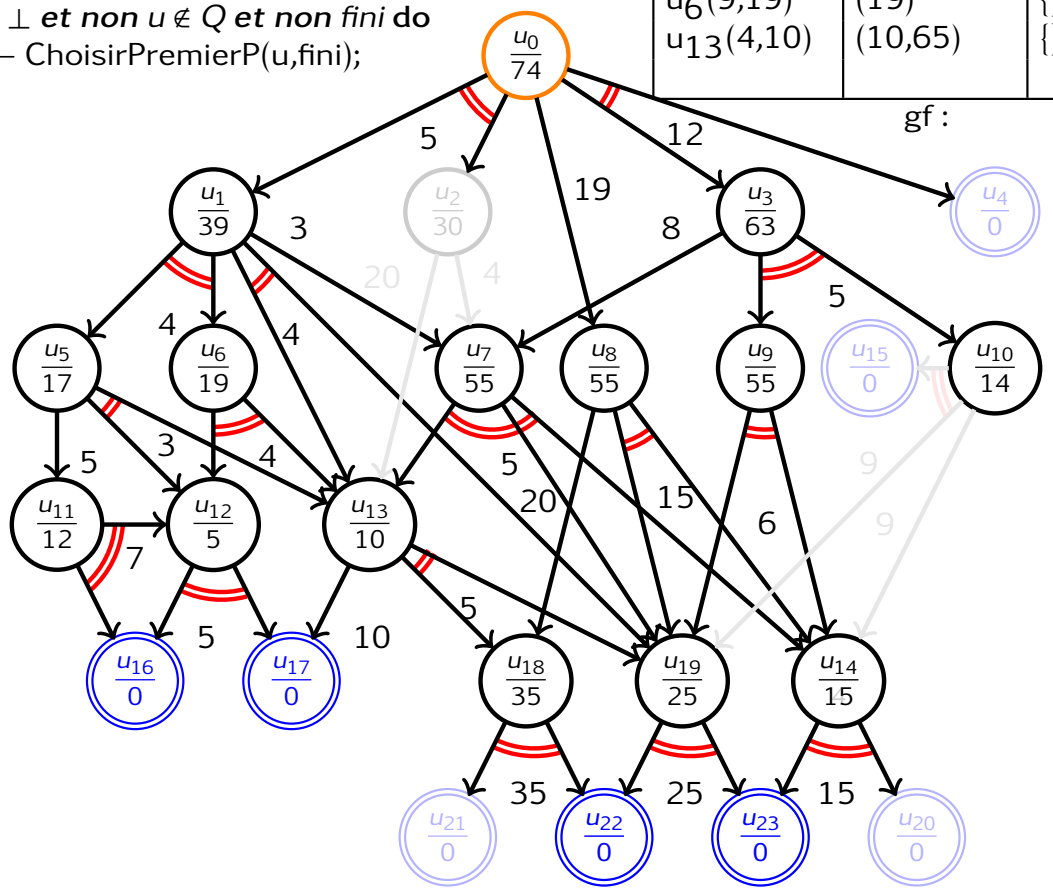


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(10,65)	{}	u_2(5,30)

gf :

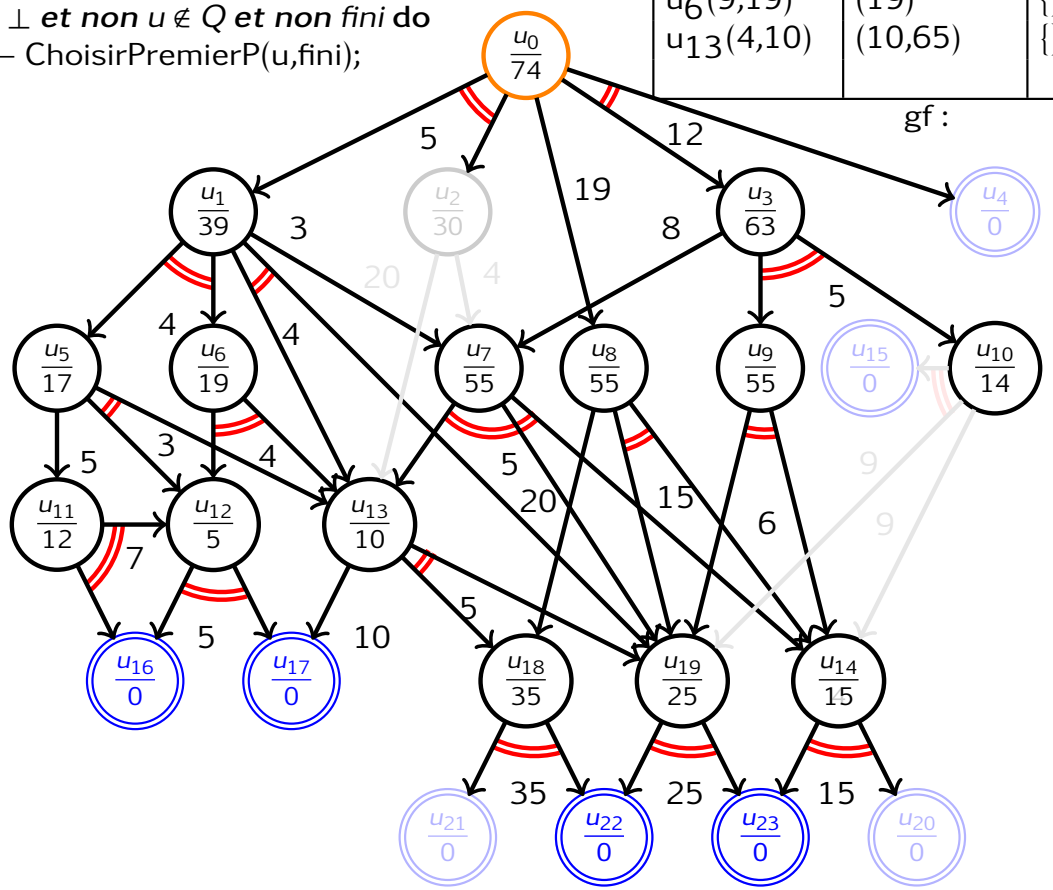


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(10,65)	{}	u_2(5,30)

gf :

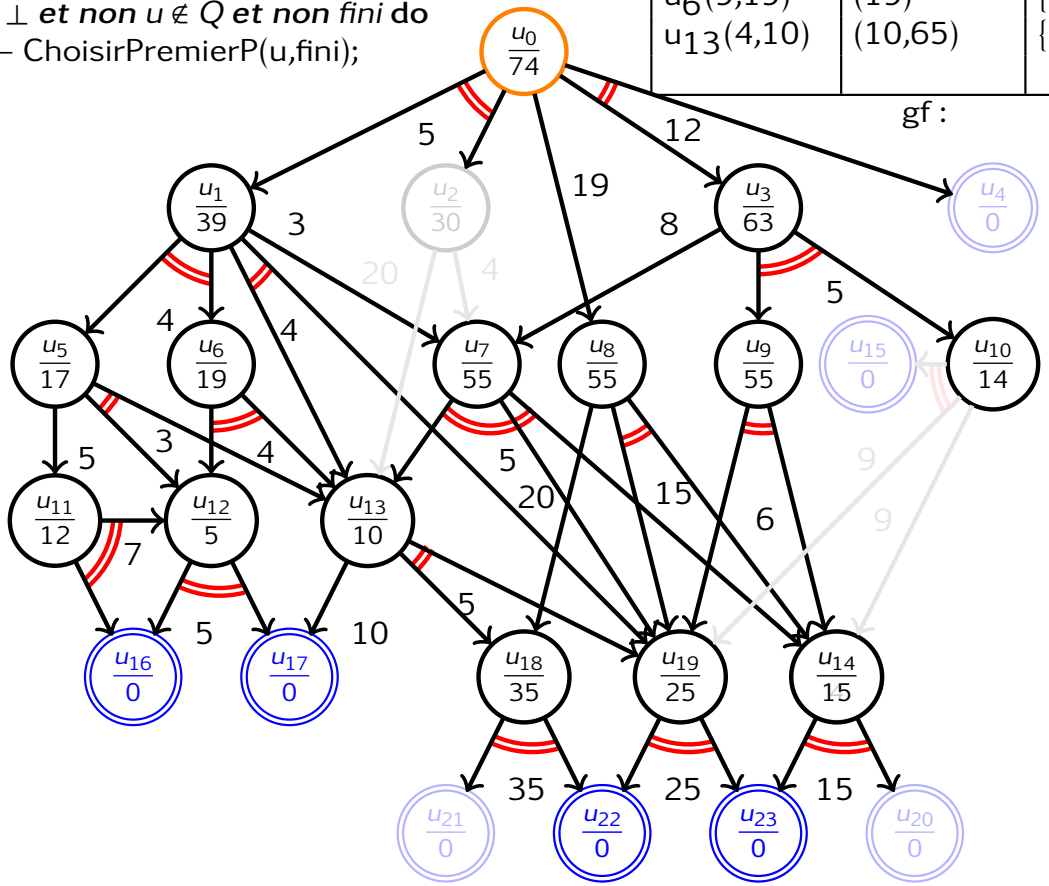


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(10,65)	{}	u_2(5,30)

gf :

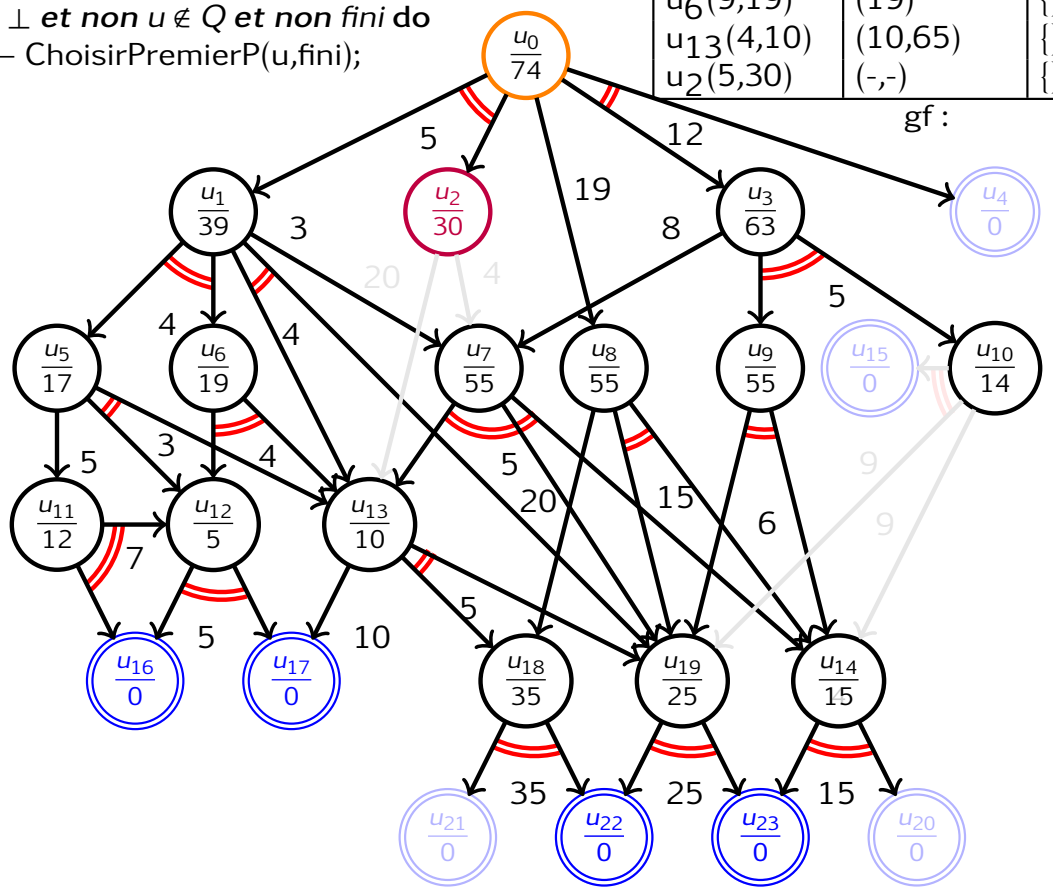


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(10,65)	{}	u_2(5,30)
u_2(5,30)	(-,-)	{}	

gf :

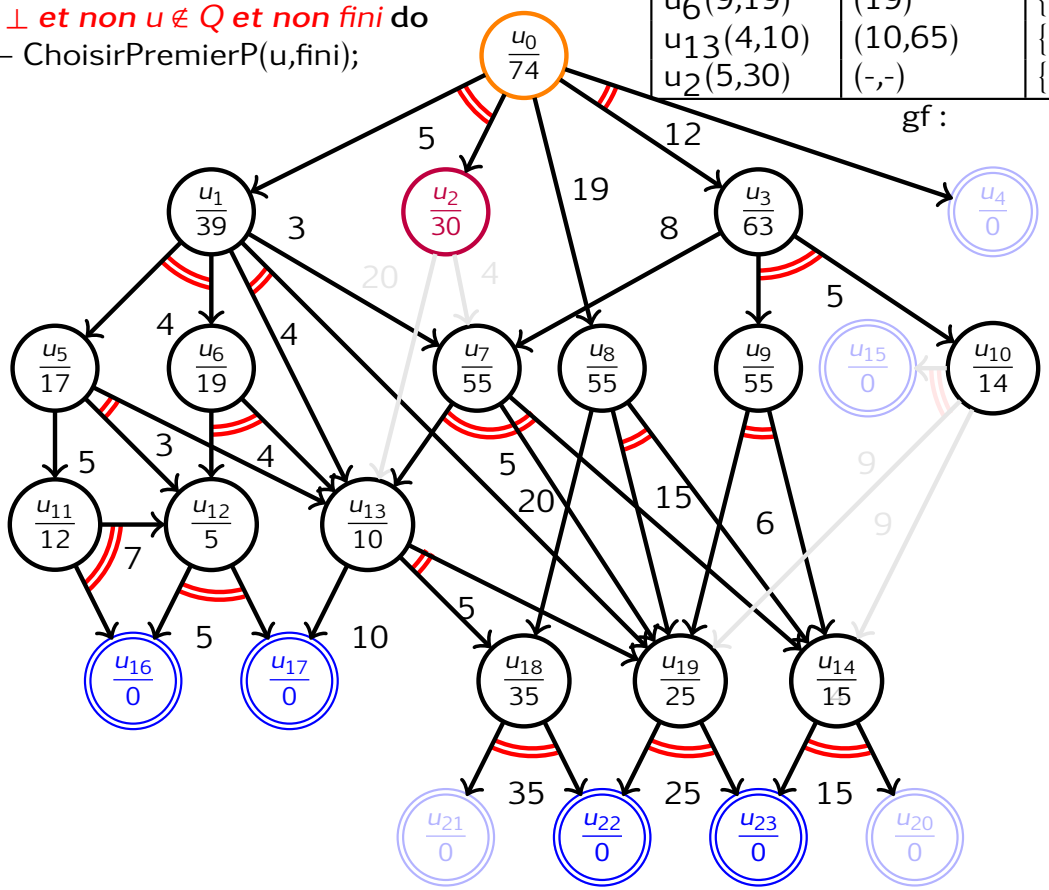


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(10,65)	{}	u_2(5,30)
u_2(5,30)	(-,-)	{}	

gf :

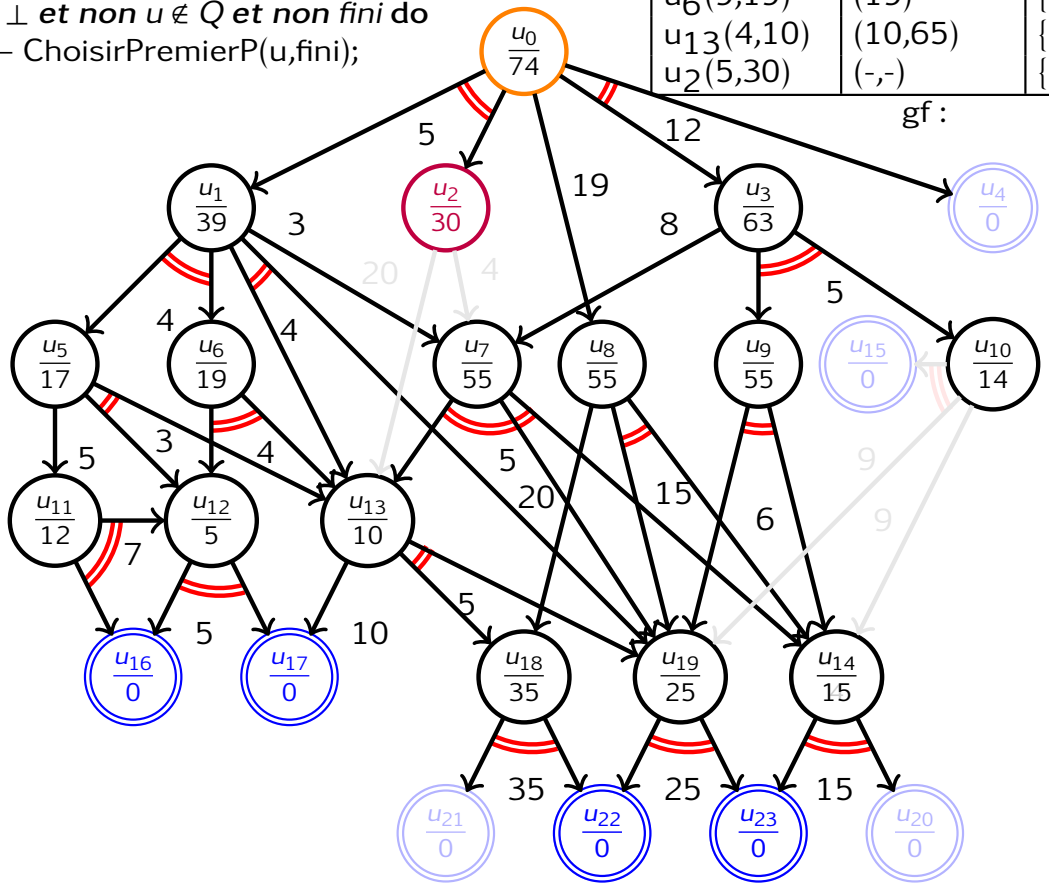


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(10,65)	{}	u_2(5,30)
u_2(5,30)	(-,-)	{}	

gf :

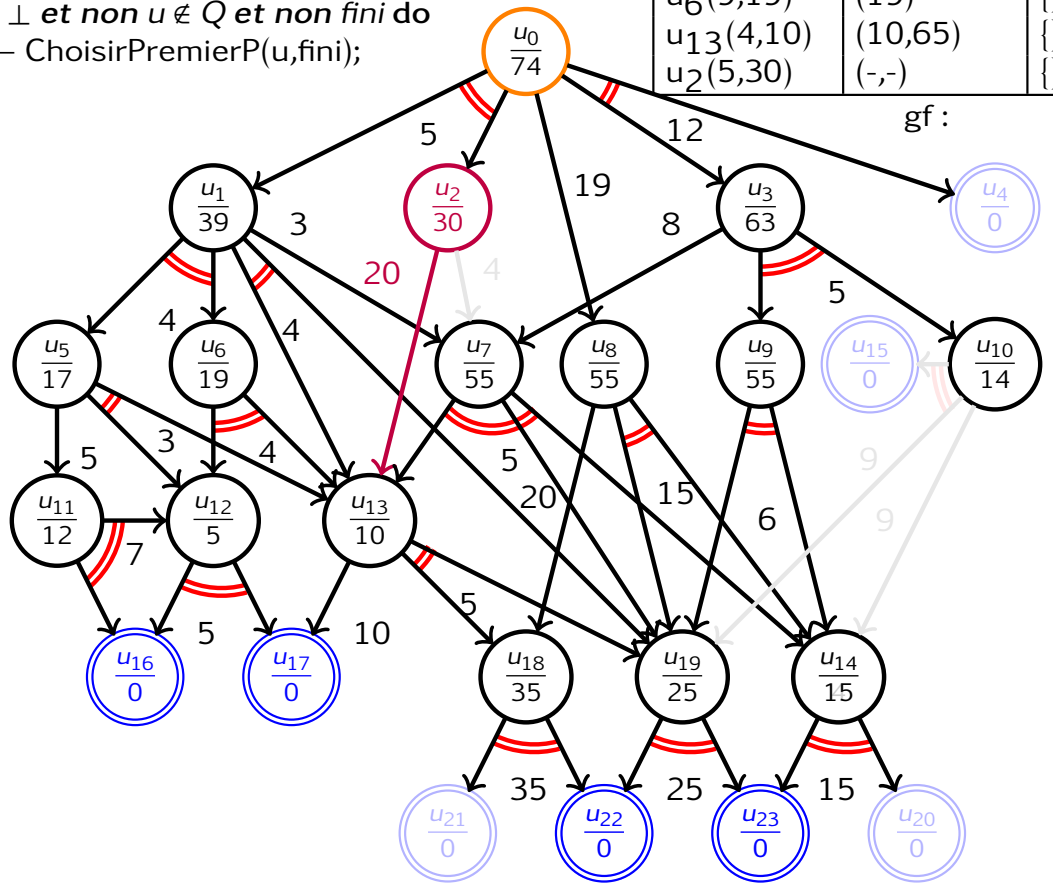


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(10,65)	{}	u_2(5,30)
u_2(5,30)	(-,-)	{}	

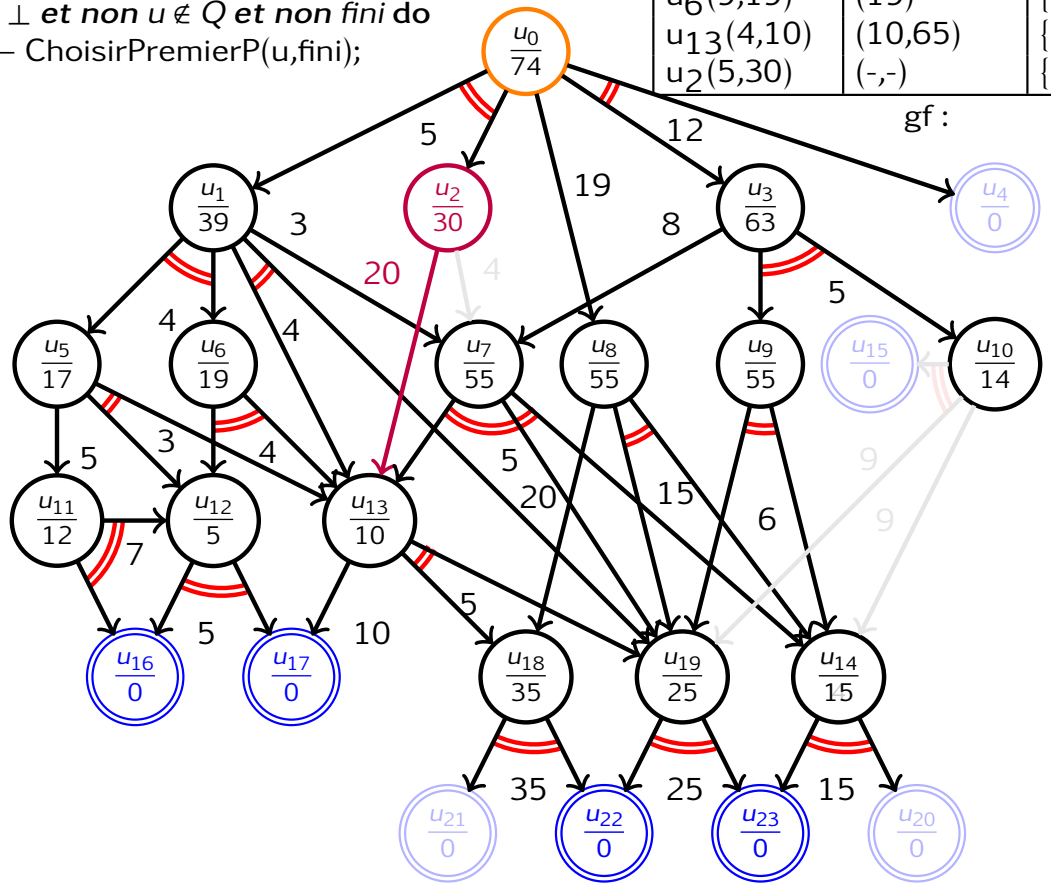
gf : $\{u_{13}\}$



Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while $u \neq \perp$ ***et non*** *fini* **do**
  gf ← groupeFils(u) ;
  while $gf \neq \emptyset$ **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while $A \neq \emptyset$ **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while $u \neq \perp$ ***et non*** $u \notin Q$ ***et non*** *fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(10,65)	{}	u_2(5,30)
u_2(5,30)	(-,-)	{}	
	gf :		$\{u_{13}\}$

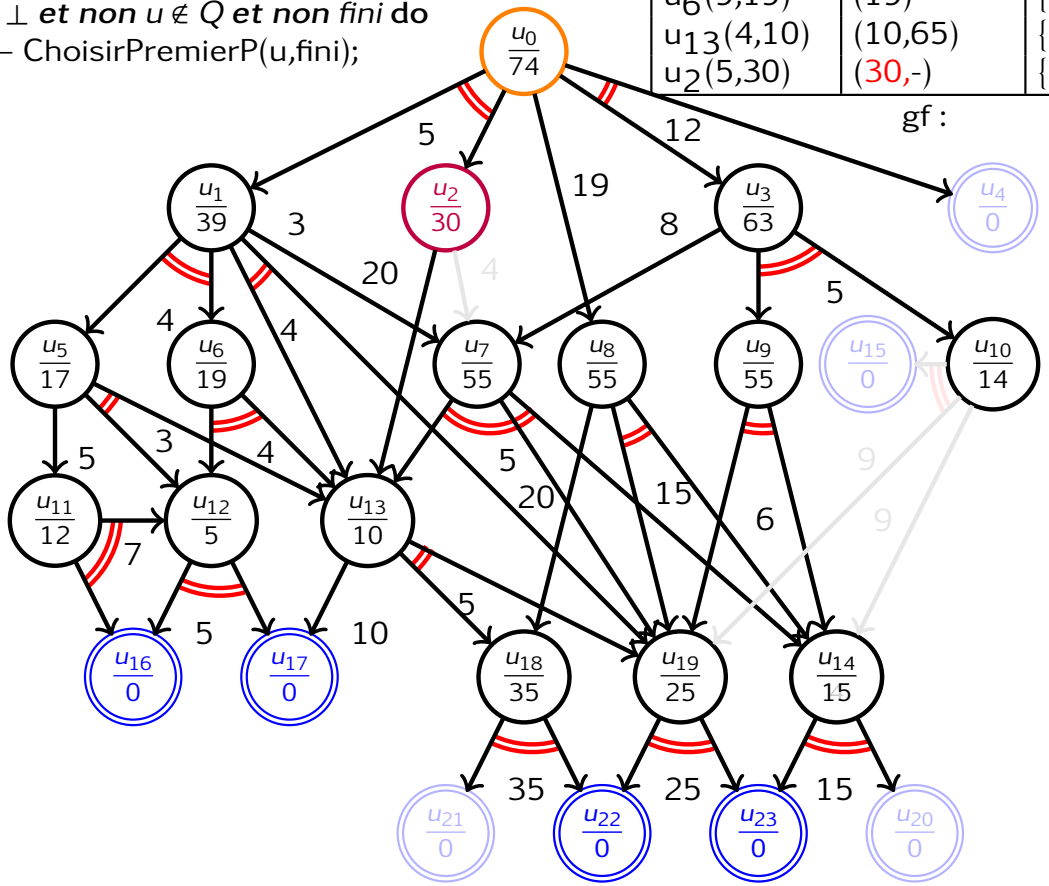


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(10,65)	{}	u_2(5,30)
u_2(5,30)	(30,-)	{}	

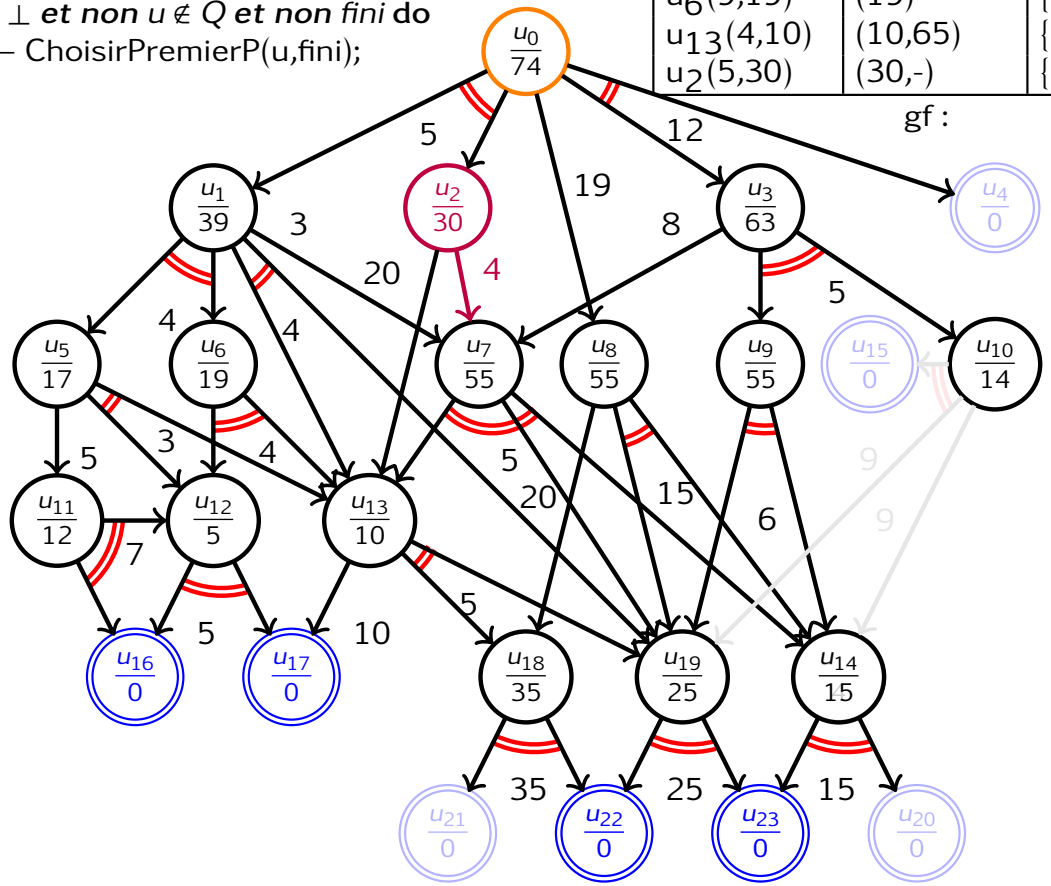
gf : $\{u_{13}\}$ = 20+10 = 30



Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

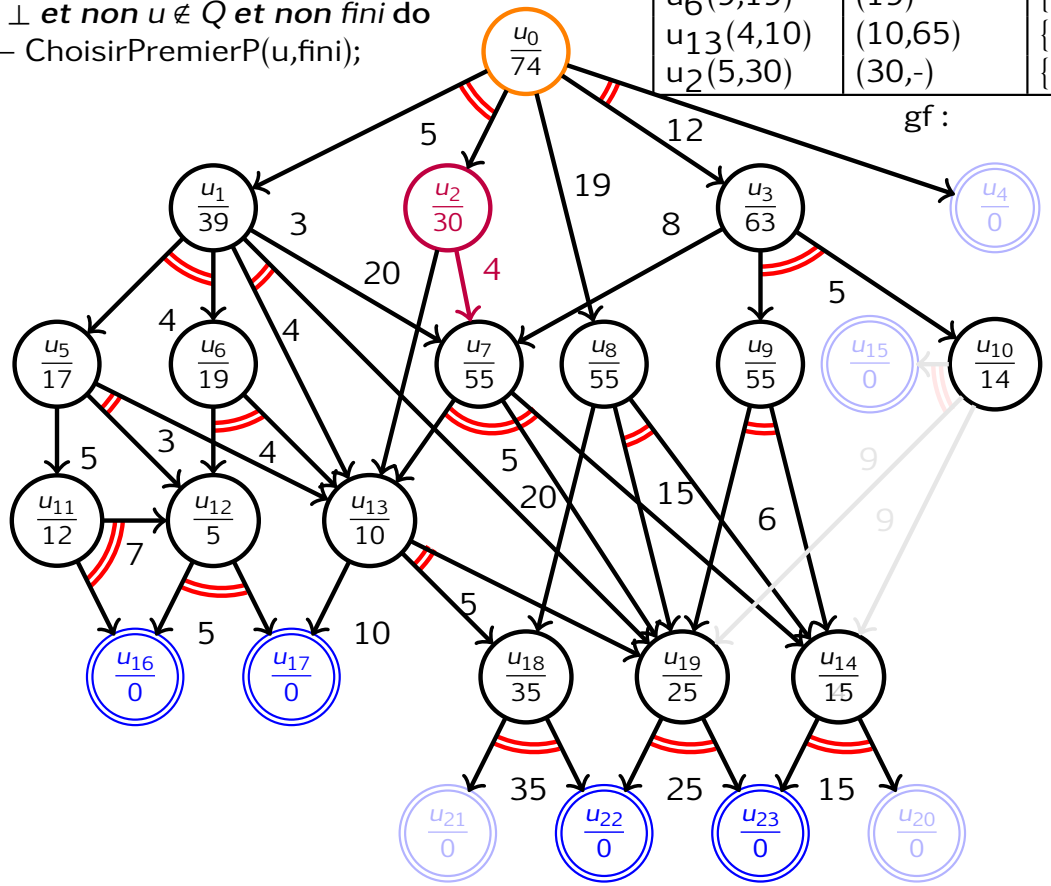
u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(10,65)	{}	u_2(5,30)
u_2(5,30)	(30,-)	{}	
	gf :		$\{u_7\}$



Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(10,65)	{}	u_2(5,30)
u_2(5,30)	(30,-)	{}	
	gf :		$\{u_7\}$

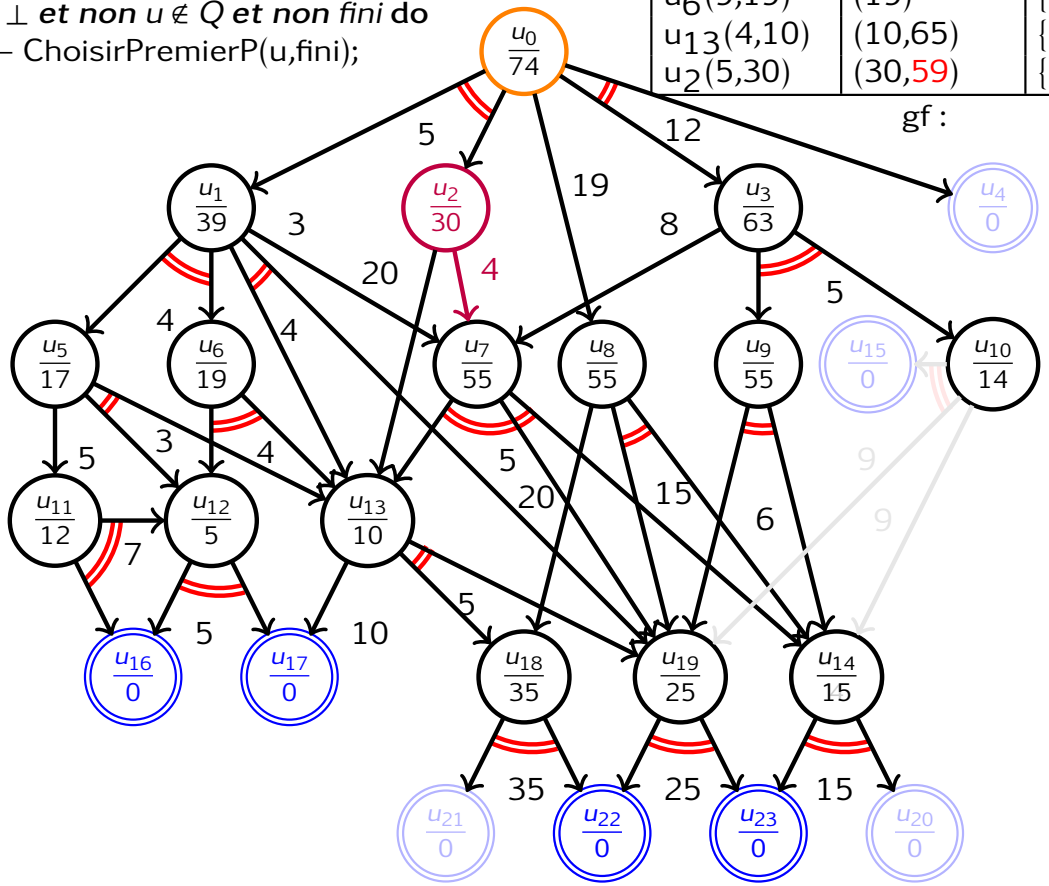


Algorithme OA*(Liste,Sol)

u ← Premier(Liste) ;
while *u ≠ ⊥ et non fini* **do**
  gf ← groupeFils(u) ;
  while *gf ≠ ∅* **do**
    calculerCout(gf,u) ; garderPlusPetit(gf,u);
    gf ← gfSuivant(u);
  while *A ≠ ∅* **do**
    u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
  u ← ChoisirPremierP(u,fini);
  while *u ≠ ⊥ et non u ∉ Q et non fini* **do**
    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(10,65)	{}	u_2(5,30)
u_2(5,30)	(30,59)	{}	

gf : $\{u_7\}$ = 4+55 = 59

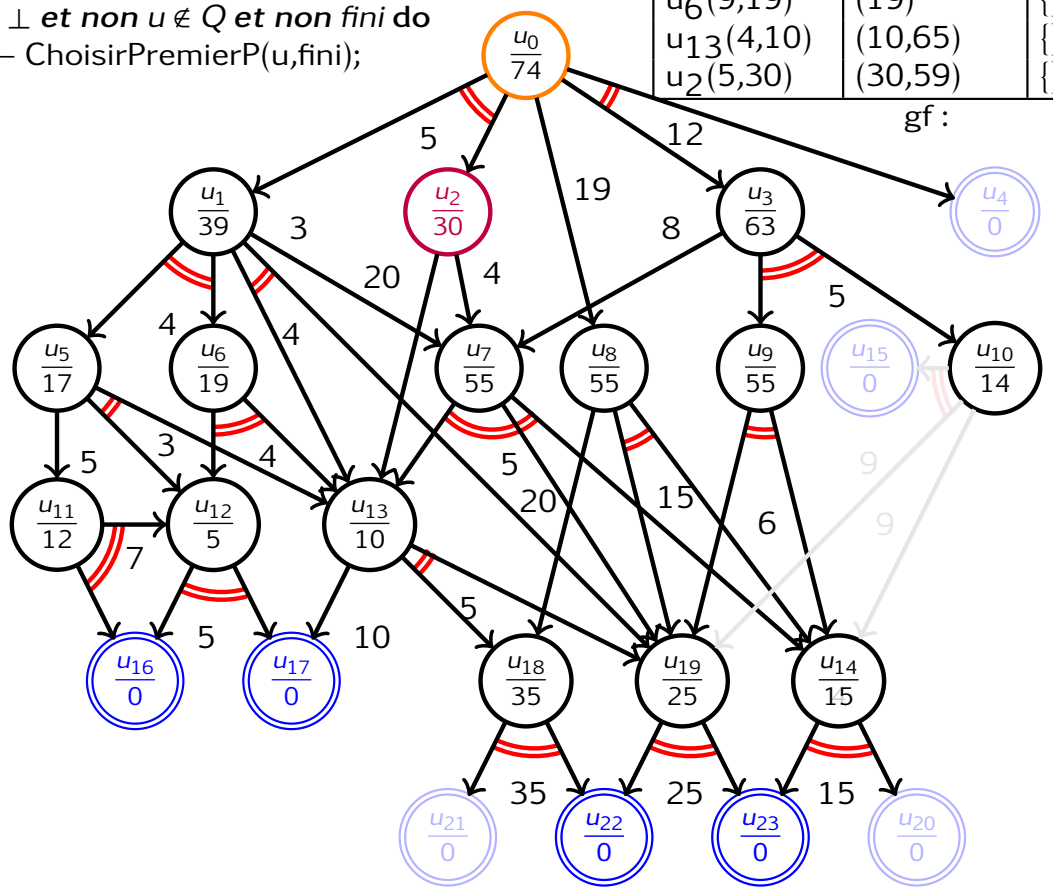


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(10,65)	{}	u_2(5,30)
u_2(5,30)	(30,59)	{}	

gf :

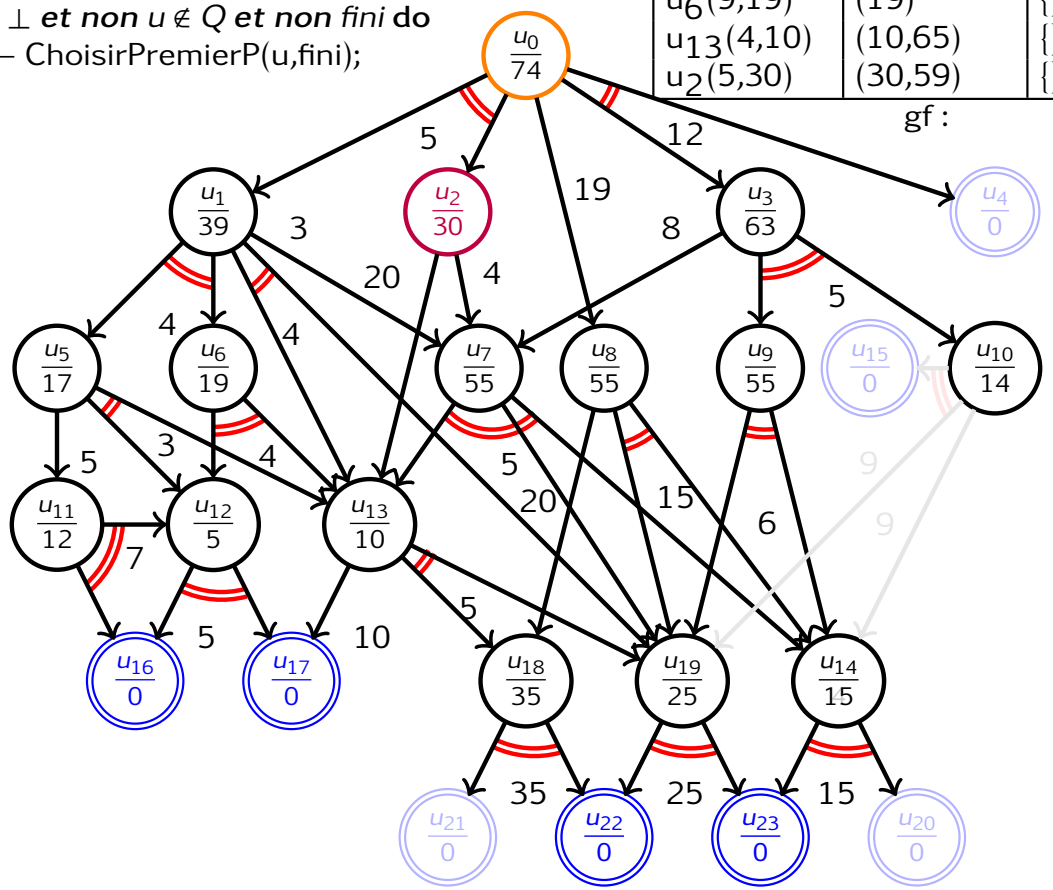


Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(10,65)	{}	u_2(5,30)
u_2(5,30)	(30,59)	{}	

gf :



Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
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    while gf ≠ ∅ do
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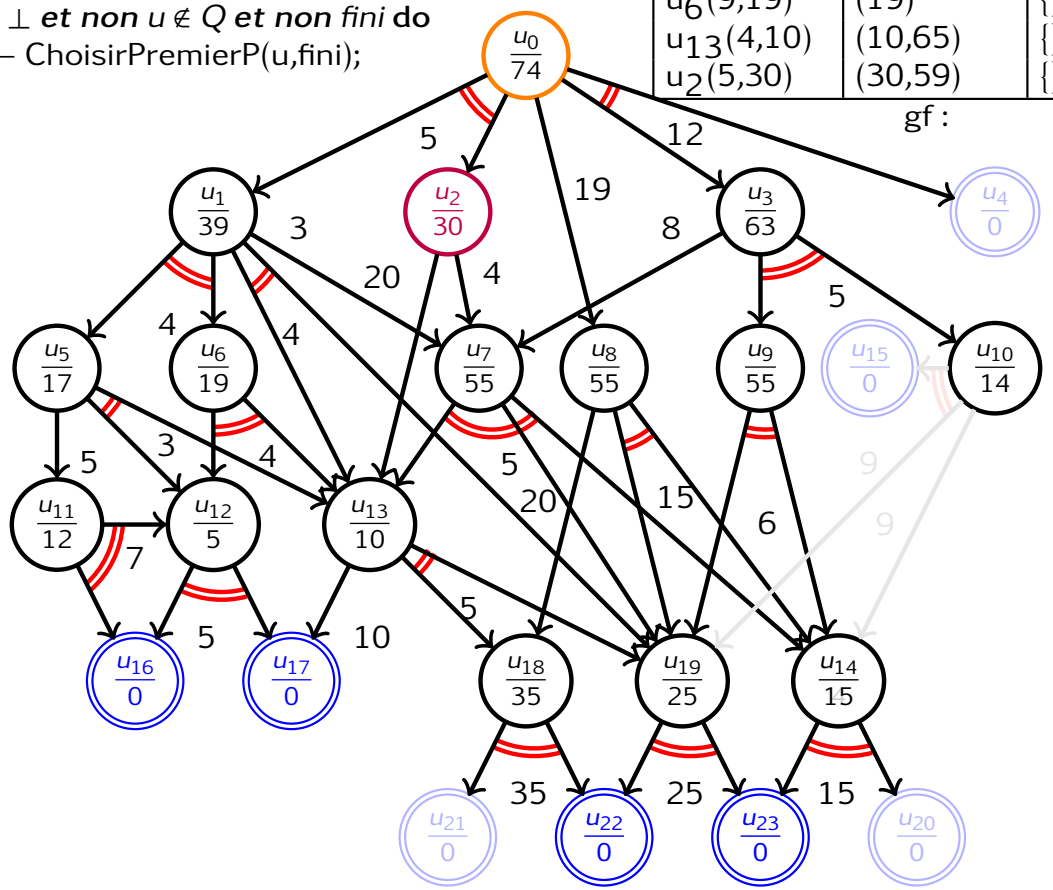
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(10,65)	{}	u_2(5,30)
u_2(5,30)	(30,59)	{}	

gf :



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  u ← ChoisirPremierP(u,fini);
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    u ← ChoisirPremierP(u,fini);

u(g(u),h(u))	fils(u)	A	P
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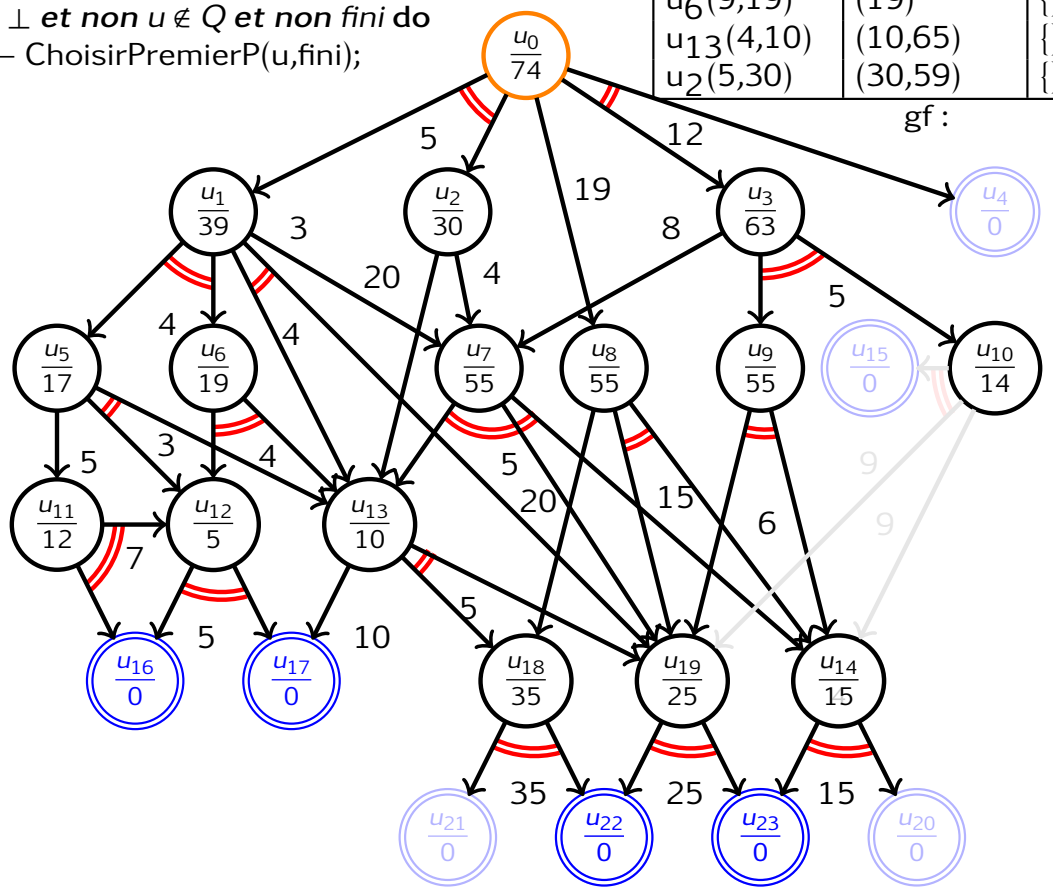
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(10,65)	{}	u_2(5,30)
u_2(5,30)	(30,59)	{}	

gf :



Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
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        gf ← gfSuivant(u);
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        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
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    while u ≠ ⊥ et non u ∉ Q et non fini do
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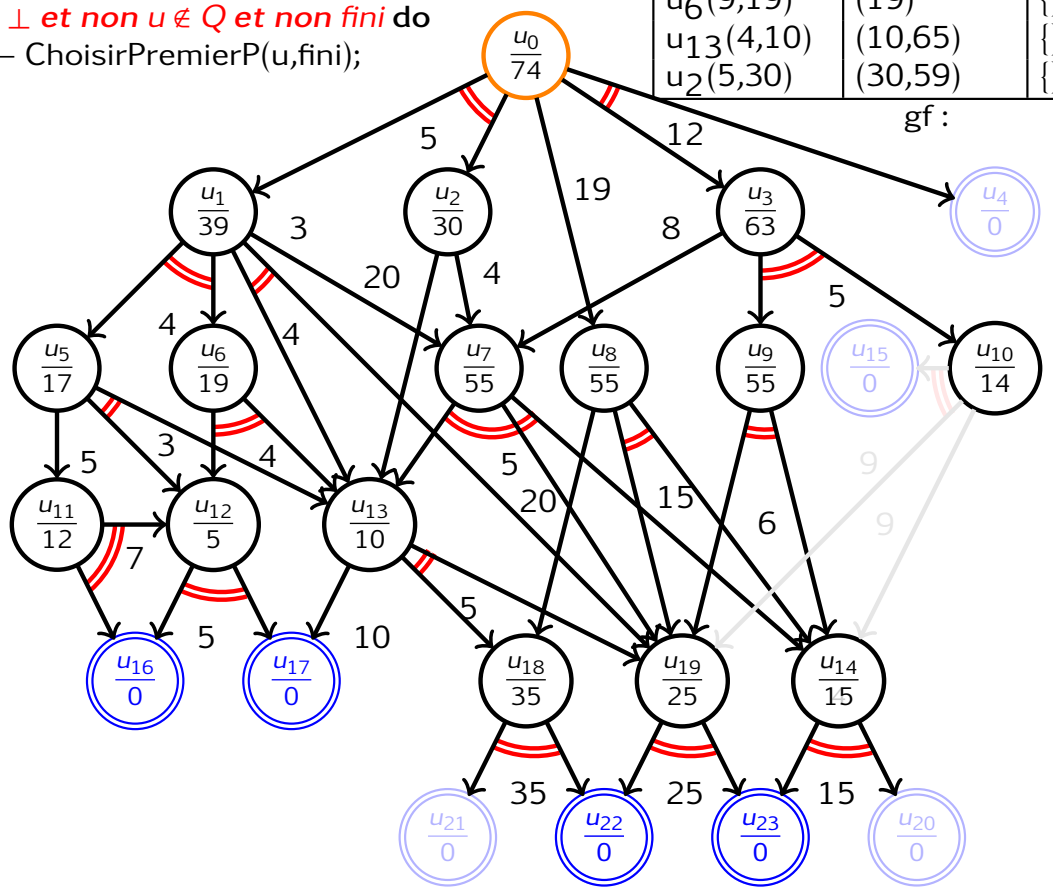
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(10,65)	{}	u_2(5,30)
u_2(5,30)	(30,59)	{}	

gf :



Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
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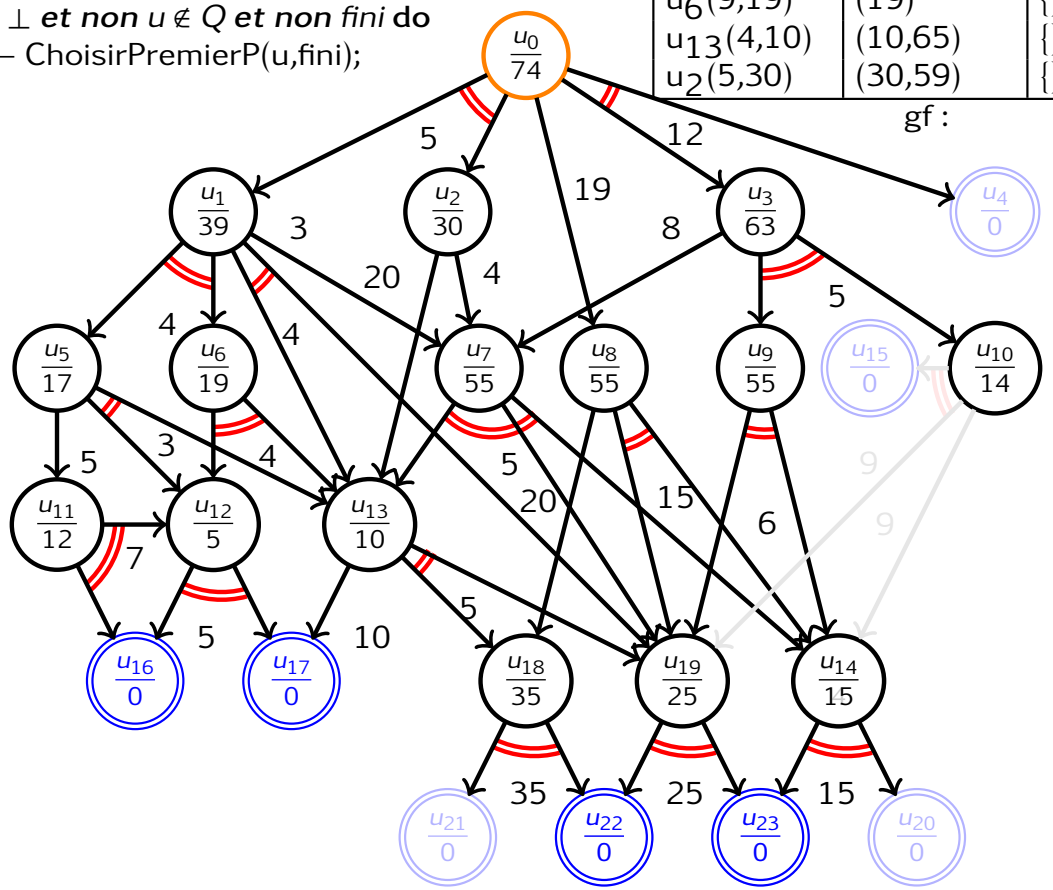
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

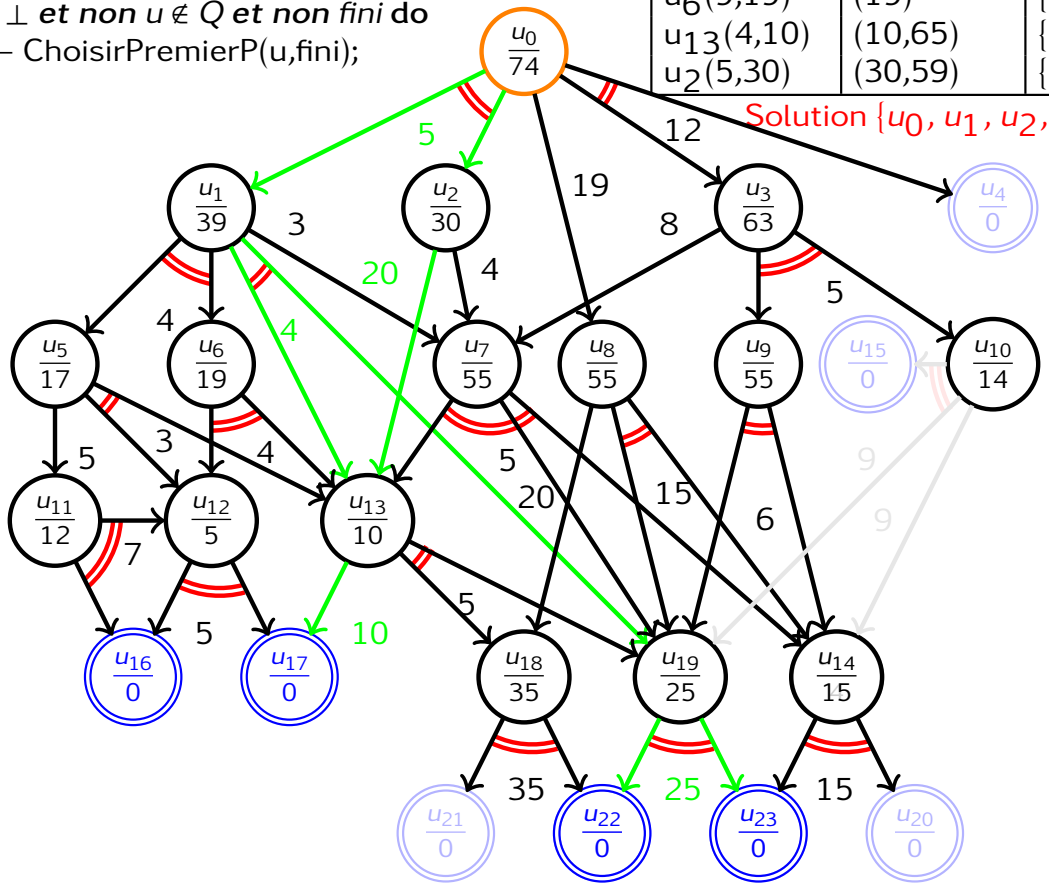
u(g(u),h(u))	fils(u)	A	P
$u_0(0,74)$	(74,74,75)	{}	$u_1(5,39)$ $u_2(5,30)$
$u_8(19,55)$	(55,55)	{}	$u_{19}(44,11)$ $u_{14}(38,4)$
$u_{18}(39,35)$	(35)	{}	$u_{21}(74,0)$ $u_{22}(74,0)$
$u_{19}(34,25)$	(25)	{}	$u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)$
$u_3(12,63)$	(63,74)	{}	$u_9(17,35)$ $u_{10}(17,14)u_4(12,0)$
$u_7(20,55)$	(55)	{}	$u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)$
$u_1(5,39)$	(40,39,58)	{}	$u_{13}(4,10)u_{19}(4,25)u_2(5,30)$
$u_9(17,46)$	(46)	{}	$u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)$
$u_{14}(34,15)$	(15)	{}	$u_{23}(49,0)$ $u_{20}(49,0)$
$u_5(9,17)$	(17,18)	{}	$u_{11}(14,15)u_6(9,19)$ $u_2(5,30)$
$u_{11}(14,12)$	(12)	{}	$u_{12}(7,5)u_6(9,19)u_2(5,30)$
$u_{12}(21,5)$	(5)	{}	$u_{17}(5,0)$ $u_6(9,19)u_2(5,30)$
$u_6(9,19)$	(19)	{}	$u_{13}(4,10)u_2(5,30)$
$u_{13}(4,10)$	(10,65)	{}	$u_2(5,30)$
$u_2(5,30)$	(30,59)	{}	

gf :



Algorithme OA*(Liste,Sol)

```
u ← Premier(Liste) ;
while u ≠ ⊥ et non fini do
    gf ← groupeFils(u) ;
    while gf ≠ ∅ do
        calculerCout(gf,u) ; garderPlusPetit(gf,u);
        gf ← gfSuivant(u);
    while A ≠ ∅ do
        u ← Choisir(A) ; A ← A \ {u} ; miseAJour(u);
    u ← ChoisirPremierP(u,fini);
    while u ≠ ⊥ et non u ∉ Q et non fini do
        u ← ChoisirPremierP(u,fini);
```

u(g(u),h(u))	fils(u)	A	P
u_0(0,74)	(74,74,75)	{}	u_1(5,39) u_2(5,30)
u_8(19,55)	(55,55)	{}	u_{19}(44,11) u_{14}(38,4)
u_{18}(39,35)	(35)	{}	u_{21}(74,0) u_{22}(74,0)
u_{19}(34,25)	(25)	{}	u_{22}(59,0)u_{23}(59,0)u_{14}(34,38)
u_3(12,63)	(63,74)	{}	u_9(17,35) u_{10}(17,14)u_4(12,0)
u_7(20,55)	(55)	{}	u_{13}(25,10)u_{19}(25,25)u_{14}(25,4)
u_1(5,39)	(40,39,58)	{}	u_{13}(4,10)u_{19}(4,25)u_2(5,30)
u_9(17,46)	(46)	{}	u_{19}(23,25)u_{14}(23,4)u_{10}(17,31)u_{14}(34,38)
u_{14}(34,15)	(15)	{}	u_{23}(49,0) u_{20}(49,0)
u_5(9,17)	(17,18)	{}	u_{11}(14,15)u_6(9,19) u_2(5,30)
u_{11}(14,12)	(12)	{}	u_{12}(7,5)u_6(9,19)u_2(5,30)
u_{12}(21,5)	(5)	{}	u_{17}(5,0) u_6(9,19)u_2(5,30)
u_6(9,19)	(19)	{}	u_{13}(4,10)u_2(5,30)
u_{13}(4,10)	(10,65)	{}	u_2(5,30)
u_2(5,30)	(30,59)	{}	

Solution $\{u_0, u_1, u_2, u_{13}, u_{17}, u_{19}, u_{22}, u_{23}\}$ coût de 74